C-PHAST: Compositional Port-Hamiltonian World Models for Structured Dynamics Transfer

Shubham Bhardwaj [1 2] **Chandrajit Bajaj** [1]

Abstract

Before connecting a new generation unit to an electrical network, an operator needs to know how currents will redistribute and whether any line will overload. Testing every proposed connection on the real network is impractical and could cause the failure we want to avoid. The proposed network has not yet produced measurements, and measurements from the existing network may leave some currents unobserved. Prediction therefore requires reusable component laws and an estimate of the hidden state. Likewise, a model trained on one four-legged robot typically mispredicts another with different mass, leg lengths and motors, so each new robot needs fresh data and retraining.

We introduce C-PHAST, a compositional port-Hamiltonian world model for predicting the physical consequences of proposed configurations. Extending our single-system model PHAST, it learns how each component type stores and dissipates energy, connects components through power-preserving ports, and estimates unobserved states from measurement histories. Given proposed wiring and initial states for added components, it predicts the resulting dynamics without retraining. Connections can also create behavior no component shows alone. For known linear systems, we show when one component's damping drains energy from another and when an existing sensor can better estimate a hidden state elsewhere. We also derive error margins under which predictions certify which of two proposed configurations scores better.

In simulated DC microgrids, given the same data and starting states as prior compositional port-Hamiltonian neural networks, C-PHAST predicts unseen five-unit networks with **over 40-fold** lower voltage error, in a comparison of complete training procedures that differ in model family, initialization, objective and integrator. From noisy voltage histories, it reduces hidden-current error by **67.6%** relative to CT-SUBNET, a black-box state-space model. In simulated zero-shot cross-embodiment transfer, a leg model learned only on ANYmal-D is reused unchanged on Unitree Go2 and never told how Go2 differs; a short history of Go2 joint positions sets only its starting state, and its error grows **1.6-fold**, against 21-fold for a recurrent network that fits ANYmal-D better. Four-joint arm models also transfer zero-shot to two to eight joints. On real Spot-with-arm logs, a predictor built on its physical readouts retrospectively estimates joint work and battery draw (Pearson $r \geq 0.94$). Each capability is tested separately; ranking configurations with learned laws from partial measurements remains open. Learning laws for components, not whole systems, is a step toward world models that predict a configuration before it is built.

1. Introduction

Before adding a distributed generation unit (DGU) to a direct-current (DC) microgrid, an operator must decide where it should attach. The existing wiring can remain unchanged, yet the new connection can redistribute currents throughout the enlarged network and overload a line (Figure 1). The unit and line may be familiar; their joint response in this connection has not been observed. Trying the connection to collect data risks the very failure the operator wants to foresee. Can we predict how the enlarged network will behave using what we have learned about its components?

[1]Department of Computer Science, Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin, Austin, TX, USA [2]Mihawk.ai. Correspondence to: Shubham Bhardwaj <shubham.bhardwaj@utexas.edu>.

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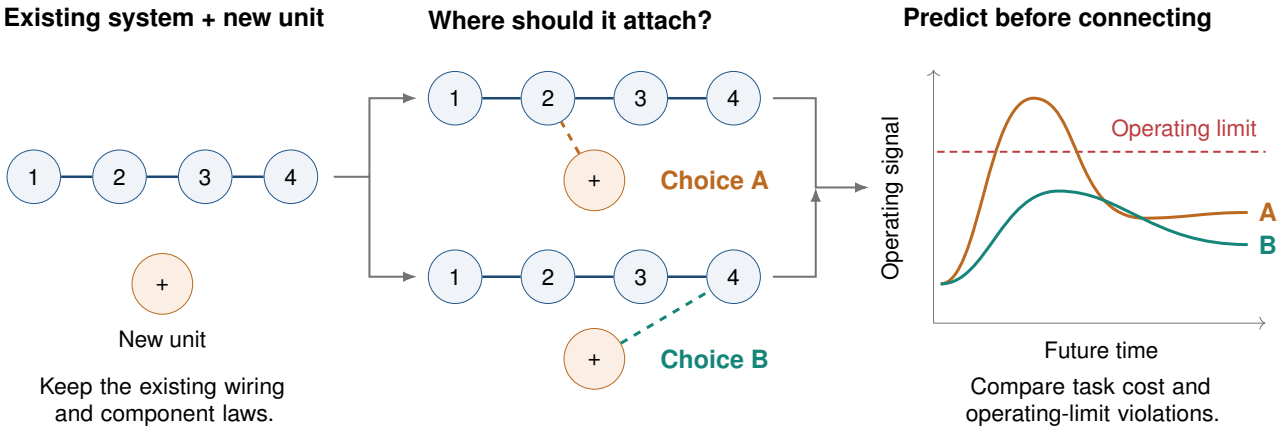


Figure 1. **Same component laws, different connections, different possible futures.** A new unit can attach at different points of a supplied network. The query asks how each attachment and its proposed inputs affect the subsequent motion or electrical response, including possible failures, before the connection is made. The curves illustrate an operating-limit violation; they are schematic, not experimental results. The electrical realization in Figure 4 instantiates this general question.

Reusing component laws is only part of the prediction. The network already carries currents and stores energy, but its sensors may reveal only voltages. A forecast must infer its starting state from these measurements and return physical quantities that can be checked against operating limits. Underestimating a current can hide an overload; an overly pessimistic forecast can reject a useful connection. The operator needs a world model, a learned simulator of these candidate futures.

Changing a connection is one way to change a physical system. A robot may retain its joint layout while its mass and actuation change, or an arm may contain more joints. How much of a learned model remains useful when the body itself changes?

Existing methods already learn reusable dynamics or infer hidden states. Compositional port-Hamiltonian (pH) neural networks learn subsystem models and combine them through known or learned interconnections, integrating from supplied subsystem states (Neary & Topcu, 2023). N-PHDAE embeds learned electrical laws in differential-algebraic equations and composes subsystems through known circuit constraints (Neary et al., 2024). PHAST forecasts a single system with structured energy and damping and estimates its starting state from measurement histories (Bhardwaj & Bajaj, 2026), and CT-SUBNET estimates the state of a fixed system from its input and output history with a learned encoder (Beintema et al., 2023). Local models and state estimates also serve voltage control (Tucci et al., 2018; Sadabadi et al., 2018), current sharing (Cucuzzella et al., 2019), and protection (Liu et al., 2018). Comparing proposed configurations needs these pieces in one predictor: laws reused across instances of a type, a starting state estimated through those same laws, and a test of whether its remaining errors permit a reliable choice.

We introduce **C-PHAST** (Compositional Port-Hamiltonian Architecture for Structured Temporal Dynamics), which extends PHAST to systems assembled from reusable component models. Components of the same type, such as DGUs or lines, share one law for energy storage and dissipation; we supply known laws and learn the uncertain ones from training trajectories (Table 14). Components exchange power through ports, pairs of variables such as voltage and current whose product is power. In a power-preserving connection, power leaving one component enters others, and for a fixed connection the assembled system has an equivalent monolithic pH representation (Ehrhardt et al., 2026). What C-PHAST adds is the query of a proposed configuration. It estimates the existing system's present state from partial measurement histories through the learned laws, keeps that estimate for each candidate's existing components, and takes supplied starting states for added components (Figure 4). Each candidate is simulated with the laws fixed, and voltages and currents are derived from the learned energy, so an operator or controller can reject candidates that exceed limits and compare the rest (Figure 2; Section 3.4).

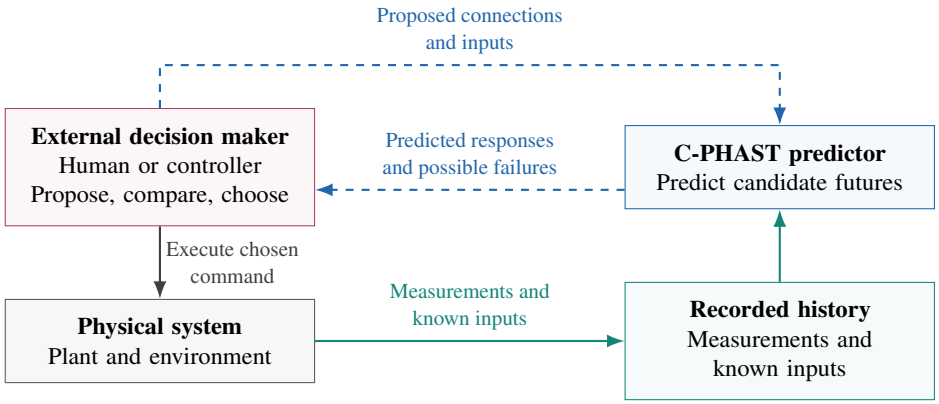


Figure 2. **Prediction supports decisions; execution remains external.** C-PHAST uses measurement histories and known inputs to predict the effects of proposed configurations and inputs. A human or external controller compares those predictions and determines what to execute. Dashed arrows carry hypothetical queries and predictions; solid arrows carry measurements or commands.

1.1. Contributions

1. **Predicting proposed systems from reusable laws.** We extend PHAST from a single-system model to a predictor of proposed configurations, combining history-based state inference with reusable component laws and physical readouts. The laws stay fixed during a query: existing components retain their estimated states, while additions receive supplied starting states. The predictions expose candidate responses and possible operating-limit violations before intervention (Section 3).

2. **When interconnection creates emergent properties.** For known linear–quadratic systems, we establish conditions under which a lossless connection increases expected energy dissipation through existing damping, or improves estimation of a hidden initial state using existing sensors (Sections 3.6.1–3.6.2). The connection can route disturbances to damping elsewhere or let an existing sensor reveal a previously hidden state, while the component laws remain unchanged.

3. **Reliable comparisons under prediction uncertainty.** Under stated model, integration and estimation assumptions, we propagate prediction errors into margins for candidate rankings and operating-limit checks (Section 3.7). For affine state assignments, known affine dynamics and quadratic scores, we retain shared starting-state uncertainty when bounding score differences. This resolves comparisons that separate candidate bounds can leave undecided (Appendix F.4).

4. **Evidence of reuse across changed systems.** Electrical composition provides the main test: from supplied starting states, fixed learned nonlinear DGU laws predict unseen five-unit networks with **over 40-fold** lower voltage RMSE than compositional pHNNs in each complete paired fit (Table 4). A separate sensing study improves hidden-current prediction from noisy voltage histories and known inputs on a fixed network relative to CT-SUBNET (Table 7). Supporting robotics experiments transfer learned leg dynamics from ANYmal-D to Go2 without retraining (Table 9).

These capabilities are tested separately. Learned nonlinear electrical laws are composed from supplied starting states, voltage-only initialization is tested on a fixed network and on held-out five-node networks, and certified rankings use known laws. No study yet ranks proposed connections with learned laws from voltage-only initialization, or tests physical switching or repeated closed-loop operation. The comparisons evaluate complete procedures and do not isolate the effect of architecture. Reuse covers component types already in the model; adding a DGU reuses the DGU law, but its wiring and starting state are supplied. Section 3.7 gives the formal assumptions, and Appendix A.9 the modeling resolution in each domain.

2. Related Work

Learning and composing physical models. Hamiltonian and Lagrangian neural models encode energy or variational structure in learned dynamics (Greydanus et al., 2019; Cranmer et al., 2020; Lutter et al., 2019). Port-Hamiltonian (pH) models additionally represent dissipation and external power exchange (Desai et al., 2021; Zhong et al., 2020; Duong et al., 2024). Generalized Hamiltonian energies, including port-Hamiltonian ones, can be learned from noisy, sparse trajectories (McLennan et al., 2025), and pH structure also shapes event-triggered learned optimizers (Wang & Bajaj, 2026). PHAST (Bhardwaj & Bajaj, 2026) combines structured energy and damping with state initialization from measurement histories. C-PHAST reuses them within component types and connects their instances to model proposed systems.

Power-preserving composition is classical pH theory (Van Der Schaft & Jeltsema, 2014; Cervera et al., 2007). More directly, Neary & Topcu (2023) learn compositional pHNNs from observed subsystem state–input trajectories and combine them through known or learned interconnections. With known coupling, composition requires no composite-system training data; unknown coupling can be fitted while retaining the subsystem models. Their analysis bounds composite vector-field error through subsystem, energy-gradient, and coupling errors. Their predictor integrates from supplied subsystem states. We compare this construction with C-PHAST in both component learning and electrical composition (Tables 3 and 4A). C-PHAST builds on this foundation: it combines shared component laws with state initialization from incomplete measurements and physical readouts for supplied candidate configurations. Under the stated assumptions, our analysis asks when errors in the initial state and predicted dynamics are small enough to preserve the ordering of proposed configurations or certify satisfaction of their stated limits (Section 3.7). Learning a subsystem while it remains embedded in a larger system, as studied by van Otterdijk et al. (2024), is complementary to reusing fitted laws under new connections.

Poisson–Dirac neural networks (PoDiNNs) learn component characteristics and their interconnection in a reduced dynamical representation (Khosrovian et al., 2025). Dirac-interconnected neural elements (DINEs) instead retain algebraic constraints and jointly learn constitutive relations and a constant separable Dirac interconnection; they also study partial observation and reuse of learned elements (Li et al., 2026b). Thus neither reusable physical components nor partial observation alone distinguishes C-PHAST across this literature. The comparison must specify what is learned about the interconnection, which measurements initialize prediction, and what calibration is permitted after composition. Our main electrical studies supply the wiring and line laws, withhold numerical DGU laws, and evaluate physical readouts under that contract.

N-PHDAE and electrical composition. N-PHDAE (Neary et al., 2024) is the closest electrical predecessor. Its state includes storage variables and algebraic node voltages and source currents. C-PHAST's electrical model evolves charge and flux and derives voltage and current from its learned energy (Section 3.1). N-PHDAE derives circuit structure from supplied incidence matrices and learns inductor energy, capacitor charge–voltage relations, and resistor voltage–current relations. N-PHDAE composes subsystems by adding interface currents and voltage-equality constraints, retaining trained subsystem laws without further fitting. C-PHAST connects storage states through power-preserving incidence ports. This avoids independent algebraic coordinates for the modeled circuits, while N-PHDAE explicitly accommodates algebraic constraints.

Both approaches use supplied electrical topology; our studies also supply line resistance and inductance. Our DGU energy has positive effective inductance and is quadratic in flux, a restriction absent from an unrestricted learned inductor-energy function; damping is nonnegative. N-PHDAE can also use ODE integration: for regular index-1 systems with an invertible algebraic Jacobian, differentiating its constraints produces an ODE for training and simulation. The distinction is therefore the representation of component laws and their interconnection. Table 1 separates these choices from how each model obtains its starting state.

Table 1. **Closest model families.** Model representation, interconnection, and initialization are separate design choices. C-PHAST shares the pH composition principle with prior work. These entries describe the reported methods; predictive accuracy is compared in the experiments.

Method	Model	Component interconnection	How prediction starts
PHAST (Bhardwaj & Bajaj, 2026)	Energy, damping, and phase-state maps	No separate component interconnection	Position history initializes a phase state
Compositional pHNN (Neary & Topcu, 2023)	Subsystem energy, dissipation, and input maps	Supplied or learned coupling	Supplied subsystem states
N-PHDAE (Neary et al., 2024)	Circuit laws in a differential–algebraic model	Supplied incidence and interface constraints	Supplied storage states; consistent algebraic variables
PoDiNNs (Khosrovian et al., 2025)	Component laws in a reduced ODE	Learned Dirac interconnection	Observed component states
DINEs (Li et al., 2026b)	Component laws in an unreduced DAE	Learned constant separable Dirac constraints	State fitted to initial observations or their history
CT-SUBNET (Beintema et al., 2023)	Continuous-time latent dynamics and output map; fixed state/input/output dimensions	No separate component interconnection	Encoder uses past inputs and measured outputs
C-PHAST (ours)	pH laws shared by type; physical readouts	Supplied candidate layout; power-preserving ports	Measured or history-estimated state; supplied states for additions

Prediction from incomplete measurements. Algebraic consistency and state estimation solve different problems. Given storage states, a consistency solve recovers compatible algebraic variables; it does not itself estimate missing storage states from a measurement history. CT-SUBNET (Beintema et al., 2023) jointly learns an encoder, continuous-time latent dynamics, and an output decoder. Past inputs and measured outputs initialize the latent state; future inputs drive the predicted trajectory. The model learns from input/output sequences without requiring measured physical states or state derivatives. OE-pHNN (Moradi et al., 2025) jointly learns a continuous-time pH model and a subspace encoder that initializes it from past inputs and outputs under measurement noise. CIPHER (Li et al., 2026a) first learns a causal history encoder and latent dynamics contrastively, then fits pH dynamics through a learned invertible coordinate transformation. These works develop state inference and structured identification for a given system. C-PHAST studies history-based initialization of a reusable composition on held-out configurations.

Our electrical comparison with CT-SUBNET tests prediction of unmeasured currents on a fixed three-unit network (Table 7). A separate C-PHAST study tests transfer to larger networks (Table 8). In the CT-SUBNET comparison, both procedures receive the same richer calibration measurements, then use only voltage histories and known inputs at deployment. We restrict CT-SUBNET's encoder to those observed channels and supervise its decoder with the shared calibration labels. C-PHAST additionally uses supplied wiring and line physics; the comparison includes its numerical state estimator and CT-SUBNET's learned encoder.

Graph simulators and world models. Graph Network-based Simulators and MeshGraphNets share learned local updates across particles or mesh nodes and generalize to larger systems (Sanchez-Gonzalez et al., 2020; Pfaff et al., 2021). This provides a strong precedent for variable-size prediction. Message passing alone, however, does not impose an energy balance on exchanged messages. C-PHAST specifies such a balance through energy-gradient port variables and skew interconnection, with numerical and estimation errors treated separately. Our graph baselines test this modeling choice empirically; continuous-time power cancellation alone does not guarantee accurate or bounded numerical rollouts.

World models such as Dreamer and TD-MPC2 learn predictive representations for policy learning or trajectory optimization (Hafner et al., 2023; Hansen et al., 2024). TD-MPC2 also supports multiple embodiments and action spaces within one agent, so cross-embodiment use is not unique to compositional physical models. Closer to our cross-robot study, QWM conditions a quadrupedal world model on scale-invariant morphology features, such as actuator dynamics and limb lengths, and trains policies inside it that transfer zero-shot to unseen quadrupeds (Danesh et al., 2026); GenLoco instead trains locomotion controllers on procedurally randomized quadruped morphologies (Feng et al., 2022). C-PHAST receives no parameters or morphology description of the target robot: the four-leg layout is shared, leg laws learned on ANYmal-D are reused unchanged, and the target's joint-position history sets only its starting state (Section 4.3). The distinction here is reuse of explicit component laws under a supplied reconfiguration, with physical readouts that can reveal predicted overloads or

other failure modes before an operator or controller selects an action.

Control and protection. Plug-and-play microgrid control addresses voltage stabilization as units are connected (Tucci et al., 2018); dynamic-state-estimation protection uses model–measurement residuals to detect faults (Liu et al., 2018). On the learning side, distributed pH controllers provide closed-loop stability or finite-gain guarantees under their plant and interconnection assumptions (Furieri et al., 2022; Zakwan & Ferrari-Trecate, 2024). Pontryagin-based Hamiltonian learning (Bajaj & Nguyen, 2024) and differential policy optimization (Nguyen & Bajaj, 2025) instead learn controllers from Hamiltonian formulations of optimal control. C-PHAST learns a reusable plant model for evaluating candidate futures; it does not replace these controller or protection designs. Its predicted voltages, currents, losses, and constraint violations support comparison before acting, while safe execution requires the downstream control and protection layer.

3. Method: C-PHAST Architecture

Where should a new unit connect to an existing system, and what inputs should it receive? Figure 1 poses this question without choosing a physical domain. The existing connections stay fixed in this example; the alternatives change the new unit's attachment and may also change its inputs. A candidate $a = (\kappa, u_{0:N_{\text{pred}}-1})$ pairs a configuration with N_{pred} future physical inputs, spaced by Δt; N_{pred} is the prediction horizon in steps. The system specification $\mathcal{S}_\kappa$ records the components, wiring, and observation and actuation maps for configuration κ. Candidate starting states combine estimates for existing components with supplied initial states for additions. These choices determine the future to be predicted, not new component parameters to be fitted.

The decision needs a dynamics model. A useful candidate must reach its goal without unacceptable consequences along the way. The task score J_w combines a terminal-goal cost with accumulated costs, weighted by priorities w; operating limits are checked separately. For smooth costs and a fixed configuration κ, the continuous-time control formulation is

$$\begin{aligned} \min_{u(\cdot)} \quad & \Psi_G(x(T)) + \int_0^T \ell_w(x(t), u(t))\, dt, \\ \text{subject to} \quad & \dot{x} = f^{(\mathcal{S}_\kappa)}(x, u), \qquad x(0) = x_0. \end{aligned} \tag{1}$$

Here $T = N_{\text{pred}}\Delta t$, x_0 is the candidate's assigned starting state, Ψ_G measures terminal-goal error, and ℓ_w is a weighted running cost. The physical vector field $f^{(\mathcal{S}_\kappa)}$ gives the instantaneous state rate. In this control calculation, u denotes only adjustable inputs; prescribed loads are held fixed as known forcing and suppressed from the arguments of f. Equation (20) will specify the predicted score $\hat{J}_w$ using the physical consequences returned by the model, including their units and normalization.

To see how the dynamics enter the decision, suppress the fixed configuration in f and introduce a time-dependent multiplier λ, the *costate*. The augmented action Lagrangian, action, and control Hamiltonian are

$$\begin{aligned} \mathcal{L}_{\text{aug}}(x, \dot{x}, u, \lambda) &= \ell_w(x, u) + \lambda^\top (f(x, u) - \dot{x}), \\ \mathcal{I}_{\text{ctrl}} &= \Psi_G(x(T)) + \int_0^T \mathcal{L}_{\text{aug}}\, dt, \\ \mathcal{H}_{\text{ctrl}}(x, \lambda, u) &= \ell_w(x, u) + \lambda^\top f(x, u). \end{aligned} \tag{2}$$

Thus $\mathcal{L}_{\text{aug}} = \mathcal{H}_{\text{ctrl}} - \lambda^\top \dot{x}$; on a feasible trajectory, the multiplier term vanishes and the action equals the task cost. Varying λ enforces the state equation; varying x and integrating by parts gives the costate equation and terminal condition. For the smooth normal problem with fixed initial state, free terminal state, and no active state constraints, Pontryagin's minimum principle (PMP) (Pontryagin et al., 1962; Liberzon, 2012) gives

$$\dot{x} = \partial_\lambda \mathcal{H}_{\text{ctrl}}, \qquad \dot{\lambda} = -\partial_x \mathcal{H}_{\text{ctrl}}, \qquad \lambda(T) = \nabla \Psi_G(x(T)). \tag{3}$$

The optimal input minimizes $\mathcal{H}_{\text{ctrl}}$ over its admissible values; an interior minimizer satisfies $\partial_u \mathcal{H}_{\text{ctrl}} = 0$. These are necessary conditions, not a global-optimality certificate (Tedrake, 2024). Appendix N gives the variation step by step; Appendix N.2 derives the Legendre–Fenchel form and its relation to this minimization convention. The control Hamiltonian combines task cost and dynamics; it is not stored physical energy, and λ is not mechanical momentum.

What remains is to construct f for each proposed connection. C-PHAST supplies the dynamics model from component energies and their interactions. Given candidate connections and input sequences, it predicts their consequences for comparison by an external decision maker. We next build the component laws, interconnection, and numerical transition, then specify calibration, initialization, and physical readout. Algorithm 1 collects the prediction operations; Section 3.5 gives their electrical realization.

3.1. Reusable Component Models

A component type fixes its state coordinates, ports, and shared physical laws. Each instance i has state x_i, differentiable stored energy H_i, lossless structure $J_i = -J_i^\top$, dissipation $R_i = R_i^\top \succeq 0$, and port matrix B_i. Suppressing the component index, the common port-Hamiltonian (pH) law is

$$\dot{x} = (J - R)\nabla H(x) + \mathbf{B}u, \quad y^{\text{port}} = \mathbf{B}^\top \nabla H(x). \tag{4}$$

Here H is energy, J exchanges it without loss, R dissipates it, and $u^\top y^{\text{port}}$ is the supplied power. The power-port output y^{port} need not coincide with the channels measured by the sensors. This physical Hamiltonian specifies part of f; it is distinct from $\mathcal{H}_{\text{ctrl}}$, which uses f to express an optimization problem.

For a conservative mechanical block, the action is $\mathcal{S}_{\text{plant}}[q] = \int_0^T L_{\text{plant}}(q, \dot{q})\, dt$, where $L_{\text{plant}}(q, v) = \frac{1}{2} v^\top M(q) v - V(q)$. With $M(q) \succ 0$, its Legendre–Fenchel transform in velocity gives the stored energy (Liberzon, 2012):

$$H(q, p) = \sup_v \left[p^\top v - L_{\text{plant}}(q, v)\right] = \tfrac{1}{2} p^\top M(q)^{-1} p + V(q). \tag{5}$$

The maximizing velocity satisfies $p = \partial_v L_{\text{plant}} = M(q)v$; here p is physical momentum. Appendix N.3 derives the resulting conservative state equations before adding loss and input. PHAST (Bhardwaj & Bajaj, 2026) uses this physical form, $H_i(q_i, p_i) = V_i(q_i) + \frac{1}{2} p_i^\top M_i(q_i)^{-1} p_i$ and $R_i = \text{diag}(0, D_i)$, with $M_i \succ 0$ and $D_i \succeq 0$. Thus H_i and R_i are constructed from the local physical functions, not additional independent networks. PHAST's port matrix G has the role of B_i here. The coordinate and numerical realizations are given in Appendix A. Parameter sharing is within a type, not across unrelated electrical and mechanical components.

3.2. Compose the Supplied Interconnection

Connecting components changes how they exchange energy, while preserving their local physical laws. The exchange takes place through *ports*: an input and its conjugate output form a power pair. A force paired with the velocity at its application point gives mechanical power; voltage paired with current gives electrical power. Neither input nor output alone is a rate of energy change. Two objects specify the exchange: B_i determines where an interaction acts on component i, and $\hat{C}_{ij}$ determines how its ports connect to component j.

For $x_i \in \mathbb{R}^{n_i}$ and m_i port channels, the matrix $B_i(x_i) \in \mathbb{R}^{n_i \times m_i}$ maps a port input into the state dynamics. Its transpose selects the conjugate output:

$$\dot{x}_i = (J_i - R_i)\nabla H_i + B_i \hat{u}_i, \qquad \hat{y}_i = B_i^\top \nabla H_i.$$

Here $\hat{y}_i = y_i^{\text{port}}$ in the declared coupling-port coordinates, and $\hat{u}_i^\top \hat{y}_i$ is the power entering through these ports. The hat identifies an interconnection variable, not an estimate. The gradient ∇H_i differentiates energy with respect to the state, not time: for canonical mechanical coordinates, its momentum derivative is velocity. The port output selects the combination paired with the input; it need not be a measured sensor channel. State dependence is suppressed below for readability.

For two components without external inputs, connect their ports by

$$\hat{u}_1 = -\hat{C}_{12}\hat{y}_2, \qquad \hat{u}_2 = \hat{C}_{12}^\top \hat{y}_1.$$

Substitution makes the interaction explicit:

$$\begin{aligned} \dot{x}_1 &= \underbrace{(J_1 - R_1)\nabla H_1}_{\text{local dynamics}} - \underbrace{B_1 \hat{C}_{12} B_2^\top \nabla H_2}_{\text{interaction with component 2}}, \\ \dot{x}_2 &= \underbrace{(J_2 - R_2)\nabla H_2}_{\text{local dynamics}} + \underbrace{B_2 \hat{C}_{12}^\top B_1^\top \nabla H_1}_{\text{interaction with component 1}}. \end{aligned} \tag{6}$$

Changing an attachment or connection therefore changes the joint trajectory without requiring a change to either component's energy or dissipation law. In the cross-component term, $B_j^\top \nabla H_j$ produces the port output, $\hat{C}_{ij}$ pairs it with an input at i, and B_i inserts that input into component i's dynamics.

For homogeneous components, a common port layout gives equally sized coupling blocks and permits parameter sharing across repeated component types. For heterogeneous components, the state and port dimensions may differ: $\hat{C}_{ij} \in \mathbb{R}^{m_i \times m_j}$ may be rectangular. Both cases use the same construction:

$$\hat{C}_{ji} = -\hat{C}_{ij}^\top, \qquad \hat{C}_{ii} = 0, \tag{7}$$

with zero blocks for absent connections. This makes the assembled $\hat{C}$ skew-symmetric; an individual off-diagonal block need not be skew. The graph specifies which connections exist. Port conventions and the connection model specify their signs and coefficients, pairing physical variables in consistent units.

For the full network, external inputs u_i^{ext} include applied controls and prescribed loads. Interconnection inputs are supplied by the other modeled components. External-only channels can be included by appending port columns with zero rows and columns in $\hat{C}$ (Appendix A.1). Through the declared port columns, these give

$$\dot{x}_i = \underbrace{(J_i - R_i)\nabla H_i}_{\text{reusable local dynamics}} - \underbrace{B_i \sum_j \hat{C}_{ij}\hat{y}_j}_{\text{interconnection}} + \overbrace{B_i u_i^{\text{ext}}}^{\text{external input}}. \tag{8}$$

Here the stacked internal inputs and outputs satisfy

$$\hat{u} = -\hat{\mathbf{C}}\,\hat{y}, \qquad \hat{\mathbf{C}} = -\hat{\mathbf{C}}^\top. \tag{9}$$

with $\hat{u} = [\hat{u}_1; \ldots; \hat{u}_s]$ and $\hat{y} = [\hat{y}_1; \ldots; \hat{y}_s]$. Here s is the number of modeled components. The system specification $\mathcal{S}$ records which components are present and how they are connected; s is one field of this specification (Appendix A.1).

The port maps may be rectangular, state-dependent, or rank-deficient. Their columns specify the directions in which interactions act, while their rank determines how many independent directions are accessible instantaneously. Using the same B_i in the input and output preserves the power pairing:

$$\nabla H_i^\top B_i \hat{u}_i = \hat{y}_i^\top \hat{u}_i.$$

Together with skew pairing, this gives the classical continuous-time cancellation of internal power (Cervera et al., 2007; Van Der Schaft & Jeltsema, 2014; Neary & Topcu, 2023). We assume the modeled total energy is the sum of the component energies, $H_{\text{total}}(x) = \sum_i H_i(x_i)$, with $\hat{R} = \text{diag}(R_1, \ldots, R_s)$. Appendix A.1.2 gives mechanical port examples and explains when this energy decomposition is an approximation.

Theorem 3.1 (Power-preserving interconnection)**.** *If $\hat{C} = -\hat{C}^\top$, internal coupling contributes zero net power:*

$$\sum_i \hat{u}_i^\top \hat{y}_i = -\hat{y}^\top \hat{C}\hat{y} = 0.$$

Writing $e_i = \nabla H_i$ and taking $R_i \succeq 0$, the assembled model satisfies

$$\frac{dH_{\text{total}}}{dt} = -\sum_i e_i^\top R_i e_i + \sum_i \hat{y}_i^\top u_i^{\text{ext}}. \tag{10}$$

Without external inputs, Equation (10) gives $\dot{H}_{\text{total}} \leq 0$: the interconnection redistributes energy rather than creating it. Connections with their own storage or dissipation contribute those laws separately. A line's inductance is a storage component; a resistive link is a memoryless element whose elimination adds its dissipation to the total. Appendix A.1 gives the equivalent block-matrix form.

These choices also determine how disturbances reach damping. For well-posed unforced dynamics, a strict local minimum of total energy gives a stable equilibrium. Convergence depends on whether motion can persist outside the damped directions.

The linear–quadratic analysis in Section 3.6.1 makes this dependence explicit: changing the interconnection can carry energy into existing damping without adding damping to either component.

The cancellation identity supplies none of the interface laws: compliant contact enters through its storage and loss, while rigid constraints and impacts need constraint and event equations in addition. For the state-only port outputs and explicit pairing used here, assembly introduces no algebraic loop. Regularity and bounded-trajectory hypotheses are stated separately in Appendix D.

Equation (8) combines the local dynamics, interconnection, and external inputs into continuous-time state rates. For a proposed configuration κ, the system specification $\mathcal{S}_\kappa$ identifies the components and their wiring. Substituting that specification gives the candidate's full differential equations. Denote their learned vector field by $f_\theta^{(\mathcal{S}_\kappa)}$:

$$\dot{x} = f_\theta^{(\mathcal{S}_\kappa)}(x, u) := \big(\hat{J} - \hat{B}\hat{C}_\kappa\hat{B}^\top - \hat{R}\big)\nabla H_{\mathrm{total},\theta}(x) + \hat{B}u. \tag{11}$$

Here $\hat{J} = \mathrm{diag}(J_i)$ and $\hat{B} = \mathrm{diag}(B_i)$, u stacks the external inputs u_i^{ext}, including controls and prescribed loads, and $\hat{C}_\kappa$ is the pairing for the proposed configuration. The subscript θ records the fitted component laws; their parameters remain fixed. The other state, parameter, and configuration dependencies on the right-hand side are suppressed. This is the learned counterpart of the physical $f^{(\mathcal{S}_\kappa)}$ in Equation (1). A model-based control Hamiltonian would therefore use $\mathcal{H}_{\mathrm{ctrl},\theta} = \ell_w + \lambda^\top f_\theta^{(\mathcal{S}_\kappa)}$: component energies enter through the dynamics, not through an identification of task cost with stored energy.

To predict forward, we advance these equations numerically. Given the stacked component state x and external input u held for an interval Δt, the declared numerical scheme computes one predicted step:

$$x^+ = F_{\theta,\Delta t}^{(\mathcal{S}_\kappa)}(x, u). \tag{12}$$

Here x^+ approximates the state Δt later under the assembled model. The lowercase f_θ returns a state rate; the uppercase F returns a next state after numerical integration. They are not two independently learned models: F packages the same field with a step size and integrator. We denote the construction of this update rule by

$$F_{\theta,\Delta t}^{(\mathcal{S}_\kappa)} = \mathrm{Compose}_\theta(\mathcal{S}_\kappa, \Delta t). \tag{13}$$

Thus $\mathrm{Compose}_\theta$ constructs F for the proposed system; applying F advances a supplied state. Repeated applications under the proposed inputs generate a candidate trajectory. The numerical realizations and study-specific schemes are given in Appendices A.2 and A.4. Finite-step energy properties depend on that scheme, not on continuous-time power cancellation alone.

State evolution and measurements. The evolving state and the measured channels need not coincide. For a fixed observed configuration $\mathcal{S}$, let $x = [x_1; \ldots; x_s]$ be the stacked state and u the known physical external input. Write $f^{(\mathcal{S})}$ for the physical dynamics and $h_{\mathrm{meas}}^{(\mathcal{S})}$ for the selected sensor response. At physical sampling times τ_k, the two processes take the form

$$\begin{aligned} \dot{x}(\tau) &= f^{(\mathcal{S})}(x(\tau), u(\tau)), && \text{state evolution,} \\ \tilde{q}_k &= h_{\mathrm{meas}}^{(\mathcal{S})}(x(\tau_k), u(\tau_k)) + \epsilon_k, && \text{measurement.} \end{aligned} \tag{14}$$

Here $\tilde{q}_k \in \mathbb{R}^{d_{\mathrm{obs}}}$ contains the measured channels, not necessarily mechanical positions or electrical charges. The noise ϵ_k is additive in the declared measurement coordinates; its law is specified by the experiment. This does not introduce process noise into the state equation.

C-PHAST predicts both the state and what the sensors would report from it. The readout $h_{\mathrm{obs}}^{(\mathcal{S})}$ converts a model state and physical input into the physical channels declared by that realization. The selector $\Pi_{\mathrm{obs}}^{(\mathcal{S})}$ retains the sensor channels in their measurement order; it is the identity when all readout channels are observed. Given a starting-state estimate for candidate a, the model is

$$\begin{aligned} \hat{x}_{\ell+1}(a) &= F_{\theta,\Delta t}^{(\mathcal{S}_\kappa)}(\hat{x}_\ell(a), u_\ell), && \text{predicted state,} \\ \hat{q}_{\ell+1}^{\mathrm{obs}}(a) &= \Pi_{\mathrm{obs}}^{(\mathcal{S}_\kappa)} h_{\mathrm{obs}}^{(\mathcal{S}_\kappa)}(\hat{x}_{\ell+1}(a), u_\ell), && \text{predicted sensor channels.} \end{aligned} \tag{15}$$

Here ℓ counts steps of duration Δt from the query, and u_ℓ is held over the step and used for its endpoint readout. The transition F numerically advances the assembled learned field $f_\theta^{(\mathcal{S}_\kappa)}$ in Equation (11), which models the physical dynamics

$f^{(\mathcal{S}_\kappa)}$. The selected readout $\Pi_{\text{obs}} h_{\text{obs}}$ models the sensor response h_{meas} in Equation (14). Thus $\hat{q}^{\text{obs}}$ predicts sensor channels without adding measurement noise; it is not a future observation. Model error and measurement noise remain distinct, and the power-port output y^{port} has a separate role.

The readout depends on the physical realization. For mechanical positions,

$$h_{\text{obs}}^{(\mathcal{S})}(x, u) = \chi_{\mathcal{S}}^{-1}(\Pi_q x),$$

where Π_q extracts model-coordinate positions and $\chi_{\mathcal{S}}^{-1}$ decodes them into observation coordinates. The electrical readout is given in Section 3.5. Appendix A.1 gives the coordinate and channel conventions, including Equation (55).

The full consequence readout in Equation (19) retains h_{obs} and adds energy and power quantities used to compare candidate futures.

3.3. Calibrate the Reusable Laws

PHAST's information contract $\mathcal{K}$ determines which local laws are supplied, learned, or corrected. C-PHAST's specification $\mathcal{S}$ determines where those laws are used. In the fully law-unknown comparison, numerical constitutive laws are withheld while component roles and wiring remain supplied. Table 14 records the benchmark-specific information regimes. The amount of supplied physics is distinct from the measurements available at deployment.

The fitted parameters comprise local laws θ_{loc}, any learned coupling parameters θ_C, and the observer/canonicalizer parameters ϕ, ψ when those maps are trained. With $\mathcal{D} = \{(\tilde{q}_{0:T_i}^{(i)}, u_{0:T_i-1}^{(i)}, \mathcal{S}^{(i)})\}$ the source trajectories and $\mathcal{I} = \{(i,t) : L_h - 1 \le t < T_i\}$ the valid one-step windows on the data grid, where the history length L_h counts observations and spans $L_h - 1$ intervals, the observation-supervised term is

$$\mathcal{L}_{\text{data}} = \frac{1}{|\mathcal{I}|} \sum_{(i,t)\in\mathcal{I}} \left\| \delta_q\left(\tilde{q}_{t+1}^{(i)}, \hat{q}_{t+1}^{\text{obs},(i)}\right) \right\|^2 . \tag{16}$$

Here $\hat{q}_{t+1}^{\text{obs},(i)}$ is the decoded prediction on the declared supervised channels. The residual δ_q is ordinary subtraction in Euclidean coordinates or the declared chart-aware difference otherwise. Partial observations used at deployment need not be the full set of source calibration labels.

The electrical studies use domain-specific prediction and constitutive-consistency losses, detailed with their calibration data in Section 3.5.

Matched mechanical forecasting uses observation prediction loss. Optional energy, observer, passivity, and rollout terms are defined separately in Appendix A.6; they are not assumed active in every benchmark. Observer fitting for electrical deployment likewise has its own source-only objectives (Appendix I). After calibration, a candidate query changes states, wiring, and inputs, not the fitted parameters.

3.4. Predict Candidate Trajectories and Consequences

With the calibrated component laws fixed, we now specify how to predict the alternatives in Figure 1. State estimation must describe the system that produced the measurements, whereas each rollout describes the system proposed by its candidate.

At query index t, let $\mathcal{S}_{\text{now}}$ denote the actual-system specification and $\mathcal{H}_{\text{now}}$ its observation–input history space, with available history $h_t \in \mathcal{H}_{\text{now}}$. Let $\mathcal{A}$ be the finite set of proposed candidates $a = (\kappa, u_{0:N_{\text{pred}}-1})$, where κ specifies a configuration and $u_{0:N_{\text{pred}}-1}$ supplies physical inputs for $N_{\text{pred}} \ge 1$ future steps of duration $\Delta t > 0$. For each configuration, let $\mathcal{S}_\kappa$ denote the proposed-system specification and ζ_κ its state-assignment data: retained-component identities, initial states of added components, and any prescribed reset. Both specifications record components, wiring, and observation and actuation maps (Appendix A.1); $\mathcal{S}_{\text{now}}$ describes the system that produced the history, whereas $\mathcal{S}_\kappa$ describes the system to be simulated. The component-model parameters θ remain fixed throughout the query. In the microgrid query, converter commands may vary between candidates, while load currents follow a supplied demand scenario.

Initialize and assign. All candidates start from one estimate of the actual state at the query time. The index t locates this time on the integration grid; ℓ counts future simulation steps from the query, with $\ell = 0$ at its start. The recorded samples $\tilde{q}_k$ in Equation (14) belong to the actual configuration $\mathcal{S} = \mathcal{S}_{\text{now}}$. The history $h_t = (Y_t, U_t^{\text{now}})$ contains these observation rows, arranged as $Y_t \in \mathbb{R}^{L_h \times d_{\text{obs}}}$, together with known actual-system input rows $U_t^{\text{now}} \in \mathbb{R}^{(L_{\text{sim}}+1)\times d_u}$. Here L_h is the number of observations, L_{sim} is the number of past integration intervals, and d_{obs} and d_u count measured channels and physical inputs. Appendix A.1.5 specifies the sampling indices and their physical times. The last input is known before candidate selection, not a proposed action. Observations need not include every state variable; for voltage-only sensing, currents remain unmeasured.

$\mathrm{InitializeState}$ converts this information into the starting state. Its realization is chosen for the available measurements: complete states undergo a coordinate conversion, whereas partial observations use an observer or a numerical history fit. The mechanical observer estimates velocity and maps it to a phase state (Eq. (52)). The electrical history fit simulates the *actual* system to match its recorded voltages, then returns the inferred present state (Eq. (204)); this is why it requires $\mathrm{Compose}_\theta(\mathcal{S}_{\text{now}}, \Delta t)$ before initialization. The learned electrical alternative uses smoothed voltage and its rate (Eqs. (205)–(206)). All model and estimator parameters stay fixed during the query.

The proposed system may contain components absent from the actual one. For example, predicting a new DGU connection also requires initial storage states for the added DGU and line. $\mathrm{AssignState}_\kappa$ uses the supplied ζ_κ to construct the candidate's starting state (Eq. (53)):

$$\hat{x}_t = \mathrm{InitializeState}(h_t; \mathcal{S}_{\text{now}}), \qquad \hat{x}_0(a) = \mathrm{AssignState}_\kappa(\hat{x}_t; \zeta_\kappa). \tag{17}$$

Here $\hat{x}$ denotes predicted state. Initialization is performed once; assignment is performed for each candidate, without refitting the laws.

Simulate and read out. For each candidate, $\mathrm{Compose}_\theta(\mathcal{S}_\kappa, \Delta t)$ assembles the fixed component laws under its proposed wiring into the transition $F_{\theta,\Delta t}^{(\mathcal{S}_\kappa)}$ (Eq. (13)). At step ℓ, the candidate state $\hat{x}_\ell(a)$ is the predicted state $\ell\Delta t$ seconds after the query. The input $u_\ell \in \mathcal{U}_\kappa$ is held constant for the next Δt seconds. Apply the paired state and sensor-prediction equations in Equation (15) for $\ell = 0, \ldots, N_{\text{pred}} - 1$. Repeating the state update produces the trajectory; applying the selected readout at each endpoint predicts the sensor channels without future measurements. Physical actuation maps convert raw commands to port inputs; a proxy command can condition a forecast but is not itself a physical effort for work accounting (Appendix A.1).

For comparing candidate futures, the model returns the physical channels in h_{obs} together with six energy-accounting quantities. The sensor prediction in Equation (15) selects from those physical channels; the additional quantities describe stored energy, power, and energy transferred over each step. For transition endpoints $x = \hat{x}_\ell(a)$, $x^+ = \hat{x}_{\ell+1}(a)$ and input $u = u_\ell$, let W_ℓ be external work and $E_{\text{loss},\ell}$ dissipated energy over the interval. Their accounting defect is

$$\underbrace{\epsilon_{H,\ell}}_{\text{accounting residual}} = \underbrace{H_{\text{total}}(x^+) - H_{\text{total}}(x)}_{\text{stored-energy change}} - \overbrace{W_\ell}^{\text{external work}} + \underbrace{E_{\text{loss},\ell}}_{\text{dissipated energy}}. \tag{18}$$

The powers are $P_{\text{loss}} = \nabla H_{\text{total}}^\top \hat{R} \nabla H_{\text{total}}$ and $P_{\text{ext}} = \hat{y}^\top u^{\text{ext}}$, where $u^{\text{ext}} = [u_1^{\text{ext}}; \ldots; u_s^{\text{ext}}]$ collects the component external inputs. Endpoint trapezoidal quadrature defines W_ℓ and $E_{\text{loss},\ell}$ (Eq. (54)), so $\epsilon_{H,\ell}$ includes integration and quadrature error. The returned record is

$$\begin{aligned}\rho_{\ell+1}(a) &= \mathcal{O}_{\text{target}}(x, x^+, u; \mathcal{S}_\kappa, \Delta t) \\ &:= \Big(\underbrace{h_{\text{obs}}^{(\mathcal{S}_\kappa)}(x^+, u)}_{\substack{\text{physical readouts} \\ \text{select sensor channels for } \hat{q}_{\ell+1}^{\text{obs}}(a)}}, \underbrace{H_{\text{total}}(x^+)}_{\text{stored energy}}, \\ &\qquad \underbrace{P_{\text{loss}}(x^+),\ P_{\text{ext}}(x^+, u)}_{\text{instantaneous powers}}, \underbrace{W_\ell,\ E_{\text{loss},\ell}}_{\text{interval work and loss}}, \underbrace{\epsilon_{H,\ell}}_{\text{accounting residual}} \Big).\end{aligned} \tag{19}$$

The sensor prediction $\hat{q}_{\ell+1}^{\text{obs}}(a)$ in Equation (15) is obtained by selecting channels from the first block of the returned record $\rho_{\ell+1}(a)$. The next three entries describe energy and power at the endpoint; W_ℓ and $E_{\text{loss},\ell}$ accumulate work and loss over the interval. The last entry checks the numerical energy balance: $\epsilon_{H,\ell}$ is not the sensor noise ϵ_k in Equation (14).

Algorithm 1 C-PHAST candidate prediction with fixed component models

Inputs: h_t, $\mathcal{S}_{\text{now}}$; candidates $a = (\kappa, u_{0:N_{\text{pred}}-1}) \in \mathcal{A}$ with $(\mathcal{S}_\kappa, \zeta_\kappa)$; horizon N_{pred}, step size Δt.
Fixed: Component-model parameters θ after calibration (Section 3.3); state initializer InitializeState.
Outputs: Predicted states $\hat{x}_{0:N_{\text{pred}}}(a)$ and consequences $\rho_{1:N_{\text{pred}}}(a)$ for each $a \in \mathcal{A}$.

Q01 $\hat{x}_t \leftarrow \text{InitializeState}(h_t; \mathcal{S}_{\text{now}})$
Estimate the present state from recorded history.
Electrical history fit: adjust the past state to match recorded sensor readings, then propagate to t.
State entry; sensor map Eq. (15); mechanical Eq. (52); electrical Eq. (204).

for each $a = (\kappa, u_{0:N_{\text{pred}}-1}) \in \mathcal{A}$ **do**

Q02 $\hat{x}_0(a) \leftarrow \text{AssignState}_\kappa(\hat{x}_t; \zeta_\kappa)$
Retain existing states; apply supplied additions and resets.
Eq. (53).

Q03 $F_{\theta,\Delta t}^{(\mathcal{S}_\kappa)} \leftarrow \text{Compose}_\theta(\mathcal{S}_\kappa, \Delta t)$
Assemble f_θ from the fixed laws and wiring; construct its numerical step F.
Dynamics: Eq. (11); construction: Eq. (13).

for $\ell = 0, \ldots, N_{\text{pred}} - 1$ **do**

Q04 $\hat{x}_{\ell+1}(a) \leftarrow F_{\theta,\Delta t}^{(\mathcal{S}_\kappa)}(\hat{x}_\ell(a), u_\ell)$
Apply the update rule to advance the state by Δt.
One-step map: Eq. (12); mechanical: Algorithm 2; electrical: Eq. (71) and the study-specific schemes in Appendix A.4.

Q05 $\rho_{\ell+1}(a) \leftarrow \Big(\underbrace{h_{\text{obs}}^{(\mathcal{S}_\kappa)}(\hat{x}_{\ell+1}(a), u_\ell)}_{\text{physical channels}},$
$\underbrace{H_{\text{total}}(\hat{x}_{\ell+1}(a)),\ P_{\text{loss}}(\hat{x}_{\ell+1}(a)),\ P_{\text{ext}}(\hat{x}_{\ell+1}(a), u_\ell)}_{\text{stored energy, loss power, input power}},$
$\underbrace{W_\ell,\ E_{\text{loss},\ell},\ \epsilon_{H,\ell}}_{\text{interval work, dissipated energy, balance residual}} \Big)$
Read physical quantities at the new state; account for work, loss, and energy balance over the step.
Eqs. (18)–(19).

end for
end for

Q06 **return** $\{(a, \hat{x}_{0:N_{\text{pred}}}(a), \rho_{1:N_{\text{pred}}}(a)) : a \in \mathcal{A}\}$
Return states and consequences for external constraint checks and scoring.
Section 3.4; Eq. (20); Figure 3.

Here a *typed* readout has a declared meaning and units. The energy and power channels computed from the pH state obey the model's accounting relation under its port assumptions. Telemetry-derived quantities, such as joint work computed from measured torque and velocity, can instead serve as targets for a separate predictor. Their predictions, and operational proxies such as sensing quality, require empirical validation; naming a channel does not give it a Hamiltonian balance law. Each candidate returns its trajectory $\hat{x}_{0:N_{\text{pred}}}(a)$ and consequence sequence $\rho_{1:N_{\text{pred}}}(a)$ for external comparison; the predictor does not choose or execute an action.

Algorithm 1 collects these operations into the predictor in Figure 2; Figure 4 later instantiates them electrically. It specifies a partial map: its successful domain consists of histories and candidates for which all displayed calls return finite, valid states and readouts. Numerical failure and estimator diagnostics are specified in Appendices A and I; decision admission is a separate restriction, not a forecast filter.

Compare the returned futures. The transition F has produced candidate trajectories; the readout has converted them into physical consequences ρ. We can now evaluate the task score introduced with Equation (1). In the electrical example developed below, an operator can compare losses among candidates whose predicted voltages and currents meet supplied limits. An *action card* summarizes these consequences for a candidate; it is not another predictor. Figure 3 previews the comparison of the two connections shown later in Figure 4: peak current and the voltage range determine predicted feasibility, while accumulated loss supplies an operating cost. These summaries retain their physical units for comparison with limits; cost normalization then makes the chosen priorities explicit. Let $\widehat{\mathcal{A}}_{\text{feas}} \subseteq \mathcal{A}$ contain candidates satisfying the specified constraints in prediction. For nonnegative priority weights $w = (w_k)_k$, define the predicted score and its minimizing candidate by

$$\hat{J}_w(a) = \underbrace{\Psi_G(\hat{x}_{N_{\text{pred}}}(a))}_{\text{terminal-goal cost}} + \underbrace{\Delta t \sum_{\ell=0}^{N_{\text{pred}}-1} \sum_k w_k \bar{\ell}_k(\rho_{\ell+1}(a))}_{\text{accumulated consequence cost}}, \qquad a_w^\star \in \arg\min_{a \in \widehat{\mathcal{A}}_{\text{feas}}} \hat{J}_w(a). \tag{20}$$

Here Ψ_G is a terminal-goal cost and $\bar{\ell}_k$ is a normalized consequence cost, such as loss or voltage deviation, weighted by w_k. The hat distinguishes the model's predicted score $\hat{J}_w$ from the corresponding physical score J_w used in the error analysis. The goal, input, and operating constraints are specified in Appendix A.8; an empty predicted-feasible set leads to abstention. Predicted feasibility alone is not a safety certificate; the error analysis distinguishes the true feasible set from its uncertainty-tightened subset. Changing the weights reuses the predictions. Selection and execution remain external: the electrical study tests partial-sensing prediction, not executed reconfiguration, whereas the Go2 study evaluates one selected recovery command.

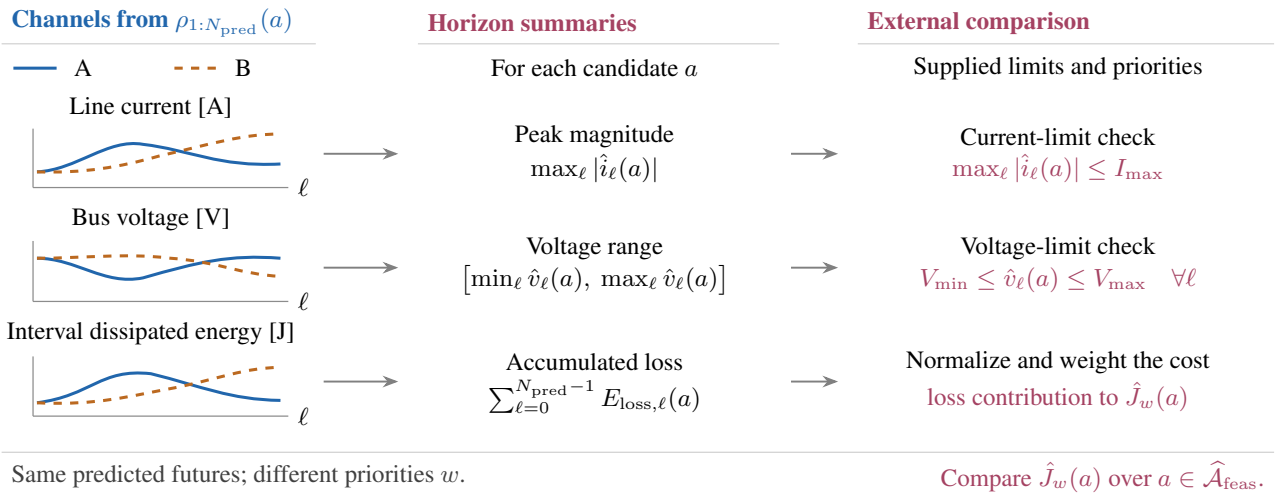


Figure 3. **From predicted consequences to candidate comparison.** For connections A and B in Figure 4, current $\hat{i}_\ell(a)$, voltage $\hat{v}_\ell(a)$, and loss channels from the returned sequence $\rho_{1:N_{\text{pred}}}(a)$ provide summaries with distinct roles: operating-limit checks and an objective cost. Checks apply to every relevant bus and line at the predicted steps; one of each is shown. The summaries form an action card for each candidate. Changing priorities reuses the predictions, while selection remains external to C-PHAST. Traces are schematic, not measured results; passing predicted limits is not an uncertainty certificate.

3.5. Electrical Realization

The microgrid in Figure 4 makes the preceding construction concrete: DGU and line models supply the local laws, and the candidate wiring specifies their exchange. We now collect their states, ports, observations, and calibration in one realization.

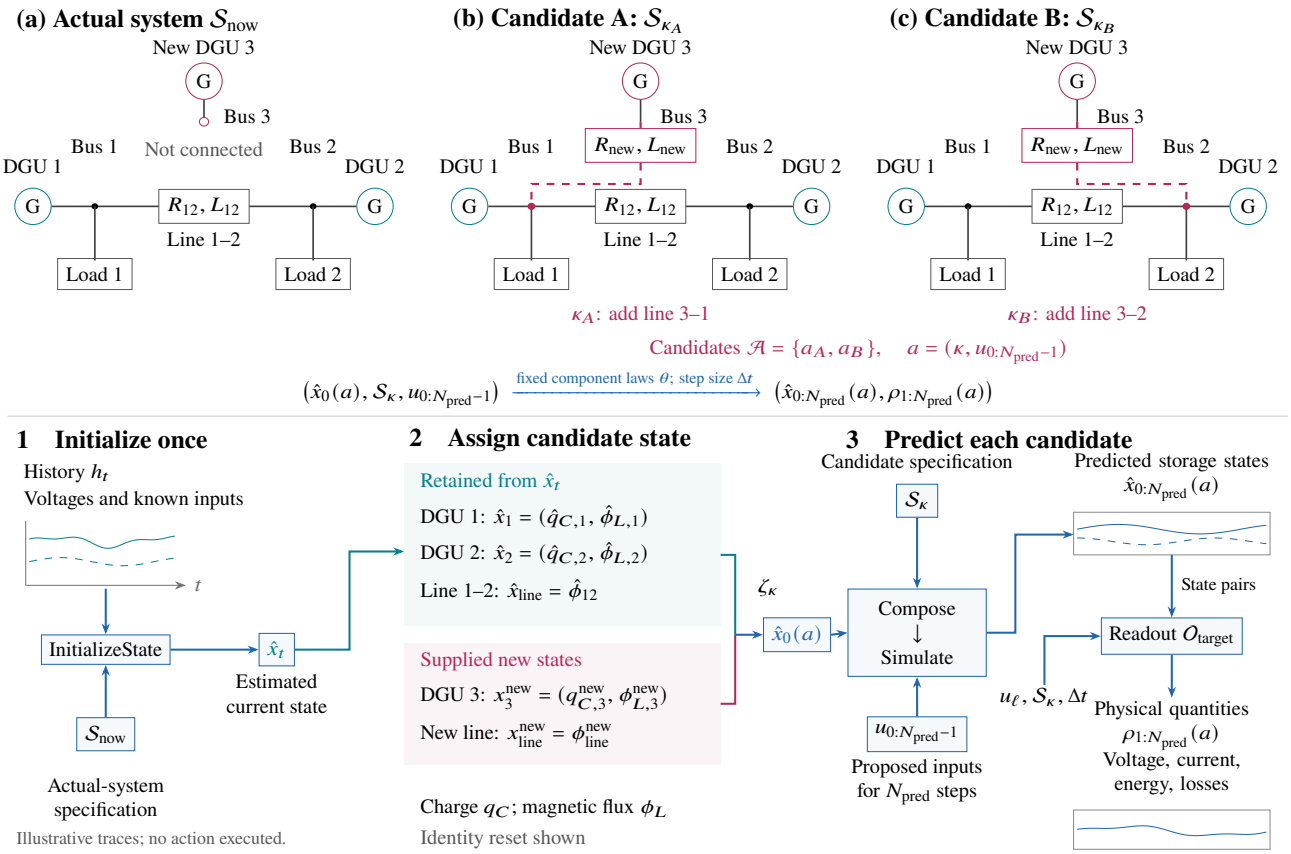


Figure 4. **Inside the C-PHAST predictor: one actual system, two candidate futures.** The upper row grounds the specifications in a microgrid: the new DGU is disconnected in $\mathcal{S}_{\text{now}}$ and joins bus 1 or bus 2 in $\mathcal{S}_{\kappa_A}$ or $\mathcal{S}_{\kappa_B}$. The lower flow expands the predictor in Figure 2. Initialization estimates the actual state from history; assignment retains existing storage states and adds supplied initial states according to ζ_κ (identity reset shown). Rollout step 0 begins at the present time t. Fixed component laws then generate the candidate trajectory. The readout map uses successive state pairs and inputs to report physical quantities, including interval energy accounting. The traces illustrate these roles, not experimental results. Algorithm 1 formalizes the procedure.

Storage and observations. For an electrical DGU, the storage state is $x = (q_C, \phi_L)$: capacitor charge and inductor flux. Charge changes with net current, and flux changes with net voltage; the energy derivatives supply the voltage and current in these balances. With input $u = (I_{\text{in}}, V_{\text{src}})$, the effective laws U_θ, $M_\theta > 0$, and $D_\theta > 0$ give

$$
\begin{aligned}
H_\theta(q_C, \phi_L) &= U_\theta(q_C) + \frac{\phi_L^2}{2M_\theta(q_C)}, \\
i_\theta = \partial_{\phi_L} H_\theta &= \frac{\phi_L}{M_\theta(q_C)}, \qquad v_\theta = \partial_{q_C} H_\theta = U_\theta'(q_C) - \frac{\phi_L^2 M_\theta'(q_C)}{2M_\theta(q_C)^2}, \\
\dot{q}_C &= i_\theta + I_{\text{in}}, \qquad \dot{\phi}_L = V_{\text{src}} - v_\theta - D_\theta(q_C) i_\theta.
\end{aligned}
\tag{21}
$$

Constant M_θ is inductance, while $D_\theta i_\theta$ is a resistive voltage drop. A line is a separate component type with energy $\phi_e^2/(2L_e)$ and current $i_e = \phi_e/L_e$. In the electrical studies, line resistance and inductance are supplied; the DGU storage and damping laws are fitted.

Under voltage-only sensing, the observation at physical time τ_k is

$$\tilde{v}_k = v(\tau_k) + \epsilon_k.$$

Here v is the physical voltage and ϵ_k is measurement noise; v_θ in Equation (21) predicts the voltage from the model state. The storage variables q_C and ϕ_L remain unmeasured. This voltage observation is the electrical instance of the general measurement model in Equation (14).

For the electrical network realization with n DGUs and m lines,

$$h^{(\mathcal{S})}_{\text{obs}}(x,u) = (v_1, i_1, \ldots, v_n, i_n, i_{e,1}, \ldots, i_{e,m}).$$

Here $v_j = v_{\theta,j}$ and $i_j = i_{\theta,j}$ follow the energy derivatives in Equation (21), and $i_{e,k} = \phi_{e,k}/L_{e,k}$. Voltage-only sensing selects $\Pi_{\text{obs}} h_{\text{obs}} = (v_1, \ldots, v_n)$; it does not remove charge, flux, or the resulting currents from the model.

Ports and wiring. The microgrid instantiates these same port maps and pairings. At a DGU, an injected current changes capacitor charge and is paired with bus voltage. At a line, a voltage difference changes flux and is paired with line current. For the states in Equation (21) and the line energy $H_e = \phi_e^2/(2L_e)$, the coupling-port maps are

$$\begin{aligned} B_i &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{y}_i = B_i^\top \nabla H_i = \partial_{q_C} H_i = v_i \quad \text{(DGU)}, \\ B_e &= 1, \qquad \hat{y}_e = \partial_{\phi_e} H_e = i_e \qquad\qquad\quad \text{(line)}. \end{aligned} \tag{22}$$

Only coupling-port columns are shown; the DGU's external source-voltage input uses an additional column. Thus B_i specifies where an input acts, not which other component supplies it.

Wiring supplies that second choice. Let A_κ have one row per bus and one column per line, with -1 at a line's sending bus, $+1$ at its receiving bus, and zero elsewhere. Grouping the coupling outputs into bus voltages v and line currents i_e gives the pairing

$$\hat{y} = \begin{bmatrix} v \\ i_e \end{bmatrix}, \qquad \hat{C}_\kappa = \begin{bmatrix} 0 & -A_\kappa \\ A_\kappa^\top & 0 \end{bmatrix}, \qquad -\hat{C}_\kappa \hat{y} = \begin{bmatrix} A_\kappa i_e \\ -A_\kappa^\top v \end{bmatrix}. \tag{23}$$

Equation (23) gives the input at each port. The local map B_i then places that input into the state equation. For bus k and line j, the coupling contributions are

$$\begin{aligned} \begin{bmatrix} \dot{q}_{C,k} \\ \dot{\phi}_{L,k} \end{bmatrix}\Big|_{\text{coupling}} &= \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{B_k} \underbrace{(A_\kappa i_e)_k}_{\text{net injected current}} = \begin{bmatrix} (A_\kappa i_e)_k \\ 0 \end{bmatrix}, \\ \dot{\phi}_{e,j}\Big|_{\text{coupling}} &= \underbrace{1}_{B_e} \underbrace{(-A_\kappa^\top v)_j}_{\text{voltage difference}}. \end{aligned} \tag{24}$$

The first line acts on the DGU's capacitor charge, not directly on its internal inductor flux $\phi_{L,k}$; the second acts on the connecting line's flux $\phi_{e,j}$. For a line directed from bus 1 to bus 2, positive current leaves bus 1 and enters bus 2, while $v_1 - v_2$ drives the line flux. Stacking the bus charges as q and line fluxes as ϕ_e gives

$$\dot{q}_{\text{coupling}} = A_\kappa i_e, \qquad \dot{\phi}_{e,\text{coupling}} = -A_\kappa^\top v. \tag{25}$$

The power entering the buses is $v^\top A_\kappa i_e$; the power entering the lines through these connections is its negative. Hence

$$v^\top A_\kappa i_e - i_e^\top A_\kappa^\top v = 0.$$

This cancellation concerns the connection power; each component still has its own storage and loss dynamics. For connections A and B in Figure 4, the receiving bus changes one incidence column. The same component laws then predict different voltage and current trajectories.

Physical anchors and calibration. Known port geometry and units give learned coupling coefficients a fixed physical meaning. Without that reference, different attachment and coupling scalings can reproduce the same motion (Appendix A.1.3). In the electrical studies, the port maps, wiring, and line laws are supplied; the DGU laws are fitted. Other settings may require learning coupling coefficients or interface laws.

The electrical studies use domain-specific prediction and constitutive-consistency losses. Let $\mathcal{L}^{\mathrm{loc}}_{\mathrm{constraint}}$ penalize local current, voltage, and resistive-drop residuals, and $\mathcal{L}^{\mathrm{loc}}_{\mathrm{step}}$ compare the DGU's next readout with the next measurement. The residuals are evaluated at measured storage states. The corresponding source-network terms evaluate these relations and predictions on connected calibration systems. The tested objectives are

$$\begin{aligned}\mathcal{L}_{\mathrm{pre}} &= \mathcal{L}^{\mathrm{loc}}_{\mathrm{constraint}} + \mathcal{L}_{\mathrm{reg}},\\ \mathcal{L}^{\mathrm{loc}}_{\mathrm{joint}} &= \mathcal{L}^{\mathrm{loc}}_{\mathrm{step}} + 0.01\mathcal{L}^{\mathrm{loc}}_{\mathrm{constraint}} + \mathcal{L}_{\mathrm{reg}},\\ \mathcal{L}^{\mathrm{src}}_{\mathrm{joint}} &= \mathcal{L}^{\mathrm{src}}_{\mathrm{step}} + 0.01\mathcal{L}^{\mathrm{src}}_{\mathrm{constraint}} + \mathcal{L}_{\mathrm{reg}}.\end{aligned} \tag{26}$$

Appendix A.5 defines every residual, normalization, and regularizer. The isolated-DGU study compares combined-loss training throughout with constraint-first training followed by combined-loss fitting. Separately, the network study further calibrates shared DGU laws on connected source systems before freezing them. Sections 4.1 and 4.2 report these separate studies.

3.6. How Interconnection Changes Dissipation and Inference

The following results first study exact dynamics and then specialize Equation (14) to known linear sensor responses in Equation (35).

3.6.1. Dissipation under a fixed interconnection

A lossless connection can carry disturbances into directions where damping acts. We compare the energy removed over the same interval, holding the component energies, damping, and initial distribution fixed.

Consider the fixed, unforced, linear–quadratic specialization of Equation (8), with the same s components in each configuration. Each component has constant matrices $J_i = -J_i^\top$, $R_i \succeq 0$, $Q_i \succ 0$, a compatible coupling-port matrix B_i, and energy $H_i(x_i) = \frac{1}{2}x_i^\top Q_i x_i$. There are no external inputs, switching, or state resets. For $x = [x_1; \dots; x_s] \in \mathbb{R}^{n_x}$ and $Q = \mathrm{diag}(Q_1, \dots, Q_s)$, the energy coordinates $x_E = Q^{1/2}x \in \mathbb{R}^{n_x}$ give $H_E(x_E) := H_{\mathrm{total}}(Q^{-1/2}x_E) = \frac{1}{2}\|x_E\|_2^2$. With $\hat{J} = \mathrm{diag}(J_i)$, $\hat{R} = \mathrm{diag}(R_i)$, and $\hat{B} = \mathrm{diag}(B_i)$, define

$$\begin{aligned} J_{E,0} &= Q^{1/2}\hat{J}Q^{1/2}, & R_E &= Q^{1/2}\hat{R}Q^{1/2},\\ K_\kappa &= -Q^{1/2}\hat{B}\hat{C}_\kappa\hat{B}^\top Q^{1/2}, & J_{E,\kappa} &= J_{E,0} + K_\kappa. \end{aligned} \tag{27}$$

The supplied skew interconnection $\hat{C}_\kappa$ in Equation (9) is selected by $\mathcal{S}_\kappa$ and has zero diagonal component blocks. Isolation is $\hat{C}_0 = 0$. Thus $J_{E,\kappa}, R_E, K_\kappa \in \mathbb{R}^{n_x \times n_x}$, $J_{E,\kappa}^\top = -J_{E,\kappa}$, and $R_E \succeq 0$. The exact state evolution is

$$\begin{aligned} \dot{x}_E &= (J_{E,\kappa} - R_E)x_E,\\ E_\kappa(t) &= e^{(J_{E,\kappa}-R_E)t}, \qquad x_E(t) = E_\kappa(t)x_{E,0}. \end{aligned} \tag{28}$$

Here $x_{E,0} = Q^{1/2}x(0)$; $E_\kappa(t) \in \mathbb{R}^{n_x \times n_x}$ is the evolution matrix. For a finite horizon $T > 0$, define the damping Gramian and its smallest eigenvalue:

$$W_\kappa(T) = \int_0^T E_\kappa(t)^\top \overbrace{R_E}^{\text{fixed damping}} \underbrace{E_\kappa(t)}_{\text{flow changed by coupling}}\, dt, \tag{29}$$

$$\eta_\kappa(T) = \lambda_{\min}(W_\kappa(T)).$$

The quantity $v^\top W_\kappa(T)v$ is the energy removed from initial state $v \in \mathbb{R}^{n_x}$. We use the same initial-state distribution for every connection and assume its second moment is finite:

$$M_0 := \mathbb{E}[x_{E,0}x_{E,0}^\top] \in \mathbb{R}^{n_x \times n_x}.$$

Its entries record average squared initial coordinates and their pairwise products. To compare connections, let $\mathcal{D}_\kappa(T)$ denote the energy dissipated over the same interval $[0, T]$, averaged over this initial-state distribution:

$$\mathcal{D}_\kappa(T) = \mathbb{E}[H_E(x_{E,0}) - H_E(E_\kappa(T)x_{E,0})] \tag{30}$$

Neither Gaussianity nor independence between component initial states is assumed.

Theorem 3.2 (Finite-time dissipation under composition)**.** *Under the preceding fixed, unforced, linear–quadratic assumptions, for every $T > 0$,*

$$E_\kappa(T)^\top E_\kappa(T) = I - 2W_\kappa(T), \qquad 0 \preceq W_\kappa(T) \prec \tfrac{1}{2}I. \tag{31}$$

Moreover, $\eta_\kappa(T) > 0$ if and only if no nonzero $J_{E,\kappa}$-invariant subspace lies in $\ker R_E$. In that case, for every initial state and integer $\ell \geq 0$,

$$H_E(x_E(\ell T)) \leq [1 - 2\eta_\kappa(T)]^\ell H_E(x_{E,0}). \tag{32}$$

The exact comparison with isolation is

$$\underbrace{\mathcal{D}_\kappa(T) - \mathcal{D}_0(T)}_{\text{connected loss minus isolated losses}} = \overbrace{\operatorname{tr}((W_\kappa(T) - W_0(T))M_0)}^{\text{positive: greater expected dissipation}}. \tag{33}$$

Here $\mathcal{D}_0$ is the sum of the expected losses of the isolated components, initialized with the same component marginals.

Consider two systems that dissipate energy while operating separately. Connecting them allows energy to move between them, including from weakly damped directions toward stronger damping. The connected system can therefore dissipate more energy on average over the same interval, even though neither component acquires additional damping. The theorem quantifies this comparison. Its strict-decay condition says that every nonzero disturbance reaches a damped direction, as in the standard hypocoercivity criterion (Achleitner et al., 2021, Proposition 1 and Lemma 3). The proof is in Appendix B.2. The following corollary gives conditions under which the connection strictly increases the average energy dissipated.

Corollary 3.3 (Strict expected benefit from mixing damping directions)**.** *In addition to the theorem's assumptions, suppose $M_0 = \sigma_0^2 I$ with $\sigma_0 > 0$, and $J_{E,0}R_E = R_E J_{E,0}$. Then every admissible lossless interconnection satisfies $\mathcal{D}_\kappa(T) \geq \mathcal{D}_0(T)$ for every $T > 0$. The inequality is strict for every $T > 0$ if and only if $K_\kappa R_E \neq R_E K_\kappa$.*

Under these assumptions, **coupling different damping rates increases finite-time expected dissipation without adding damping.** The condition $M_0 = \sigma_0^2 I$ is an additional assumption for this corollary, not the general definition of M_0. It imposes equal mean squared amplitudes and zero cross-second moments in energy coordinates. The isolated dynamics must also preserve the damping eigenspaces. The benefit need not hold for every initial state, and strict expected benefit does not require $\eta_\kappa(T) > 0$. There is no universal improvement for arbitrary isolated dynamics. These are exact linear–quadratic results; learned nonlinear models, numerical rollouts, and objectives with additional costs require separate analysis.

The corollary is proved in Appendix B.3 using the classical trace inequality and equality case of Bernstein (1988); So (1992). Appendix B also gives the rank test, all-time decay bounds, two-component calculation, and counterexample.

3.6.2. State inference under a fixed interconnection

A connection can make an unmeasured state leave a trace in the measurements. We ask when that trace improves estimation, using the known, fixed linear–quadratic dynamics and energy coordinates of Equation (28). Time zero is the beginning of the observed interval. Assume

$$x_{E,0} \sim \mathcal{N}(m_0, P_0), \qquad P_0 \succ 0. \tag{34}$$

where $m_0 \in \mathbb{R}^{n_x}$ is the mean and $P_0 \in \mathbb{R}^{n_x \times n_x}$ is the covariance (its inverse is the prior precision). The same second moment used in Theorem 3.2 can therefore be written as

$$M_0 = \mathbb{E}[x_{E,0}x_{E,0}^\top] = P_0 + m_0 m_0^\top.$$

It equals the covariance P_0 only when the mean m_0 is zero. Initial coordinates may be correlated. There is no process noise, switching, reset, or unknown input. Known prescribed additive inputs are allowed after subtracting their known response.

Let $L_h \geq 1$ and $d_{\text{obs}} \geq 1$ be the numbers of samples and measured channels. The known linear measurement map $C_{\text{obs}} \in \mathbb{R}^{d_{\text{obs}} \times n_x}$ converts the state in energy coordinates into noiseless sensor readings. Each row specifies which combination

of state coordinates one measured channel observes. For readings $C_{\text{phys}}x$ in physical coordinates, $C_{\text{obs}} = C_{\text{phys}}Q^{-1/2}$; these measurements are distinct from power-port outputs. At times $0 \le s_0 < \cdots < s_{L_h-1} \le T < \infty$, $T > 0$, fixed before observing the data, the unforced specialization of Equation (14) is

$$\begin{aligned} \dot{x}_E(s) &= (J_{E,\kappa} - R_E)x_E(s), && \text{state evolution,} \\ y_{\kappa,r} &= C_{\text{obs}}x_E(s_r) + \epsilon_{\kappa,r}, && \text{measurement.} \end{aligned} \tag{35}$$

Stacking these observations in chronological order gives

$$\begin{aligned} \boldsymbol{y}_\kappa &= b_\kappa + G_\kappa x_{E,0} + \epsilon_\kappa, \\ G_\kappa &= \begin{bmatrix} C_{\text{obs}}E_\kappa(s_0) \\ \vdots \\ C_{\text{obs}}E_\kappa(s_{L_h-1}) \end{bmatrix}, \qquad \epsilon_\kappa \sim \mathcal{N}(0, \Gamma). \end{aligned} \tag{36}$$

Here $\boldsymbol{y}_\kappa, b_\kappa, \epsilon_\kappa \in \mathbb{R}^{L_h d_{\text{obs}}}$, $G_\kappa \in \mathbb{R}^{L_h d_{\text{obs}} \times n_x}$, and $\Gamma \in \mathbb{R}^{L_h d_{\text{obs}} \times L_h d_{\text{obs}}}$ is positive definite. The noise is independent of $x_{E,0}$; b_κ is the known input response and is zero in the unforced case. Compared configurations use the same components, prior, measurement map, sampling times, and noise law, with each history generated under its own configuration. The observation information matrix is

$$W_\kappa^{\text{obs}} = \overbrace{G_\kappa^\top}^{\text{noise weighting}} \overbrace{\Gamma^{-1}}^{} \underbrace{G_\kappa}_{\text{sampled sensor response}}. \tag{37}$$

It measures the sampled sensor response relative to noise; the damping Gramian $W_\kappa(T)$ instead measures energy loss.

Theorem 3.4 (Initial-state inference under composition)**.** *Under the preceding assumptions, the posterior covariance is*

$$\begin{aligned} \Sigma_\kappa &= \operatorname{Cov}(x_{E,0} \mid \boldsymbol{y}_\kappa) \\ &= \Big(\underbrace{P_0^{-1}}_{\substack{\text{prior} \\ \text{precision}}} + \underbrace{W_\kappa^{\text{obs}}}_{\substack{\text{information} \\ \text{from samples}}} \Big)^{-1}. \end{aligned} \tag{38}$$

For a fixed nonzero target vector $c_{\text{tar}} \in \mathbb{R}^{n_x}$, the minimum mean-squared error is

$$\begin{aligned} \operatorname{MSE}_\kappa(c_{\text{tar}}) &:= \inf_{\widehat{z}} \mathbb{E}\big[(\widehat{z}(\boldsymbol{y}_\kappa) - c_{\text{tar}}^\top x_{E,0})^2\big] \\ &= c_{\text{tar}}^\top \Sigma_\kappa c_{\text{tar}}, \end{aligned} \tag{39}$$

where the infimum is over square-integrable estimators using these samples. It is attained by the conditional mean, and the expectation includes both initial-state uncertainty and measurement noise. Moreover,

$$\begin{aligned} \operatorname{MSE}_\kappa(c_{\text{tar}}) < c_{\text{tar}}^\top P_0 c_{\text{tar}} \\ \iff \quad G_\kappa P_0 c_{\text{tar}} \neq 0. \end{aligned} \tag{40}$$

The exact comparison with isolation is

$$\begin{aligned} &\operatorname{MSE}_0(c_{\text{tar}}) - \operatorname{MSE}_\kappa(c_{\text{tar}}) \\ &\quad = c_{\text{tar}}^\top (\Sigma_0 - \Sigma_\kappa) c_{\text{tar}}. \end{aligned} \tag{41}$$

Thus a connection improves this target's estimation precisely when the right-hand side is positive.

Consider two systems whose states are only partly known from their measurements. When the systems are physically connected, a change in one can affect measurements of the other. Those measurements can now reveal something about an initial state that was difficult to estimate in isolation. The theorem determines when this reduces the minimum mean-squared error for a chosen initial-state quantity, with the same prior, sensors, sampling times, and noise law. Appendix C.1 proves these standard Gaussian-conditioning results (Särkkä, 2013; Qian et al., 2021). The following corollary gives a concrete example: observing one system reduces uncertainty about an independently initialized hidden state of the other.

Corollary 3.5 (Strict benefit for an independently initialized hidden mode)**.** *Consider two scalar components with known finite $d_1, d_2 \geq 0$ and k:*

$$\dot{x}_E = L_k x_E, \qquad L_k = \begin{bmatrix} -d_1 & -k \\ k & -d_2 \end{bmatrix}, \\ C_{\text{obs}} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad P_0 = \sigma_0^2 I_2, \quad \sigma_0 > 0. \tag{42}$$

Write $E_k(s) = e^{L_k s}$ and form G_k from this flow using Equation (36). Under the sampled observation assumptions above, for $e_2 = (0,1)^\top$,

$$0 < \text{MSE}_k(e_2) < \text{MSE}_0(e_2) = \sigma_0^2 \\ \iff \quad G_k e_2 \neq 0. \tag{43}$$

In isolation, the first component's measurements reveal nothing about the independently initialized hidden coordinate. With a connection, **its error decreases exactly when at least one sampled measurement responds to it.** Unequal damping and full-state observability are unnecessary, but nonzero coupling alone is insufficient: sampling can miss the response. With correlated initial states, isolation can already be informative, and coupling can increase the error. Independence is an assumption in energy coordinates.

Appendix C.2 proves the corollary. Appendix C also treats missed responses, continuous noisy observations, and correlated priors; Appendix C.4 gives the comparison condition and a counterexample. The link to C-PHAST's initialization in Algorithm 1, step Q01, is detailed in Appendix C.6. Candidate predictions share the actual recorded history; hypothetical connections provide no new observations. The result is an optimal known-linear benchmark; achieved-error guarantees for nonlinear initializers require separate analysis and validation.

Together, these results show how unchanged components can gain capabilities through their interactions: a connection can increase expected energy dissipation and make hidden states easier to estimate. These collective benefits depend on how the components are connected and need not be present when they operate separately.

3.7. When Are Predicted Comparisons Reliable?

Sections 3.6.1–3.6.2 describe how interconnection affects dissipation and state inference in known linear systems. With learned laws, estimated starting states, and numerical integration, prediction errors may hide a limit violation or reverse a ranking. We ask when a comparison remains reliable despite those errors.

C-PHAST reuses a fixed set of component types. A new system may have more or fewer components, different connections, or different instance properties, such as a robot's mass or geometry. Each type keeps the same number of state variables and shared model laws. Further state-estimator training must be reported separately. Appendix A.10 lists the states modeled in each domain and the guarantees, including the separately trained recovery model (Section 4.5).

From initialization to forecast error. Use matched inputs and mode schedules, express true and predicted states in common coordinates, and assume both trajectories stay in the specified invariant region. A model step must multiply state separation by at most $q \in (0,1]$ in a fixed Lyapunov metric (a weighted state norm). Let ϵ_{lift} bound starting error and $\bar{\delta}$ the one-step mismatch between physical and modeled dynamics. Theorem E.2 then bounds the state error e_h after h steps:

$$e_h \;\leq\; \overbrace{q^h\,\epsilon_{\text{lift}}}^{\text{propagated initialization error}} + \underbrace{S_h(q)\,\bar{\delta}}_{\text{accumulated dynamics error}}, \qquad S_h(q) = \tfrac{1-q^h}{1-q} \;\; (q<1), \quad h \;\; (q=1), \tag{44}$$

In Equation (44), the first term carries starting error forward; the second accumulates errors in component laws, coupling, applied-input response, and integration. For $q < 1$, the first decays and the total bound stays finite as the horizon grows. The numerical term requires third-order state consistency: one step approximates the exact model evolution to $\mathcal{O}(\Delta t^3)$. Contraction must be checked in the operating region; passivity alone does not imply it. This gives no experimental guarantee across arbitrary network sizes.

For a second-order mechanical step without a mode switch and with matching coupling and Hamiltonian masses, Theorem D.1 gives $H(x_{t+1}) - H(x_t) = W_t^{\text{port}} - D_t + \mathcal{O}(\Delta t^3)$, so model-energy change equals supplied work W_t^{port} minus dissipation D_t, up to numerical error. Partial sensing adds state-estimation and coordinate-conversion errors. Dissipation

must cover the accounting error for energy increase not to exceed supplied work (discrete passivity). For electrical midpoint integration, Proposition D.2 bounds third-order one-step error plus error from stopping the implicit solves after finitely many iterations; exact energy preservation is not claimed.

From forecast error to a reliable comparison. State errors affect physical readouts and their summaries over the prediction horizon. Bounds on these operations give consequence errors (Proposition F.1). For the score in Eq. (20), let $\hat{J}_w(a)$ and $J_w(a)$ be predicted and physical costs, with $\Delta_J(a)$ bounding $|\hat{J}_w(a) - J_w(a)|$. Write a limit as $g_j(a) \le 0$, such as peak current minus its allowed maximum. Its predicted value is $\hat{g}_j(a)$, with $\Delta_{g_j}(a)$ bounding $|\hat{g}_j(a) - g_j(a)|$. Corollary F.6 gives

$$\begin{aligned} \overbrace{\hat{g}_j(a) + \Delta_{g_j}(a)}^{\text{prediction + error bound}} \le 0 \ \forall j \quad &\Longrightarrow \quad a \text{ satisfies the stated limits,} \\ \underbrace{\hat{J}_w(a) + \Delta_J(a)}_{\text{upper score bound}} < \underbrace{\hat{J}_w(b) - \Delta_J(b)}_{\text{lower score bound}} \quad &\Longrightarrow \quad J_w(a) < J_w(b). \end{aligned} \tag{45}$$

The first test keeps candidates within every limit after allowing for error. The second requires one candidate's largest possible cost to be below the other's smallest. Include errors in assigned starting states, dynamics, integration, and readouts; probabilistic bounds must hold simultaneously across relevant candidates and times. Inconclusive tests give no verdict; an empty retained set gives no certified choice.

If any candidates remain, the one with lowest predicted cost has physical cost at most $2 \max_a \Delta_J(a)$ above the best in that set. Comparison with all feasible candidates additionally requires retaining a true optimum (Proposition F.5). These are guarantees for the stated region, constraints, and horizon, not repeated closed-loop safety. The known-law three-DGU example certifies six of nine candidates under a sampled-current limit despite prescribed initialization and resistance errors; larger uncertainty makes the test inconclusive (Appendix F.3).

Candidates inherit uncertainty from the same starting state. Their costs depend on common unknown quantities, whose effects may cancel in the difference. For affine state assignments, known affine dynamics, and quadratic scores, Proposition F.8 bounds that difference while retaining the effects of rewiring. This can resolve rankings despite overlapping individual intervals. The known-law example certifies 74 of 81 comparisons versus 62 using separate polynomial score bounds, under the same current-limit checks at sampled times (Appendix F.4). The predictor and each candidate's feasibility checks stay unchanged. A small average error can hide a large error in one component. Proposition F.7 therefore bounds each component's error separately. A bound independent of component count requires uniform bounds on starting errors, one-step sensitivity, and model mismatch; pH structure alone does not supply these conditions or contraction.

Appendices D–F give the hypotheses, proofs, and constants. The experiments next test component learning, reuse, incomplete sensing, and decisions, beginning with an isolated electrical component before composition. Learned-model certification additionally needs valid error bounds across proposed configurations; the prediction and recovery experiments do not establish them.

4. Experiments and Results

The electrical attachment study anchors the evaluation in the question of Figure 1: keep the existing wiring fixed, add a DGU and line at one of the proposed locations, and predict the resulting response under prescribed inputs. We also examine initialization from voltage-only histories. Table 2 connects this setting to the supporting robotics questions introduced in the introduction. First we change the embodiment or component count; then we keep the robot fixed and vary command inputs. Spot logs test physical readouts on recorded hardware behavior.

These tests use distinct model realizations. Electrical networks compose DGU and line states; the arm, robot-transfer, and FHN studies reuse joint, leg, and cell models, respectively. The recovery predictor is trained separately and also advances the robot body, whereas the Spot model predicts consequences directly from telemetry and behavior descriptors. Thus the studies share the component-model construction or consequence interface, not a single checkpoint. Appendix A.9 maps each realization to its evidence; Appendices L and M specify the two decision-facing predictors.

Table 2. **What stays fixed, what changes, and why each study is included.** Electrical attachment is the anchor; the robotics studies examine supporting questions about reuse, decisions, and physical readouts.

Study	What stays fixed?	What changes?	Question
Electrical attachment Section 4.1	Existing wiring and learned component laws	Where the new DGU and line attach	Can we predict the augmented system before making the connection?
ANYmal-D to Go2 Section 4.3	Learned predictor and four-leg joint layout	Physical embodiment: geometry, inertia, actuation, and contact response	How well does the model transfer when the body changes?
Arm-count transfer Section 4.4	Shared local models	Number of joint components	Can those models predict a larger or smaller assembly without retraining?
Go2 recovery Section 4.5	Robot body and interconnection	Candidate command inputs	Can predicted responses help choose a useful recovery command?
Spot logs Section 4.6	Hardware platform	Recorded behaviors and scoring priorities	Can we predict and score physical consequences recorded on real hardware?

Each study reports the error or outcome relevant to its question: electrical voltage/current RMSE, mechanical joint-position MSE, recovery and survival, or agreement with measured consequences. The electrical studies evaluate prediction without executing a topology switch; the Go2 decision study executes one selected command. Reported rollout stability is empirical, not a theorem-level guarantee. Appendix G collects the observation contracts and evaluation conventions; the following appendix studies mirror the result order below and keep each protocol with its supporting diagnostics.

4.1. Learning Electrical Component Laws

We first test whether learning improves the local responses that will be composed. An isolated distributed generation unit (DGU) has a nonlinear capacitor; all learners receive the circuit structure and fully observed calibration trajectories, but not the numerical constitutive laws. The comparison includes a fitted-linear control, two C-PHAST training schedules, and fresh fits of the authors' compositional pHNN (Neary & Topcu, 2023) and N-PHDAE (Neary et al., 2024) implementations. C-PHAST uses a single-DGU configuration, testing component learning before composition.

Five independent fits use fresh data and random streams. Each has 32 training, eight validation, and 16 test trajectories. Calibration is noiseless and contains charge, flux, node voltages, source current, and applied inputs.

The two losses test different aspects of the learned laws. The constraint loss compares their voltage, current, and resistive relations with measurements at the same instant, without time integration. The prediction loss advances the model one timestep and compares its readout with the next measured sample. Matching the instantaneous relations need not give accurate predictions, so we test whether fitting those relations first helps subsequent prediction training. *Combined-loss training* (joint-only) optimizes both terms together for all 100,000 updates. *Constraint-first training* (staged) uses 25,000 constraint-only updates followed by 75,000 combined-loss updates. Equation (26) gives the loss weights and regularization. Both schedules train the same C-PHAST architecture and all eligible local-law parameters, sharing data-fitted starting weights and the minibatch sequence within each fit. N-PHDAE uses the same two-stage update counts with its own constraint and prediction losses. The local model of compositional pHNN uses its original charge/flux transition loss for all 100,000 updates; Appendix H specifies its architecture and optimizer.

We evaluate final checkpoints on five-second open-loop rollouts under fixed inputs, reporting voltage and inductor-current RMSE. A failed path makes the fit's all-path score unavailable. Appendix H gives the component laws, optimizer settings, and linear-DGU control.

Both C-PHAST schedules have lower voltage and current errors than the fitted-linear control and compositional pHNN in all five fits, and than N-PHDAE in all four complete comparisons (Table 3). Constraint-first C-PHAST reduces voltage RMSE by 24.3–90.3% and current RMSE by 24.0–97.5% relative to N-PHDAE in those comparisons. N-PHDAE also improves on linear in its four complete fits; its remaining fit is unavailable because the initial algebraic consistency solve fails on one test path. The failure is retained and reproduced by checkpoint replay. Compositional pHNN completes all 80 test paths. Its voltage RMSE ranges from 46.98 to 67.56 mV and its current RMSE from 25.22 to 44.27 mA; both improve on

Table 3. **Learning unknown nonlinear DGU laws.** C-PHAST is evaluated in its single-DGU configuration; all learners receive circuit structure and full-state calibration data, but not numerical constitutive laws. Combined loss fits physical-consistency and prediction terms together; constraint-first begins with physical-consistency pretraining. Each cell is capacitor-voltage (e_3) RMSE (mV) / inductor-current RMSE (mA); lower is better. Errors pool the 16 held-out, five-second rollouts within each fit and exclude time zero. A fit is the repetition unit. The median and range include all five fits; a failed path makes the entire N-PHDAE fit and its five-fit summary unavailable. Compositional pHNN retains its original storage-transition loss; training procedures differ as detailed in Appendix H. Best available errors are bold within each row and metric.

Fit	Fitted linear	Fresh N-PHDAE	Compositional pHNN	C-PHAST Combined loss	C-PHAST Constraint-first
1	100.6 / 54.38	1.353 / 2.327	58.83 / 44.27	**0.1282 / 0.05257**	0.131 / 0.0581
2	102.2 / 57.12	1.427 / 2.211	51.7 / 36.27	0.3435 / 0.2245	**0.3428 / 0.1857**
3	81.99 / 42.61	unavailable[†]	46.98 / 25.22	1.666 / 1.103	**1.665 / 1.101**
4	94.71 / 54.89	4.224 / 2.507	67.56 / 34.8	**3.192 / 1.855**	3.197 / 1.905
5	103.2 / 63.24	10.01 / 6.172	53.78 / 29.39	1.225 / **0.7399**	**1.216** / 0.746
Median	100.6 / 54.89	unavailable[†]	53.78 / 34.8	1.225 / **0.7399**	**1.216** / 0.746
Minimum	81.99 / 42.61	unavailable[†]	46.98 / 25.22	**0.1282 / 0.05257**	0.131 / 0.0581
Maximum	103.2 / 63.24	unavailable[†]	67.56 / 44.27	**3.192 / 1.855**	3.197 / 1.905

[†]Fit 3: the initial algebraic consistency solve failed on one of 16 paths. Across all fits, N-PHDAE completed 79/80 test paths; each C-PHAST arm and compositional pHNN completed 80/80. Both C-PHAST schedules have **2,407 total parameters**, of which 2,404 are optimized; N-PHDAE has **31,203 total parameters**, all optimized; compositional pHNN has **1,188**, all optimized.

the fitted-linear control in every fit. Its 1,188 parameters are fewer than C-PHAST's 2,407; the approximately thirteenfold parameter reduction applies specifically to N-PHDAE's 31,203.

Constraint-first training has no consistent final-accuracy advantage over combined-loss training; the paired changes are reported in Appendix H. These comparisons support predictive improvement beyond the fitted-linear approximation under full-state calibration and fixed inputs. They compare complete fitting procedures with different initializations, function families, losses, optimizers, and integrators, rather than isolating which design choice causes the cross-method gain.

Nonlinear composition under proposed connections. We next compose learned nonlinear DGUs under two proposed additions to a four-unit network. Each candidate $a = (\kappa, u_{0:N_{\text{pred}}-1})$ adds the same DGU and line at a different receiving bus, with supplied wiring and known line laws. Each method converts the same full physical starting state through its learned laws, retains the existing components' storage states, and assigns the supplied states of the added components. The laws remain fixed for 200 prediction steps of 0.01 s under held inputs. Sixteen starting states for each of five fitted models give 160 candidate trajectories per method. This evaluation uses supplied present states rather than measurement histories.

Table 4 separates two training settings. In panel A, compositional pHNNs (Neary & Topcu, 2023) and C-PHAST each receive 32 fully observed local DGU trajectories and 100,000 updates per fit, with no network calibration. Compositional pHNN retains the authors' state-transition objective; C-PHAST uses constraint-first training with its fitted initialization and constitutive and prediction objectives. C-PHAST integrates the inter-component coupling by explicit midpoint. Their initializations, function families, and integrators also differ, so the comparison evaluates complete learning procedures. The compositional pHNN and C-PHAST local models have 1,188 and 2,407 parameters, respectively. Appendix H.1 describes the electrical adapters and numerical checks.

C-PHAST has lower voltage, DGU-current, line-current, and sampled peak-current errors in all four complete comparisons. Its voltage RMSE is 0.124–0.251 mV, compared with 10.43–13.02 mV for compositional pHNNs in those fits. In fit 3, compositional pHNN's learned-law initialization fails for ten of 32 candidate trajectories; its remaining errors are conditional on successful initialization and are not paired with C-PHAST's complete-fit mean. Across the planned trajectories, compositional pHNNs complete 150/160 and C-PHAST completes 160/160. The methods correctly order 74/75 and 80/80 valid candidate pairs, respectively; five compositional pHNN pairs remain unavailable. These previously examined starting states support a descriptive comparison, and the failures remain in the denominator.

Panel B reports a separate comparison with N-PHDAE after both methods receive additional calibration on three- and four-node source networks; C-PHAST uses Euler coupling throughout that study. C-PHAST reduces mean voltage, DGU-current, and line-current RMSE by 64.06%, 70.25%, and 28.50%, respectively, relative to corrected N-PHDAE. The uncorrected

N-PHDAE integrator produces finite trajectories but fails the declared algebraic residual check on all 160; correcting its algebraic state after each step yields 160 qualified trajectories. Both N-PHDAE variants use the study's initialization and source-network fitting extensions; the corrected variant additionally changes the integration procedure. Despite the trajectory-RMSE gains, C-PHAST does not meet this study's prespecified peak-current improvement criterion: the mean reduction is 3.62%, with mixed effects across fits. Both qualified methods correctly order all 80 candidate pairs. The different calibration and numerical protocols prevent a pooled ranking with panel A.

Table 4. **Nonlinear composition with fixed learned laws under two training settings.** The first three error columns are voltage and current RMSE; peak error is the absolute error in the largest sampled absolute line current. All exclude time zero. Panel A averages per-trajectory errors within each fit; panel B averages the five fit means after additional source-network calibration. C-PHAST uses explicit-midpoint coupling in A and Euler coupling in B. Each fit has 16 starting states and two candidates (32 trajectories). Ordering counts correct orderings among valid, reference-resolved pairs; reference gaps at most 0.002 mA are unresolved. Bold marks lower errors only within complete panel-A comparisons. The panels do not support a pooled ranking.

Fit / method	Voltage (mV)	DGU current (mA)	Line current (mA)	Peak error (mA)	Qualified branches	Correct orderings
A. Component-only training: five separate fits						
1 / Compositional pHNN	13.02	21.13	15.15	8.929	32/32	16/16
1 / C-PHAST	**0.1237**	**0.04041**	**0.1337**	**0.05838**	32/32	16/16
2 / Compositional pHNN	12.48	20.5	15.13	8.036	32/32	16/16
2 / C-PHAST	**0.2512**	**0.08510**	**0.2744**	**0.09036**	32/32	16/16
3 / Compositional pHNN[†]	8.033	11.57	8.544	5.109	22/32	11/11
3 / C-PHAST	0.3859	0.1393	0.3652	0.3918	32/32	16/16
4 / Compositional pHNN	10.43	16.01	14.48	6.92	32/32	16/16
4 / C-PHAST	**0.1388**	**0.05030**	**0.1610**	**0.1068**	32/32	16/16
5 / Compositional pHNN	11.19	17.68	14.65	7.842	32/32	15/16
5 / C-PHAST	**0.2122**	**0.07449**	**0.1624**	**0.08842**	32/32	16/16
B. Additional source-network calibration: means across five fits						
N-PHDAE, uncorrected[‡]	0.4123	0.1665	0.3549	0.2763	0/160	–
N-PHDAE, corrected	0.4116	0.166	0.3549	0.2763	160/160	80/80
C-PHAST	0.1479	0.0494	0.2538	0.2663	160/160	80/80

[†]The compositional pHNN fit-3 errors use only 22 successful trajectories; ten initialization failures exclude five of its 16 candidate pairs. This row is conditional, not a paired comparison with the complete C-PHAST fit. No panel-A five-fit aggregate is reported. [‡]The gray row summarizes finite but algebraically unqualified N-PHDAE trajectories; its 80/80 finite-path orderings are diagnostic and are not counted as qualified orderings. In panel B, the prespecified peak-current improvement criterion is *not met*, and both qualified methods order 80/80 pairs correctly.

Supplied-state forecasts with original baselines. Panels A and B evaluate the baselines inside the candidate protocol, and panel B adds study-specific N-PHDAE extensions. A further comparison keeps both baselines' original objectives and solvers. On the fixed three-DGU network of Section 4.2, five fits each of compositional pHNN (Neary & Topcu, 2023), N-PHDAE (Neary et al., 2024) and C-PHAST share each fit's calibration bank and 10,000 training updates, then forecast 2 s from a supplied state under recorded inputs. The supplied state is either the true charges and fluxes or the true voltages and currents, which each method converts through its learned laws; the two conditions are scored separately. The baselines use their authors' RK4 solvers, and N-PHDAE uses its authors' repeated-consistency option, the only tested N-PHDAE setting whose completed forecasts pass the algebraic residual check (Appendix H.2). The C-PHAST fits are those of the CT-SUBNET comparison and use Euler coupling.

From supplied charges and fluxes, C-PHAST has lower voltage and current error than compositional pHNN in all five fits, and than N-PHDAE in the four fits where N-PHDAE completes every forecast (Table 5). Its five-fit mean current RMSE is 0.68 mA, compared with 81.21 mA for compositional pHNN; N-PHDAE has no complete five-fit mean because fit 2 completes one of eight forecasts. C-PHAST completes all 80 forecasts across both conditions, compared with 59 for each baseline, and every baseline failure with supplied voltages and currents occurs during state entry. Shared data and update counts do not match objectives, compute or convergence, and the test bank was inspected beforehand, so this comparison is descriptive.

Table 5. **Supplied-state forecasts with the baselines' original objectives and solvers.** Fixed three-DGU network, 2 s forecasts from the true charges and fluxes under recorded inputs. Each cell is the mean episode RMSE within a fit; voltage pools the three DGU voltages and current pools three DGU and two line currents. N-PHDAE uses its authors' repeated-consistency option. Each cell averages eight completed forecasts except the daggered N-PHDAE fit-2 cells, which use one. Bold marks the lowest error in each fit among methods that complete all eight forecasts; the five-fit mean is the mean ± sample standard deviation of fit means. The lower block counts completed forecasts out of 40 per method and supplied-state condition. The test bank was inspected before this analysis, so the comparison is descriptive.

	Compositional pHNN		**N-PHDAE**		**C-PHAST**	
Fit	V (mV)	I (mA)	V (mV)	I (mA)	V (mV)	I (mA)
1	169.71	45.64	3.84	11.02	**0.90**	**0.63**
2	144.92	48.31	1062.90†	289.75†	**1.35**	**0.89**
3	183.61	91.85	3.67	3.35	**0.81**	**0.35**
4	212.46	77.81	3.97	9.82	**1.33**	**0.90**
5	254.14	142.45	6.83	16.14	**1.06**	**0.63**
Five-fit mean	192.97 ± 42.01	81.21 ± 39.44	–	–	**1.09 ± 0.25**	**0.68 ± 0.23**
Completed forecasts						
True charge/flux	40/40		33/40		**40/40**	
True voltage/current	19/40		26/40		**40/40**	

Linear ten-DGU composition. We separately compare composition on the linear-DGU microgrid task of Neary et al. (2024). Each method copies one frozen local model to ten DGU instances and connects them through 45 known RL lines. No parameters are fitted on the network, and no intermediate reference states are supplied during its five-second rollout. Table 6 reports capacitor-voltage RMSE over the ten nodes. The composed PHAST model reduces error from N-PHDAE's 14.27 mV to 0.475 mV, approximately thirtyfold, and closely matches the known-law control.

Table 6. **Ten-DGU composition without network retraining.** Voltage RMSE pools ten capacitor-voltage channels and 499 future samples from one five-second trajectory. Each learned method reuses one frozen component checkpoint across all nodes. The known-law PHAST row is a numerical control, not a learned comparator. Methods retain their own integrators and different substep counts; this is not a matched-compute comparison.

Model	**Voltage RMSE (mV) ↓**
N-PHDAE	14.26693
Composed PHAST	**0.475286**
PHAST, known laws	0.475253

The error of both PHAST realizations decreases with finer integration, while N-PHDAE's changes little (Appendix H). This numerical check supports the measured reuse result for one network and checkpoint per method, without isolating an architectural advantage. The nonlinear composition evidence is reported separately in Table 4.

4.2. Prediction from Incomplete Measurements

Comparison with CT-SUBNET. Can the models predict currents from voltage measurements alone at deployment? We compare C-PHAST with CT-SUBNET (Beintema et al., 2023) on a fixed three-DGU chain. Each of five paired fits receives the same 16 training and eight validation trajectories, with voltage, current, storage, and resistor-drop labels available for calibration. Both train for 10,000 updates from fresh models, without local pretrained weights or four-node source data. At deployment, both receive 21 bus-voltage samples spanning one second, plus known source/load inputs. They predict three voltages and five hidden currents for two seconds without further measurements. Training is clean; testing uses clean histories and histories with 0.02 V Gaussian noise.

CT-SUBNET encodes the voltage and input histories into eight latent coordinates, advances them with its continuous-time dynamics, and decodes the physical outputs. Its decoder also predicts storage and resistor-drop labels during calibration; those channels never enter the history encoder. C-PHAST instead fits the initial state to the voltage history through its frozen component laws. The two procedures share source data and deployment observations, but C-PHAST explicitly uses supplied wiring and line physics. Their objectives, numerical integration, and state estimators differ (Appendix I.5).

Table 7. **Partial-sensing comparison on a fixed three-DGU network.** Two-second RMSE, mean ± sample standard deviation across five fit means, each averaging eight fresh episodes. Hidden-current RMSE pools three DGU and two line currents; the noisy condition is the primary outcome. All 40 forecasts per row are finite. Fits vary source data, fitted models, and test episodes; clean and noisy rows reuse episodes. Both methods use richer calibration labels and voltage-only deployment histories. Best RMSEs are bold within each condition.

σ_v (V)	Method	Voltage (mV)	DGU current (mA)	Line current (mA)	Hidden current (mA)
0.00	CT-SUBNET	61.67 ± 12.69	63.93 ± 11.31	106.77 ± 16.31	85.96 ± 4.01
0.00	C-PHAST	$\mathbf{1.89 \pm 0.74}$	$\mathbf{15.17 \pm 1.88}$	$\mathbf{16.03 \pm 3.17}$	$\mathbf{15.64 \pm 2.35}$
0.02	CT-SUBNET	61.58 ± 12.61	63.97 ± 11.25	107.50 ± 16.13	86.25 ± 4.55
0.02	C-PHAST	$\mathbf{10.89 \pm 1.49}$	$\mathbf{27.83 \pm 1.94}$	$\mathbf{27.06 \pm 2.47}$	$\mathbf{27.98 \pm 1.98}$

With noisy histories, CT-SUBNET has pooled hidden-current RMSE of 86.25 mA, compared with C-PHAST's 27.98 mA, a 67.6% reduction. C-PHAST has lower error in all five paired fits and completes all 80 forecasts across both noise conditions, as does CT-SUBNET (Table 7). With the true initial physical state supplied, C-PHAST's hidden-current RMSE is 0.90 mA; the remaining error therefore largely enters through state estimation. This comparison tests accuracy at fixed topology; it does not test graph-size transfer or isolate an architectural cause for the improvement.

Transfer to larger networks. A separate study asks whether calibrated components can predict a larger network when current sensors are absent. Five independently fitted dynamics models are calibrated on a three-node chain and four-node star, then frozen. Each is tested on four fresh episodes of each five-node target: a chain, star, and heterogeneous cycle. Deployment receives 21 samples of every bus voltage spanning one second, plus known source voltages, load currents, wiring, and line coefficients. DGU inductor currents and line currents are withheld, as are charge and flux: the five-node chain exposes five voltage channels while hiding nine current channels. Source calibration uses richer state and rate labels; this is voltage-only deployment, not learning the laws from voltages alone.

We compare two initializers for the same frozen dynamics. Numerical history estimation fits a starting state whose simulated voltages match the measured history. The learned observer is fitted only on source data with endpoint and ten-step prediction losses. It estimates voltage rates and line currents from the history, then uses the component laws to recover the storage state (Appendix I). A supplied-state diagnostic instead starts from exact simulator voltage/current values, isolating prediction error without noisy state estimation.

Independent Gaussian measurement noise has standard deviation $\sigma_v \in \{0, 0.02\}$ V, where 0.02 V is 2% of the benchmark's 1 V reference. All initializers receive the same measurement-noise draws when using observations; the simulator and scoring targets remain uncorrupted. After initialization, each model predicts for 0.5 and 2 s without further measurements, under known inputs held constant within an episode.

Table 8. **Voltage-only prediction on held-out electrical networks.** Mean RMSE ± sample standard deviation across five fit-level means, each averaging 12 episodes over the five-node chain, star, and heterogeneous cycle. All 60 planned target cases per row are finite; both horizons use the same rollouts. The supplied-state diagnostic bypasses noisy sensing and therefore repeats across noise levels. Numerical history estimation has lower mean error than the learned predictive observer in every displayed condition. Best history-based RMSEs are bold per condition; gray rows are supplied-state diagnostics.

σ_v (V)	Horizon (s)	Initialization	Voltage (mV) ↓	DGU current (mA) ↓	Line current (mA) ↓
0	0.5	Numerical history	$\mathbf{0.361 \pm 0.0400}$	$\mathbf{26.1 \pm 1.76}$	$\mathbf{25.4 \pm 2.33}$
		Learned + prediction	9.89 ± 2.69	77.3 ± 5.89	57.6 ± 19.3
		Supplied state (diagnostic)	0.0530 ± 0.0235	0.00628 ± 0.00204	0.0920 ± 0.0246
0	2	Numerical history	$\mathbf{1.02 \pm 0.143}$	$\mathbf{18.2 \pm 1.23}$	$\mathbf{18.3 \pm 1.76}$
		Learned + prediction	20.3 ± 5.29	50.9 ± 3.78	44.2 ± 11.6
		Supplied state (diagnostic)	0.141 ± 0.0261	0.0443 ± 0.0102	0.219 ± 0.0466
0.02	0.5	Numerical history	$\mathbf{8.42 \pm 0.787}$	$\mathbf{42.9 \pm 3.38}$	$\mathbf{36.6 \pm 2.65}$
		Learned + prediction	17.8 ± 2.87	120 ± 10.3	57.0 ± 17.7
		Supplied state (diagnostic)	0.0530 ± 0.0235	0.00628 ± 0.00204	0.0920 ± 0.0246
0.02	2	Numerical history	$\mathbf{10.0 \pm 0.937}$	$\mathbf{29.3 \pm 2.05}$	$\mathbf{27.7 \pm 2.37}$
		Learned + prediction	28.7 ± 5.03	78.6 ± 6.97	51.8 ± 8.19
		Supplied state (diagnostic)	0.141 ± 0.0261	0.0443 ± 0.0102	0.219 ± 0.0466

With noisy measurements and a two-second horizon, numerical history estimation achieves 10.0 mV voltage RMSE, 29.3 mA DGU-current RMSE, and 27.7 mA line-current RMSE; the learned predictive observer gives 28.7 mV, 78.6 mA, and 51.8 mA (Table 8). The much smaller supplied-state errors indicate that initialization is the main remaining limitation. The learned-observer ablation compares rate-only, endpoint-supervised, and endpoint-plus-prediction training. Relative to rate-only supervision, the endpoint-plus-prediction recipe reduces noisy two-second DGU-current and line-current errors by 32.7% and 12.8%, but increases voltage error by 27.3%. Relative to matched endpoint-only training, adding the ten-step prediction loss changes the primary mean errors by at most 0.233%. Appendix I gives these ablations and all 1,200 planned, finite evaluations, including the source and symmetric-cycle diagnostics.

Current recovery also depends on observability: a separate known-law control admits a circulating current on a symmetric cycle that leaves all bus voltages unchanged (Appendix I). The primary target cycle is heterogeneous and is evaluated separately from that control. The larger-network results measure transfer under the specified inputs and topologies. They compare C-PHAST initializers; the external-baseline comparison in Table 7 uses the fixed three-node network. The known-law examples in Appendices F.3 and F.4 test decision certificates with prescribed uncertainty, not uncertainty inferred by these learned observers. Section 4.5 evaluates an executed choice separately on Go2; learned, voltage-initialized electrical candidate selection is not tested here.

4.3. Cross-Robot Transfer

Can a model trained on one body predict another without target retraining? Instead of changing an attachment, we change the physical body while keeping the joint layout and learned predictor fixed. ANYmal-D and Unitree Go2 share four legs with three actuated joints per leg, but differ in mass distribution, geometry, inertia, contact response, and actuation. The model receives joint-position history and normalized joint-position commands, not measured torques; the robots' action scales and PD gains differ. This tests fixed-layout embodiment transfer under a shared command interface, rather than subsystem-count transfer.

Each robot supplies 500 policy-driven trajectories of 200 steps at $\Delta t = 0.02\,\mathrm{s}$. Models train on 350 ANYmal-D trajectories, validate on 50, and transfer to 100 held-out Go2 trajectories. All learned rows in Table 9 use five seeds, 50 epochs, the same data split and AdamW schedule, and one-step MSE without energy or observer auxiliary losses. Shared-GRU, MPNN, and DeepSets test alternative per-leg models; monolithic PHAST, a fixed-dimension GRU, and a Neural ODE provide fixed-layout comparisons. Model capacities differ. Appendix J.1 specifies data collection and Appendix J.2 records the artifact provenance and alternate training recipe.

We evaluate fifty-step joint-position predictions in two modes. *Takeover* infers the state from five observed frames and then runs open-loop, as required when scoring a hypothetical future. *Refreshed* evaluation uses a ten-frame context and new measurements every five steps, as in monitoring an actual system. These are distinct observation protocols, not an isolated ablation of refresh frequency. Both report coordinate-averaged MSE in rad^2. Takeover means exclude per-robot seed errors at or above 10^3; refreshed means retain all five seeds. Exclusion counts accompany the results.

C-PHAST attains the lowest Go2 takeover error, 0.073, compared with 0.112 for the fixed-dimension GRU, 0.145 for monolithic PHAST, and 0.178 for Shared-GRU. Persistence gives 0.152. The target advantage does not come from better source fit: the fixed-dimension GRU achieves 0.005 on ANYmal-D versus C-PHAST's 0.045. The transfer error ratio is $1.6\times$ for C-PHAST, compared with $21\times$ and $3.7\times$ for the fixed-dimension GRU and monolithic PHAST, and $13\times$ for Shared-GRU. Figure 5 follows these comparisons across prediction horizons. MPNN and DeepSets have much larger Go2 rollout errors; their results and the excluded seeds remain visible in the table.

With refreshed observations, C-PHAST also has the lowest Go2 MSE, 0.015, versus 0.021 for DeepSets, 0.022 for monolithic PHAST and MPNN, and 0.025 for persistence. C-PHAST therefore transfers most accurately in both open-loop prediction and measurement-refreshed tracking. Its advantage over monolithic PHAST and unstructured per-leg models supports combining physical structure with reusable components, although these architectural comparisons do not isolate the contribution of each design choice. Earlier long-horizon diagnostics exposed over-smoothed locomotion predictions (Appendix J.2).

Table 9. **ANYmal-D to Go2 transfer without retraining.** Matched training across five seeds; fifty-step joint-position MSE in rad^2, mean $\pm$ population standard deviation. Takeover uses five context frames and no subsequent observations; refreshed evaluation uses ten context frames and observations every $K{=}5$ steps. Persistence follows the corresponding observation schedule. Superscript pairs give excluded takeover seeds out of five on ANYmal-D/Go2, respectively (MSE $\geq 10^3$); unmarked rows exclude none. Refreshed means include all seeds. These are rollout exclusions, not counts of failed training runs. Lowest MSE per column is bold, including ties at displayed precision.

Model	**Params**	**Takeover rollout** ($N_{\mathrm{pred}}{=}50$)			**Refreshed rollout** ($K{=}5$)		
		ANYmal-D	Go2	gap	ANYmal-D	Go2	gap
Compositional (per-leg):							
C-PHAST	40,034	0.045 ± 0.000	$\mathbf{0.073 \pm 0.001}$	1.6×	0.0066 ± 0.000	$\mathbf{0.015 \pm 0.000}$	2.3×
Shared-GRU $^{1/5,\,1/5}$	60,804	0.014 ± 0.003	0.178 ± 0.025	13×	0.048 ± 0.087	0.047 ± 0.048	1.0×
MPNN (per-leg)	42,692	0.638 ± 0.686	263 ± 361	—	0.0069 ± 0.000	0.022 ± 0.001	3.3×
DeepSets $^{0/5,\,1/5}$	26,116	0.071 ± 0.050	28.1 ± 45.7	—	0.0083 ± 0.000	0.021 ± 0.001	2.6×
Non-compositional (fixed 12-DOF input):							
Monolithic PHAST	34,601	0.039 ± 0.002	0.145 ± 0.016	3.7×	$\mathbf{0.0035 \pm 0.000}$	0.022 ± 0.000	6.4×
Fixed-dim GRU	43,020	$\mathbf{0.005 \pm 0.000}$	0.112 ± 0.005	21×	$\mathbf{0.0035 \pm 0.000}$	0.082 ± 0.003	23×
Neural ODE $^{1/5,\,3/5}$	14,860	357 ± 391	18.9 ± 13.9	—	0.080 ± 0.016	1.04 ± 0.35	13×
Reference (no learned parameters):							
Persist ($\hat{q}_{t+1}{=}q_t$)	0	0.045	0.152	3.4×	0.012	0.025	2.0×

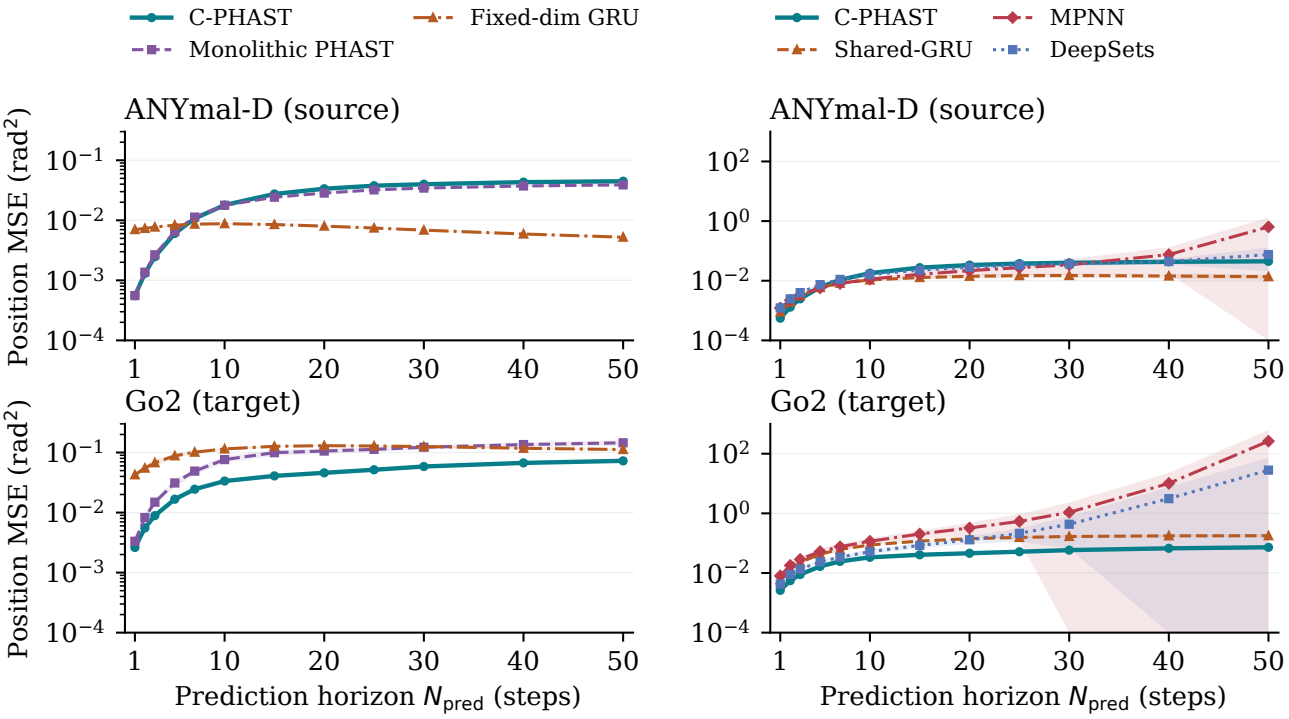


Figure 5. **Takeover error versus horizon.** C-PHAST has the lowest Go2 error at fifty steps, despite the fixed-dimension GRU's lower source error. Left: C-PHAST and the fixed-layout comparisons. Right: C-PHAST and the alternative per-leg models. Lines show means and shading shows $\pm$ one population standard deviation; shading below the logarithmic axis range is clipped. Each model uses one fixed cohort across horizons and robots: five seeds for C-PHAST, monolithic PHAST, the fixed-dimension GRU, and MPNN; four for Shared-GRU and DeepSets. A seed is omitted from both robot panels if its fifty-step error reaches 10^3 on either robot. Table 9 instead applies exclusions per robot.

4.4. Composition Across Subsystem Counts

Can the same local models be reused when the number of components changes? The preceding robot-transfer test keeps the joint layout fixed; here we change the number of repeated components. The arm models train at $N{=}4$ and deploy without retraining at other counts; a circuit example then tests the same reuse principle outside robotics. Fixed-dimension models require a changed input projection; the relevant comparisons are therefore models that share parameters across parts.

Planar arms. The controlled simulator has diagonal inertia, viscous damping, separable gravity, and a fixed skew-symmetric inter-joint coupling. Models observe joint positions and sinusoidal torque inputs, with 200 training trajectories and 50 test trajectories generated per count. Table 10 compares variants that differ in which equations governing each joint's dynamics are specified and which are learned, as well as in their training procedure. The KNOWN and PARTIAL labels refer to these local physics models, rather than supplied inter-joint coupling. At $N{=}8$, both give MSE 0.0014; the local-law-unknown variants give 0.2943 with one-step training and 0.0584 with rollout-aligned training. The latter two training procedures also differ in their observation history and energy and passivity penalties, so the comparison does not isolate the effect of rollout loss. In the dense learned-coupling construction, entries activated beyond the four-joint training block receive no source gradients. The result therefore tests reuse of local dynamics with new coupling entries not learned from source trajectories, not a learned rule for new interactions (Eq. (208)).

Table 10. **Robot-arm count transfer across local-physics and training choices.** Train at $N{=}4$ and test without retraining at each displayed count. The model estimates its initial state from five joint-position observations, then predicts $N_{\text{pred}}{=}50$ steps under the supplied torque inputs without further position measurements. Joint-position MSE is evaluated on 20 test trajectories per count and seed; values are means $\pm$ population standard deviations over three seeds. Lowest MSE per count is bold, including ties at displayed precision. Appendix K.1.3 specifies the training differences and coupling parameterization.

Params	Known local physics 13,015	Partial local physics 21,977	Unknown local physics one-step 23,386	Unknown local physics rollout-aligned 23,386
$N{=}2$	**0.0029 ± 0.0001**	0.0031 ± 0.0001	0.3208 ± 0.0008	0.0684 ± 0.0038
$N{=}3$	**0.0021 ± 0.0000**	0.0022 ± 0.0001	0.3191 ± 0.0012	0.0624 ± 0.0075
$N{=}4$	**0.0017 ± 0.0000**	0.0018 ± 0.0001	0.2920 ± 0.0012	0.0645 ± 0.0053
$N{=}5$	**0.0016 ± 0.0000**	0.0018 ± 0.0000	0.2474 ± 0.0015	0.0528 ± 0.0071
$N{=}6$	**0.0018 ± 0.0000**	0.0019 ± 0.0000	0.2961 ± 0.0012	0.0663 ± 0.0046
$N{=}8$	**0.0014 ± 0.0000**	**0.0014 ± 0.0000**	0.2943 ± 0.0003	0.0584 ± 0.0044

Repeated-block reuse also extends to coupled FitzHugh–Nagumo (FHN) excitable circuits (FitzHugh, 1961). A shared cell trained at $N{=}3$ is copied to $N \in \{2, 4, 5, 6, 8\}$ and the diffusive chain coupling is rebuilt. The partially specified C-PHAST model has nearly flat one-step MSE (6.6–8.2×10^{-6}), but N-PHDAE is more accurate at the fixed $N{=}3$ comparison (8.0×10^{-7} versus 8.2×10^{-6}). Appendix K.2 gives all regimes and rollout results, including the weaker fully learned model. FHN cells are active and their interconnection is dissipative, so this is evidence of recomposition outside robotics, not a test of the skew, power-preserving guarantee.

These studies establish reuse across changing component counts while showing where accuracy depends on the learned local laws and coupling. Appendix K.1 collects the controlled coupling, normalization, mass-prior, and power-residual diagnostics, together with the illustrative C-PHAST/MPNN trajectory comparison in Figure 11.

4.5. Forecasts for an Executed Recovery Decision

Do predicted consequences help choose a recovery command? After testing attachment, embodiment, and count changes, we fix the robot body and interconnection, vary command inputs, and independently test the selected command. On a policy-controlled Go2 under lateral pushes, we evaluate early warning and command selection separately. Figure 6 shows the policy's recovery boundary. Unlike the joint-position-only transfer benchmark, these decision queries receive current root and joint telemetry, contact/slip geometry, and declared terrain friction. Command selection uses a separately trained, frozen root-aware C-PHAST runtime with a body-plus-legs model and support-conditioned damping, rather than the joint-only transfer checkpoint. It advances body pose and velocity together with joint positions and velocities. For each candidate, the frozen policy converts a held velocity command into successive low-level actions using predicted observations; support and terrain context stay fixed during the forecast (Appendix L.2). The physical-critic interface evaluates supplied command choices, while the existing locomotion policy executes the selected command.

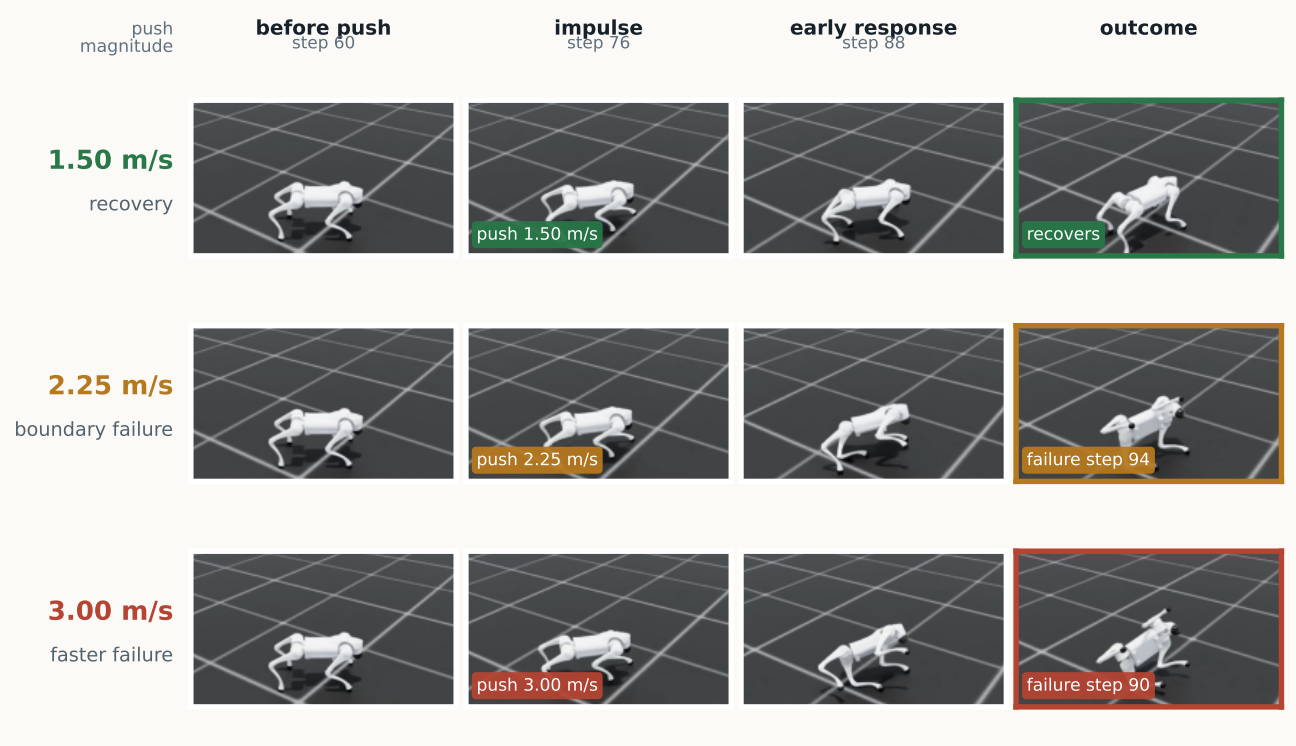


Figure 6. **Go2 response boundary under lateral pushes.** The same policy-controlled Go2 recovers from a $1.50\,\mathrm{m/s}$ lateral impulse, reaches a boundary-failure regime at $2.25\,\mathrm{m/s}$, and fails sooner at $3.00\,\mathrm{m/s}$. This frozen-policy response provides the perturbation envelope for the early-warning and single-decision command-selection tests.

Early warning. Across four push directions and five magnitudes, 14 of 20 frozen-policy runs terminate in simulator-level failure. C-PHAST warns before all 14, with 7/15.6/29 control steps of minimum/mean/maximum lead, and gives two false warnings among six recoverable runs. Scalar alarms are competitive: leave-one-direction-out calibration yields 14 detections and one false warning for both an IMU-tilt threshold and a linear measured-state alarm, with mean leads of 7.4 and 15.3 steps. Table 11 reports the full comparison. Detection alone therefore does not show an advantage for the richer forecast; the next test evaluates its use in choosing a command.

Table 11. **Early warning under Go2 lateral pushes.** Four directions and five magnitudes, under a frozen locomotion policy. Failure boundary is the smallest tested failing magnitude; lead is minimum/mean/maximum control steps before failure. Scalar and learned baseline thresholds use leave-one-direction-out calibration; C-PHAST uses its existing typed-veto signal. All 20 runs are retained. Best panel-B values are bold, including ties.

A. C-PHAST directional warning sweep

Push dir.	Failure boundary	Detected	False warnings	Lead steps
$+x$	≥ 1.75 m/s	3/3	1/2	14/15.3/16
$-x$	≥ 1.50 m/s	4/4	1/1	7/12.8/22
$+y$	≥ 1.75 m/s	3/3	0/2	11/15.3/18
$-y$	≥ 1.50 m/s	4/4	0/1	13/19.0/29
All	–	14/14	2/6	7/15.6/29

B. Aggregate alarm comparison

Alarm	Calibration	Detected ↑	False warnings ↓	Mean lead ↑
C-PHAST	existing typed veto	**14/14**	2/6	**15.6**
IMU tilt	leave-one-direction-out	**14/14**	**1/6**	7.4
Linear measured-state	leave-one-direction-out	**14/14**	**1/6**	15.3

Single-decision command selection. A lateral velocity impulse $\Delta v_y = \pm 2.75\,\mathrm{m/s}$ occurs at step 75; the command is selected at step 76. C-PHAST receives the measured state and nine supplied command labels, but neither the push label/magnitude nor future simulator state. It produces an eight-step forecast for each candidate. A regularized pairwise selector trained on eight development scenarios chooses once, and the command is executed for 24 control steps ($0.48\,\mathrm{s}$), after which the nominal command resumes without another selection. Appendix L.2 gives the command values, including the nominal/forward duplicate and the separate raw default-stance action.

The held-out set contains 20 scenarios: five seeds, two push signs, and normal or low-friction terrain. Among commands applied unchanged across all 20 scenarios, *right* has the highest mean survival score, 116.55 steps. The online selector achieves 121.80, a paired gain of 5.25 steps. Resampling the five seeds, with each seed's four scenarios kept together, gives a 95% bootstrap interval $[2.50, 7.05]$ for this gain. The trained selector remains fixed; the right-command comparator, chosen from these evaluation results, is also held fixed across resamples (Appendix L.2). Recovery is $10/20$, tying the right command, versus $8/20$ for nominal continuation. The outcome oracle over the same library reaches $12/20$. The online selector therefore improves the survival score while leaving recovery count unchanged relative to the best fixed command.

Table 12. **Single-decision command selection after a Go2 push.** Panel A independently executes the selected commands. The survival score counts control steps completed without triggering failure, stopping at the first failure; runs with no failure receive the maximum score of 140. The 140-step evaluation includes 20 settling steps before failure checks begin. Recovery means completing the evaluation without triggering failure. Panel B ranks executed branches on the same 20 held-out scenarios. Both selectors receive the measured state, candidate identity, predicted motion summaries, and maximum predicted energy growth; the full selector adds ledger, dissipation, port, support, and modal readouts. Longest counts choices tying the longest-surviving branch; shortfall is mean survival steps lost relative to that branch. Best non-oracle values are bold per panel, including ties; gray marks the oracle.

A. Independently executed decision rules

Decision rule	Recovery ↑	Mean survival ↑
Nominal continuation	$8/20$	114.65
Best fixed command	$\mathbf{10/20}$	116.55
C-PHAST typed selector	$\mathbf{10/20}$	**121.80**
Outcome oracle	$12/20$	129.55

B. Held-out branch-ranking ablation

Selector information	Longest ↑	Mean shortfall ↓
Reduced readouts	$12/20$	10.8
Full readouts	$\mathbf{14/20}$	**7.7**

The branch-ranking ablation isolates the additional readouts at the selector input: longest-survival selection rises from $12/20$ to $14/20$, and mean shortfall falls from 10.8 to 7.7 steps (Table 12). This is distinct from the independently executed full-selector comparison in panel A. The gap to the same-library oracle measures selection shortfall; the oracle's $12/20$ recovery count separately exposes the limits of the supplied commands. The full selector improves executed survival over the best fixed command; adding typed readouts improves branch ranking over the reduced-readout selector. The command library limits the recoveries that either selector can achieve in this one-decision setting.

4.6. Physical Consequences from Real Robot Logs

Can physical consequences be predicted for different behaviors on the same hardware? After the executed Go2 choice in simulation, Spot-with-arm logs test the physical readout and task-scoring interface on recorded hardware behavior. Six teleoperated ROS-bag episodes provide 1,070 overlapping windows, with 897 training and 173 held-out windows. Each query pairs two seconds of observed history with descriptors of a five-second behavior and predicts its recorded consequences. The energetic-channel results use a five-seed neural ensemble; the objective-score comparison uses a separate regression predictor of typed consequences (Appendix M). Both map window summaries directly to consequences, rather than advancing PHASTCore states. This is retrospective: some input descriptors, including realized yaw change and hand motion, are measured from the executed future. A live query would need those fields supplied or predicted before execution. The result tests the typed consequence interface on hardware data, not a closed-loop PHASTCore rollout or executed hardware selection.

A five-seed typed query model predicts positive, absolute, and locomotion work with Pearson correlations 0.973, 0.975, and 0.971, and battery draw with 0.943 (Table 13). Work targets integrate jointwise positive or absolute torque–velocity products; battery draw is mean discharge current, not electrical energy. These are telemetry-grounded targets, distinct from

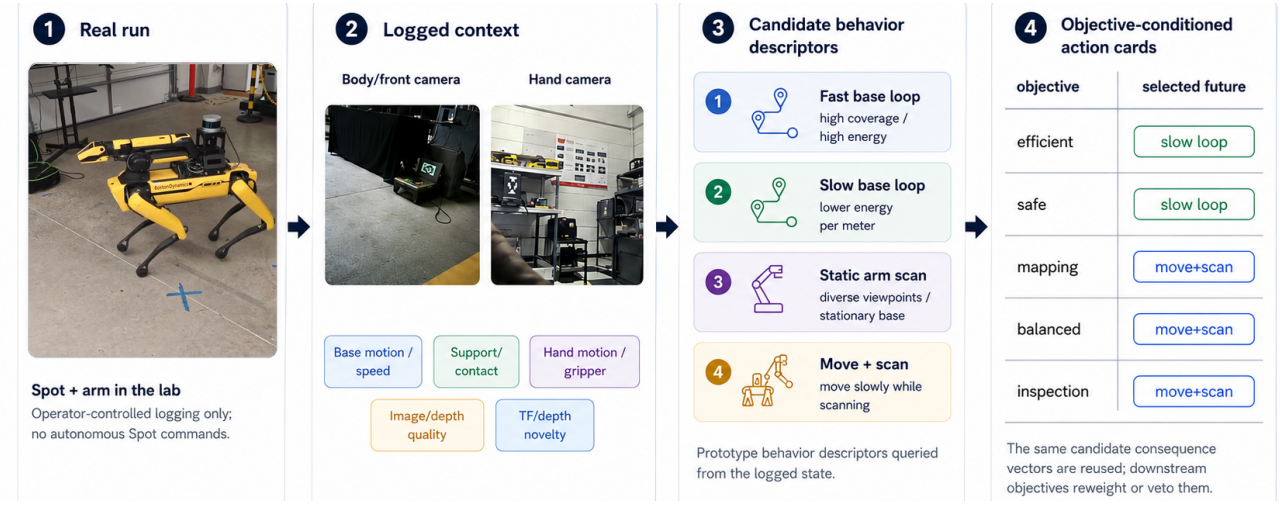


Figure 7. **Spot-with-arm logged consequence query.** Telemetry history and behavior descriptors condition predictions of work, battery draw, support/contact, and sensing consequences. The displayed action cards reuse one history with prototype behavior descriptors estimated from training episodes; reweighting predictions favors a slow loop for efficient/safe objectives and move-and-scan for mapping, balanced, and inspection objectives. This illustrates objective-dependent selection in the model, not measured outcomes for all alternatives from that state. Reported logged-window inputs include future-derived descriptors; no commands are sent to Spot.

the model-native pH accounting channels.

For action-card evaluation, the regression predictor's consequences are scaled using training-split 10–90 percentile ranges and combined with fixed objective weights. The resulting scores have mean Pearson/Spearman agreement $0.880/0.860$ across efficiency, safety, mapping, balance, and inspection objectives; per-objective ranges are 0.744–0.948 and 0.753–0.939. The fixed split uses 897 windows from four training bags and 173 from two test bags; it does not compare executed alternatives from a common state. Six-fold leave-one-episode-out evaluation tests all six bags and gives $0.652/0.545$.

Table 13. **Spot consequence and score agreement under logged replay.** Panel A reports direct energetic targets from the five-seed neural ensemble. Panel B reports the separate regression study's mean objective-score agreement across five objectives. The first four rows of panel B use the fixed four-bag/two-bag split; the final row uses six-fold leave-one-episode-out evaluation. Inputs include logged future-derived behavior descriptors. Higher is better. Best fixed-split agreement is bold; gray marks the separate six-fold evaluation.

A. Direct energetic targets		
Target	Pearson	R^2
Positive joint work	0.973	0.944
Absolute joint work	0.975	0.947
Locomotion work	0.971	0.939
Battery draw	0.943	0.878
B. Objective-score agreement		
Input/model or split	Pearson ↑	Spearman ↑
History only	0.767	0.737
Behavior descriptors only	0.780	0.778
History + descriptors, typed output	**0.880**	0.860
History + descriptors, scalar output	0.871	**0.875**
Leave-one-episode-out, typed output	0.652	0.545

Using history and descriptors improves on either input alone in the fixed held-out-bag evaluation. A direct scalar head is rank-competitive ($0.871/0.875$), so the demonstrated benefit of typed outputs is access to separate physical channels, not superior scalar ranking. Figure 7 illustrates how their weights can change the preferred behavior. Appendix M gives the descriptor audit, target definitions, objective weights, and ablations.

The decision and telemetry studies show two uses of the same output design: an external selector can compare predicted consequences, and its objective can change without collapsing those consequences into a fixed learned score.

5. Conclusion

Predicting a reconfigured system before the change is made requires component models that stay valid under new connections and an estimate of the existing system's current state. C-PHAST learns one storage-and-dissipation law per component type, estimates that state from partial measurements through those laws, and predicts proposed configurations without retraining. The strongest evidence is transfer to systems unseen in training. With the same data and starting states, it predicts unseen five-unit simulated microgrids with over 40-fold lower voltage error than prior compositional port-Hamiltonian networks. In simulation, a leg model trained only on ANYmal-D predicts Go2 zero-shot; its error grows 1.6-fold from ANYmal-D, versus 21-fold for a recurrent network. For known linear systems, we show when a connection lets one component's damping drain another's energy or sharpens a hidden-state estimate, and we give error margins for certifying the better configuration.

These capabilities were tested separately; we have not yet ranked proposed configurations, or checked line-overload limits, with learned laws initialized from voltages alone, and certified rankings assume known laws. Future work addresses both by carrying joint uncertainty over states and learned laws through connections, and by enlarging the decision from where a new unit joins a fixed network to the topology itself, tested independently. Transfer results favor component laws over whole-system models; that work would show whether such models can choose a change before it is made.

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How to read this appendix. For the operations behind Algorithm 1, start with Appendix A: it specifies state inference, numerical transitions, readouts, and calibration. Appendix B proves the continuous-time dissipation results for fixed linear–quadratic compositions and gives their limitations. Appendix C proves the corresponding state-inference results, including sampling limits and the correlated-prior counterexample. The guarantees in Appendices D–F then address finite-step energy accounting, deployment prediction error, and reliable candidate comparisons under their respective assumptions. For a particular experimental result, go directly to its supporting protocol: electrical component learning (Appendix H), voltage-only network prediction (Appendix I), cross-robot transfer (Appendix J), subsystem-count transfer (Appendix K), recovery decisions (Appendix L), or real-robot logs (Appendix M). These follow the order of Section 4; shared measurement and evaluation conventions are in Appendix G.

A. Domain Realizations and Reproduction Details

Algorithm 1 initializes the actual state once, assigns a starting state to each candidate, and returns its simulated consequences. This appendix specifies the operations left abstract there. We first give the port and state-assignment conventions, then the mechanical transition and observer, and the electrical transition and readout. Calibration and decision factors follow the operations that they train or evaluate. In particular, Algorithm 2 specifies a symmetric mechanical realization of Q04; Algorithm 3 gives the isolated DGU update. They are domain-specific realizations, not successive stages of the query. The electrical midpoint network transition is defined below; voltage-only observers and their fixed-dynamics evaluation appear in Appendix I.

A.1. Port assembly and query conventions

A.1.1. Equivalent port-assembly forms

Section 3.2 gives the full forced dynamics. With external ports closed, the component law is

$$\dot{x}_i = (J_i - R_i)\nabla H_i(x_i) + \mathbf{B}_i \hat{u}_i, \qquad \hat{y}_i = \mathbf{B}_i^\top \nabla H_i(x_i), \tag{46}$$

Here B_i denotes the declared coupling-port columns. Local/control and domain inputs are expressed in compatible port coordinates; channels may be concatenated with zero padding when necessary. The total input is

$$u_i = u_i^{\text{int}} + u_i^{\text{ext}}, \qquad u_i^{\text{int}} := \hat{u}_i, \tag{47}$$

with $u_i^{\text{ext}} = u_i^{\text{loc}} + u_i^{\text{dom}}$. For repeated components under a specified wiring rule, $\hat{C}^{(s)}$ denotes the coupling matrix assembled for s components. The superscript records the component count; it does not specify the topology on its own. The general notation in Section 3.2 is $\hat{C}$. Appendix K.1.3 gives the dense learned rule used in the count-transfer study. Let $x = [x_1; \dots; x_s]$ and let $\hat{B}, \hat{J}, \hat{R}$ be the block-diagonal matrices of B_i, J_i, R_i. For $H_{\text{total}} = \sum_i H_i$, the closed-external-port dynamics are equivalently

$$\dot{x} = \left(\hat{J} - \hat{\mathbf{B}}\hat{\mathbf{C}}^{(s)}\hat{\mathbf{B}}^\top - \hat{R}\right)\nabla H_{\text{total}}(x). \tag{48}$$

The internal coupling is skew and contributes no net power. The resulting power balance with external inputs is Eq. (10); for a mechanical momentum port, its loss term is $(\partial_{p_i} H_i)^\top D_i (\partial_{p_i} H_i)$.

A.1.2. Mechanical attachment maps

Changing a joint's angle changes the leverage of an applied force. $B_i(q_i)$ records that change. The mechanical counterpart of the electrical selectors in Equation (22) follows from attachment geometry. Let $x_i = (q_i, p_i)$ be canonical mechanical coordinates and let $r_i(q_i)$ give the position of an attachment in the coordinates of an applied force F_i. Virtual work gives $F_i^\top \delta r_i = \tau_i^\top \delta q_i$, so the generalized force is $\tau_i = (\partial r_i/\partial q_i)^\top F_i$. The collocated port construction is therefore

$$B_i(q_i) = \begin{bmatrix} 0 \\ (\partial r_i/\partial q_i)^\top \end{bmatrix}, \qquad \hat{u}_i = F_i, \qquad \hat{y}_i = B_i^\top \nabla H_i = \frac{\partial r_i}{\partial q_i}\dot{q}_i = \dot{r}_i. \tag{49}$$

Here $\partial_{p_i} H_i = M_i(q_i)^{-1} p_i = \dot{q}_i$ for the canonical mechanical model. The input is force, the output is attachment velocity, and $\hat{u}_i^\top \hat{y}_i = F_i^\top \dot{r}_i = \tau_i^\top \dot{q}_i$ is power. Known geometry determines B_i even when the local storage or damping law is learned. Rectangular maps and loss of rank at particular postures do not invalidate this power identity; the map is not inverted.

For example, a link of length ℓ rotating through an angle q from horizontal has tip height $r(q) = \ell \sin q$. For a vertical tip force F, $B(q) = [0; \ell \cos q]$, the joint torque is $\ell \cos q\, F$, and the port output is the vertical tip velocity $\ell \cos q\, \dot{q}$. This specifies the force's leverage, not the law generating the force. Contact stiffness, damping, and contact events require their own interface model. The identity momentum ports used in the robotics studies are the special case of directly applied generalized forces, not an implementation of this full attachment/contact construction. Their learned skew coupling is gyroscopic: it produces velocity-dependent forces with zero total coupling power, but has no interface storage or friction. The dense rule allows every pair of components to couple; entries beyond the training count receive no training gradients (Appendix K.1.3).

The assumption $H_{\text{total}}(x) = \sum_i H_i(x_i)$ also depends on the chosen component states. For example, a forearm's speed depends on both shoulder and elbow motion. Its kinetic energy generally cannot be split into one function of the shoulder state and another of the elbow state. Thus a model with independent joint energies approximates this mechanical coupling. This does not prevent summing physical body energies; those energies can depend on several joints. Assumption A5 and the robotics studies state the approximation used here.

A.1.3. Anchoring attachment and coupling scales

The internal dynamics contain products $B_i \hat{C}_{ij} B_j^\top$, not the factors separately. For any constant $\alpha > 0$, replacing B_1 by αB_1 and $\hat{C}_{12}$ by $\hat{C}_{12}/\alpha$ leaves both interaction terms in Equation (6) unchanged:

$$(\alpha B_1)(\hat{C}_{12}/\alpha) B_2^\top = B_1 \hat{C}_{12} B_2^\top, \qquad B_2 (\hat{C}_{12}/\alpha)^\top (\alpha B_1)^\top = B_2 \hat{C}_{12}^\top B_1^\top.$$

Internal-coupling trajectories alone therefore cannot distinguish these scalings. Known geometry or separately calibrated forcing can fix this ambiguity: rescaling B_1 also rescales its response to a fixed external input through the same port. This concerns the physical meaning of the factors, not the prediction of their unchanged product.

A.1.4. System specification

A system specification records the component types, their graph, measurement and readout interface, coordinate and command maps, actuator map, and instance metadata:

$$\mathcal{S} = (s, \{\alpha_i\}_{i=1}^s, G_{\text{target}}, \mathcal{O}_{\text{target}}, \chi^q_{\text{target}}, \chi^u_{\text{target}}, \mathcal{A}_{\text{target}}, \Xi_{\text{target}}).$$

These fields describe both the actual system $\mathcal{S}_{\text{now}}$ and each proposed $\mathcal{S}_\kappa$. A type α has a fixed local dimension; the mechanical examples have $N = \sum_i d_{\alpha_i}$ actuated coordinates. If $\mathcal{X}_i \subseteq \mathbb{R}^{n_i}$ is component i's state space, then $\mathcal{X}_{\mathcal{S}} = \prod_{i=1}^s \mathcal{X}_i$. The graph fixes exchange, not unknown storage states. In the observation interface, $\tilde{q}$ denotes measured coordinates or readout channels; electrical voltage observations are not capacitor charge.

A.1.5. State initialization and assignment

The query index $t \geq L_{\text{sim}}$ belongs to the integration grid $\tau_k = \tau_0 + k\Delta t$. A configured history has L_h observations at offsets $0 = j_0 < \cdots < j_{L_h - 1} = L_{\text{sim}}$. Define

$$\begin{aligned} Y_t &= (\tilde{q}_{t - L_{\text{sim}} + j_r})_{r=0}^{L_h - 1}, \qquad U_t^{\text{now}} = (u_k^{\text{now}})_{k=t-L_{\text{sim}}}^{t}, \\ h_t &= (Y_t, U_t^{\text{now}}) \in \mathcal{H}_{\text{now}} := \mathbb{R}^{L_h \times d_{\text{obs}}} \times \mathbb{R}^{(L_{\text{sim}}+1) \times d_u}. \end{aligned} \tag{50}$$

Rows store observation or input vectors. The d_{obs} measured channels and d_u physical inputs belong to $\mathcal{S}_{\text{now}}$. The first L_{sim} input rows drive the historical intervals; the last is the known input at the query endpoint, used by the electrical voltage-rate lift. It is distinct from a candidate's u_0. The physical query time is τ_t. Candidate step ℓ starts at $\tau_t + \ell\Delta t$ and holds u_ℓ on $[\tau_t + \ell\Delta t, \tau_t + (\ell+1)\Delta t)$; candidate index zero therefore refers to the same physical time as history index t. Uniform mechanical histories have $L_{\text{sim}} = L_h - 1$, $j_r = r$; the electrical study uses $L_h = 21$, $L_{\text{sim}} = 100$, $j_r = 5r$.

The configured state initializer is a partial map $\text{InitializeState} : \mathcal{H}_{\text{now}} \rightharpoonup \mathcal{X}_{\mathcal{S}_{\text{now}}}$, defined when its numerical state construction succeeds. For full-state entry, x_t^{meas} is the last row of Y_t, and $\text{InitializeState}(h_t; \mathcal{S}_{\text{now}}) = \chi^x_{\text{now}}(x_t^{\text{meas}})$. With missing variables, the initializer estimates the state from the same history; $\hat{x}$ denotes a predicted state.

For mechanical position-only sensing, the chart first converts observations, then the observer and canonicalizer supply the missing phase coordinates:

$$z_\tau = \chi_{\text{now}}(\tilde{q}_\tau), \qquad \tilde{q}_\tau = \chi_{\text{now}}^{-1}(z_\tau). \tag{51}$$

$$\hat{x}_t = \mathrm{LiftState}_{\phi,\psi}(z_{t-L_h+1:t}) = C_\psi\big(z_t, \widehat{\dot{z}}_t\big). \tag{52}$$

The observer supplies chart velocity $\widehat{\dot{z}}_t$ and C_ψ supplies a phase state. This mechanical specialization does not use past inputs. In the identity chart, a supplied mass gives $\hat{p} = M\widehat{\dot{q}}$; the identity canonicalizer instead uses a velocity-like latent coordinate. The causal finite-difference/TCN realization and position decoding are in Appendix A.3. For voltage-only electrical sensing, initialization fits the past state through the frozen network and returns the state at time t (Eq. (204)); the learned alternative uses smoothed voltage and its rate (Eqs. (205)–(206)). Neither receives measured DGU or line currents at deployment.

Assignment data ζ_κ identify retained components through $m_\kappa(i)$, supply initial states x_i^{new} for additions, and specify any reset $\mathcal{R}_\kappa$. In compatible component coordinates,

$$\bar{x}_i = \begin{cases} (\hat{x}_t)_{m_\kappa(i)}, & i \text{ retained}, \\ x_i^{\mathrm{new}}, & i \text{ added}, \end{cases} \qquad \mathrm{AssignState}_\kappa(\hat{x}_t;\zeta_\kappa) = \mathcal{R}_\kappa(\mathrm{stack}_{i=1}^{s_\kappa}\,\bar{x}_i). \tag{53}$$

Here s_κ is the candidate's component count and m_κ maps retained candidate indices to compatible actual-component indices. Removed components are omitted and $\mathcal{R}_\kappa$ is identity when no reset is specified. The same inferred actual state is used for all candidates; their assigned initial states can differ when components or resets differ. With ζ_κ fixed, assignment maps $\mathcal{X}_{\mathcal{S}_{\mathrm{now}}}$ to $\mathcal{X}_{\mathcal{S}_\kappa}$. The candidate transition has type $F : \mathcal{X}_{\mathcal{S}_\kappa} \times \mathcal{U}_\kappa \rightharpoonup \mathcal{X}_{\mathcal{S}_\kappa}$, where $\mathcal{U}_\kappa \subseteq \mathbb{R}^{d_u(\kappa)}$ is its physical input space; its dimension need not equal that of the actual system.

A.1.6. Actuation and physical readouts

A raw command c first becomes a physical input through $c \xrightarrow{\chi^u} \bar{c} \xrightarrow{\mathcal{A}(x,\Xi)} u$. Already supplied physical inputs bypass this conversion. A proxy command may condition a forecast, but its product with a port output is not physical work. This distinction matters for the normalized position-offset commands in the Isaac transfer protocol.

For the transition endpoints $x = \hat{x}_\ell(a)$ and $x^+ = \hat{x}_{\ell+1}(a)$, and input $u = u_\ell$ held over the interval, the accounting convention in Eq. (18) uses

$$\begin{aligned} P_{\mathrm{loss}}(x) &= \nabla H_{\mathrm{total}}^\top \hat{R} \nabla H_{\mathrm{total}}, \\ P_{\mathrm{ext}}(x,u) &= \hat{y}(x)^\top\big(u^{\mathrm{loc}} + u^{\mathrm{dom}}(x)\big), \\ W_\ell &= \tfrac{\Delta t}{2}\big(P_{\mathrm{ext}}(x,u) + P_{\mathrm{ext}}(x^+,u)\big), \\ E_{\mathrm{loss},\ell} &= \tfrac{\Delta t}{2}\big(P_{\mathrm{loss}}(x) + P_{\mathrm{loss}}(x^+)\big), \\ \epsilon_{H,\ell} &= H_{\mathrm{total}}(x^+) - H_{\mathrm{total}}(x) - W_\ell + E_{\mathrm{loss},\ell}. \end{aligned} \tag{54}$$

Both integration and endpoint-quadrature error contribute to the defect. A deployment selects the available channels and their units and aggregation conventions. The state and sensor-prediction pair is Equation (15); its selector Π_{obs} retains channels from the full physical readout defined here. Mechanical position readout is $h_{\mathrm{obs}}^{(\mathcal{S})}(x,u) = \chi_{\mathcal{S}}^{-1}(\Pi_q x)$. For an electrical network it is

$$h_{\mathrm{obs}}^{(\mathcal{S})}(x,u) = (v_1, i_1, \ldots, v_n, i_n, i_{e,1}, \ldots, i_{e,m}), \qquad i_{e,k} = \phi_{e,k}/L_{e,k}, \tag{55}$$

where DGU voltages and currents follow Eq. (21). For the full physical-readout space $\mathcal{Y}_\kappa$, the native consequence vector in Eq. (19) belongs to $\mathcal{Y}_\kappa \times \mathbb{R}^6$: one physical-readout vector and six scalar accounting quantities. The selected sensor prediction is $\Pi_{\mathrm{obs}}^{(\mathcal{S}_\kappa)} h_{\mathrm{obs}}^{(\mathcal{S}_\kappa)}(x^+,u)$. Selecting voltage entries changes the observed channels, not the underlying state or the full consequence vector. The isolated-DGU measurement ordering is Eq. (72). Telemetry targets and learned operational proxies use additional readout maps; their provenance differs from the model-native accounting channels, as explained after Eq. (19). The finite-step accounting defect is analyzed in Theorem D.1.

A.2. Mechanical coordinates and numerical realization

The deployment chart in Eq. (51) maps raw measurements into model-facing coordinates. PHASTCore expects one additional alignment step: it advances local pH states, not raw telemetry. The chart says which local coordinate patch of the deployed system the model is moving in; the observer or canonicalizer says which storage and port variables should be

advanced inside that patch. For FHN and microgrids, the benchmark declares these local variables directly, such as repeated excitable-cell states or DGU/line storage and port variables after the chosen normalization. For mechanical systems, the local pH state is canonical phase space (q, p).

Mechanical canonicalization. Many robot pipelines provide joint angles q, while velocities $\dot{q}$ may be noisy or unavailable; Section A.3 later gives the causal observer used in that deployment setting. The only mechanical fact needed here is that momentum need not equal velocity: the canonical momentum is $p = M(q)\dot{q}$, where $M(q)$ is the configuration-dependent mass matrix. This moves the state from $(q, \dot{q})$ coordinates, where mass and velocity estimation are tied to the particular embodiment, to canonical phase-space coordinates (q, p) in which the pH structure takes a standard form:

$$\begin{pmatrix} \dot{q} \\ \dot{p} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}}_{J_{\text{canonical}}} \nabla H(q, p) + \text{dissipation} + \text{control} \tag{56}$$

The key point is that the symplectic structure $J_{\text{canonical}}$ is independent of robot geometry: a 2-joint arm and a 12-joint quadruped share the same canonical pH form once written in (q, p) coordinates. Across fixed-layout robot instances, target metadata Ξ_{target} then enters as a local response mismatch rather than as a topology change: mass, geometry, actuator response, and friction change the map from model-coordinate q histories to (q, p), even when the typed layout and deployed coupling graph are unchanged.

Why this matters for composition. The preceding coordinate step is not cosmetic: a shared block only makes sense if the state can be written in local variables whose stored energy is mostly local. The decoupling theory of (Ehrhardt et al., 2026) shows that a monolithic pHODE can be split into subsystems when a coordinate transformation exposes an additive Hamiltonian, $H(x) = \sum_i H_i(w_i)$, over local coordinate blocks w_i. C-PHAST uses this as a modeling bias rather than as an exact guarantee: choose local coordinates that expose repeated storage terms, learn a shared local block in those coordinates, and leave remaining power exchange to the interconnection and domain ports. In robotics this means that approximately block-local mechanical energy makes shared joint or limb blocks meaningful; in FHN and microgrids the repeated excitable cells, DGUs, and lines supply the local typed blocks directly.

The bridge used by the rest of the method is therefore

raw observations $\rightarrow$ normalized/wrapped coordinates $\rightarrow$ local pH state $\rightarrow$ shared PHASTCore update.

The local learnable block is specified next; the practical q-only realization of the observer and canonicalizer is deferred to Section A.3.

A.2.1. MECHANICAL PHASTCORE AND SPLIT TRANSITION

PHAST blocks and knowledge regimes. Once each subsystem has been represented in local pH coordinates, C-PHAST applies the same PHASTCore update to every repeated block. Each per-subsystem dynamics block is an explicit port-Hamiltonian step with learnable potential $V_{\theta_V}(q)$, mass $M_{\theta_M}(q)$, and damping $D_{\theta_D}(q)$ modules. The same block template supports three *knowledge regimes*: KNOWN instantiates specified physics modules on the supplied target layout, PARTIAL augments typed modules with learnable residuals or observation interfaces, and LAW-UNKNOWN keeps the declared subsystem layout but learns the local dynamics and coupling subject to physical constraints. These regime labels are domain-grounded rather than perfectly uniform: in robotics, the maintained PARTIAL regime is defined by what the asset and IO stack actually expose, including grouped articulation, observation/action semantics, and coarse instance metadata, whereas an electrical specification can supply typed interconnection and local ports. Each experiment specifies which quantities are supplied or learned, along with its numerical realization and training objective. This is also where weight sharing enters concretely: the same local update rule is reused across all instantiated subsystems of a given type.

The amount of supplied physics is separate from the deployment change being tested. In particular, PARTIAL does not mean topology-unaware: known and partially specified microgrid models can receive the same feeder graph. In a voltage-only query, the DGU/line layout and incidence pattern remain supplied, while hidden-state initialization uses the declared estimator in Appendix I. LAW-UNKNOWN means declared-layout / law-unknown; it does not denote discovery of the decomposition or topology. Table 14 records what is supplied and learned.

Table 14. **C-PHAST knowledge regimes.** The labels describe supplied physical knowledge. In the planar-arm comparison, KNOWN and PARTIAL refer to local physics; coupling and training choices are specified separately in Appendix K.1.3.

Regime	Specified or supplied	Learned from data
KNOWN	specified local physics modules and target layout/topology; coupling where provided	calibration or interfaces; coupling parameters when not supplied
PARTIAL	typed block structure and available target layout/topology	uncertain local physics, damping, coupling corrections, or interfaces
LAW-UNKNOWN	compositional scaffold and declared subsystem layout	local physics and coupling almost entirely

In the mechanical setting with state $x = (q, p)$ (configuration q and momentum p, each in $\mathbb{R}^d$ for a d-DOF subsystem), we use the mechanical Hamiltonian

$$H_{\theta_{\mathrm{loc}}}(q,p) = V_{\theta_V}(q) + \tfrac{1}{2} p^\top M_{\theta_M}(q)^{-1} p, \tag{57}$$

with $M_{\theta_M}(q) \succ 0$ parameterized as $M_{\theta_M}(q) = L_{\theta_M}(q) L_{\theta_M}(q)^\top + \varepsilon I$ ($\varepsilon > 0$) so positive definiteness and a uniform eigenvalue floor $\lambda_{\min}(M_{\theta_M}) \geq \varepsilon$ are architectural; bounded conditioning additionally requires a cap or parameterization controlling $\lambda_{\max}(M_{\theta_M})$. Equation (57) is a *mechanical*, not a standard separable, Hamiltonian once M_{θ_M} depends on q. The explicit-leapfrog finite-step result of Theorem D.1 applies to constant-mass Hamiltonians when the complete assembled transition satisfies its second-order, matched-mass, and remaining hypotheses. A configuration-dependent $M(q)$ requires a suitable nonseparable integrator and lies outside the stated explicit-leapfrog certificate. We then choose a dissipation matrix that acts on momenta, $R_{\theta_D} = \mathrm{diag}(0, D_{\theta_D}(q))$ with $D_{\theta_D}(q) \succeq 0$. With total port input $u_i^{\mathrm{tot}} = u_i^{\mathrm{loc}} + \hat{u}_i + u_i^{\mathrm{dom}}$, this yields the standard passivity identity

$$\frac{dH_{\theta_{\mathrm{loc}}}}{dt} = -v^\top D_{\theta_D}(q)\, v + (y_i^{\mathrm{port}})^\top u_i^{\mathrm{tot}} \leq (y_i^{\mathrm{port}})^\top u_i^{\mathrm{tot}}, \qquad v := \frac{\partial H_{\theta_{\mathrm{loc}}}}{\partial p} = M_{\theta_M}(q)^{-1} p. \tag{58}$$

When all ports are closed, this reduces to monotone energy dissipation. Figure 8 shows this single-block interface explicitly: the same PHASTCore block exposes local/control, internal coupling, and optional domain ports.

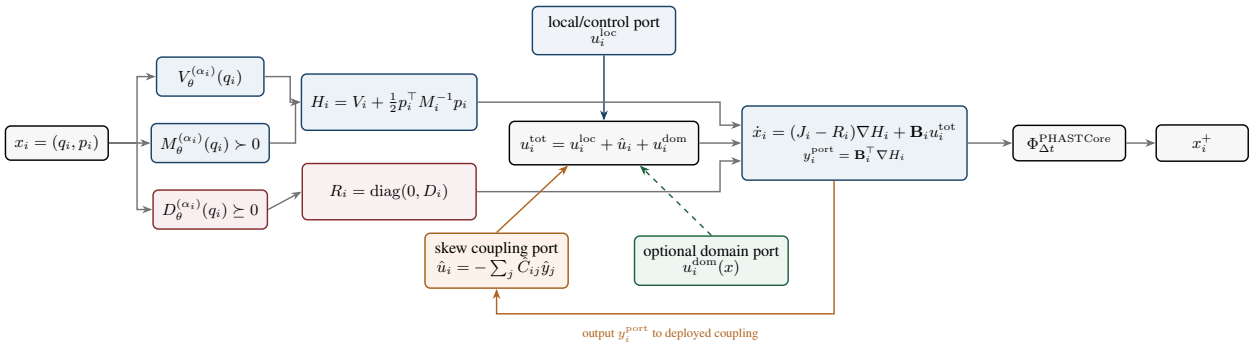


Figure 8. **One reusable PHASTCore block with ports.** For a typed subsystem α_i, the local modules $V_\theta^{(\alpha_i)}$, $M_\theta^{(\alpha_i)}$, and $D_\theta^{(\alpha_i)} \succeq 0$ assemble into a mechanical port-Hamiltonian block with port matrix $\mathbf{B}_i$. Local/control inputs enter through u_i^{loc}, C-PHAST composes blocks through the orange skew-coupling port $\hat{u}_i$, and optional benchmark-specific forcing enters through u_i^{dom}. The continuous-time storage balance is $\dot{H}_i = -v_i^\top D_i v_i + (y_i^{\mathrm{port}})^\top (u_i^{\mathrm{loc}} + \hat{u}_i + u_i^{\mathrm{dom}})$. This is Eq. (58) instantiated inside the block-level interface of Eq. (8). Under skew-symmetric coupling, the orange power terms cancel after summing over blocks, while the blue and green ports remain explicit. Algorithm 2 calls this block in its blue PHASTCore stage.

Householder-style damping. Following PHAST, "Householder-style" means a low-rank additive PSD parameterization, not a Householder-reflection dynamics model. We parameterize configuration-dependent damping as a low-rank outer-product expansion:

$$D(q) = d_0 I + \sum_{i=1}^{r} \beta_i(q)\, k_i(q) k_i(q)^\top, \qquad d_0 \geq 0,\ \beta_i(q) \geq 0,\ \|k_i(q)\|_2 = 1, \tag{59}$$

This form guarantees $D(q) \succeq 0$ for all q while keeping matrix-vector products $O(nr)$. The essential requirement for the present paper is the PSD damping operator, because this is what preserves the pH energy-dissipation contract.

Bounding damping strength. The optional PHAST damping cap

$$\sum_{i=1}^{r} \beta_i(q) \le \bar{\beta} \quad \Rightarrow \quad \lambda_{\max}(D(q)) \le d_0 + \bar{\beta} \tag{60}$$

has two roles. First, it prevents damping from becoming an error sink for mistakes in V, M, the observer, or the input model. Second, with the Cayley damping step—unconditionally nonexpansive in the kinetic-energy metric—the cap improves accuracy and conditioning rather than being required for finite-step nonexpansion. Stiff sign-flipping is governed by the dimensionless rate $\Delta t\, \lambda_{\max}(M^{-1/2} D M^{-1/2})$, not damping magnitude alone. With a mass floor $M \succeq m_{\min} I$, this rate is at most $\Delta t(d_0 + \bar{\beta})/m_{\min}$. When the cap is active, the damping block is an attribution channel for irreversible loss rather than a generic numerical stabilizer.

The Hamiltonian mass $M(q)$ belongs to the local kinetic energy; the effective inverse-mass metric M_{eff}^{-1} in Algorithm 2 converts skew exchange into a momentum update.

Structure-preserving integration (Strang splitting). To advance dissipative dynamics we use Strang splitting:

$$\Phi_{\Delta t} = \Phi_D^{\Delta t/2} \circ \Phi_H^{\Delta t} \circ \Phi_D^{\Delta t/2}, \tag{61}$$

where $\Phi_H^{\Delta t}$ is a conservative (symplectic) update and $\Phi_D^{\Delta t/2}$ is a half-step of damping. For the constant-mass mechanical kernel, the conservative map is the kick–drift–kick Störmer–Verlet update:

$$\begin{aligned} p^{1/2} &= p - \tfrac{\Delta t}{2} \nabla V(q), \\ q^{+} &= q + \Delta t M^{-1} p^{1/2}, \\ p^{+} &= p^{1/2} - \tfrac{\Delta t}{2} \nabla V(q^{+}), \qquad \Phi_H^{\Delta t}(q,p) = (q^{+}, p^{+}). \end{aligned} \tag{62}$$

We write $\Phi_{\Delta t}^{(i)}$ when the component identity matters and omit (i) for the shared local routine. In Algorithm 2, this routine advances local storage and damping; port kicks and interconnection updates are applied separately around it. Their assembled step is $F_{\theta,\Delta t}^{(\mathcal{S})}$, not $\Phi_{\Delta t}$ alone. The dissipative half-step integrates $\dot{\boldsymbol{p}} = -A_D \boldsymbol{p}$ with $A_D = D(q)M(q)^{-1}$ and position frozen, whose exact flow is $\exp(-A_D\tau)\boldsymbol{p}$ ($\tau = \Delta t/2$). We use the Cayley (trapezoidal / $[1/1]$-Padé) step

$$\Phi_D^{\tau} : \ \boldsymbol{p} \ \longmapsto \ (I + c\, A_D)^{-1}(I - c\, A_D)\, \boldsymbol{p}, \qquad A_D = D(q)M(q)^{-1}, \quad c = \tfrac{\tau}{2} = \tfrac{\Delta t}{4}, \tag{63}$$

which matches $\exp(-A_D\tau)$ to $\mathcal{O}(\tau^3)$—so the full split step is genuinely second order—and is *unconditionally non-expansive in the kinetic-energy metric*. Because $A_D = DM^{-1}$ is generally nonnormal, a spectral-radius argument does not by itself give a norm contraction; the correct statement is that A_D is self-adjoint and positive semidefinite in the inverse-mass inner product $\langle \boldsymbol{p}, \boldsymbol{q} \rangle_{M^{-1}} = \boldsymbol{p}^\top M^{-1} \boldsymbol{q}$ (indeed $M^{-1}A_D = M^{-1}DM^{-1}$ is symmetric and, since $D \succeq 0$, positive semidefinite). It is therefore orthogonally diagonalizable in that metric with real eigenvalues $\lambda \ge 0$, and its Cayley transform contracts the induced norm, $|(1 - c\lambda)/(1 + c\lambda)| \le 1$ for every $\lambda \ge 0$ and every Δt, so the half-step is non-expansive in the kinetic energy $\boldsymbol{p}^\top M^{-1} \boldsymbol{p}$ and never injects energy (A-stability). The step is A-stable but not L-stable: as $\tau\lambda \to \infty$ the amplification factor tends to -1, so very stiff modes are not strongly damped—the momentum can flip sign rather than decay—while the energy stays non-increasing, which is all the finite-step defect requires. This is the improvement over the second-order Taylor truncation $I - \tau A_D + \frac{1}{2}\tau^2 A_D^2$, which amplifies once $\tau\, \lambda_{\max}(A_D) > 2$; and over a forward-Euler half, which is only first order. The step needs the small per-block solve $(I + cA_D)^{-1}$; the operator A_D is assembled from the damping/mass matrix–vector products, so state-dependent and learned $D(q)$ are covered. Algorithm 2 assembles these local updates with the port and coupling steps to realize Q04 of Algorithm 1. Here $\bar{\boldsymbol{u}}^{\text{loc}} \in \mathbb{R}^N$ denotes the stacked local/control port inputs and $\bar{\boldsymbol{u}}^{\text{dom}} \in \mathbb{R}^N$ denotes optional domain-port inputs (zero when absent). The matrix M_{eff}^{-1} is not a second learned Hamiltonian mass module: $M_{\theta_M}(q)$ appears in the local kinetic energy in Eq. (57), while M_{eff}^{-1} is the fixed inverse-mass metric used only by the Cayley coupling step to convert deployed skew exchange into a momentum update. For the finite-step certificate of Theorem D.1 to model the same coupling law as the continuous port output $\hat{\boldsymbol{y}} = M_{\theta_M}(q)^{-1}\boldsymbol{p}$, the effective metric is taken to match the Hamiltonian mass, $M_{\text{eff}} = M_{\theta_M}$, in the constant, scalar, or blockwise-scalar regime where the two coincide; a distinct $M_{\text{eff}} \neq M_{\theta_M}$ would define a different modeled coupling law and falls outside the certified set. For readability, define the Cayley half-step operator

$$\mathcal{C}_{\Delta t/2}(A)\boldsymbol{p} := \left(I + \tfrac{\Delta t}{4} A\right)^{-1} \left(I - \tfrac{\Delta t}{4} A\right) \boldsymbol{p}, \qquad A_p := \hat{\mathbf{C}}^{(s)} M_{\text{eff}}^{-1}. \tag{64}$$

The generator $A_p = \hat{\mathbf{C}}^{(s)} M_{\text{eff}}^{-1}$ is the momentum form of the coupling law: the port output is a velocity, $\hat{\boldsymbol{y}} = M_{\text{eff}}^{-1}\boldsymbol{p}$, and the internal input $\hat{\boldsymbol{u}} = -\hat{\mathbf{C}}^{(s)}\hat{\boldsymbol{y}}$ acts on momentum, so the flow being integrated is $\dot{\boldsymbol{p}} = -\hat{\mathbf{C}}^{(s)} M_{\text{eff}}^{-1}\boldsymbol{p}$. Because $\hat{\mathbf{C}}^{(s)}$ is skew, A_p is skew in the M_{eff}^{-1}-inner product, the same inverse-mass geometry in which the damping generator A_D above is self-adjoint, and $\mathcal{C}_{\Delta t/2}(A_p)$ therefore preserves the kinetic form $\boldsymbol{p}^\top M_{\text{eff}}^{-1}\boldsymbol{p}$ for any admissible effective mass. When M_{eff} is a scalar multiple of the identity, or blockwise scalar on every coupled subspace, A_p coincides with the velocity-coordinate generator $M_{\text{eff}}^{-1}\hat{\mathbf{C}}^{(s)}$. A diagonal but *non*-scalar mass need not commute with a dense skew $\hat{\mathbf{C}}^{(s)}$, so diagonal mass alone would not suffice, and for a noncommuting effective mass one must keep the momentum form above or, equivalently, apply the Cayley map in velocity coordinates ($\boldsymbol{v} = M_{\text{eff}}^{-1}\boldsymbol{p}$, update $\boldsymbol{v}$, then $\boldsymbol{p}^+ = M_{\text{eff}}\boldsymbol{v}^+$). The step is metric-preserving, not canonically symplectic: the momentum-only map $(q, \boldsymbol{p}) \mapsto (q, Q\boldsymbol{p})$ has Jacobian $\operatorname{diag}(I, Q)$ and is symplectic only when $Q = I$. Algorithm 2 assumes that the observation front end has already produced phase-space subsystem states. It then applies the same logical sequence introduced above: local/control ports, optional domain ports, power-preserving coupling, shared PHASTCore updates, and the symmetric closing half-steps.

The coupling constructor $\mathrm{SkewCouple}_{\theta_C}(\mathcal{S})$ returns a supplied operator in compatible momentum-port coordinates, or the homogeneous count-transfer construction in Eq. (208) with its stated $\otimes I_d$ lift. The local maps and M_{eff}^{-1} are configured before the transition; the certified path uses constant local masses and their matching block-diagonal metric. Skew-symmetry alone does not specify a constructor.

Algorithm 2 Symmetric mechanical realization of Q04 of Algorithm 1

Inputs: Subsystem states $\{(q_i, p_i)\}_{i=1}^s$, stacked local/control inputs $\bar{\boldsymbol{u}}^{\text{loc}} \in \mathbb{R}^N$, optional domain-port inputs $\bar{\boldsymbol{u}}^{\text{dom}} \in \mathbb{R}^N$ (zero if absent).
Fixed: Deployed skew coupling $\hat{\mathbf{C}}^{(s)} = -(\hat{\mathbf{C}}^{(s)})^\top$ from Eq. (9), PHASTCore step $\Phi_{\Delta t}$ from Fig. 8 and Eq. (61), inverse mass proxy M_{eff}^{-1}, time step Δt.
Outputs: Updated subsystem states $\{(q_i, p_i)\}_{i=1}^s$.

M01 $\boldsymbol{p} \leftarrow [p_1, \ldots, p_s]$, $A_p \leftarrow \hat{\mathbf{C}}^{(s)} M_{\text{eff}}^{-1}$
Assemble the state and coupling generator.
Eq. (64).

M02 $\boldsymbol{p} \leftarrow \boldsymbol{p} + \frac{\Delta t}{2}\bar{\boldsymbol{u}}^{\text{loc}}$
Opening local/control-port half-kick.

M03 $\boldsymbol{p} \leftarrow \boldsymbol{p} + \frac{\Delta t}{2}\bar{\boldsymbol{u}}^{\text{dom}}$
Opening domain-port half-kick (zero if absent).

M04 $\boldsymbol{p} \leftarrow \mathcal{C}_{\Delta t/2}(A_p)\boldsymbol{p}$
$(p_1, \ldots, p_s) \leftarrow \boldsymbol{p}$
Opening coupling half-step.

for $i = 1$ to s **do**
  M05 $(q_i, p_i) \leftarrow \Phi_{\Delta t}(q_i, p_i)$
  Local PHASTCore evolution.
end for

M06 $\boldsymbol{p} \leftarrow [p_1, \ldots, p_s]$
$\boldsymbol{p} \leftarrow \mathcal{C}_{\Delta t/2}(A_p)\boldsymbol{p}$
Closing coupling half-step.

M07 $\boldsymbol{p} \leftarrow \boldsymbol{p} + \frac{\Delta t}{2}\bar{\boldsymbol{u}}^{\text{dom}}$
Closing domain-port half-kick.

M08 $\boldsymbol{p} \leftarrow \boldsymbol{p} + \frac{\Delta t}{2}\bar{\boldsymbol{u}}^{\text{loc}}$
$(p_1, \ldots, p_s) \leftarrow \boldsymbol{p}$
Closing local/control-port half-kick.

return updated $\{(q_i, p_i)\}_{i=1}^s$

Algorithm 2 displays the certified zero-order-held, fixed-coupling path: one generator A_p and one $\bar{\boldsymbol{u}}^{\text{dom}}$ are reused in both half-steps. A state-dependent domain port or a configuration-dependent coupling requires symmetric endpoint reevaluation

(A_{open}=$A_p(q_k)$, A_{close}=$A_p(q_{k+1})$, and likewise for $\boldsymbol{u}^{\text{dom}}$) and, when evaluated only at the opening state, falls outside the second-order certificate of Theorem D.1.

The bound concerns this assembled numerical realization. The ANYmal–Go2 per-leg predictor (Appendix J.2) uses aggregate-velocity coupling and a different ordering of coupling and input updates, while the single-decision recovery predictor (Appendix L.2) adds a support-conditioned momentum correction after its base update. We therefore report the robotics comparisons as empirical prediction and decision evaluations, without claiming this finite-step certificate for their learned predictors.

A.3. Mechanical State Inference and Position Decoding

The mechanical initialization equation, Eq. (52), first charts the measured history and then applies the following causal observer and canonicalizer. This realizes the mechanical Q01 initialization in Algorithm 1; Q02 subsequently assigns the estimate to each candidate. A candidate rollout initializes once and repeats its declared latent transition, without rerunning the observer. Measurement-refreshed tracking instead uses new observed histories.

Observer for velocity estimation. We infer velocities from position history using a causal observer. Following the state pipeline, after the deployment chart is applied we write $q_\tau = z_\tau$ for the generalized coordinate used by the local mechanical pH block. For the window ending at t, write $a = t - L_h + 1$. We first compute finite-difference velocities over that window:

$$\dot{q}_a^{\text{fd}} = 0, \qquad \dot{q}_\tau^{\text{fd}} = \frac{\Delta(q_\tau, q_{\tau-1})}{\Delta t}, \quad \tau = a+1, \dots, t, \tag{65}$$

where $\Delta(\cdot,\cdot)$ is the coordinate-aware difference induced by χ_{target}: standard subtraction for Euclidean charts and the wrapped difference for periodic charts. A learned residual corrects finite-difference errors:

$$\hat{\dot{q}}_{a:t} = \dot{q}_{a:t}^{\text{fd}} + o_\phi\big(q_{a:t}, \dot{q}_{a:t}^{\text{fd}}\big)\,, \tag{66}$$

where o_ϕ is a temporal convolutional network (TCN) that outputs a per-timestep correction; its value at index τ is causal in the prefix ending at τ. The observer is *shared per subsystem*: the same TCN weights process each joint's (or leg's) position history independently, making the observer compositional and compatible with variable N.

Canonicalization and rollout. The canonicalizer C_ψ maps $(q, \hat{\dot{q}}) \mapsto (q, \hat{p})$. In practice we use either a diagonal-mass canonicalizer, $\hat{p} = M\hat{\dot{q}}$, when an effective mass model is specified, or the identity map $\hat{p} = \hat{\dot{q}}$ when no reliable mass model is available. This is the lift from model-coordinate evidence to the latent pH state $\hat{x}_t = (q_t, \hat{p}_t)$ that the dynamics can advance. Given the inferred initial state at the end of burn-in, write that end time as t_0. We roll out open-loop in the latent state while applying the known input sequence:

$$\hat{x}_{t_0} = C_\psi(q_{t_0}, \hat{\dot{q}}_{t_0}), \quad \hat{x}_{\tau+1} = F_{\theta,\Delta t}^{(\mathcal{S}_{\text{target}})}(\hat{x}_\tau, u_\tau), \quad \hat{q}_\tau^{\text{obs}} = \chi_{\text{target}}^{-1}(\Pi_q(\hat{x}_\tau))\,, \quad \tau \geq t_0, \tag{67}$$

where $\Pi_q(q,p) = q$ extracts model-coordinate positions and $\hat{q}_\tau^{\text{obs}}$ is the decoded observation-space prediction.

At inference, the primary mode is *takeover*: infer initial state from a burn-in window, then integrate open-loop. This q-only interface is the relevant transfer setting in robotics: when deploying to a new morphology, we can usually reuse encoders and commanded inputs, but we cannot assume calibrated velocity sensors on the target system. The observer generalizes across morphologies because it operates per-joint in canonical coordinates.

For this fixed-layout prediction, the specification determines the subsystem partition of $\hat{x}_t$ and maps the supplied inputs to $\bar{\boldsymbol{u}}_t^{\text{loc}}$ and $\bar{\boldsymbol{u}}_t^{\text{dom}}$. The declared mechanical transition advances these subsystem states, which are restacked before the position decoding in Eq. (67). Changing the component layout additionally requires the Q02 assignment of Algorithm 1; the observer is not applied to an unobserved candidate history.

A.4. Electrical DGU numerical update and readout

For $x = (q_C, \phi_L)$, the fitted energy and state rates are Eq. (21). The following operations implement one isolated DGU sampling interval. Appendix I uses this local update within the assembled network transition.

Algorithm 1, step Q04: the electrical numerical calls. The implementation advances Eq. (21) using two substeps per sampling interval, with the inputs held fixed. For a substep $\delta = \Delta t/2$, define an input translation and a damping map:

$$\begin{aligned} \mathrm{Kick}_\tau(q_C, \phi_L; u) &= (q_C + \tau I_{\mathrm{in}},\ \phi_L + \tau V_{\mathrm{src}}), \\ \mathrm{Damp}_\tau(q_C, \phi_L) &= \left(q_C, \frac{1 - \tau D_\theta(q_C)/(2M_\theta(q_C))}{1 + \tau D_\theta(q_C)/(2M_\theta(q_C))} \phi_L \right). \end{aligned} \tag{68}$$

The damping call freezes charge and applies the scalar Cayley update to flux. The conservative call $\mathrm{Midpoint}_\delta$ approximates the simultaneous equations

$$q_C^+ = q_C + \delta\, \partial_{\phi_L} H_\theta(\bar{q}_C, \bar{\phi}_L), \qquad \phi_L^+ = \phi_L - \delta\, \partial_{q_C} H_\theta(\bar{q}_C, \bar{\phi}_L), \quad (\bar{q}_C, \bar{\phi}_L) = \tfrac{1}{2}(q_C + q_C^+, \phi_L + \phi_L^+). \tag{69}$$

Its initial iterate is the explicit-Euler prediction from the input state. Each Picard iteration evaluates the right-hand sides at the midpoint formed with the preceding iterate. The implementation stops when successive charge and flux iterates each change by less than 10^{-6} in their stored coordinates, or after ten iterations; it returns the last iterate at the cap without a convergence exception. Thus this is an approximate implicit step, not an assumption of an exact nonlinear solve.

Algorithm 3 Isolated DGU transition used in Algorithm 1, step Q04: $\mathrm{DGUStep}_\theta(x, u; \Delta t)$

Require: $x = (q_C, \phi_L)$, held $u = (I_{\mathrm{in}}, V_{\mathrm{src}})$, fixed $U_\theta, M_\theta, D_\theta$; sampling interval Δt
Ensure: Next charge–flux state x^+
$\delta \leftarrow \Delta t/2$
for $r = 1, 2$ **do**
D01 $x \leftarrow \mathrm{Kick}_{\delta/2}(x; u)$
Opening input half-step: Eq. (68).
D02 $x \leftarrow \mathrm{Damp}_{\delta/2}(x)$
Opening dissipative half-step: Eq. (68).
D03 $x \leftarrow \mathrm{Midpoint}_\delta(x)$
Full conservative step: Eq. (69) and the stated Picard stopping rule.
D04 $x \leftarrow \mathrm{Damp}_{\delta/2}(x)$
Eq. (68); reevaluate M_θ, D_θ at the updated charge.
D05 $x \leftarrow \mathrm{Kick}_{\delta/2}(x; u)$
Closing input half-step: Eq. (68).
end for
return $x^+ \leftarrow x$

The local-learning comparison uses this DGU transition. Its finite-iteration midpoint evaluation need not retain the symplecticity of an exact implicit solve, and neither evaluation conserves a general nonlinear Hamiltonian exactly.

Assembled electrical transition. For n DGUs and m lines, write $x = (q, \phi, \phi_e) \in \mathbb{R}^{2n+m}$. The supplied incidence matrix $A_\kappa \in \mathbb{R}^{n \times m}$ is negative at a line's sending bus and positive at its receiving bus. The voltage and current efforts are $v_i = \partial_{q_i} H_\theta(q_i, \phi_i)$, $i_i = \partial_{\phi_i} H_\theta(q_i, \phi_i)$, and $i_{e,k} = \phi_{e,k}/L_{e,k}$. Define

$$\begin{aligned} H(x) &= \sum_{i=1}^{n} H_\theta(q_i, \phi_i) + \sum_{k=1}^{m} \frac{\phi_{e,k}^2}{2L_{e,k}}, \\ f_{\mathrm{c},\kappa}(x) &= (A_\kappa i_e,\ 0,\ -A_\kappa^\top v), \\ \mathcal{E}_\tau^{\mathrm{mp}}(x) &= x + \tau f_{\mathrm{c},\kappa}\left(x + \tfrac{\tau}{2} f_{\mathrm{c},\kappa}(x)\right). \end{aligned} \tag{70}$$

This explicit-midpoint coupling step recomputes the full Hamiltonian efforts at the intermediate state, including any kinetic contribution to $\partial_q H_\theta$. The continuous coupling power cancels, $v^\top A_\kappa i_e - i_e^\top A_\kappa^\top v = 0$; this cancellation does not imply exact energy preservation by the finite step.

For held inputs, let $\mathcal{L}_\delta$ apply *one* DGU pass D01–D05 of Algorithm 3, rather than its full two-pass loop, and multiply each line flux by $[(1-\delta R_{e,k}/(4L_{e,k}))/(1+\delta R_{e,k}/(4L_{e,k}))]^2$. The midpoint network transition is

$$F_{\theta,\Delta t}^{(\mathcal{S}_\kappa),\mathrm{mp}} = \left(\mathcal{E}_{\delta/2}^{\mathrm{mp}} \circ \mathcal{L}_\delta \circ \mathcal{E}_{\delta/2}^{\mathrm{mp}}\right)^{\circ 2}, \qquad \delta = \Delta t/2, \tag{71}$$

where the exponent denotes two repetitions. The nonlinear component-only composition study uses this midpoint transition. Network calibration and partial-sensing experiments use the Euler coupling variant specified in their protocols. Proposition D.2 states the midpoint transition's energy-accounting bound, including the local-solver error.

Algorithm 1, step Q05: read electrical quantities from the same state. For one DGU with state $x = (q_C, \phi_L)$, the observation part of $\mathcal{O}_{\mathrm{target}}$ in the benchmark's incidence orientation is

$$\mathcal{O}_{\mathrm{DGU}}(x,u) = \Big(q_C, \phi_L, \underbrace{V_{\mathrm{src}}}_{e_1}, \underbrace{V_{\mathrm{src}} - D_\theta(q_C) i_\theta}_{e_2}, \underbrace{v_\theta}_{e_3}, \underbrace{-i_\theta}_{j_V}\Big). \tag{72}$$

Here e_1, e_2, e_3 are node voltages relative to ground, and j_V is the signed voltage-source current. These are constitutive readouts, not extra networks or independently advanced states. The corresponding instantaneous energy accounting is $\dot{H}_\theta = v_\theta I_{\mathrm{in}} + i_\theta V_{\mathrm{src}} - D_\theta i_\theta^2$. When read out along a discrete trajectory, its integrated balance can have a numerical defect. Equation (19) specifies how such domain channels enter the generic consequence vector returned by Q05.

A.5. Local electrical calibration

An isolated DGU provides two complementary tests of its laws: do the measured storage and electrical quantities agree at one instant, and does a numerical step predict the next observation? Write a calibration sample as $(\tilde{y}_j, u_j, \tilde{y}_j^+)$, where

$$\tilde{y}_j = (\tilde{q}_{C,j}, \tilde{\phi}_{L,j}, \tilde{e}_{1,j}, \tilde{e}_{2,j}, \tilde{e}_{3,j}, \tilde{j}_{V,j}), \qquad \tilde{x}_j = (\tilde{q}_{C,j}, \tilde{\phi}_{L,j}).$$

Tildes denote calibration measurements and $+$ denotes the next sample. All six channels and both inputs are available during this calibration. They need not all be measured at deployment.

At the measured storage state, evaluate i_θ, v_θ from Eq. (21) and D_θ at the measured charge. The instantaneous residual is

$$r_\theta^{\mathrm{loc}}(\tilde{y}) = \begin{pmatrix} i_\theta(\tilde{x}) + \tilde{j}_V \\ (\tilde{e}_2 - \tilde{e}_1)/D_\theta(\tilde{q}_C) - \tilde{j}_V \\ v_\theta(\tilde{x}) - \tilde{e}_3 \end{pmatrix}. \tag{73}$$

The entries test current balance at the source, the resistive relation, and capacitor-node voltage, respectively. They use measured channels, not channels synthesized by $\mathcal{O}_{\mathrm{DGU}}$; otherwise the constructed readout relations would cancel by definition.

For N samples, the two data terms are

$$\begin{aligned} \mathcal{L}_{\mathrm{constraint}}^{\mathrm{loc}} &:= \frac{1}{N}\sum_{j=1}^{N} \|r_\theta^{\mathrm{loc}}(\tilde{y}_j)\|_2^2, \\ \hat{y}_j^+ &:= \mathcal{O}_{\mathrm{DGU}}\big(\mathrm{DGUStep}_\theta(\tilde{x}_j, u_j; \Delta t), u_j\big), \\ \mathcal{L}_{\mathrm{step}}^{\mathrm{loc}} &:= \frac{1}{N}\sum_{j=1}^{N} \|\hat{y}_j^+ - \tilde{y}_j^+\|_2^2. \end{aligned} \tag{74}$$

The first term fits local laws without a timestep; the second differentiates through Algorithm 3 and its readout in Eq. (72). Both sum squared channels before averaging samples. The residual channels in Eq. (73) have units A, A, V; the transition term mixes charge, flux, voltage, and current in the stored calibration units. These are the executed unwhitened objectives, so their relative channel weighting depends on those units.

Let θ_{opt} contain the optimized parameters of U_θ, M_θ, and D_θ, excluding fixed scalar settings. The two schedules use

$$\begin{aligned}\mathcal{L}_{\mathrm{reg}} &:= 10^{-8}\|\theta_{\mathrm{opt}}\|_2^2,\\ \mathcal{L}_{\mathrm{pre}} &:= \mathcal{L}^{\mathrm{loc}}_{\mathrm{constraint}} + \mathcal{L}_{\mathrm{reg}},\\ \mathcal{L}^{\mathrm{loc}}_{\mathrm{joint}} &:= \mathcal{L}^{\mathrm{loc}}_{\mathrm{step}} + 0.01\mathcal{L}^{\mathrm{loc}}_{\mathrm{constraint}} + \mathcal{L}_{\mathrm{reg}}.\end{aligned} \tag{75}$$

Constraint-first fitting uses 25,000 updates of $\mathcal{L}_{\mathrm{pre}}$ followed by 75,000 combined-loss updates. Combined-loss fitting uses that objective for all 100,000 updates. Both optimize all three eligible local function groups from the same data-fitted linear warm start, with Adam at 10^{-4} and batch size 128. The warm start fits parameters to training data without reference constitutive coefficients. The nonlinear family remains active: a learned quadratic potential plus a neural residual, a positive charge-dependent storage factor, and positive neural resistance. Fixed final checkpoints are evaluated; validation is diagnostic rather than a checkpoint selector. This comparison isolates the training schedule.

A.5.1. Calibration on source networks

The call $\mathrm{NetworkStep}_{\Delta t}$ below uses the network transition F defined in Appendix I, with the DGU laws trainable during calibration and fixed at deployment.

The source-network stage asks whether the same DGU laws remain predictive when components exchange power. It initializes from a local checkpoint and shares one DGU parameter set across the nodes of two supplied source graphs. Known line laws and incidence matrices are not optimized.

For graph $g \in \{1,2\}$, let n_g and m_g count DGUs and lines and set $d_g = 2n_g + m_g$. Each of its N_g calibration samples provides measured storage $\tilde{x}_{g,j}$, input $u_{g,j}$, present and next physical voltage/current vectors $\tilde{y}^{\mathrm{phys}}_{g,j}$ and $\tilde{y}^{\mathrm{phys},+}_{g,j}$, and the measured resistor drop $\Delta\tilde{e}_{g,j,i} = \tilde{e}_{2,g,j,i} - V_{\mathrm{src},g,j,i}$. The physical vectors follow the network ordering in Eq. (55), with d_g channels. Define the source prediction and present-state residual by

$$\begin{aligned}\hat{y}^{\mathrm{phys},+}_{g,j} &:= h^{(\mathcal{S}_g)}_{\mathrm{obs}}\big(\mathrm{NetworkStep}_{\Delta t}(\tilde{x}_{g,j}, u_{g,j}; \mathcal{S}_g), u_{g,j}\big),\\ r^{\mathrm{src}}_{g,j} &:= \mathrm{stack}^{n_g}_{i=1}\begin{pmatrix} v_{\theta,i}(\tilde{x}_{g,j}) - \tilde{v}_{g,j,i}\\ i_{\theta,i}(\tilde{x}_{g,j}) - \tilde{i}_{g,j,i}\\ \Delta\tilde{e}_{g,j,i} + D_\theta(\tilde{q}_{g,j,i})\tilde{i}_{g,j,i}\end{pmatrix}.\end{aligned} \tag{76}$$

The last residual uses the *measured* DGU current in the resistive drop. Unlike the local residual in Eq. (73), it is a voltage-drop error, not a resistor-current error. The source objective also averages over channels rather than summing them:

$$\begin{aligned}\mathcal{L}^{\mathrm{src}}_{\mathrm{step}} &:= \frac{1}{2}\sum_{g=1}^{2}\frac{1}{N_g d_g}\sum_{j=1}^{N_g}\|\hat{y}^{\mathrm{phys},+}_{g,j} - \tilde{y}^{\mathrm{phys},+}_{g,j}\|_2^2,\\ \mathcal{L}^{\mathrm{src}}_{\mathrm{constraint}} &:= \frac{1}{2}\sum_{g=1}^{2}\frac{1}{3N_g n_g}\sum_{j=1}^{N_g}\|r^{\mathrm{src}}_{g,j}\|_2^2,\\ \mathcal{L}^{\mathrm{src}}_{\mathrm{joint}} &:= \mathcal{L}^{\mathrm{src}}_{\mathrm{step}} + 0.01\mathcal{L}^{\mathrm{src}}_{\mathrm{constraint}} + \mathcal{L}_{\mathrm{reg}}.\end{aligned} \tag{77}$$

The factor $1/2$ weights the two graphs equally. The stage performs 10,000 updates of the shared DGU parameters with Adam at 10^{-4} and batch size 128, and retains the fixed final checkpoint. Both local-only and source-calibrated variants use rich source measurements for calibration and fixed laws for voltage-only target-state estimation.

A.6. Observation objectives and optional auxiliary losses

For the valid source windows $\mathcal{I}$ and decoded predictions defined in Section 3.3, the Euclidean base objective is

$$\min_{\theta_{\mathrm{loc}},\theta_C,\phi,\psi}\ \frac{1}{|\mathcal{I}|}\sum_{(i,t)\in\mathcal{I}}\left\|\tilde{q}^{(i)}_{t+1} - \hat{q}^{\mathrm{obs},(i)}_{t+1}\right\|^2, \tag{78}$$

Equation (16) uses the corresponding chart-aware residual when coordinates are periodic or constrained. Optional structured training augments this term as

$$\mathcal{L} = \lambda_{\mathrm{data}}\mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{energy}}\mathcal{L}_{\mathrm{energy}} + \lambda_{\mathrm{observer}}\mathcal{L}_{\mathrm{observer}} + \lambda_{\mathrm{pass}}\mathcal{L}_{\mathrm{pass}} + \lambda_{\mathrm{roll}}\mathcal{L}_{\mathrm{roll}}, \tag{79}$$

The optional pH terms are evaluated on the predicted latent transition using start-of-step power estimates. For a sample (i,t), let $\hat{H}_t^{(i)} = H(\hat{x}_t^{(i)})$, $\Delta\hat{H}_t^{(i)} = \hat{H}_{t+1}^{(i)} - \hat{H}_t^{(i)}$, $\hat{v}_t^{(i)} = \partial H/\partial p(\hat{x}_t^{(i)})$, $\hat{y}_t^{\mathrm{port},(i)} = \hat{v}_t^{(i)}$ for mechanical input ports, $\hat{P}_t^{\mathrm{ext},(i)} = (u_t^{(i)})^\top \hat{y}_t^{\mathrm{port},(i)}$, and $\hat{P}_t^{\mathrm{diss},(i)} = -(\hat{v}_t^{(i)})^\top D(\Pi_q(\hat{x}_t^{(i)}))\hat{v}_t^{(i)} \le 0$, where Π_q extracts the model-coordinate configuration component used by the damping module. For typed C-PHAST deployments, these powers are computed per block or declared port and summed before entering the scalar loss. The passivity term is a one-sided hinge on surplus stored energy: it enforces the discrete inequality $\Delta\hat{H}_t \le \Delta t\, \hat{P}_t^{\mathrm{ext}}$,

$$\mathcal{L}_{\mathrm{pass}} = \frac{1}{|\mathcal{I}|} \sum_{(i,t)\in\mathcal{I}} \mathrm{ReLU}\Big(\Delta\hat{H}_t^{(i)} - \Delta t\, \hat{P}_t^{\mathrm{ext},(i)}\Big), \tag{80}$$

which reduces to $\mathrm{ReLU}(\Delta\hat{H}_t)$ for closed ports. If an external port extracts power ($\hat{P}_t^{\mathrm{ext}} < 0$), the same hinge penalizes predicted trajectories that fail to lose enough stored energy. The stricter energy-budget term then matches the discrete Hamiltonian rate to the full pH power balance, including modeled dissipation,

$$\mathcal{L}_{\mathrm{energy}} = \frac{1}{|\mathcal{I}|} \sum_{(i,t)\in\mathcal{I}} \left| \frac{\Delta\hat{H}_t^{(i)}}{\Delta t} - \Big(\hat{P}_t^{\mathrm{diss},(i)} + \hat{P}_t^{\mathrm{ext},(i)}\Big) \right|. \tag{81}$$

This is a first-order training surrogate evaluated at start-of-step power; it is *not* the certified finite-step residual of Theorem D.1, whose $\mathcal{O}(\Delta t^3)$ ledger uses trapezoidal boundary accounting, and the two agree only to first order in Δt. When simulator velocities are available, the q-only observer can also be supervised directly,

$$\mathcal{L}_{\mathrm{observer}} = \frac{1}{|\mathcal{I}|} \sum_{(i,t)\in\mathcal{I}} \left\| \hat{\dot{q}}_t^{(i)} - \dot{q}_t^{(i)} \right\|^2. \tag{82}$$

Finally, the optional rollout term uses a nonempty horizon set $\mathcal{H}_{\mathrm{train}} \subset \mathbb{N}_{>0}$ and only windows with every required target available:

$$\mathcal{I}_{\mathrm{roll}} = \{(i,t) : L_h - 1 \le t, \quad t + \max \mathcal{H}_{\mathrm{train}} \le T_i\}.$$

For nonempty $\mathcal{I}_{\mathrm{roll}}$, it unrolls from the same burn-in state:

$$\mathcal{L}_{\mathrm{roll}} = \frac{1}{|\mathcal{I}_{\mathrm{roll}}|\,|\mathcal{H}_{\mathrm{train}}|} \sum_{(i,t)\in\mathcal{I}_{\mathrm{roll}}} \sum_{h\in\mathcal{H}_{\mathrm{train}}} \left\| \delta_q\Big(\tilde{q}_{t+h}^{(i)}, \hat{q}_{t+h|t}^{\mathrm{obs},(i)}\Big) \right\|^2. \tag{83}$$

Matched-protocol comparisons use only $\mathcal{L}_{\mathrm{data}}$ unless stated otherwise, while the full PHAST-family recipe may additionally activate the energy and observer terms when the corresponding labels or pH diagnostics are available; Appendix A.7 records the shared training settings, with benchmark-specific overrides in the corresponding protocols.

A.7. Mechanical Forecasting: Training Conventions

For mechanical forecasting, the following defaults apply unless a benchmark protocol overrides them. Electrical calibration instead uses the objectives and schedules in Appendix H and Section 3.3. The measurement and evaluation conventions in Appendix G apply across studies; the following benchmark appendices state their study-specific data, losses, and optimizer overrides.

Optimization. AdamW optimizer with learning rate 10^{-3}, weight decay 10^{-5}, cosine annealing over 50 epochs.

Loss function. The mechanical predictors use the next-step position term in Eq. (16). For C-PHAST and monolithic PHAST, the full structured recipe additionally uses the energy-budget term in Eq. (81) with $\lambda_{\mathrm{energy}}$=1.0 and observer supervision in Eq. (82) with $\lambda_{\mathrm{observer}}$=1.0 when simulator $\dot{q}$ is available. These auxiliary terms train internal pH variables and q-only observers, so they are not applied to non-pH baselines. The matched protocol trains all compared models for 50 epochs with only $\mathcal{L}_{\mathrm{data}}$; the full structured PHAST-family rows are trained for 200 epochs with the active pH/observer terms stated in the relevant table caption.

Implementation. Mechanical results are generated on CPU for the robot arm and GPU for Isaac Lab (RTX 5090, 32 GB). A single C-PHAST training run (42,722 params, 200 epochs, Isaac Lab) takes $\sim$25 minutes; inference (rollout N_{pred}=50 on the full test set) takes < 1 s. Fixed-dimension baselines are trained and evaluated at their fixed input dimension; C-PHAST is trained once on $N = 4$ and evaluated on all N. We report means and standard deviations over the completed independent seeds, with counts specified in Appendix G.1 and the individual studies.

A.8. External constraints and objective-conditioned factors

Section 3.4 introduces candidate comparison; Figure 3 shows how current, voltage, and loss predictions acquire different decision roles. Here we specify additional envelopes and aggregation rules for the returned consequence sequence $\rho_{1:N_{\text{pred}}}(a)$. The candidate a, its assigned starting state, and its rollout remain those of Algorithm 1.

Constraints on a candidate future. When the corresponding validated channels are available, a predicted-feasible set can combine a terminal goal, input limits, and operating envelopes:

$$\begin{aligned}\widehat{\mathcal{A}}_{\text{feas}} = \Big\{ a \in \mathcal{A} : \hat{x}_{N_{\text{pred}}}(a) \in \mathcal{X}_G, \quad u_\ell \in \mathcal{U}, \\ c_j\big(\rho_{\ell+1}(a)\big) \le 0 \quad \forall j, \ell, \\ P_\ell(a) \le P_{\max}, \quad \textstyle\sum_\ell P_\ell(a)\,\Delta t \le E_{\max}, \\ \textstyle\sum_\ell r_\ell(a)\,\Delta t \le R_{\max} \Big\}.\end{aligned}$$

Here P_ℓ and r_ℓ are the declared power and risk channels; $P_{\max}$, $E_{\max}$, and $R_{\max}$ are their peak-power, energy, and cumulative-risk budgets. Only constraints relevant to the deployment are included. These are tests on predictions, not guarantees about the physical outcome; Section 3.7 gives the uncertainty-margin conditions. Numerical failure or an empty admitted set requires an external abstention or fallback policy.

From channels to factors. A planner factor reduces selected channels to a scalar cost, reward, margin, or diagnostic. For example, the current check in Figure 3 takes a maximum over the horizon and subtracts a current limit; its loss summary sums interval dissipated energy. Writing $\rho_{m,\ell}(a)$ for channel m at predicted step ℓ, define

$$\phi_r(a) = \eta_r(h_r(\{\rho_{m,\ell}(a) : m \in \mathcal{M}_r,\ \ell = 1, \dots, N_{\text{pred}}\}) - b_r), \qquad r = 1, \dots, d_\phi. \tag{84}$$

The channel set $\mathcal{M}_r$ fixes what is summarized; h_r specifies the horizon reduction, such as a maximum, a sum, or a terminal value. The reference b_r is a limit, budget, or target, and η_r specifies the sign and scaling of the resulting margin. Thus a current margin is formed in amperes before any normalization, whereas an energy budget is compared with an energy sum. These conventions are fixed before candidate comparison.

Collect the d_ϕ factors as $\phi(a) = (\phi_1(a), \dots, \phi_{d_\phi}(a))$. A linear objective and factor-based constraints take the form

$$\hat{J}_w(a) = w^\top \phi(a), \qquad g_j(\phi(a)) \le 0, \quad j = 1, \dots, J. \tag{85}$$

Choosing factors to match the trajectory costs in Eq. (20) recovers its stage-cost score; a terminal cost, when used, is included as an additional factor with weight one. Maximum or residual factors allow other external objectives without changing the predictor. Cost normalization and weights set the trade-off, while hard constraints determine admission. The minimization remains over the supplied candidate set, as in Eq. (20); no continuous action optimizer is assumed.

How the studies use the interface. The electrical example illustrates comparison of proposed connections; the learned electrical study evaluates prediction, not executed reconfiguration. The Go2 early-warning study thresholds predicted envelopes, whereas its single-decision recovery study uses a learned pairwise selector with current state and forecast features. That selector is not the fixed weighted score above, and it selects once without repeated replanning (Appendix L.2). Spot's objective-score study instead uses a direct regression predictor of consequences from logged history and behavior descriptors, then applies five fixed objective weightings. Its concrete channel definitions, normalization, and weights appear in Tables 34 and 35; it is not a PHASTCore hardware rollout. These studies share the separation between predicted consequences and their downstream use, not one universal predictor or selector.

A.9. Benchmark decompositions

Each benchmark fixes its modeling resolution: joints or limbs for mechanical rollouts, excitable cells for FHN, and DGU/line storage blocks for microgrids. Changing this resolution is not an evaluated transfer operation. Spot uses telemetry and consequence interfaces rather than a PHASTCore hardware rollout.

Table 15. **Hierarchy used by each benchmark.** Each row states the physical block, observation or readout interface, and evidence role; parameter sharing is within block type.

Domain	Hierarchy	pH rollout block	Interface / readout level	Evidence role
Planar arm	arm $\rightarrow N$ joints	one 1-DOF joint	joint torque, q-only rollout, energy and coupling diagnostics	clean mechanical count-scaling check
Robot transfer	robot $\rightarrow$ four HAA/HFE/KFE legs	one 3-DOF leg, floating base excluded	q-only leg state and normalized position commands	fixed-layout cross-robot transfer
Go2 recovery	robot $\rightarrow$ body and four legs	separate root-aware body/leg model	root/joint telemetry, contact geometry, terrain, candidate commands	one executed recovery-command choice
FHN circuits	chain of repeated excitable cells	one voltage/recovery cell	source current, cell readout, diffusive graph port	non-robot repeated-block transfer check
Electrical (DGU)	isolated DGU and composed DGU/line networks	fitted DGU storage and damping; supplied RL lines	rich source calibration; voltage-only deployment and voltage/current prediction	five-fit component learning, network prediction, and observability qualification
Spot with arm	robot $\rightarrow$ base, leg support/contact, arm, camera/depth, battery/power interface blocks	telemetry interface blocks; no PHASTCore hardware rollout	telemetry histories, candidate descriptors, typed consequences, and action cards	real-robot typed consequence forecasting

A.10. Transfer assumptions and guarantee status

Any additional training of an observer for a changed sensing interface must be disclosed separately from reuse of the component laws. The transfer assumptions below do not replace those of Section 3.6.1, whose fixed linear–quadratic dissipation results also require the stated damping and initial-distribution conditions. Section 3.6.2 gives optimal initial-state estimation error for known linear dynamics, a Gaussian prior, and sampled observations. Its configuration comparison depends on the prior and sensor response; it is not an achieved-error bound for the learned observer.

The method covers two dynamics-redeployment patterns studied in the paper: *homogeneous repeated-block transfer*, where one local block type is instantiated many times (planar arms and FitzHugh–Nagumo excitable cells), and *heterogeneous typed-block transfer*, where the system is assembled from a fixed finite library of block types (microgrid DGU and line blocks). These assumptions scope the deployment contract for the dynamics model: they state what may change at transfer, what must remain fixed within a block type, and where the continuous-time pH idealization becomes an approximation under finite-step observed-coordinate rollouts. The Spot-with-arm result uses the same typed consequence interface for real robot telemetry and action cards, but it is not a subsystem-count or topology-transfer benchmark.

1. **A1. Finite typed block library.** Each system instance is composed from either one repeated block type or a fixed finite library $\mathcal{T}$ of typed local blocks. Parameter sharing is within type, not across unrelated types.

2. **A2. Within-type local dimension is fixed.** Every block type $\alpha \in \mathcal{T}$ has a fixed local state dimension n_α; a mechanical block with d_α configuration coordinates has $n_\alpha = 2d_\alpha$. Transfer may change components of the deployment specification $\mathcal{S}_{\text{target}}$, such as subsystem count, typed layout, topology, observation interface, or instance metadata, but it does not change the internal dimension of an existing type.

3. **A3. Transfer respects the typed decomposition.** In repeated-block settings, transfer resizes a family of same-type blocks, as in planar arms and FHN cell-count transfer. In fixed-layout embodiment transfer, the typed layout is reused while instance metadata changes, as in ANYmal-D to Go2. In heterogeneous settings, transfer reuses the same type library while changing only declared components of $\mathcal{S}_{\text{target}}$, such as typed layout, topology, observation interface, or instance metadata.

4. **A4. Mechanical rollout state restriction.** In the arm and joint-position transfer benchmarks, the learned pH rollout state contains actuated coordinates only; floating-base pose is excluded. Identity canonicalization gives velocity-like latent coordinates rather than identified physical momenta. The Go2 recovery study includes body pose and velocity in a separately trained body-plus-legs model (Section 4.5; Appendix L). The Spot-with-arm result uses the typed consequence interface on real telemetry rather than this q-only compositional rollout state.

5. **A5. Approximate separability.** The per-subsystem decomposition is exact when the Hamiltonian is separable after canonicalization ($H = \sum_i H_i$); for systems with strongly configuration-dependent cross-coupling mass, it is an approximation whose quality we verify empirically.

6. **A6. Continuous-time vs. deployed rollout gap.** Internal power cancellation is exact in continuous time for collocated ports and skew interconnection. A metric-preserving coupling substep preserves its associated kinetic energy, but the complete discrete rollout retains the integration, boundary-accounting, and state-estimation defects quantified in the finite-step analysis (Appendix D); we treat the balance as a structural prior and report coupling-power diagnostics (Table 27). Joint constraints, impacts, and interface storage or dissipation require their corresponding physical laws.

These assumptions complete the transfer contract for the dynamics model.

Table 16 records the assumptions and certification status of the guarantees summarized in Section 3.7. The detailed proofs and constant constructions are in Appendices B, C, D, E, and F.

Table 16. Guarantee status under the stated assumptions. Internal power cancellation is architectural. The remaining results are conditional theorems, instantiated in selected analytic cases; the trained prediction and recovery experiments do not establish the candidate-wide error envelopes required for certified learned reconfiguration.

Claim	Result	Required assumptions	KNOWN cases	Trained cases
Internal power cancellation	Thm 3.1	exact ports, skew coupling	exact	architectural
Interval energy decay	Thm 3.2	fixed quadratic, unforced; damping rank (93)	analytic	not established
Expected dissipation benefit	Cor 3.3	fixed quadratic, unforced; $J_{E,0}R_E = R_E J_{E,0}$, $K_\kappa R_E \neq R_E K_\kappa$, isotropy	analytic	not established
Initial-state inference	Thm 3.4	known linear dynamics; Gaussian prior and sample noise	analytic	not established
Hidden-mode inference benefit	Cor 3.5	two scalar modes; $P_0 = \sigma_0^2 I_2$, $G_k e_2 \neq 0$	analytic	not established
Finite-step energy balance	Thm D.1	smooth fixed mode, 2nd-order step	verified/certified	diagnostic
Horizon tube	Thm E.2/E.4	regional contraction, model defect	selected certificates	bounds required
Decision margin	Thm F.2	valid state/readout/map bounds	conditional certificate	envelopes required

B. Dissipation under Fixed Linear–Quadratic Composition

This appendix proves Theorem 3.2 and Corollary 3.3. The comparison holds the component energies, damping, state space, and initial distribution fixed; only the supplied lossless interconnection changes. We first relate the normalized dynamics to the paper's port equations, then prove the general energy and decay identities, the sufficient condition for strict expected benefit, and its limitation. All flows below are exact continuous-time flows.

B.1. From port interconnection to energy coordinates

Stack the physical component states as $x = [x_1; \dots; x_s]$. For constant quadratic component energies, $\nabla H_{\text{total}}(x) = Qx$, with $Q = \operatorname{diag}(Q_1, \dots, Q_s) \succ 0$. Close the external ports in Equation (48), and let $\hat{C}_\kappa$ be the supplied skew interconnection

selected by $\mathcal{S}_\kappa$. Substituting $x_E = Q^{1/2}x$ gives

$$\begin{aligned}\dot{x}_E &= Q^{1/2}(\hat{J} - \hat{B}\hat{C}_\kappa\hat{B}^\top - \hat{R})Q^{1/2}x_E \\ &= (J_{E,0} + K_\kappa - R_E)x_E = (J_{E,\kappa} - R_E)x_E,\end{aligned} \tag{86}$$

using the definitions in Equation (27). Congruence by the symmetric matrix $Q^{1/2}$ preserves skew symmetry and positive semidefiniteness, so $J_{E,\kappa}^\top = -J_{E,\kappa}$ and $R_E \succeq 0$. The transformation also preserves the physical energy:

$$H_{\text{total}}(x) = \tfrac{1}{2}x^\top Q x = \tfrac{1}{2}x_E^\top x_E = H_E(x_E).$$

The initial states satisfy $x_{E,0} = Q^{1/2}x_0$, where $x_0 = x(0)$. Substituting $x_E(t) = E_\kappa(t)x_{E,0}$ into $x(t) = Q^{-1/2}x_E(t)$ gives the physical-coordinate flow

$$x(t) = Q^{-1/2}E_\kappa(t)Q^{1/2}x_0. \tag{87}$$

Within this unforced quadratic specialization, the numerical transition F in Equation (13) approximates this flow over a step Δt; Algorithm 1 assembles it at step Q03 and applies it at step Q04. The energy identities below use the exact flow E_κ in energy coordinates. The matrices $J_{E,0}$ and R_E are block diagonal in the component partition; K_κ has zero diagonal component blocks. Isolation sets $\hat{C}_0 = K_0 = 0$. No arbitrary skew matrix is presumed realizable through fixed ports: the corollary applies to the K_κ admitted by Equation (27).

B.2. Energy accounting, strict decay, and the exact comparison

Proof of Theorem 3.2. Fix a configuration κ and a horizon $T > 0$. Within the proof, abbreviate $J_E = J_{E,\kappa}$, $E(t) = E_\kappa(t)$, and $W(T) = W_\kappa(T)$.

For an initial state v, the trajectory is $x_E(t) = E(t)v$. Its energy satisfies

$$\frac{d}{dt}H_E(x_E(t)) = x_E(t)^\top(J_E - R_E)x_E(t) \tag{88}$$

$$= -x_E(t)^\top R_E x_E(t). \tag{89}$$

The second equality uses skew symmetry: the scalar $a^\top J_E a$ equals its transpose, which is $-a^\top J_E a$, and hence vanishes. Substituting the trajectory and integrating yields

$$H_E(v) - H_E(E(T)v) = \int_0^T v^\top E(t)^\top \overbrace{R_E}^{\text{fixed damping}} \underbrace{E(t)}_{\text{flow changed by coupling}} v\, dt \tag{90}$$

$$\begin{aligned}&= v^\top W(T)v \\ &= \int_0^T \|R_E^{1/2}E(t)v\|^2\, dt,\end{aligned} \tag{91}$$

where $R_E^{1/2}$ is the symmetric positive semidefinite square root of R_E.

Since $H_E(v) = \frac{1}{2}\|v\|^2$, this equality holds for every v precisely when

$$E(T)^\top E(T) = I - 2W(T).$$

Each integrand defining $W(T)$ is positive semidefinite, so $W(T) \succeq 0$. The matrix exponential $E(T)$ is invertible, giving $E(T)^\top E(T) \succ 0$ and therefore $0 \preceq W(T) \prec \frac{1}{2}I$.

To determine when every nonzero disturbance loses energy, consider a vector satisfying $v^\top W(T)v = 0$. The energy-loss integrand is continuous and nonnegative. If it were positive at any time, it would have a positive integral over a neighboring interval. Thus

$$v^\top W(T)v = 0 \quad \Longleftrightarrow \quad R_E^{1/2}E(t)v = 0 \quad \text{for every } t \in [0, T].$$

Because R_E and $R_E^{1/2}$ have the same kernel, such a trajectory remains in $\ker R_E$. Along it, the dynamics reduce to $\dot{x}_E = J_E x_E$, so uniqueness gives $E(t)v = e^{J_E t}v$. Conversely, if $R_E^{1/2} e^{J_E t} v = 0$ throughout the interval, then $e^{J_E t}v$ also solves the full dynamics and equals $E(t)v$.

We can therefore test for zero loss using the lossless trajectory. Analyticity implies that $R_E^{1/2} e^{J_E t} v = 0$ throughout $[0, T]$ if and only if every Taylor coefficient $R_E^{1/2} J_E^j v$ vanishes. By the Cayley–Hamilton theorem, powers with $j \geq n_x$ are linear combinations of the first n_x powers. Consequently,

$$\ker W(T) = \bigcap_{j=0}^{n_x - 1} \ker(R_E^{1/2} J_E^j). \tag{92}$$

Equivalently, $W(T) \succ 0$ if and only if

$$\operatorname{rank} \mathcal{O}_{R_E}(J_E) = n_x, \qquad \mathcal{O}_{R_E}(J_E) := \begin{bmatrix} R_E^{1/2} \\ R_E^{1/2} J_E \\ \vdots \\ R_E^{1/2} J_E^{n_x - 1} \end{bmatrix}. \tag{93}$$

This is the observability form of the classical Kalman rank condition in Achleitner et al. (2021, Lemma 3); their Proposition 1 connects the absence of undamped modes to hypocoercive decay.

A nonzero vector v in this kernel generates the subspace

$$\operatorname{span}\{v, J_E v, \ldots, J_E^{n_x - 1} v\}.$$

Every vector in the span lies in $\ker R_E$, and Cayley–Hamilton makes the span invariant under J_E. Conversely, a nonzero J_E-invariant subspace contained in $\ker R_E$ supplies a nonzero vector whose lossless trajectory never leaves that kernel. This proves the invariant-subspace criterion. The criterion does not depend on the positive horizon: $W(T)$ is positive definite for one $T > 0$ if and only if it is positive definite for every $T > 0$.

Let $\eta = \lambda_{\min}(W(T))$. The energy identity now gives, for every v,

$$\begin{aligned} H_E(E(T)v) &= H_E(v) - v^\top W(T) v \\ &\leq H_E(v) - \eta \|v\|^2 \\ &= (1 - 2\eta) H_E(v). \end{aligned} \tag{94}$$

When $\eta > 0$, the previously established bounds on $W(T)$ imply $0 < \rho := 1 - 2\eta < 1$. Thus every interval of duration T removes at least the fraction $1 - \rho$ of the energy present at its start. The estimate holds for every starting state, including the state reached after an earlier interval. Since the dynamics remain fixed,

$$\begin{aligned} H_E(x_E((\ell + 1)T)) &= H_E(E(T) x_E(\ell T)) \\ &\leq \rho H_E(x_E(\ell T)). \end{aligned}$$

Induction therefore yields

$$H_E(x_E(\ell T)) \leq \rho^\ell H_E(x_{E,0}), \qquad \ell = 0, 1, 2, \ldots.$$

This proves the interval-decay conclusion. Positive definite instantaneous damping is unnecessary: lossless transport can make $W(T)$ positive definite even when R_E is singular.

The same argument gives a bound at every time. Write

$$t = \ell T + r, \qquad \ell = \lfloor t/T \rfloor, \qquad 0 \leq r < T.$$

Energy is nonincreasing, so

$$H_E(x_E(t)) \leq H_E(x_E(\ell T)) \leq \rho^{\lfloor t/T \rfloor} H_E(x_{E,0}).$$

For this fixed configuration and T, define $\alpha = -\log(\rho)/T > 0$. Then

$$\rho^{\ell} = e^{-\alpha(t-r)} \le e^{\alpha T} e^{-\alpha t} = \rho^{-1} e^{-\alpha t},$$

and hence

$$H_E(x_E(t)) \le \rho^{-1} e^{-\alpha t} H_E(x_{E,0}), \qquad t \ge 0. \tag{95}$$

The prefactor accounts for the incomplete interval; the contraction estimate does not require energy to decrease at a constant rate within each interval.

Taking square roots gives

$$\|x_E(t)\| \le \rho^{-1/2} e^{-\alpha t/2} \|x_{E,0}\|. \tag{96}$$

In physical coordinates, use

$$\lambda_{\min}(Q)\|x\|^2 \le x^\top Q x \le \lambda_{\max}(Q)\|x\|^2$$

to obtain

$$\|x(t)\| \le \sqrt{\frac{\lambda_{\max}(Q)}{\rho\,\lambda_{\min}(Q)}}\, e^{-\alpha t/2} \|x_0\|. \tag{97}$$

These are bounds on the energy and state of a single configuration. A comparison with isolation requires the corresponding energy losses in both configurations.

Restore the configuration subscripts and let the initial state be random with finite second moment M_0. Since $W_\kappa(T)$ is fixed, the identity

$$x_{E,0}^\top W_\kappa(T) x_{E,0} = \operatorname{tr}\big(W_\kappa(T) x_{E,0} x_{E,0}^\top\big)$$

places all dependence on the random initial state in its outer product. Linearity of expectation and trace gives

$$\begin{aligned} \mathcal{D}_\kappa(T) &= \mathbb{E}[x_{E,0}^\top W_\kappa(T) x_{E,0}] \\ &= \operatorname{tr}\big(W_\kappa(T)\mathbb{E}[x_{E,0} x_{E,0}^\top]\big) \\ &= \operatorname{tr}(W_\kappa(T) M_0). \end{aligned} \tag{98}$$

Subtracting the same expression for isolation proves Equation (33). In isolation, $J_{E,0} - R_E$, $E_0(t)$, and $W_0(T)$ are block diagonal. The quadratic form $x_{E,0}^\top W_0(T) x_{E,0}$ is therefore the sum of the isolated component losses. Taking its expectation gives their sum under the same component marginals, even when the component initial states are correlated. □

The initial distribution determines how the comparison weights different disturbance directions. If u_i are orthonormal eigenvectors of $W_\kappa(T) - W_0(T)$, with eigenvalues λ_i, then

$$\mathcal{D}_\kappa(T) - \mathcal{D}_0(T) = \sum_{i=1}^{n_x} \lambda_i\, \mathbb{E}[(u_i^\top x_{E,0})^2]. \tag{99}$$

When the Gramian difference has eigenvalues of both signs, changing the initial distribution can reverse the expected benefit. Its second moment includes both variation and a systematic initial displacement: if $m_0 = \mathbb{E}[x_{E,0}]$ and $P_0 = \operatorname{Cov}(x_{E,0})$, then $M_0 = P_0 + m_0 m_0^\top$.

A nonnegative benefit for every deterministic initial state requires $W_\kappa(T) - W_0(T) \succeq 0$; strict benefit for every nonzero initial state requires positive definiteness. A positive expected benefit does not imply either matrix condition. Likewise, $W_\kappa(T) \succ 0$ establishes decay of the connected configuration without ordering it relative to isolation.

B.3. A structural condition for strict expected benefit

We use the following classical matrix-exponential trace inequality and its equality case (Bernstein, 1988; So, 1992): for a real square matrix X,

$$\operatorname{tr}(e^{X^\top} e^{X}) \le \operatorname{tr}(e^{X^\top + X}), \tag{100}$$

with equality if and only if X is normal, meaning $X^\top X = X X^\top$. We apply this classical bound to the port-assembled generator.

Proof of Corollary 3.3. For any skew matrix J_E, apply Equation (100) with $X = T(J_E - R_E)$. Since $X^\top + X = -2TR_E$,

$$\operatorname{tr}\big(e^{(J_E-R_E)^\top T} e^{(J_E-R_E)T}\big) \le \operatorname{tr}(e^{-2R_E T}). \tag{101}$$

The equality condition has a direct damping interpretation:

$$(J_E - R_E)^\top (J_E - R_E) = -J_E^2 + J_E R_E - R_E J_E + R_E^2,$$
$$(J_E - R_E)(J_E - R_E)^\top = -J_E^2 - J_E R_E + R_E J_E + R_E^2.$$

Their difference is $2(J_E R_E - R_E J_E)$. Because $T > 0$, X is normal if and only if $J_E R_E = R_E J_E$.

The assumption $J_{E,0} R_E = R_E J_{E,0}$ therefore makes the isolated flow attain the upper bound in Equation (101):

$$\operatorname{tr}(E_0(T)^\top E_0(T)) = \operatorname{tr}(e^{-2R_E T}).$$

The connected flow is bounded above by the same quantity. Since $M_0 = \sigma_0^2 I$, the expected residual energy is

$$\mathbb{E} H_E(E_\kappa(T) x_{E,0}) = \frac{\sigma_0^2}{2} \operatorname{tr}(E_\kappa(T)^\top E_\kappa(T)). \tag{102}$$

The initial expected energy is common to both configurations, hence

$$\mathcal{D}_\kappa(T) - \mathcal{D}_0(T) = \frac{\sigma_0^2}{2} \Big[\underbrace{\operatorname{tr}(E_0(T)^\top E_0(T))}_{\text{isolated}} - \underbrace{\operatorname{tr}(E_\kappa(T)^\top E_\kappa(T))}_{\text{connected}} \Big] \ge 0. \tag{103}$$

Equality holds exactly when $J_{E,\kappa} R_E = R_E J_{E,\kappa}$. Since $J_{E,\kappa} = J_{E,0} + K_\kappa$ and $J_{E,0} R_E = R_E J_{E,0}$, this is equivalent to $K_\kappa R_E = R_E K_\kappa$. Thus a noncommuting connection gives strict benefit for every $T > 0$; a commuting connection gives equality for every $T > 0$. □

The hypothesis $M_0 = \sigma_0^2 I$ specifies the second moment, not necessarily a centered Gaussian distribution. If a covariance is used instead, it equals this second moment only for zero mean. The commutation condition on the baseline is essential to this argument: when isolated components already mix different damping directions, both traces may be strictly below the common upper bound, which does not order them. Section B.5 gives an explicit reversal.

The isotropic second moment in energy coordinates is equivalent to $\mathbb{E}[x_0 x_0^\top] = \sigma_0^2 Q^{-1}$ in physical coordinates. The condition $J_{E,0} R_E = R_E J_{E,0}$ means that isolated dynamics preserve the damping eigenspaces. In a basis where $R_E = \operatorname{diag}(d_1, \dots, d_{n_x})$, $(K_\kappa R_E - R_E K_\kappa)_{ij} = (d_j - d_i)(K_\kappa)_{ij}$; strict benefit therefore requires coupling different damping rates. For example, two scalar components with $d_1 > d_2 \ge 0$ and nonzero skew coupling satisfy the corollary at every positive horizon, as shown next. The all-time energy and state bounds are Equations (95)–(97).

B.4. Two scalar components: the complete finite-time comparison

Consider two scalar energy coordinates, with damping rates $d_1 > d_2 \ge 0$ and coupling strength $k \in \mathbb{R}$. Set

$$L_k = \begin{bmatrix} -d_1 & -k \\ k & -d_2 \end{bmatrix}, \qquad E_k(t) = e^{L_k t}, \qquad M_0 = \sigma_0^2 I_2, \tag{104}$$

where $\sigma_0 > 0$. Here L_k is the continuous generator, while $E_k(t)$ is its exact flow. Let $\overline{H}_k(t) = \mathbb{E} H_E(E_k(t) x_{E,0})$. The isolated baseline is $k = 0$.

Proposition B.1 (Two-mode expected dissipation)**.** *For $d_1 > d_2 \ge 0$, $k \ne 0$, and every $t > 0$, $\overline{H}_k(t) < \overline{H}_0(t)$. Equivalently, the connected system has strictly greater expected dissipated energy over $[0, t]$.*

Proof. Put $m = (d_1 + d_2)/2$ and $\delta = (d_1 - d_2)/2 > 0$. Then

$$L_k = -mI + B_k, \qquad B_k = \begin{bmatrix} -\delta & -k \\ k & \delta \end{bmatrix}, \qquad B_k^2 = (\delta^2 - k^2) I. \tag{105}$$

Separating the even and odd powers in the exponential series gives

$$E_k(t) = e^{-mt}(c_k(t)I + s_k(t)B_k), \tag{106}$$

where

regime	$c_k(t)$	$s_k(t)$
$\lvert k\rvert < \delta$	$\cosh(\beta t)$	$\sinh(\beta t)/\beta$
$\lvert k\rvert = \delta$	1	t
$\lvert k\rvert > \delta$	$\cos(\omega t)$	$\sin(\omega t)/\omega$

(107)

with $\beta = \sqrt{\delta^2 - k^2}$ in the first row and $\omega = \sqrt{k^2 - \delta^2}$ in the last. Each row satisfies

$$c_k(t)^2 + (k^2 - \delta^2)s_k(t)^2 = 1. \tag{108}$$

Since $\operatorname{tr}(B_k + B_k^\top) = 0$ and $\operatorname{tr}(B_k^\top B_k) = 2(\delta^2 + k^2)$,

$$\begin{aligned}\operatorname{tr}(E_k(t)^\top E_k(t)) &= e^{-2mt}\operatorname{tr}\big[c_k^2 I + c_k s_k(B_k + B_k^\top) + s_k^2 B_k^\top B_k\big] \\ &= 2e^{-2mt}\big[c_k^2 + (\delta^2 + k^2)s_k^2\big] \\ &= 2e^{-2mt}\big[1 + 2\delta^2 s_k(t)^2\big].\end{aligned} \tag{109}$$

The last equality substitutes $c_k^2 = 1 - (k^2 - \delta^2)s_k^2$ from Equation (108). Equation (102) now gives

$$\overline{H}_k(t) = \sigma_0^2 e^{-2mt}\big[1 + 2\delta^2 s_k(t)^2\big]. \tag{110}$$

For isolation $s_0(t) = \sinh(\delta t)/\delta$, so

$$\begin{aligned}\overline{H}_0(t) &= \sigma_0^2 e^{-2mt}\big[1 + 2\sinh^2(\delta t)\big] \\ &= \tfrac{\sigma_0^2}{2}(e^{-2d_1 t} + e^{-2d_2 t}).\end{aligned} \tag{111}$$

We compare $s_k(t)^2$ with $s_0(t)^2$ in all three regimes.

If $0 < |k| < \delta$, then $0 < \beta < \delta$. For fixed $t > 0$, the function $a \mapsto \sinh(at)/a$ is strictly increasing for $a > 0$. Its derivative has numerator $at\cosh(at) - \sinh(at)$; this numerator vanishes at $a = 0$ and has derivative $at^2\sinh(at) > 0$ for $a > 0$. Therefore $0 < s_k(t) < s_0(t)$.

If $|k| = \delta$, then $s_k(t) = t$. Since $\sinh r > r$ for $r > 0$, $t < \sinh(\delta t)/\delta = s_0(t)$.

If $|k| > \delta$, the inequality $|\sin r| \le |r|$ gives $|s_k(t)| \le t < s_0(t)$. In every case with $k \neq 0$, $s_k(t)^2 < s_0(t)^2$, so

$$\overline{H}_0(t) - \overline{H}_k(t) = 2\sigma_0^2 e^{-2mt}\big[\sinh^2(\delta t) - \delta^2 s_k(t)^2\big] > 0. \tag{112}$$

The expected initial energy is the same in both configurations. Subtracting residual energy from it proves the dissipation claim. □

The strict comparison disappears for $k = 0$ or $t = 0$. If $d_1 = d_2 = d$, then $E_k(t) = e^{-dt}e^{kJ_2 t}$ with $J_2 = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$; the second factor is orthogonal, so every trajectory has the same energy as in isolation. This is the commuting case of the corollary.

For $|k| > \delta$, the horizon

$$T_* = \frac{\pi}{\sqrt{k^2 - \delta^2}} \tag{113}$$

makes $s_k(T_*) = 0$ and $E_k(T_*) = -e^{-mT_*}I$. Thus the strict gap takes the particularly simple form

$$\overline{H}_0(T_*) - \overline{H}_k(T_*) = \frac{\sigma_0^2}{2}\big(e^{-d_1 T_*} - e^{-d_2 T_*}\big)^2. \tag{114}$$

This special horizon is not needed for the all-horizon result. It also shows why an expected comparison is different from a statewise one. For the deterministic initial state $x_{E,0} = (1, 0)^\top$, the isolated residual energy at T_* is $\frac{1}{2}e^{-2d_1 T_*}$, whereas the connected residual energy is $\frac{1}{2}e^{-2mT_*}$. Because $m < d_1$, coupling dissipates *less* for this initial state.

When $d_2 = 0$, isolation retains energy in its second coordinate. For $k \neq 0$, no nonzero kJ_2-invariant subspace lies in $\ker\operatorname{diag}(d_1, 0)$, so the theorem implies decay to zero for the connected system. When $d_2 > 0$, both systems decay to zero; the positive finite-time gap tends to zero as $t \to \infty$. The proposition compares finite-time energy removal, not a universal strict gain in total energy dissipated over an infinite horizon.

B.5. Counterexample when isolated components already mix damping directions

The condition $J_{E,0}R_E = R_E J_{E,0}$ in the corollary cannot be dropped. Consider two identical two-coordinate components $a, b \in \mathbb{R}^2$, already in energy coordinates, with

$$J_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad R_{E,*} = \operatorname{diag}(d_1, d_2), \qquad d_1 > d_2 \geq 0. \tag{115}$$

Their common energy is $H_E(a,b) = \frac{1}{2}(\|a\|^2 + \|b\|^2)$. Let $\delta = (d_1 - d_2)/2$, $m = (d_1 + d_2)/2$, and choose $k > \delta$. The isolated and added interconnection matrices are

$$J_{E,0} = \begin{bmatrix} kJ_2 & 0 \\ 0 & kJ_2 \end{bmatrix}, \quad R_E = \begin{bmatrix} R_{E,*} & 0 \\ 0 & R_{E,*} \end{bmatrix}, \quad K = \begin{bmatrix} 0 & kJ_2 \\ kJ_2 & 0 \end{bmatrix}. \tag{116}$$

Here $K^\top = -K$ and its diagonal component blocks are zero, so it is a genuine lossless connection between the unchanged components. It is realizable by the paper's port equation with $Q = I_4$, $\hat{B} = I_4$, and $\hat{C}_\kappa = -K$. The connected equations are

$$\begin{aligned} \dot{a} &= (kJ_2 - R_{E,*})a + kJ_2 b, \\ \dot{b} &= (kJ_2 - R_{E,*})b + kJ_2 a. \end{aligned} \tag{117}$$

The cross power is $a^\top kJ_2 b + b^\top kJ_2 a = 0$. Neither the component energies nor their damping has changed.

Proposition B.2 (Reduced dissipation under lossless coupling)**.** *For the preceding assembly, let $\mathbb{E}[(a_0, b_0)(a_0, b_0)^\top] = \sigma_0^2 I_4$ with $\sigma_0 > 0$, where (a_0, b_0) denotes the stacked vector. At the positive horizon T_* in Equation (113), the connected system has strictly higher expected residual energy and strictly lower expected dissipated energy than the isolated components.*

Proof. Introduce the sum and difference coordinates

$$u = \frac{a+b}{\sqrt{2}}, \qquad v = \frac{a-b}{\sqrt{2}}, \qquad P = \frac{1}{\sqrt{2}} \begin{bmatrix} I_2 & I_2 \\ I_2 & -I_2 \end{bmatrix}. \tag{118}$$

Since $P^\top P = I_4$, this transformation preserves both energy and the isotropic second moment. Adding and subtracting Equation (117) yields

$$\dot{u} = (2kJ_2 - R_{E,*})u, \qquad \dot{v} = -R_{E,*}v. \tag{119}$$

Thus the connection doubles mixing in the sum mode and cancels it in the difference mode. The connected residual energy is the sum of a two-mode block with strength $2k$ and one with strength 0. Isolation has two identical blocks with strength k. Use $s_q(t)$ from Equation (107) with coupling argument q. At T_*, $s_k(T_*) = 0$, so Equation (110) gives

$$\begin{aligned} \overline{H}_{\mathrm{iso}}(T_*) &= 2\overline{H}_k(T_*) = 2\sigma_0^2 e^{-2mT_*}, \\ \overline{H}_{\mathrm{conn}}(T_*) &= \overline{H}_{2k}(T_*) + \overline{H}_0(T_*) \\ &= 2\sigma_0^2 e^{-2mT_*} \left[1 + \delta^2 \big(s_{2k}(T_*)^2 + s_0(T_*)^2\big)\right]. \end{aligned} \tag{120}$$

Here $\overline{H}_{\mathrm{iso}}$ and $\overline{H}_{\mathrm{conn}}$ denote the expected four-coordinate residual energies. Because $\delta > 0$ and $s_0(T_*) = \sinh(\delta T_*)/\delta > 0$,

$$\overline{H}_{\mathrm{conn}}(T_*) - \overline{H}_{\mathrm{iso}}(T_*) = 2\sigma_0^2 e^{-2mT_*} \delta^2 \big(s_{2k}(T_*)^2 + s_0(T_*)^2\big) > 0. \tag{121}$$

The initial expected energy is common, so the difference in expected dissipated energy is the negative of this strictly positive quantity. □

In this example $J_{E,0}R_E \neq R_E J_{E,0}$: each isolated component already transports energy between unequal damping rates. A lossless connection can undo some of that transport. The example refutes a universal connected-versus-isolated benefit, while preserving the energy identity, the decay criterion, and the exact trace comparison of Theorem 3.2. It makes no claim that the reversal occurs at every horizon.

C. State Inference under Fixed Composition

This appendix proves Theorem 3.4 and Corollary 3.5. It separates information gained from a measurement history from improvement relative to a different configuration. The distinction matters when prior correlations already make an unmeasured coordinate partly predictable. All observation maps below use known dynamics and exact continuous-time flows; the conclusions do not certify the error of a fitted nonlinear initializer.

C.1. Sampled observations and the Gaussian posterior

Use the energy coordinates and flow E_κ from Equation (28). A known prescribed additive input extends the unforced dynamics to

$$\dot{x}_E = (J_{E,\kappa} - R_E)x_E + f_{E,\kappa}(s). \tag{122}$$

Here $f_{E,\kappa}$ is deterministic and integrable over the observation interval. Variation of constants gives

$$\begin{aligned} x_E(s) &= E_\kappa(s)x_{E,0} + r_\kappa(s), \\ r_\kappa(s) &= \int_0^s E_\kappa(s-v)f_{E,\kappa}(v)\,dv. \end{aligned} \tag{123}$$

Thus the known input changes the mean response without changing the linear map from the initial state. At the prescribed elapsed times s_r, $r = 0, \dots, L_h - 1$, the block rows of G_κ are $C_{\mathrm{obs}}E_\kappa(s_r)$, and the corresponding blocks of b_κ are $C_{\mathrm{obs}}r_\kappa(s_r)$. This gives Equation (36). The matrix C_{obs} measures energy coordinates; a physical linear sensor $C_{\mathrm{phys}}x$ has $C_{\mathrm{obs}} = C_{\mathrm{phys}}Q^{-1/2}$. The prior describes the state at the start of this observed interval, and the sample model must be the one that generated the observations.

Proof of Theorem 3.4. Fix a configuration κ and abbreviate $G = G_\kappa$, $b = b_\kappa$, $\boldsymbol{y} = \boldsymbol{y}_\kappa$, and $\epsilon = \epsilon_\kappa$ within this proof. The assumptions are

$$x_{E,0} \sim \mathcal{N}(m_0, P_0), \qquad P_0 \succ 0, \qquad \boldsymbol{y} = b + Gx_{E,0} + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \Gamma), \quad \Gamma \succ 0,$$

with ϵ independent of $x_{E,0}$. The matrices, known response, and sampling times are fixed rather than selected from the realized observations. For $v \in \mathbb{R}^{n_x}$, the quadratic form $v^\top W_\kappa^{\mathrm{obs}} v = \|\Gamma^{-1/2}G_\kappa v\|^2$ is the squared sensor response relative to noise. The Gaussian likelihood times the prior density has exponent, up to a term independent of a candidate initial state x,

$$-\tfrac{1}{2}(x - m_0)^\top P_0^{-1}(x - m_0) - \tfrac{1}{2}(\boldsymbol{y} - b - Gx)^\top \Gamma^{-1}(\boldsymbol{y} - b - Gx).$$

Set $u = x - m_0$ and $d = \boldsymbol{y} - b - Gm_0$. Expanding the second quadratic gives

$$-\tfrac{1}{2}u^\top \Big(\underbrace{P_0^{-1}}_{\substack{\text{prior} \\ \text{precision}}} + \underbrace{G^\top\Gamma^{-1}G}_{\substack{\text{information} \\ \text{from samples}}} \Big) u + u^\top G^\top \Gamma^{-1} d + \text{constant}. \tag{124}$$

Since $P_0^{-1} \succ 0$ and $G^\top\Gamma^{-1}G \succeq 0$, the precision matrix is positive definite even if G is rank deficient. Completing the square proves that $x_{E,0} \mid \boldsymbol{y}_\kappa$ is Gaussian with

$$\begin{aligned} \Sigma_\kappa &= (P_0^{-1} + W_\kappa^{\mathrm{obs}})^{-1}, \\ m_\kappa &= m_0 + \Sigma_\kappa G_\kappa^\top \Gamma^{-1}(\boldsymbol{y}_\kappa - b_\kappa - G_\kappa m_0). \end{aligned} \tag{125}$$

These are the usual Gaussian-conditioning formulas (Särkkä, 2013; Qian et al., 2021). In particular, the observed values enter the posterior mean, whereas the posterior covariance depends only on the specified prior, observation map, and noise covariance.

For a fixed target $c_{\mathrm{tar}}^\top x_{E,0}$, let $a(\boldsymbol{y}_\kappa)$ be any square-integrable estimator. The posterior residual has conditional mean zero, so expanding the squared error and taking expectations gives

$$\begin{aligned} &\mathbb{E}\big[(a(\boldsymbol{y}_\kappa) - c_{\mathrm{tar}}^\top x_{E,0})^2\big] \\ &\quad = \mathbb{E}\big[(a(\boldsymbol{y}_\kappa) - c_{\mathrm{tar}}^\top m_\kappa)^2\big] + c_{\mathrm{tar}}^\top \Sigma_\kappa c_{\mathrm{tar}}. \end{aligned} \tag{126}$$

Both expectations include initial-state and measurement-noise randomness. Thus the posterior mean of the target attains the minimum in Equation (39). The proper prior and finite, positive-definite noise covariance also imply $\Sigma_\kappa \succ 0$, so every nonzero target retains positive uncertainty after these finite samples.

To characterize strict improvement over the prior, put

$$\mathcal{B} = P_0^{1/2} W_\kappa^{\mathrm{obs}} P_0^{1/2} \succeq 0, \qquad v = P_0^{1/2} c_{\mathrm{tar}}.$$

Factoring the posterior precision by $P_0^{-1/2}$ yields

$$\begin{aligned} c_{\mathrm{tar}}^\top (P_0 - \Sigma_\kappa) c_{\mathrm{tar}} &= v^\top \left[I - (I + \mathcal{B})^{-1} \right] v \\ &= v^\top \mathcal{B} (I + \mathcal{B})^{-1} v. \end{aligned} \tag{127}$$

In an eigenbasis of $\mathcal{B}$, each coefficient is $\lambda/(1+\lambda)$ with $\lambda \geq 0$. The expression is therefore nonnegative, and it is positive exactly when $\mathcal{B}v \neq 0$. This condition is equivalent to $W_\kappa^{\mathrm{obs}} P_0 c_{\mathrm{tar}} \neq 0$. Moreover,

$$\ker(G_\kappa^\top \Gamma^{-1} G_\kappa) = \ker G_\kappa,$$

because the corresponding quadratic form is $\|\Gamma^{-1/2} G_\kappa v\|^2$. Consequently strict prior-risk reduction is equivalent to $G_\kappa P_0 c_{\mathrm{tar}} \neq 0$, as asserted in Equation (40).

Finally, applying Equation (39) to both configurations under the same prior and target gives Equation (41) by subtraction. The criterion for gain relative to the prior does not establish the sign of this configuration comparison. □

C.2. The two-mode benefit and the sampling condition

Use L_k and $E_k(s) = e^{L_k s}$ from Equation (104), with $C_{\mathrm{obs}} = [1, 0]$ and $e_2 = (0, 1)^\top$. For this inference result, any finite $d_1, d_2 \geq 0$ are allowed; unequal damping is not required.

Proof of Corollary 3.5. Set $P_0 = \sigma_0^2 I_2$, $\sigma_0 > 0$, and let the Gaussian sample noise have the specified covariance $\Gamma \succ 0$, independently of the initial state. Theorem 3.4 gives the posterior covariance and the target risk for $c_{\mathrm{tar}} = e_2$. In isolation,

$$C_{\mathrm{obs}} E_0(s_r) = [e^{-d_1 s_r}, 0],$$

so $G_0 e_2 = 0$. Independence of the two initial coordinates then gives $\mathrm{MSE}_0(e_2) = \sigma_0^2$. For any k, the theorem's strict target criterion reduces to $G_k e_2 \neq 0$. Since $\Sigma_k \succ 0$, this proves the two-sided inequality in Equation (43).

The scalar inversion also quantifies the benefit. Write $w_{ij,k}$ for the entries of W_k^{obs} and set $\alpha = \sigma_0^{-2}$. Then

$$\iota_k = w_{22,k} - \frac{w_{12,k}^2}{\alpha + w_{11,k}},$$

$$\mathrm{MSE}_k(e_2) = \frac{1}{\underbrace{\alpha}_{\substack{\text{prior} \\ \text{precision}}} + \underbrace{\iota_k}_{\substack{\text{additional measurement} \\ \text{precision}}}},$$

$$\mathrm{MSE}_0(e_2) - \mathrm{MSE}_k(e_2) = \frac{\iota_k}{\alpha(\alpha + \iota_k)}. \tag{128}$$

Here ι_k is the additional marginal precision for the hidden coordinate. If the first coordinate were known exactly, the measurement contribution would instead be $w_{22,k}$; the subtracted term accounts for the uncertainty in that first coordinate when both starts affect the same measurements. Positive semidefiniteness gives $w_{12,k}^2 \leq w_{11,k} w_{22,k}$, hence

$$\iota_k \geq \frac{\alpha w_{22,k}}{\alpha + w_{11,k}}, \qquad w_{22,k} = \|\Gamma^{-1/2} G_k e_2\|^2. \tag{129}$$

Thus $\iota_k > 0$ exactly when $G_k e_2 \neq 0$. When $G_k e_2 = 0$, both $w_{22,k}$ and $w_{12,k}$ vanish. □

Nonzero coupling alone does not suffice for arbitrary sample times. Put $m = (d_1 + d_2)/2$, $\delta = (d_1 - d_2)/2$, and suppose $k^2 > \delta^2$, with $\omega = \sqrt{k^2 - \delta^2}$. The two-mode exponential in Equation (106) gives

$$C_{\mathrm{obs}} E_k(s) e_2 = -k e^{-ms} \frac{\sin(\omega s)}{\omega}. \tag{130}$$

The same exponential calculation holds regardless of the sign of $d_1 - d_2$. If every sampled s_r is an integer multiple of π/ω, then $G_k e_2 = 0$: the samples miss the hidden coordinate even though the continuous system is observable. Conversely, $C_{\mathrm{obs}} E_k(s) e_2 = -ks + O(s^2)$ near zero, so every nonzero k admits a sufficiently early positive sample time with a nonzero response. Full column rank of G_k is sufficient but not necessary for this particular target's risk to decrease. A single mixed measurement can reduce its uncertainty because the other initial coordinate already has a finite prior variance.

C.3. The continuous-observation counterpart

An entire noisy observation record has a different noise convention from finite point samples. After subtracting any known deterministic input response, let

$$dY_\kappa^c(s) = C_{\mathrm{obs}} E_\kappa(s) x_{E,0}\, ds + \Gamma_c^{1/2}\, dB_{\mathrm{obs}}(s), \qquad Y_\kappa^c(0) = 0. \tag{131}$$

Here $0 < T < \infty$, C_{obs} has d_{obs} rows, $\Gamma_c \in \mathbb{R}^{d_{\mathrm{obs}} \times d_{\mathrm{obs}}}$ is a known positive-definite noise intensity, and B_{obs} is standard d_{obs}-dimensional Brownian motion independent of $x_{E,0} \sim \mathcal{N}(m_0, P_0)$. There is no process noise or unknown input. Define

$$W_\kappa^{\mathrm{obs},c} = \int_0^T \underbrace{E_\kappa(s)^\top C_{\mathrm{obs}}^\top}\, \overbrace{\Gamma_c^{-1}}^{\text{noise-intensity weighting}}\, \underbrace{C_{\mathrm{obs}} E_\kappa(s)}_{\text{continuous sensor response}}\, ds,$$

$$\nu_\kappa^c = \int_0^T E_\kappa(s)^\top C_{\mathrm{obs}}^\top \Gamma_c^{-1}\, dY_\kappa^c(s). \tag{132}$$

The integrand in the stochastic integral is deterministic. The first quantity determines the posterior covariance; the second carries the realized record into the posterior mean.

Proposition C.1 (Inference from a continuous noisy history)**.** *Under the preceding assumptions, the initial state conditional on the entire record $Y_\kappa^c|_{[0,T]}$ is Gaussian, with*

$$\begin{aligned} \Sigma_\kappa^c &= (P_0^{-1} + W_\kappa^{\mathrm{obs},c})^{-1}, \\ m_\kappa^c &= \Sigma_\kappa^c (P_0^{-1} m_0 + \nu_\kappa^c). \end{aligned} \tag{133}$$

The minimum expected squared error for $c_{\mathrm{tar}}^\top x_{E,0}$ is $c_{\mathrm{tar}}^\top \Sigma_\kappa^c c_{\mathrm{tar}}$, with strict reduction from its prior variance if and only if $W_\kappa^{\mathrm{obs},c} P_0 c_{\mathrm{tar}} \neq 0$.

Proof. For a fixed candidate initial state x, the whitened observation $\Gamma_c^{-1/2} Y_\kappa^c$ is standard Brownian motion plus deterministic drift $\Gamma_c^{-1/2} C_{\mathrm{obs}} E_\kappa(s) x$. This drift has finite squared integral on $[0, T]$. The deterministic-drift Brownian likelihood (Särkkä & Solin, 2019, Corollary 7.6, Eq. (7.42)) gives

$$\begin{aligned} \log \mathcal{L}_\kappa(x; Y_\kappa^c) &= \int_0^T (\Gamma_c^{-1/2} C_{\mathrm{obs}} E_\kappa(s) x)^\top d(\Gamma_c^{-1/2} Y_\kappa^c(s)) \\ &\quad - \tfrac{1}{2} \int_0^T \|\Gamma_c^{-1/2} C_{\mathrm{obs}} E_\kappa(s) x\|^2\, ds \\ &= x^\top \nu_\kappa^c - \tfrac{1}{2} x^\top W_\kappa^{\mathrm{obs},c} x. \end{aligned} \tag{134}$$

The likelihood is relative to the zero-drift observation law. The finite drift energy is required for each fixed x; no additional unconditional exponential-moment assumption on the Gaussian prior is needed here. Multiplying by the prior density gives

$$-\tfrac{1}{2} x^\top (P_0^{-1} + W_\kappa^{\mathrm{obs},c}) x + x^\top (P_0^{-1} m_0 + \nu_\kappa^c) + \text{constant}.$$

Completing the square proves Equation (133). The conditional-mean decomposition in Equation (126) applies to estimators measurable with respect to this record and proves the optimal risk. Finally, the argument in Equation (127), with $W_\kappa^{\mathrm{obs},c}$ in place of W_κ^{obs}, proves the strict target criterion. □

For the two-mode model, set $C_{\mathrm{obs}} = [1, 0]$, $\Gamma_c = \sigma_y^2 > 0$, and $P_0 = \sigma_0^2 I_2$, with $\sigma_0 > 0$. Write $\mathrm{MSE}_k^c(e_2) = e_2^\top \Sigma_k^c e_2$. The continuous observability matrix is

$$\begin{bmatrix} C_{\mathrm{obs}} \\ C_{\mathrm{obs}} L_k \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -d_1 & -k \end{bmatrix}. \tag{135}$$

It has full rank exactly when $k \neq 0$. To see directly why this implies a positive-definite observation Gramian on every positive interval, suppose $v^\top W_k^{\mathrm{obs},c} v = 0$. Continuity and nonnegativity imply $C_{\mathrm{obs}} E_k(s) v = 0$ throughout $[0, T]$. The value and derivative at zero give $v_1 = 0$ and $-d_1 v_1 - k v_2 = 0$. If $k \neq 0$, then $v = 0$, proving $W_k^{\mathrm{obs},c} \succ 0$. At $k = 0$, the entire record depends only on the independent first coordinate and observation noise. Proposition C.1 therefore yields

$$0 < \mathrm{MSE}_k^c(e_2) < \mathrm{MSE}_0^c(e_2) = \sigma_0^2 \qquad (k \neq 0,\ T > 0). \tag{136}$$

The Schur formulas and positivity lower bound in Equations (128)–(129) also apply with entries of the continuous Gramian, whose hidden-coordinate diagonal entry is now

$$w_{22,k} = \sigma_y^{-2} \int_0^T (C_{\mathrm{obs}} E_k(s) e_2)^2 \, ds.$$

This is the continuous two-mode statement; finite samples additionally require the response condition in Corollary 3.5.

The process Y_κ^c records an accumulated sensor signal. An increment over duration Δs has noise covariance $\Gamma_c \Delta s$, and its interval average has covariance $\Gamma_c / \Delta s$. This is distinct from point measurements with fixed per-sample noise covariance Γ. Adding independent point samples while retaining the original samples cannot increase posterior risk, but does not by itself approximate a fixed-intensity Brownian observation. Likewise, full continuous observability does not prevent the finite-sample failure in Equation (130).

C.4. Configuration comparisons with a correlated prior

For a common prior and target, either observation convention gives the exact comparison

$$\mathrm{MSE}_0(c_{\mathrm{tar}}) - \mathrm{MSE}_\kappa(c_{\mathrm{tar}}) = c_{\mathrm{tar}}^\top (\Sigma_0 - \Sigma_\kappa) c_{\mathrm{tar}}. \tag{137}$$

For the continuous convention, all MSEs and covariances in this equation carry the superscript c. Each configuration is observed separately under its specified dynamics; proposed future configurations do not supply additional observations to the actual system's initializer.

The two-dimensional case shows explicitly why correlation changes the comparison. Write, locally in this subsection,

$$P_0^{-1} = \begin{bmatrix} \alpha & \beta \\ \beta & \gamma \end{bmatrix} \succ 0.$$

Let $w_{ij,k}$ denote the entries of the observation Gramian for the coupled system, using either W_k^{obs} for sampled data or $W_k^{\mathrm{obs},c}$ for continuous data throughout a comparison. Observing only coordinate 1 in isolation gives the corresponding observation Gramian $\mathrm{diag}(w_0, 0)$. Both posterior precision matrices are positive definite. Their hidden-coordinate marginal precisions are

$$\begin{aligned} \lambda_k &= \gamma + w_{22,k} - \frac{(\beta + w_{12,k})^2}{\alpha + w_{11,k}}, \\ \lambda_0 &= \gamma - \frac{\beta^2}{\alpha + w_0}. \end{aligned} \tag{138}$$

For completeness, if a centered posterior has precision $\left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right]$, then

$$a u^2 + 2buv + cv^2 = a \left(u + \frac{b}{a} v \right)^2 + \left(c - \frac{b^2}{a} \right) v^2.$$

Integrating over the uncertain first coordinate u leaves precision $c - b^2/a$ for the second coordinate. Thus c is its precision conditional on knowing the first coordinate exactly; the subtraction accounts for marginalizing that uncertainty. It is not a separate negative-information measurement.

Since the MSEs are the reciprocals of the positive marginal precisions, the necessary and sufficient comparison is

$$\mathrm{MSE}_k(e_2) < \mathrm{MSE}_0(e_2) \quad \Longleftrightarrow \quad w_{22,k} > \frac{(\beta + w_{12,k})^2}{\alpha + w_{11,k}} - \frac{\beta^2}{\alpha + w_0}. \tag{139}$$

Equality gives equal MSE; reversing the inequality gives higher MSE under coupling. The same equivalence holds with superscript c for continuous records. A simpler sufficient condition is

$$w_{22,k} > 0, \qquad \frac{(\beta + w_{12,k})^2}{\alpha + w_{11,k}} \le \frac{\beta^2}{\alpha + w_0}. \tag{140}$$

It is not necessary: an increase in the marginalization correction can still be outweighed by $w_{22,k}$.

For example, with $P_0 = v \left[\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right]$, $v > 0$ and $|\rho| < 1$, disconnection gives

$$\mathrm{MSE}_0(e_2) = v - \frac{\rho^2 v^2 w_0}{1 + v w_0}. \tag{141}$$

Under the jointly Gaussian prior, the second coordinate can be written as ρ times the first plus an independent residual of variance $v(1-\rho^2)$. Observing the first coordinate reduces uncertainty in the correlated part even without a physical connection. Consequently positive information about the second coordinate under coupling must be compared with this already informative baseline, not merely with its prior variance.

C.5. An observable coupling with higher target MSE

Take the known two-mode model with

$$d_1 = 1, \quad d_2 = 0, \quad T = 1, \quad \sigma_y^2 = 1, \qquad x_{E,0} \sim \mathcal{N}\left(0, \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}\right).$$

Use the continuous sensor in Equation (131) with $C_{\mathrm{obs}} = [1, 0]$, no input or process noise, and the same prior and noise law in both configurations. The prior is positive definite. For $k = 1/2$, put $r = e^{-1}$. The exponential in Equation (106) is in its critical case and gives

$$E_{1/2}(1) = e^{-1/2} \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 3/2 \end{bmatrix}. \tag{142}$$

Here $C_{\mathrm{obs}}^\top C_{\mathrm{obs}} = R_E = \mathrm{diag}(1, 0)$ and $\sigma_y^2 = 1$. The energy identity in Equation (31) therefore evaluates the observation Gramian exactly:

$$W_{1/2}^{\mathrm{obs},c} = \frac{I - E_{1/2}(1)^\top E_{1/2}(1)}{2} = \begin{bmatrix} 1/2 - r/4 & -r/4 \\ -r/4 & 1/2 - 5r/4 \end{bmatrix}. \tag{143}$$

The prior precision is $P_0^{-1} = \left[\begin{smallmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{smallmatrix}\right]$. Adding the Gramian gives

$$P_0^{-1} + W_{1/2}^{\mathrm{obs},c} = \frac{1}{12} \begin{bmatrix} 22 - 3r & -8 - 3r \\ -8 - 3r & 22 - 15r \end{bmatrix}.$$

Its determinant is $(35 - 37r + 3r^2)/12 > 0$, and inversion yields

$$\mathrm{MSE}_{1/2}^c(e_2) = \frac{22 - 3r}{35 - 37r + 3r^2}. \tag{144}$$

In isolation, $W_0^{\mathrm{obs},c} = \mathrm{diag}((1 - r^2)/2, 0)$, since $\int_0^1 e^{-2s}\, ds = (1 - r^2)/2$. Equation (141) then gives

$$\mathrm{MSE}_0^c(e_2) = \frac{11 - 3r^2}{12 - 4r^2}. \tag{145}$$

Subtracting the two exact expressions,

$$\begin{aligned} &\mathrm{MSE}_{1/2}^c(e_2) - \mathrm{MSE}_0^c(e_2) \\ &\quad = \frac{9r^4 - 99r^3 - 16r^2 + 371r - 121}{(35 - 37r + 3r^2)(12 - 4r^2)} > 0. \end{aligned} \tag{146}$$

The denominator is positive. For an analytic sign check of the numerator, the elementary bounds $8/3 < e < 11/4$ imply $4/11 < r < 3/8$, so

$$\begin{aligned} 9r^4 - 99r^3 - 16r^2 + 371r - 121 &> -121 + 371\frac{4}{11} - 16\left(\frac{3}{8}\right)^2 - 99\left(\frac{3}{8}\right)^3 \\ &= \frac{36261}{5632} > 0. \end{aligned}$$

Nevertheless the observability matrix in Equation (135) has determinant $-1/2 \neq 0$. The connected system is fully observable, but its optimal hidden-initial-state MSE at this horizon exceeds the disconnected value. Both MSEs remain below the prior variance: the coupled Gramian is positive definite, and the disconnected record informs the target through the nonzero prior correlation.

The same prior can also favor the opposite admissible coupling. Changing k to $-1/2$ while holding the original coordinates and prior fixed preserves the two diagonal Gramian entries and changes the sign of its off-diagonal entry. In the notation of Equation (138), its marginalization correction and that of isolation are

$$\frac{(8-3r)^2}{12(22-3r)} \quad \text{and} \quad \frac{8}{3(11-3r^2)},$$

respectively. The first is smaller, because cross-multiplication gives

$$32(22-3r) - (8-3r)^2(11-3r^2) = r(432 + 93r - 144r^2 + 27r^3) > 0 \quad (0 < r < 1).$$

Also $w_{22,-1/2} = 1/2 - 5r/4 > 0$ since $r < 3/8$. The sufficient condition in Equation (140) applies, and direct inversion gives

$$\mathrm{MSE}^c_{-1/2}(e_2) = \frac{22-3r}{35-29r+3r^2} < \mathrm{MSE}^c_0(e_2). \tag{147}$$

This is a physical sign change of the coupling with the prior fixed. A coordinate relabeling of the second state would also change the prior correlation. The example establishes neither an ordering for every horizon nor improvement for every target.

C.6. Relation to the state at the prediction time

The inference target above is the state at the beginning of the observed interval. For a physical initial-state target $c_{\mathrm{phys}}^\top x_0$, use $c_{\mathrm{tar}} = Q^{-1/2} c_{\mathrm{phys}}$. For Algorithm 1, step Q01, the relevant history is that of the actual system. Within this linear specialization, $\boldsymbol{y}_{\mathrm{now}}$ stacks the observation rows of Y_t. With the grid and offsets defined in Appendix A.1.5, the correspondence is

$$\begin{aligned} s_r = j_r \Delta t, \qquad T = L_{\mathrm{sim}} \Delta t, \\ x_{E,0} = Q^{1/2} x(\tau_{t - L_{\mathrm{sim}}}). \end{aligned} \tag{148}$$

Here t is the discrete query index, τ_t its physical time, and L_{sim} counts the historical integration intervals.

Under known deterministic dynamics and inputs, Equation (123) gives the endpoint posterior mean and covariance in energy coordinates as

$$E_\kappa(T) m_\kappa + r_\kappa(T), \qquad E_\kappa(T) \Sigma_\kappa E_\kappa(T)^\top.$$

Returning to physical coordinates gives

$$\Sigma^{\mathrm{phys}}_{\kappa,T} = Q^{-1/2} E_\kappa(T) \Sigma_\kappa E_\kappa(T)^\top Q^{-1/2}. \tag{149}$$

An initial-coordinate risk comparison does not automatically compare endpoint risks: the propagation maps also change with configuration. All candidate predictions use the actual recorded history; hypothetical connections supply no additional observations.

The Gaussian posterior mean is an optimal estimator under the stated known linear model. The implemented nonlinear initialization routines need not return it, and these formulas do not establish their achieved error. Unknown laws, unknown inputs, or process noise require an observation model that accounts for the resulting uncertainty; the independent-measurement-noise likelihood cannot be reused without that accounting.

D. Finite-Step Passivity Defect of the Structure-Preserving Rollout

The continuous-time interconnection satisfies the exact power balance $\dot{H} = y^\top u - e^\top R\, e$ with $e = \nabla H$ (Theorem 3.1 for the internal coupling; the external ports and dissipation supply the remaining terms). This appendix quantifies how much of that balance survives one *finite* structure-preserving step, closing the caveat that continuous-time passivity does not automatically become a finite-step, observed-coordinate guarantee. The rearranged residual $H(x_{t+1}) - H(x_t) - (W_t^{\mathrm{port}} - D_t)$—i.e. $\Delta H +$ dissipation $-$ port work, which by the theorem below is the $\mathcal{O}(\Delta t^3)$ defect—is the finite-step energy-balance residual, and it is what the energy ledger accounts each step, not $H(x_{t+1}) - H(x_t)$ alone.

Theorem D.1 (Finite-step energy-balance defect)**.** *Let one canonical transition $x_t \mapsto x_{t+1}$ be produced by the second-order structure-preserving integrator (Strang splitting (61) of the leapfrog conservative step and the metric-preserving Cayley coupling with the damping half-step (63)) at timestep Δt, and account the boundary power terms by the trapezoidal rule (under the zero-order hold below the endpoint input is u_t at both nodes)*

$$W_t^{\mathrm{port}} = \tfrac{\Delta t}{2}\big(y_t^\top u_t + y_{t+1}^\top u_t\big), \qquad D_t = \tfrac{\Delta t}{2}\big(e_t^\top R\, e_t + e_{t+1}^\top R\, e_{t+1}\big) \geq 0. \tag{150}$$

Assume $H \in C^3$ with bounded derivatives on a forward-invariant set containing the trajectory, that $s \mapsto P(x(s), u)$ has a bounded second derivative over the step (a direct hypothesis on the boundary power $P(x,u) = y^\top u - e^\top Re$, not inferred from Lipschitzness of the field alone), a zero-order-held input on $[t, t+\Delta t]$, and no mode/contact switch within the step. Assume moreover that the integrator is second order on the step; this holds for the leapfrog core with the damping half-step (63) in the constant identity, scalar, or blockwise-scalar mass regime, and excludes a configuration-dependent $M(q)$, a configuration-dependent coupling $\hat{\mathbf{C}}(q)$ (whose opening and closing coupling half-steps would otherwise use different generators $A(q_t) \neq A(q_{t+1})$ and break the second-order symmetry unless A is evaluated at both endpoints), or a start-of-step-only state-dependent domain port. Then

$$H(x_{t+1}) - H(x_t) = W_t^{\mathrm{port}} - D_t + \mathcal{O}(\Delta t^3). \tag{151}$$

Under q-only deployment, where the endpoints are replaced by observer/chart estimates $\hat{x}_t, \hat{x}_{t+1}$ with state errors $\eta_s = \|\hat{x}_s - x_s\| \leq \epsilon_{\mathrm{obs}} + \epsilon_{\mathrm{chart}}$, the estimated-state defect obeys

$$\big|H(\hat{x}_{t+1}) - H(\hat{x}_t) - (\widehat{W}_t^{\mathrm{port}} - \widehat{D}_t)\big| \;\leq\; C\,\Delta t^3 + \big(L_H + \tfrac{\Delta t}{2} L_P\big)\,(\eta_t + \eta_{t+1}), \tag{152}$$

where $\widehat{W}_t^{\mathrm{port}}, \widehat{D}_t$ are the trapezoidal terms evaluated at the estimates, L_H is a local Lipschitz constant of H, and L_P one of the boundary power P; the $\frac{\Delta t}{2} L_P$ contribution comes from perturbing the boundary-power quadrature and is absent from a naive L_H-only bound. If instead the boundary terms are accounted by a first-order (left-endpoint) rule, the $\mathcal{O}(\Delta t^3)$ remainder in (151) degrades to $\mathcal{O}(\Delta t^2)$; the third-order remainder therefore requires the second-order boundary quadrature together with *the stated smoothness, zero-order-hold, and no-switch conditions (these are sufficient, not merely necessary).*

Proof. Recall the exact power balance $\dot{H} = P(x,u)$ with continuous port power $P(x,u) = y^\top u - e^\top Re$ and effort $e = \nabla H$; the trapezoidal accounting of the theorem is $W_t^{\mathrm{port}} - D_t = \frac{\Delta t}{2}\big(P(x_t,u_t) + P(x_{t+1},u_t)\big)$ under the zero-order hold. We prove (151) by writing its defect as a quadrature error plus a state-truncation error, each $\mathcal{O}(\Delta t^3)$; the left-endpoint degradation and the q-only bound then follow.

Step 1 (exact balance). Under the zero-order hold u is constant on $[t, t+\Delta t]$, and integrating $\dot{H} = P$ along the exact flow $x(\cdot)$ from $x(t) = x_t$ gives

$$H\big(x(t{+}\Delta t)\big) - H(x_t) = \int_t^{t+\Delta t} P\big(x(s), u\big)\, ds.$$

Step 2 (quadrature error). By hypothesis $s \mapsto P(x(s),u)$ has a bounded second derivative over the step, so the trapezoidal rule applied to the Step-1 integral has local error $\mathcal{O}(\Delta t^3)$:

$$\int_t^{t+\Delta t} P\, ds = \big(W_t^{\mathrm{port}} - D_t\big) + \mathcal{O}(\Delta t^3).$$

(The trapezoidal terminal node is the numerical x_{t+1}, which the ledger uses; replacing it by the exact $x(t{+}\Delta t)$ shifts the terminal power by $\frac{\Delta t}{2} L_P \cdot \mathcal{O}(\Delta t^3) = \mathcal{O}(\Delta t^4)$, absorbed in the remainder.)

Table 17. **Finite-step defect scaling (measured).** Log–log slope of the one-step energy-balance residual versus Δt on an untrained harmonic-oscillator pH block, double precision. Slopes match Theorem D.1. The trapezoidal driven slope is the resolved-regime value ($\Delta t \in [0.02, 0.16]$; below Δt=0.02 the residual reaches the $\sim 10^{-9}$ numerical floor).

System	Boundary-power rule	Measured slope	Theorem
conservative (u=0, R=0)	— (no boundary terms)	2.99	$\mathcal{O}(\Delta t^3)$
driven + damped	left-endpoint (rectangle)	1.98	$\mathcal{O}(\Delta t^2)$
driven + damped	trapezoidal (implementation)	$\approx$3.0–3.4	$\mathcal{O}(\Delta t^3)$

Step 3 (state-truncation error). The numerical endpoint x_{t+1} differs from the exact flow $x(t+\Delta t)$ only by the integrator's local truncation error, which is $\mathcal{O}(\Delta t^3)$ because the step is second-order and time-symmetric (Strang splitting of the conservative leapfrog step with the dissipative half-steps). Since ∇H is bounded and Lipschitz on the forward-invariant set,

$$H(x_{t+1}) - H\big(x(t+\Delta t)\big) = \nabla H \cdot \mathcal{O}(\Delta t^3) = \mathcal{O}(\Delta t^3).$$

The Cayley conservative variant obeys the same bound: for a G-skew generator ($A^\top G + GA = 0$) the Cayley transform is G-orthogonal, so it preserves $\|\cdot\|_G$ exactly while approximating $e^{\Delta t A}$ to $\mathcal{O}(\Delta t^3)$, leaving its local truncation error $\mathcal{O}(\Delta t^3)$.

Step 4 (combine). Adding and subtracting $H(x(t+\Delta t))$ and using Step 1 splits the defect into exactly the two errors just bounded:

$$H(x_{t+1}) - H(x_t) - \big(W_t^{\mathrm{port}} - D_t\big) = \underbrace{H(x_{t+1}) - H\big(x(t+\Delta t)\big)}_{\mathcal{O}(\Delta t^3),\ \text{Step 3}} + \underbrace{\textstyle\int_t^{t+\Delta t} P\, ds - \big(W_t^{\mathrm{port}} - D_t\big)}_{\mathcal{O}(\Delta t^3),\ \text{Step 2}}.$$

Both terms are $\mathcal{O}(\Delta t^3)$, which proves (151).

Step 5 (left-endpoint rule). With the first-order left-endpoint rule only Step 2 changes: its local error becomes $\mathcal{O}(\Delta t^2)$, so the quadrature term above is $\mathcal{O}(\Delta t^2)$ and the remainder degrades to $\mathcal{O}(\Delta t^2)$. The state-truncation term is unaffected; hence the third-order remainder needs the second-order boundary quadrature *together with* the stated smoothness, zero-order-hold, and no-switch conditions.

Step 6 (q-only deployment). Replacing the endpoints by estimates $\hat{x}_t, \hat{x}_{t+1}$ with errors $\eta_s = \|\hat{x}_s - x_s\|$ perturbs the defect of Step 4 in two Lipschitz-controlled places: the energy difference $H(\hat{x}_{t+1}) - H(\hat{x}_t)$ changes by at most $L_H(\eta_t + \eta_{t+1})$, and the trapezoidal term $\widehat{W}_t^{\mathrm{port}} - \widehat{D}_t = \frac{\Delta t}{2}\big(P(\hat{x}_t, u_t) + P(\hat{x}_{t+1}, u_t)\big)$ changes by at most $\frac{\Delta t}{2} L_P(\eta_t + \eta_{t+1})$. Writing the $\mathcal{O}(\Delta t^3)$ of Step 4 as $C\Delta t^3$ and adding these two perturbations gives the stated estimated-state bound; the $\frac{\Delta t}{2} L_P$ term is precisely the boundary-quadrature perturbation that an L_H-only bound omits. □

Empirical validation. We verify Theorem D.1 directly on the implementation, evaluating the finite-step energy-balance residual on an analytically specified, *untrained* (KNOWN-regime) system—a harmonic-oscillator pH block—advanced one step over a geometric Δt sweep in double precision (single precision reaches its rounding floor before the scaling is resolved). Table 17 reports the measured log–log slope of the one-step defect versus Δt. The conservative case isolates the symplectic remainder ($\mathcal{O}(\Delta t^3)$); the driven, damped case exercises the boundary-power quadrature, and the measured slope confirms both that a left-endpoint rule yields $\mathcal{O}(\Delta t^2)$ and that the trapezoidal accounting of (151) restores $\mathcal{O}(\Delta t^3)$. The implementation uses the trapezoidal rule, so the logged residual is a genuine $\mathcal{O}(\Delta t^3)$ energy-balance defect. Because this defect can have either sign, $D_t \geq C\Delta t^3$ is a sufficient condition for the discrete passivity inequality $H(x_{t+1}) - H(x_t) \leq W_t^{\mathrm{port}}$. For estimated-state accounting, $\widehat{D}_t \geq C\Delta t^3 + (L_H + \frac{\Delta t}{2} L_P)(\eta_t + \eta_{t+1})$ is sufficient for $H(\hat{x}_{t+1}) - H(\hat{x}_t) \leq \widehat{W}_t^{\mathrm{port}}$. We draw no unconditional discrete-passivity conclusion for conservative steps or near zero dissipation—in a damping nullspace, or when the velocity scales with Δt—where the defect can dominate.

Electrical midpoint composition. The electrical transition in Equation (71) uses explicit midpoint for the additive coupling and approximate implicit midpoint for the local conservative dynamics. Its energy accounting depends on the accuracy of both operations.

Proposition D.2 (Electrical energy-balance defect with solver error)**.** *Fix the component laws, topology, and held inputs over one electrical step. Use the assembled Hamiltonian in Equation* (70) *and the trapezoidal work and dissipation of Theorem D.1, evaluated with the electrical boundary ports and damping. Assume the split vector fields have bounded*

derivatives through order two on a common neighborhood containing the exact trajectory, all split stages, and solver iterates. Assume H and the net boundary power P are Lipschitz there, with constants L_H and L_P, and $s \mapsto P(x(s), u_t)$ has bounded second derivative along the exact flow. There are no switches within the step.

Let x_{t+1}^{imp} be the endpoint of the same split scheme with every local implicit-midpoint equation solved exactly, on its small-step solution branch, and define the full-step solver perturbation in Euclidean norm by

$$\varepsilon_{\mathrm{solve},t} := \|x_{t+1} - x_{t+1}^{\mathrm{imp}}\|.$$

For sufficiently small Δt, a constant C independent of Δt satisfies

$$\begin{aligned} &\left|H(x_{t+1}) - H(x_t) - (W_t^{\mathrm{port}} - D_t)\right| \\ &\qquad \le C\Delta t^3 + \left(L_H + \tfrac{\Delta t}{2} L_P\right) \varepsilon_{\mathrm{solve},t}. \end{aligned} \tag{153}$$

Thus a solver perturbation of order Δt^3 preserves the third-order local energy-accounting defect. The constant depends on the fixed component laws, topology, held input, and neighborhood bounds.

Proof. The input translations are exact. The frozen-charge Cayley damping map, the exactly solved local implicit midpoint, and the explicit-midpoint coupling each approximate their respective subflows with local error $\mathcal{O}(\delta^3)$. Their Strang placement and the two fixed substeps therefore give $\|x_{t+1}^{\mathrm{imp}} - x(t+\Delta t)\| \le C_{\mathrm{st}}\Delta t^3$. This order argument does not require the explicit-midpoint map to be time symmetric or energy preserving.

Internal coupling has zero continuous power, so the exact assembled flow obeys $\dot{H} = P$. Integrating this balance and applying the trapezoidal rule at the exact endpoints incurs at most $C_P\Delta t^3$ error. Replacing the exact terminal state by the numerical endpoint changes terminal energy by at most $L_H\|x_{t+1} - x(t+\Delta t)\|$ and terminal boundary accounting by at most $\frac{\Delta t}{2} L_P\|x_{t+1} - x(t+\Delta t)\|$. Finally,

$$\|x_{t+1} - x(t+\Delta t)\| \le C_{\mathrm{st}}\Delta t^3 + \varepsilon_{\mathrm{solve},t}.$$

Substitution proves Equation (153). □

The solver term can be bounded from local equation residuals. For a conservative substep starting at z, write its vector field as f_H and the returned state as $\tilde{z}$. Its implicit-midpoint residual is $r = \tilde{z} - z - \delta f_H((z+\tilde{z})/2)$. If f_H has Lipschitz constant L_f with $\delta L_f/2 < 1$, comparison with the exact implicit solution gives a state error of at most $\|r\|/(1 - \delta L_f/2)$. The Lipschitz constants of subsequent substeps propagate these local errors to $\varepsilon_{\mathrm{solve},t}$.

The Picard routine in Equation (69) starts with Euler and performs at least one correction. Its first correction is explicit midpoint; for any of the one to ten corrections, Taylor expansion gives

$$\tilde{z} = z + \delta f_H(z) + \tfrac{\delta^2}{2} Df_H(z) f_H(z) + \mathcal{O}(\delta^3).$$

The expansion holds uniformly over this bounded iteration count under the stated smoothness and small-step assumptions, in exact arithmetic. Hence finite Picard iteration need not destroy second order. The 10^{-6} successive-iterate stopping threshold alone, however, is not a bound on the implicit-equation residual or a finite-step energy certificate. Neither the proposition nor this expansion asserts exact energy preservation; discrete passivity additionally requires dissipation to dominate the defect.

E. Observer-Aware Deployment-Transfer Bound

Section 3.7 states the horizon bound (44); here we give the hypotheses and proof. It is a comparison-map contraction estimate: it separates decay of the initial observer error from accumulation of the per-step model defect, and its usefulness rests on a contraction factor that is *measured*, not inferred from passivity.

Setup. Fix a deployment $\mathcal{S}$. Let $F_k : \mathcal{X}_{\mathcal{S}} \to \mathcal{X}_{\mathcal{S}}$ be the true one-step flow over $[t_k, t_{k+1}]$ (step Δt) under a fixed input (u_k) and mode schedule, $\widehat{F}_k : \mathcal{Z}_{\mathcal{S}} \to \mathcal{Z}_{\mathcal{S}}$ the deployed structure-preserving rollout, and $\Pi_{\mathcal{S}} : \mathcal{X}_{\mathcal{S}} \to \mathcal{Z}_{\mathcal{S}}$ an alignment map (physical and learned states may use different charts, coordinates, or dimensions). Let $\Omega_{\mathcal{S}} \subset \mathcal{Z}_{\mathcal{S}}$ be a forward-invariant

operating region and M a *fixed* symmetric positive-definite metric on $\Omega_{\mathcal{S}}$ inducing the norm $\|v\|_M = (v^\top M v)^{1/2}$—a Lyapunov metric for the deployed map, which for nonlinear systems is a region-local choice constructed as in the discrete-Lyapunov solve below. Write $e_k = \|\Pi_{\mathcal{S}} x_k - \hat{z}_k\|_M$ and $\Omega_{\mathcal{S}}^X = \Pi_{\mathcal{S}}^{-1}(\Omega_{\mathcal{S}})$.

For the C-PHAST deployment under these hypotheses, $\widehat{F}_k(z) = F^{(\mathcal{S})}_{\theta,\Delta t}(z, u_k)$: the hat distinguishes the modeled numerical step from the true physical step F_k. The generic numerical map $\Psi_{\Delta t}$ in the lemma below is this same deployed step when instantiated with C-PHAST, not a second learned transition.

The commutation defect δ_k in hypothesis (A3) below is *derived*, not assumed: the following lemma bounds it from a pointwise transported vector-field mismatch together with the state-map consistency (A2), by a Grönwall estimate in a fixed RMS-normalized metric. This makes (A2) load-bearing, states the interconnection through a generic operator $K(\mathcal{S})$ (the microgrid incidence norm being a corollary), and yields deployment-size-uniform constants. Theorem E.2 instantiates it at a single deployment, writing $M = M_{\mathcal{S}}$ and dropping the $\mathcal{S}$ subscripts.

Lemma E.1 (One-step commutation defect)**.** *Fix a deployment $\mathcal{S}$, a step size $\Delta t > 0$, and an admissible input $u_k \in \mathcal{U}_{\mathcal{S}}$ held fixed on the step. Let $f_{\mathcal{S}}$ and $\widehat{f}_{\mathcal{S}}$ generate unique flows on $\mathcal{X}_{\mathcal{S}}$ and $\mathcal{Z}_{\mathcal{S}}$, respectively. Write $\widehat{\varphi}_t(z;u) = \varphi_t^{\widehat{f}_{\mathcal{S}}}(z;u)$ and set*

$$F_k(x) = \varphi_{\Delta t}^{f_{\mathcal{S}}}(x; u_k), \qquad \widehat{F}_k(z) = \Psi_{\Delta t}(z, u_k).$$

Let $\Pi_{\mathcal{S}} : \mathcal{X}_{\mathcal{S}} \to \mathcal{Z}_{\mathcal{S}}$ be C^1, let $\Omega_{\mathcal{S}} \subset \mathcal{Z}_{\mathcal{S}}$, and write $\Omega_{\mathcal{S}}^X = \Pi_{\mathcal{S}}^{-1}(\Omega_{\mathcal{S}})$. For this deployment let $M_{\mathcal{S}} \succ 0$ be a constant (state-independent) metric and let

$$\|v\|_{M_{\mathcal{S}}} = (v^\top M_{\mathcal{S}} v)^{1/2}, \qquad \langle v, w\rangle_{M_{\mathcal{S}}} = v^\top M_{\mathcal{S}} w.$$

Assume the following.

(H1) State-map consistency (A2). *For some $K_{\Psi,\mathcal{S}} \geq 0$,*

$$\sup_{z \in \Omega_{\mathcal{S}}} \left\| \widehat{\varphi}_{\Delta t}(z; u_k) - \Psi_{\Delta t}(z, u_k) \right\|_{M_{\mathcal{S}}} \leq K_{\Psi,\mathcal{S}} \Delta t^3.$$

(H2) Pointwise transported-field mismatch. *For $x \in \Omega_{\mathcal{S}}^X$ define, without requiring an inverse of $\Pi_{\mathcal{S}}$,*

$$d_{\mathcal{S}}(x,u) := D\Pi_{\mathcal{S}}(x) f_{\mathcal{S}}(x,u) - \widehat{f}_{\mathcal{S}}(\Pi_{\mathcal{S}} x, u). \tag{154}$$

Suppose there are defect fields r_{loc}, r_{edge}, and r_{port}, an edge-residual metric $N_{\mathcal{S}} \succ 0$, and a linear deployed interconnection operator $K(\mathcal{S})$ such that

$$d_{\mathcal{S}} = r_{\mathrm{loc}} + K(\mathcal{S}) r_{\mathrm{edge}} + r_{\mathrm{port}},$$

and nonnegative constants such that, uniformly over the stated operating region and admissible inputs,

$$\|r_{\mathrm{loc}}\|_{M_{\mathcal{S}}} \leq \delta_{\mathrm{loc},\mathcal{S}}, \qquad \|r_{\mathrm{edge}}\|_{N_{\mathcal{S}}} \leq \delta_{\mathrm{edge},\mathcal{S}}, \qquad \|r_{\mathrm{port}}\|_{M_{\mathcal{S}}} \leq \delta_{\mathrm{port},\mathcal{S}}.$$

Thus, with the induced operator norm,

$$\sup_{x \in \Omega_{\mathcal{S}}^X,\, u \in \mathcal{U}_{\mathcal{S}}} \|d_{\mathcal{S}}(x,u)\|_{M_{\mathcal{S}}} \leq \delta_{\mathrm{vf},\mathcal{S}} := \delta_{\mathrm{loc},\mathcal{S}} + \|K(\mathcal{S})\|_{N_{\mathcal{S}} \to M_{\mathcal{S}}} \delta_{\mathrm{edge},\mathcal{S}} + \delta_{\mathrm{port},\mathcal{S}}. \tag{155}$$

If $\Pi_{\mathcal{S}}$ is a C^1 diffeomorphism onto its image, (154) is equivalently the Z-space field $D\Pi_{\mathcal{S}}(\Pi_{\mathcal{S}}^{-1} z) f_{\mathcal{S}}(\Pi_{\mathcal{S}}^{-1} z, u) - \widehat{f}_{\mathcal{S}}(z,u)$ used in the main-text notation.

(H3) Incremental bound. *There is a nonnegative $L_{f,\mathcal{S}}$ such that, for all relevant z, z' and $u \in \mathcal{U}_{\mathcal{S}}$,*

$$\left\langle z - z', \widehat{f}_{\mathcal{S}}(z,u) - \widehat{f}_{\mathcal{S}}(z',u) \right\rangle_{M_{\mathcal{S}}} \leq L_{f,\mathcal{S}} \|z - z'\|_{M_{\mathcal{S}}}^2. \tag{156}$$

An $M_{\mathcal{S}}$-Lipschitz bound with constant $L_{f,\mathcal{S}}$ is sufficient. When the field is differentiable, a uniform logarithmic-norm bound on a convex neighborhood containing the relevant line segments, $\mu_{M_{\mathcal{S}}}(D\widehat{f}_{\mathcal{S}}) \leq L_{f,\mathcal{S}}$, is also sufficient. A negative available upper bound may be relaxed to $L_{f,\mathcal{S}} = 0$ for the estimate below.

(H4) Within-step invariance. *For every $x_0 \in \Omega_{\mathcal{S}}^X$, both $\Pi_{\mathcal{S}}\varphi_t^{f_{\mathcal{S}}}(x_0;u_k)$ and $\widehat{\varphi}_t(\Pi_{\mathcal{S}}x_0;u_k)$ remain in $\Omega_{\mathcal{S}}$ for $0 \le t \le \Delta t$. In particular, there is no unmatched mode or contact event within the step.*

Define

$$\alpha(L,\Delta t) = \begin{cases} (e^{L\Delta t}-1)/L, & L > 0, \\ \Delta t, & L = 0. \end{cases}$$

Then the one-step commutation defect satisfies

$$\begin{aligned} \delta_k &:= \sup_{x\in\Omega_{\mathcal{S}}^X} \left\| \Pi_{\mathcal{S}} F_k(x) - \widehat{F}_k(\Pi_{\mathcal{S}}x) \right\|_{M_{\mathcal{S}}} \\ &\le \delta_{\mathrm{vf},\mathcal{S}}\alpha(L_{f,\mathcal{S}},\Delta t) + K_{\Psi,\mathcal{S}}\Delta t^3 \\ &\le \Delta t e^{L_{f,\mathcal{S}}\Delta t}\left(\delta_{\mathrm{loc},\mathcal{S}} + \|K(\mathcal{S})\|_{N_{\mathcal{S}}\to M_{\mathcal{S}}}\delta_{\mathrm{edge},\mathcal{S}} + \delta_{\mathrm{port},\mathcal{S}}\right) + K_{\Psi,\mathcal{S}}\Delta t^3. \end{aligned} \tag{157}$$

In particular, $\Delta t e^{L_{f,\mathcal{S}}\Delta t} = \Delta t(1+\mathcal{O}(\Delta t))$ for fixed $L_{f,\mathcal{S}}$.

For a family of deployments with $b_{\mathcal{S}}$ explicit blocks, take the RMS metric

$$M_{\mathcal{S}} = \frac{1}{b_{\mathcal{S}}}\,\mathrm{blockdiag}(M_{\mathcal{S},1},\dots,M_{\mathcal{S},b_{\mathcal{S}}}), \qquad 0 < m_- I \preceq M_{\mathcal{S},i} \preceq m_+ I, \tag{158}$$

with m_-, m_+ independent of $\mathcal{S}$, and use analogous per-edge and per-port RMS metrics. If the block, edge, and port RMS defects are bounded uniformly by $\bar{\delta}_{\mathrm{loc}}$, $\bar{\delta}_{\mathrm{edge}}$, and $\bar{\delta}_{\mathrm{port}}$, and if

$$\sup_{\mathcal{S}} L_{f,\mathcal{S}} \le \bar{L}_f, \quad \sup_{\mathcal{S}} K_{\Psi,\mathcal{S}} \le \bar{K}_\Psi, \quad \sup_{\mathcal{S}} \|K(\mathcal{S})\|_{N_{\mathcal{S}}\to M_{\mathcal{S}}} \le \bar{K}, \tag{159}$$

then

$$\sup_{\mathcal{S}} \delta_k \le \Delta t e^{\bar{L}_f\Delta t}\left(\bar{\delta}_{\mathrm{loc}} + \bar{K}\bar{\delta}_{\mathrm{edge}} + \bar{\delta}_{\mathrm{port}}\right) + \bar{K}_\Psi \Delta t^3, \tag{160}$$

which is independent of $b_{\mathcal{S}}$ and of the number of edges or ports.

Proof. The defect δ_k compares the exact projected flow $\Pi_{\mathcal{S}}F_k$ with the numerical map $\widehat{F}_k = \Psi_{\Delta t}$. We split it into a *flow-versus-flow* error, controlled by the vector-field mismatch through a one-sided Grönwall estimate, and a *flow-versus-map* error, controlled directly by the state-map consistency (H1); the RMS metric then removes the dependence on the block count.

Step 1 (transported dynamics). Fix $x_0 \in \Omega_{\mathcal{S}}^X$ and write

$$x(t) = \varphi_t^{f_{\mathcal{S}}}(x_0;u_k), \qquad w(t) = \Pi_{\mathcal{S}}x(t), \qquad \widehat{z}(t) = \widehat{\varphi}_t(\Pi_{\mathcal{S}}x_0;u_k)$$

for the exact trajectory, its projection, and the reduced flow from the same projected start. The chain rule and the transported-defect definition (154) give

$$\dot{w}(t) = D\Pi_{\mathcal{S}}(x(t))f_{\mathcal{S}}(x(t),u_k) = \widehat{f}_{\mathcal{S}}(w(t),u_k) + d_{\mathcal{S}}(x(t),u_k),$$

so the projected exact flow follows the reduced field *plus* the mismatch $d_{\mathcal{S}}$.

Step 2 (Grönwall on the flow-versus-flow error). Let $e(t) = w(t) - \widehat{z}(t)$; then $e(0) = 0$ and

$$\dot{e}(t) = \widehat{f}_{\mathcal{S}}(w(t),u_k) - \widehat{f}_{\mathcal{S}}(\widehat{z}(t),u_k) + d_{\mathcal{S}}(x(t),u_k).$$

Differentiating the metric norm and applying the incremental bound (H3) to the field difference and the vector-field bound (155) of (H2) to the mismatch—both valid along the step because (H4) keeps $w(t)$ and $\widehat{z}(t)$ in $\Omega_{\mathcal{S}}$—gives

$$\begin{aligned} \frac{1}{2}\frac{d}{dt}\|e(t)\|_{M_{\mathcal{S}}}^2 &= \left\langle e(t), \widehat{f}_{\mathcal{S}}(w(t),u_k) - \widehat{f}_{\mathcal{S}}(\widehat{z}(t),u_k)\right\rangle_{M_{\mathcal{S}}} + \langle e(t), d_{\mathcal{S}}(x(t),u_k)\rangle_{M_{\mathcal{S}}} \\ &\le L_{f,\mathcal{S}}\|e(t)\|_{M_{\mathcal{S}}}^2 + \delta_{\mathrm{vf},\mathcal{S}}\|e(t)\|_{M_{\mathcal{S}}}. \end{aligned}$$

Writing $r(t) = \|e(t)\|_{M_S}$ and using the upper Dini derivative D^+ where $r = 0$, this is

$$D^+ r(t) \le L_{f,\mathcal{S}} r(t) + \delta_{\mathrm{vf},\mathcal{S}}, \qquad r(0) = 0,$$

and Grönwall's comparison inequality yields

$$\|e(\Delta t)\|_{M_S} \le \delta_{\mathrm{vf},\mathcal{S}} \alpha(L_{f,\mathcal{S}}, \Delta t). \tag{161}$$

Step 3 (add the flow-versus-map error via A2). The exact projected flow and the numerical map differ only through the reduced flow $\widehat{\varphi}_{\Delta t}$ at the endpoint, so

$$\begin{aligned}&\Pi_{\mathcal{S}} F_k(x_0) - \Psi_{\Delta t}(\Pi_{\mathcal{S}} x_0, u_k) \\ &\quad = \underbrace{\Pi_{\mathcal{S}} F_k(x_0) - \widehat{\varphi}_{\Delta t}(\Pi_{\mathcal{S}} x_0; u_k)}_{e(\Delta t),\ \text{Step 2}} + \underbrace{\widehat{\varphi}_{\Delta t}(\Pi_{\mathcal{S}} x_0; u_k) - \Psi_{\Delta t}(\Pi_{\mathcal{S}} x_0, u_k)}_{\le K_{\Psi,\mathcal{S}} \Delta t^3 \ \text{by (H1)}}.\end{aligned}$$

The first bracket is $e(\Delta t)$, bounded by (161); the second is exactly the state-map consistency of (H1)/A2, at most $K_{\Psi,\mathcal{S}} \Delta t^3$. Taking norms and the supremum over x_0 gives the first inequality of (157), and the envelope follows from

$$\alpha(L, \Delta t) = \int_0^{\Delta t} e^{L(\Delta t - s)}\, ds \le \Delta t e^{L \Delta t} \qquad (L \ge 0).$$

Step 4 (size-uniformity via the RMS metric). For the RMS metric (158), a stacked defect $r = (r_1, \dots, r_{b_S})$ has

$$\|r\|_{M_S}^2 = \frac{1}{b_S} \sum_{i=1}^{b_S} \|r_i\|_{M_{S,i}}^2,$$

so a uniform per-block bound is an RMS bound of the *same* magnitude, not a $\sqrt{b_S}$ multiple; the edge and port channels are identical. Substituting the uniform bounds (159) into (157) gives the size-independent (160). □

Incidence corollary (edge-explicit node–edge composition). Let $B = B_{\mathrm{inc}}(\mathcal{S}) \in \mathbb{R}^{n \times m}$ be an oriented incidence matrix. In compatible whitened port coordinates, the two off-diagonal mismatch channels are routed by

$$\mathcal{K}_B \begin{bmatrix} r_N \\ r_E \end{bmatrix} = \begin{bmatrix} B r_E \\ -B^\top r_N \end{bmatrix}, \qquad \mathcal{K}_B = \begin{bmatrix} 0 & B \\ -B^\top & 0 \end{bmatrix}.$$

With the total-block RMS metric $(n+m)^{-1} I$ (or its compatible block version), the scalar normalization cancels in the induced norm and

$$\|\mathcal{K}_B\|_M^2 = \lambda_{\max}(\mathcal{K}_B^\top \mathcal{K}_B) = \max\{\lambda_{\max}(BB^\top), \lambda_{\max}(B^\top B)\} = \|B\|_2^2. \tag{162}$$

The same identity holds for $B \otimes I_d$; here the concatenated r_N, r_E vector is the RMS edge-channel defect appearing in (H2). If the residuals are measured before the microgrid constitutive conversions, the implemented nonzero block obeys

$$\begin{aligned}\mathcal{K}_{B,C,L} &= \begin{bmatrix} 0 & B L_E^{-1} \\ -B^\top C_N^{-1} & 0 \end{bmatrix}, \\ \|\mathcal{K}_{B,C,L}\|_M &\le \|B\|_2 \max\{\|L_E^{-1}\|_2, \|C_N^{-1}\|_2\}.\end{aligned}$$

Uniformly compatible coordinate changes add only their induced-norm factors; denote the resulting size-independent product by C_{inc}. For a simple undirected graph of maximum degree $d_{\max}$, $BB^\top$ is its graph Laplacian and $\|B\|_2 \le \sqrt{2 d_{\max}}$; hence $\|K(\mathcal{S})\|_M \le C_{\mathrm{inc}} \sqrt{2 d_{\max}}$. In the matched coordinates $C_{\mathrm{inc}} = 1$, so (157) contains the microgrid term $\|B_{\mathrm{inc}}(\mathcal{S})\|_2 \delta_{\mathrm{edge},\mathcal{S}}$ to the first power. A Laplacian power $B\,\mathrm{diag}(\cdot)B^\top$ would arise only after eliminating the explicit edge state and is not the operator considered here.

Theorem E.2 (Observer-aware deployment transfer)**.** *Assume* (A1) *incremental contraction,* $\|\widehat{F}_k(z) - \widehat{F}_k(z')\|_M \le q_k \|z - z'\|_M$ *on* $\Omega_{\mathcal{S}}$ *with* $q_k \le q \in (0,1]$*;* (A2) *state-map consistency,* $\|\Psi_{\Delta t}(z,u) - \varphi_{\Delta t}(z,u)\|_M \le K_\Psi \Delta t^3$*;* (A3) *commutation defect:* $\delta_k = \sup_{x \in \Omega_{\mathcal{S}}^X} \|\Pi_{\mathcal{S}} F_k(x) - \widehat{F}_k(\Pi_{\mathcal{S}} x)\|_M$ *obeys the bound of Lemma E.1 —* $\delta_k \le \Delta t\, e^{L_f \Delta t}(\delta_{\mathrm{loc}} +$

$\|K(\mathcal{S})\|\,\delta_{\mathrm{edge}} + \delta_{\mathrm{port}}\big) + K_\Psi \Delta t^3$ — derived *(not assumed) from a pointwise transported field mismatch (its hypotheses H2–H4) together with (A2); for the edge-explicit microgrid* $\|K(\mathcal{S})\| = \|B_{\mathrm{inc}}(\mathcal{S})\|_2$ *to the first power;* (A4) *observer initialization* $e_0 \le \epsilon_{\mathrm{lift}}$ *(see Remark E.3);* (A5) *a matched mode/contact schedule with both trajectories in* $\Omega_{\mathcal{S}}$*. Then for every horizon* h*,*

$$e_h \;\le\; \Big(\textstyle\prod_{j=0}^{h-1} q_j\Big) e_0 + \sum_{k=0}^{h-1} \Big(\prod_{j=k+1}^{h-1} q_j\Big) \delta_k \;\le\; q^h\, \epsilon_{\mathrm{lift}} + S_h(q)\, \bar{\delta}, \tag{163}$$

with $\bar{\delta} = \max_k \delta_k$ *and* $S_h(q) = \frac{1-q^h}{1-q}$ *for* $q < 1$ *(and* h *for* $q = 1$*). If* $q < 1$ *the bound is uniform in* h*:* $\limsup_h e_h \le \bar{\delta}/(1-q)$*.*

Proof. By (A5), $\Pi_{\mathcal{S}} x_k \in \Omega_{\mathcal{S}}$, so the triangle inequality and (A1) give $e_{k+1} = \|\Pi_{\mathcal{S}} F_k x_k - \widehat{F}_k \hat{z}_k\|_M \le \|\Pi_{\mathcal{S}} F_k x_k - \widehat{F}_k(\Pi_{\mathcal{S}} x_k)\|_M + \|\widehat{F}_k(\Pi_{\mathcal{S}} x_k) - \widehat{F}_k \hat{z}_k\|_M \le \delta_k + q_k e_k$. Unrolling the scalar recursion $e_{k+1} \le q_k e_k + \delta_k$ gives the product-and-sum form; $q_j \le q$, $\delta_k \le \bar{\delta}$, and $\sum_{k=0}^{h-1} q^{h-1-k} = S_h(q)$ give the closed form. □

Remark E.3 (Source of the lift error)**.** The initialization ϵ_{lift} (A4) is a hypothesis of the deterministic bound, supplied by the deployed observer; we keep Theorem E.2 deterministic and source ϵ_{lift} at the observer layer rather than folding the observer's stochastics into the transfer bound. For a bounded-error (set-membership) observer, or under a declared bounded-noise assumption, ϵ_{lift} is a genuine deterministic bound. For a linear-Gaussian observer it is not: Gaussian errors have unbounded support, so a covariance gives no almost-sure finite bound. Instead, if the zero-bias initialization error $\Pi_{\mathcal{S}} x_0 - \hat{z}_0$ is Gaussian with stationary Kalman covariance P^* (the solution of the observer's algebraic Riccati equation), then, for its vector dimension $d = \dim(\Pi_{\mathcal{S}} x_0 - \hat{z}_0)$,

$$e_0 \le r(\alpha) := \sqrt{\chi^2_{d,\,1-\alpha}\, \lambda_{\max}\big(M^{1/2} P^* M^{1/2}\big)} \quad \text{with probability } \ge 1-\alpha.$$

Note that $r(\alpha)$ scales as a standard deviation $\sqrt{\lambda_{\max}(P^*)}$, not as the covariance P^* itself: in the scalar random-walk model with identity observation, $P^* = \sqrt{Q_\theta R} + \mathcal{O}(Q_\theta)$ (process noise Q_θ, observation noise R), so the radius is $(Q_\theta R)^{1/4}$, whereas the general linear case solves a matrix Riccati equation with no universal square-root form. Under repeated q-only refresh the single-step bound does not simply carry over: the faithful object is a stochastic recursion $e_{k+1} \le q_k e_k + \delta_k + \nu_k$ with ν_k the stepwise reset error, and its bounded-in-probability analysis is Theorem E.4 below. Theorem E.2 is thus the deterministic transfer bound with an assumed initialization, and Theorem E.4 its probabilistic instantiation for a refreshing Gaussian observer.

Theorem E.4 (Probabilistic deployment transfer under refreshed observation)**.** *Assume the hypotheses of Theorem E.2 with* $q_k \le q \in (0,1)$ *and* $\delta_k \le \bar{\delta}$*, and replace the fixed initialization (A4) by a* refreshed-*observation recursion*

$$e_{k+1} \le q_k e_k + \delta_k + \nu_k, \qquad e_0 = \nu_{-1}, \tag{164}$$

in which each step injects a stochastic reset error $\nu_k = \|\xi_k\|_M$ *(the observer correction, gain included, mapped into the state metric) obeying the individual tail bound*

$$\Pr\big[\nu_k > r(\alpha')\big] \le \alpha', \qquad r(\alpha') = \sqrt{\chi^2_{d,\,1-\alpha'}\, \lambda_{\max}\big(M^{1/2} \Sigma M^{1/2}\big)},$$

for $k = -1, 0, \ldots, h-1$*, where* $\xi_k \sim \mathcal{N}(0, \Sigma)$ *with zero bias and* $d = \dim \xi_k$*. Here* Σ *describes the injected error; it is not generally the observer's stationary posterior covariance. Then for every* $\alpha \in (0,1)$ *and horizon* h*, with* $\alpha' = \alpha/(h+1)$*,*

$$\Pr\Big[\, \max_{0 \le j \le h} e_j \;\le\; S_h(q)\, \bar{\delta} + r\big(\tfrac{\alpha}{h+1}\big)\, S_{h+1}(q) \,\Big] \;\ge\; 1-\alpha. \tag{165}$$

For an infinite trajectory, assume the same recursion and marginal tail bound hold for every $k \ge -1$*. Choose* $\alpha_j > 0$ *with* $\sum_{j=0}^{\infty} \alpha_j \le \alpha$*, assigning* α_0 *to the initial error* ν_{-1} *and* α_j *to* ν_{j-1} *for* $j \ge 1$*. With probability at least* $1-\alpha$*, simultaneously for every* $k \ge 0$*,*

$$e_k \le S_k(q)\bar{\delta} + \sum_{j=0}^{k} q^{k-j} r(\alpha_j) \le S_k(q)\bar{\delta} + R_k S_{k+1}(q), \qquad R_k := \max_{0 \le j \le k} r(\alpha_j). \tag{166}$$

If α_j *is nonincreasing,* $R_k = r(\alpha_k)$*. No temporal independence is required for these bounds. The envelope holds simultaneously over time, but need not have a finite, time-independent upper bound.*

Proof. By the union (Boole) bound over the $h+1$ events $\{\nu_k > r(\alpha')\}$, $k = -1, \dots, h-1$,

$$\Pr\big[\exists\, k:\ \nu_k > r(\alpha')\big] \le (h+1)\alpha' = \alpha.$$

On the complementary event every $\nu_k \le r(\alpha')$. Unrolling (164) with $q_j \le q$, $\delta_j \le \bar{\delta}$, and $e_0 \le r(\alpha')$ gives, for each $j \le h$,

$$e_j \ \le\ q^j e_0 + \sum_{i=0}^{j-1} q^{j-1-i}\big(\delta_i + \nu_i\big) \ \le\ S_j(q)\,\bar{\delta} + r(\alpha')\Big(q^j + \textstyle\sum_{i=0}^{j-1} q^{j-1-i}\Big).$$

Since $q^j + S_j(q) = q^j + \frac{1-q^j}{1-q} = \frac{1-q^{j+1}}{1-q} = S_{j+1}(q)$ and S_j, S_{j+1} are nondecreasing in j, each $e_j \le S_h(q)\bar{\delta} + r(\alpha')S_{h+1}(q)$; the maximum over $j \le h$ obeys the same bound, which is (165). For the infinite-horizon statement, the countable union bound gives

$$\Pr[\exists j \ge 0 : \nu_{j-1} > r(\alpha_j)] \le \sum_{j\ge 0} \alpha_j \le \alpha.$$

On its complement, unrolling the recursion includes the initial contribution $q^k r(\alpha_0)$ and the refresh contributions $\sum_{j=1}^{k} q^{k-j} r(\alpha_j)$. Adding the deterministic defect sum gives the first bound in (166); replacing each radius by R_k gives the second. The marginal Gaussian assumption alone does not imply an almost-sure unbounded supremum: setting $\xi_k = \xi$ for a single Gaussian draw is a counterexample. Such a conclusion would require an additional dependence assumption and is not used here. □

Remark E.5 (Tighter constant under independent refreshes)**.** If the ξ_k are independent, $\nu_k = \|\xi_k\|_M$ is sub-Gaussian about its mean $\mu = \mathbb{E}\|\xi_0\|_M$ with variance proxy $\sigma_M^2 = \lambda_{\max}(M^{1/2}\Sigma M^{1/2})$, because $\|\cdot\|_M$ is 1-Lipschitz in the M-metric and $M^{1/2}\xi_k$ is Gaussian (Gaussian concentration). The geometric sum $W_h = \sum_{i=0}^{h-1} q^{h-1-i}\nu_i$ then concentrates,

$$\Pr\big[W_h > \mu\, S_h(q) + \sigma_M \sqrt{2\, T_h\, \log(1/\alpha)}\big] \le \alpha, \qquad T_h = \tfrac{1-q^{2h}}{1-q^2},$$

for a fixed endpoint h; the deterministic $S_h(q)\bar{\delta}$ and $q^h e_0$ terms are unchanged. This endpoint statement does not itself bound the maximum over time.

If initialization also has $e_0 = \|\xi_{-1}\|_M$, with $\xi_{-1}, \dots, \xi_{h-1}$ independent and identically distributed, a simultaneous version is

$$\Pr\left[\max_{0\le j\le h} e_j \le S_h(q)\bar{\delta} + \mu S_{h+1}(q) + \sigma_M\sqrt{2T_{h+1}}\left(\sqrt{\log(h+1)} + \sqrt{\log(1/\alpha)}\right)\right] \ge 1-\alpha.$$

Indeed, the centered weighted sums through each endpoint have variance proxy at most $\sigma_M^2 T_{h+1}$, so their expected maximum is at most $\sigma_M\sqrt{2T_{h+1}\log(h+1)}$. Their maximum is itself $\sigma_M\sqrt{T_{h+1}}$-Lipschitz in the stacked standard Gaussian variables; Gaussian concentration supplies the second deviation term. Simultaneous coverage thus retains a horizon-dependent term even under independence.

Empirical grounding of the probabilistic transfer bound. We evaluate Theorem E.4 on the affine damped oscillator with $k = 2$, $d = 3$, and step size 0.05. The numerical metric used for this coverage study is distinct from the exact witness in Remark E.6; both use the same reported contraction factor. A stationary-gain, q-only Kalman observer uses $Q_\theta = 10^{-6} I$ and $R = I$. Writing its gain as K and position-observation matrix as C, the signed error evolves through

$$A_e = (I - KC)A, \qquad \Sigma = (I-KC)Q_\theta(I-KC)^\top + KRK^\top.$$

The implementation checks $\|A_e\|_M \le q$ before applying the recursion. Initialization is an independent draw from $\mathcal{N}(0,\Sigma)$, as are the subsequent injections; it is not drawn from the stationary posterior covariance. We use $N = 3000$ independent rollouts of horizon $h = 200$. The table reports coverage of $\max_{0\le j\le h} e_j$ and the ratio of the bound to its empirical $(1-\alpha)$-quantile. The C.5-max columns use the simultaneous-max extension of Remark E.5, including both its $\sqrt{\log(h+1)}$ and $\sqrt{\log(1/\alpha)}$ terms. Every measured coverage was one, above the required Monte-Carlo lower band, but this is an empirical check and not a certificate. The C.4 envelope is conservative: its ratio is 17–18× on the oscillator. Independent-refresh concentration is tighter, reducing this range to 6–7×, while retaining the same measured coverage; neither result supports a claim of tightness.

Deployment	q	α	C.4 cov.	C.4 ratio	C.5-max cov.	C.5-max ratio
damped oscillator, $d_{\text{base}} = 3$	0.977848100	0.10	1.000	18.35×	1.000	7.10×
		0.05	1.000	17.84×	1.000	6.75×
		0.01	1.000	16.99×	1.000	6.12×

The contraction factor is verified, not structural. (A1) does *not* follow from passivity or symplecticity: preserving a two-form does not imply contraction in a positive-definite norm. For the specific conservative linear oscillator, the step has spectral radius one, precluding $q < 1$ in any fixed metric. The damped linear deployments admit the certificates in Remark E.6. For the nonlinear pendulum, the reported 0.987–0.999 values are *equilibrium-Jacobian* norms in metrics constructed from a discrete Lyapunov equation. They do not certify contraction throughout the operating region; that requires a common-metric supremum over the region.

The size-uniform result in Lemma E.1 uses the deployment-normalized RMS metric $M_{\mathcal{S}}$: uniform per-block error is $\mathcal{O}(1)$ there but $\mathcal{O}(\sqrt{b_{\mathcal{S}}})$ in the stacked Euclidean norm. For edge-explicit microgrids, incidence enters once in each node–edge coupling block, giving $\|B_{\text{inc}}\|$ rather than the Laplacian obtained by eliminating line dynamics. Both this structural scaling and the numerical consistency constant remain subject to the lemma's uniformity assumptions; a strongly configuration-dependent, nonseparable mass lies outside the separable leapfrog constants below.

Remark E.6 (Exact contraction certificates for the linear oscillators)**.** For the identity-mass oscillator $\dot{q} = p$, $\dot{p} = -kq - dp$, the Cayley–leapfrog step in Appendix A is linear, with exact matrix

$$A(h; k, d) = \begin{pmatrix} a & hr \\ br & ar^2 \end{pmatrix}, \qquad a = 1 - \frac{kh^2}{2}, \quad b = -kh + \frac{k^2h^3}{4}, \quad r = \frac{1 - dh/4}{1 + dh/4}.$$

Here h is one full integration step; the two damping stages each have duration $h/2$. For $h = 0.05$, $k = 2$, and the damping coefficients below, all matrix entries are rational. For each listed q_{cert}, we solve $q_{\text{cert}}^2 P - A^\top P A = I$ in exact arithmetic and set $M = 20P/\operatorname{tr} P$. Positivity of the leading principal minors verifies $I \prec M \prec 20I$ and

$$q_{\text{cert}}^2 M - A^\top M A \succeq \underline{\lambda}_C I \succ 0.$$

Exact upper and lower eigenvalue bounds similarly certify the displayed condition-number bounds. The same witnesses and margins also pass with the materialized binary64 values of h, k, d treated as exact coefficients; native forecast calls agree with the analytic map to rounding precision. Thus coefficient extraction is part of the certificate, not an unchecked automatic-differentiation step.

d	q_{cert}	$\operatorname{cond}(M) \le$	$\underline{\lambda}_C$
0.1	0.998751560550	2.025	0.02191703
0.3	0.996264009963	2.080	0.06473047
1.0	0.987654320988	2.730	0.18873362
3.0	0.977848100000	12.489	0.25223598

Since $\widehat{F}(z) - \widehat{F}(z') = A(z - z')$, these inequalities establish $\|\widehat{F}(z) - \widehat{F}(z')\|_M \le q_{\text{cert}} \|z - z'\|_M$ globally, discharging (A1) for these analytic deployments. They certify the mathematical split map, not state-dependent floating-point roundoff. For the conservative oscillator with $k = 1$, $h = 0.05$, $\rho(A) = 1$; no $q < 1$ is possible because contraction would imply $\rho(A) \le q$. Neither trained modules nor a regional nonlinear-pendulum contraction claim is included in this certificate.

Remark E.7 (Certified state-consistency constants)**.** For identity mass on $\Omega = [-1, 1]^2$ and $0 < h \le h_0 = 0.16$, outward-rounded interval evaluation certifies the Euclidean state-map constants $K_\Psi = 0.197$ for $\dot{q} = p,\ \dot{p} = -q$ and 0.240 for $\dot{q} = p,\ \dot{p} = -\sin q$, both undamped. For the separate damped instance $\dot{p} = -0.7q - 0.35p$, the Cayley–leapfrog map has $K_\Psi = 0.187$. To obtain this constant, first replace each damping half-step by $t = 1 - dh/2 + d^2h^2/8$: the interval Taylor-remainder enclosure is $0.184h^3$. The exact Cayley multiplier satisfies

$$r - t = -\frac{d^3h^3}{32(1 + dh/4)}.$$

On the stated box, changing both half-steps from t to r adds at most $c_q h^3$ and $c_p h^3$ in the two coordinates, with $c_q = d^3 h_0/32 = 0.000214375$ and $c_p = d^3(kh_0 + 2)/32 = 0.00282975$. Thus $\sqrt{c_q^2 + c_p^2} < 0.002838$ and $0.184 + 0.002838 <$

0.187. These constants are specific to the three stated vector fields, not to the $k = 2$ contraction examples or a gravity-scaled pendulum. Configuration-dependent mass and trained modules are not covered.

Metric reconciliation. These K_Ψ are certified in the Euclidean metric of their analytic instances ($M = I_2$), whereas (A2) uses the deployment metric $M_\mathcal{S}$; the conversion is $K_{\Psi,M_\mathcal{S}} \le \sqrt{\lambda_{\max}(M_\mathcal{S})}\, K_{\Psi,2}$. Likewise, Theorem E.2 bounds the state error in $M_\mathcal{S}$ while the readout constants of Lemma F.3 use a reliability metric M_{rel}; composing them carries the factor $c_{M_{\mathrm{rel}}\leftarrow M_\mathcal{S}} = \sqrt{\lambda_{\max}(M_\mathcal{S}^{-1/2} M_{\mathrm{rel}} M_\mathcal{S}^{-1/2})}$, so the per-channel bound of Proposition F.1 reads $|\hat{r} - r| \le L_m^x\, c_{M_{\mathrm{rel}}\leftarrow M_\mathcal{S}}\, \beta_k + \epsilon_m^{\mathrm{map}}$. Choosing $M_{\mathrm{rel}} = M_\mathcal{S}$ removes only the readout conversion. The Euclidean consistency factor remains unless the constant is certified directly in the deployment metric or that factor also equals one.

F. Typed-Consequence Decision Reliability

This appendix develops the two decision tests in Eq. (45). The channel and score bounds below lead to feasibility and same-set regret; Appendix F.1 collects the full-query conditions. Readers seeking worked electrical examples can go directly to Appendix F.3 for absolute limits and Appendix F.4 for comparisons that retain shared uncertainty across alternative connections.

Given the state tube of Theorem E.2, each typed consequence—and the planner decisions built on it—inherits a computable margin whenever the state-tube, readout, map-error, and aggregation constants are valid upper bounds on the operating region. Certified constants yield analytic certificates, as instantiated in the known-law examples below. For trained or hybrid deployments, empirically estimated envelopes must establish the required joint coverage over candidates and times before these tests support a probabilistic guarantee. Prediction accuracy or a calibrated warning threshold alone does not meet that requirement.

Setup. For candidate a, channel m, step k: truth $r_{m,k}^a = o_{m,\mathcal{S}}(x_k^a, u_k^a)$ and prediction $\hat{r}_{m,k}^a = \hat{o}_{m,\mathcal{S}}(\hat{z}_k^a, u_k^a)$; Theorem E.2 gives $\|\Pi_\mathcal{S} x_k^a - \hat{z}_k^a\|_M \le \beta_k^a$.

Proposition F.1 (Channel and horizon reliability)**.** *Assume* (B1) *per-channel* $|\hat{r}_{m,k}^a - r_{m,k}^a| \le L_m^x \beta_k^a + \epsilon_m^{\mathrm{map}}$ *for the paired states* $x_k^a \in \Pi_\mathcal{S}^{-1}(\Omega_\mathcal{S})$ *and* $\hat{z}_k^a \in \Omega_\mathcal{S}$ *(with* $\|\Pi_\mathcal{S} x_k^a - \hat{z}_k^a\|_M \le \beta_k^a$*), where* L_m^x *is a certified readout-Lipschitz constant (Lemma F.3) and* $\epsilon_m^{\mathrm{map}}$ *the learned-vs-physical readout error;* (B2) *channels smooth or mode-conditioned (energy, dissipation, port power, voltage/current directly; interval work uses the two-endpoint extension below; contact/switching identities only within a fixed mode or via a smooth surrogate);* (B3) *a* $\kappa_{m,N_{\mathrm{pred}}}$*-Lipschitz* ($\ell_p$) *horizon aggregator* $\rho_m^a = A_{m,N_{\mathrm{pred}}}(r_{m,0:N_{\mathrm{pred}}}^a)$*. Then*

$$\left|\hat{\rho}_m^a - \rho_m^a\right| \;\le\; \kappa_{m,N_{\mathrm{pred}}} \left\| \left(L_m^x \beta_k^a + \epsilon_m^{\mathrm{map}}\right)_{k=0}^{N_{\mathrm{pred}}} \right\|_p \;=:\; \Delta_m(a). \tag{167}$$

Proof. Apply (B1) pointwise in k, then the $\kappa_{m,N_{\mathrm{pred}}}$-Lipschitz aggregator (B3) to the horizon vector; β_k^a is the state tube of Theorem E.2. □

Theorem F.2 (Decision margins)**.** *Let* $\mathcal{A}$ *be a nonempty candidate set on which the displayed extrema and minimizers exist. Suppose Proposition F.1 gives, for every* $a \in \mathcal{A}$ *and channel* $m = 1, \dots, M$*,*

$$|\hat{\rho}_m(a) - \rho_m(a)| \le \Delta_m(a). \tag{168}$$

Write

$$\Delta_\rho(a) := (\Delta_m(a))_{m=1}^M \in \mathbb{R}_+^M.$$

Set

$$J_w(a) = w^\top \rho(a), \qquad \hat{J}_w(a) = w^\top \hat{\rho}(a), \qquad \Delta_J(a) = \sum_{m=1}^M |w_m|\, \Delta_m(a).$$

Assume in addition (B4) *for every hard-constraint index* j *there is a candidate-level map* $\mathrm{g}_j(a, \cdot)$ *such that*

$$g_j(a) = \mathrm{g}_j(a, \rho(a)), \qquad \hat{g}_j(a) = \mathrm{g}_j(a, \hat{\rho}(a)),$$

and, uniformly over $a \in \mathcal{A}$*, this map is* L_{g_j}*-Lipschitz in its consequence argument in* ℓ_p *on a set containing the relevant true and predicted consequence vectors:*

$$|\mathrm{g}_j(a, r) - \mathrm{g}_j(a, r')| \le L_{g_j} \|r - r'\|_p. \tag{169}$$

Define

$$\Delta_{g_j}(a) := L_{g_j}\|\Delta_\rho(a)\|_p. \tag{170}$$

Then $|\hat{J}_w(a) - J_w(a)| \le \Delta_J(a)$*, and* (i) *the ranking of* a, b *is preserved if* $|\hat{J}_w(a) - \hat{J}_w(b)| > \Delta_J(a) + \Delta_J(b)$*;* (ii) *the predicted minimizer has true regret at most* $2\max_a \Delta_J(a)$*;* (iii) *on a finite candidate set the predicted and true argmin coincide if the true best-vs-second gap exceeds* $2\max_a \Delta_J(a)$*;* (iv) *a constraint* $g_j(a) \le 0$ *is certified by* $\hat{g}_j(a) \le -\Delta_{g_j}(a)$.

Proof. The coordinatewise envelope (168) gives

$$\|\hat{\rho}(a) - \rho(a)\|_p \le \|\Delta_\rho(a)\|_p. \tag{171}$$

Consequently,

$$|\hat{J}_w(a) - J_w(a)| \le \sum_{m=1}^{M} |w_m|\,|\hat{\rho}_m(a) - \rho_m(a)| \le \Delta_J(a).$$

For (i), set $\hat{d} = \hat{J}_w(a) - \hat{J}_w(b)$ and $d = J_w(a) - J_w(b)$. The score envelope implies

$$|d - \hat{d}| \le \Delta_J(a) + \Delta_J(b).$$

Thus $|\hat{d}| > \Delta_J(a) + \Delta_J(b)$ forces d and $\hat{d}$ to have the same sign.

For (ii), let $\hat{a} \in \arg\min_{a\in\mathcal{A}} \hat{J}_w(a)$ and $a^\star \in \arg\min_{a\in\mathcal{A}} J_w(a)$. Predicted optimality and the two endpoint envelopes give

$$\begin{aligned} J_w(\hat{a}) - J_w(a^\star) &\le \hat{J}_w(\hat{a}) + \Delta_J(\hat{a}) - \hat{J}_w(a^\star) + \Delta_J(a^\star) \\ &\le \Delta_J(\hat{a}) + \Delta_J(a^\star) \le 2\max_{a\in\mathcal{A}} \Delta_J(a). \end{aligned}$$

For (iii), let $a^\star$ be the unique true minimizer and let the true best-vs-second gap exceed $2D$, where $D = \max_{a\in\mathcal{A}} \Delta_J(a)$. For every $a \ne a^\star$,

$$\hat{J}_w(a) - \hat{J}_w(a^\star) \ge J_w(a) - J_w(a^\star) - \Delta_J(a) - \Delta_J(a^\star) > 0,$$

so $a^\star$ is also the unique predicted minimizer.

Finally, (B4) and (171) give the missing constraint-error bound

$$|\hat{g}_j(a) - g_j(a)| \le L_{g_j}\|\hat{\rho}(a) - \rho(a)\|_p \le \Delta_{g_j}(a). \tag{172}$$

Hence $\hat{g}_j(a) \le -\Delta_{g_j}(a)$ implies $g_j(a) \le \hat{g}_j(a) + \Delta_{g_j}(a) \le 0$, proving (iv). □

Readout requirements. For each smooth consequence channel o_m, the quantity used in Proposition F.1 is a uniform certified upper bound

$$L_m^x \;\ge\; \sup_{(z,u)\in\Omega\times U} \|\nabla_z o_m(z,u)\|_{M^{-1}},$$

on a stated compact convex operating region Ω and, for port power, a stated compact input set U; it is not a test-set average. Thus $|o_m(z,u) - o_m(z',u)| \le L_m^x\|z - z'\|_M$ for all $z, z' \in \Omega$ at the same $u \in U$. Here $M \succ 0$ is the fixed metric used by the reliability bound, not the Hamiltonian's physical mass. For a quadratic Hamiltonian $H(z) = \frac{1}{2}z^\top Q z$, with $Q = \nabla^2 H$, the correct statements are

$$\begin{aligned} L_{H,\star}^x &:= \sup_{z\in\Omega} \|Qz\|_{M^{-1}}, \\ L_{H,\star}^x &\le \|M^{-1/2} Q M^{-1/2}\|_2 \sup_{z\in\Omega} \|z\|_M. \end{aligned}$$

L_H^x used in the proposition must satisfy $L_H^x \ge L_{H,\star}^x$. $\|Q\|_2 := \sup_{v\ne 0} \|Qv\|_2/\|v\|_2$ is the spectral norm, and a Hessian norm by itself is not L_H^x. If $e = \nabla H$ and $d = e^\top R(z)e$, then $\nabla d = 2(\nabla^2 H)^\top Re + r_R$ with $(r_R)_i = e^\top(\partial_i R)e$, so certification must include $\nabla^2 H$, R, and ∇R. Likewise, for $P = e^\top B(z)u$, one has $\nabla_z P = (\nabla^2 H)^\top B(z)u + b_B$ with $(b_B)_i = e^\top(\partial_i B)u$, and the bound must include B, ∇B, and the full set U. Linear electrical readouts $c^\top z + b(u)$ have the uniform constant $\|c\|_{M^{-1}}$, but this does not make a generally nonlinear known-physics model affine. Lemma F.3 records these formulas and the exact horizon-aggregator constants. The separate quantity $\varepsilon^{\mathrm{map}}$ must likewise be supplied as a uniform supremum bound over its declared region; semi-synthetic perturbations can diagnose it but do not certify it, and no $\varepsilon^{\mathrm{map}}$ certificate is made here. Contact, switching, and discrete support-count channels are excluded from this smooth statement and require mode-conditional constants.

Lemma F.3 (Certified Lipschitz constants for smooth readouts and horizon aggregators)**.** *Let $\Omega \subset \mathbb{R}^n$ be a nonempty compact convex box, let $U \subset \mathbb{R}^r$ be a compact input box, and let $M_{\mathrm{rel}} \succ 0$ be a fixed reliability metric. Write*

$$\|v\|_{M_{\mathrm{rel}}} := (v^\top M_{\mathrm{rel}} v)^{1/2}, \qquad \|a\|_{M_{\mathrm{rel}}^{-1}} := (a^\top M_{\mathrm{rel}}^{-1} a)^{1/2}.$$

This M_{rel} is not the configuration-dependent physical mass $\mathcal{M}(q)$ appearing in a Hamiltonian. If $o(z,u)$ is continuously differentiable on a neighbourhood of $\Omega \times U$, then any outward-rounded interval enclosure

$$L^x_{o,\mathrm{cert}} \;\geq\; \sup_{(z,u)\in\Omega\times U} \|\nabla_z o(z,u)\|_{M_{\mathrm{rel}}^{-1}} \tag{173}$$

is a uniform constant satisfying, for every $z, z' \in \Omega$ and every $u \in U$,

$$|o(z,u) - o(z',u)| \leq L^x_{o,\mathrm{cert}} \|z - z'\|_{M_{\mathrm{rel}}}.$$

In particular, the following are valid smooth-channel certificates.

1. *If $H(z) = \frac{1}{2} z^\top Q z$ with $Q = Q^\top = \nabla^2 H$, then*

$$L^x_{H,\mathrm{cert}} \geq L^x_{H,\star} := \sup_{z\in\Omega} \|Qz\|_{M_{\mathrm{rel}}^{-1}}, \qquad L^x_{H,\star} \leq \|M_{\mathrm{rel}}^{-1/2} Q M_{\mathrm{rel}}^{-1/2}\|_2 \sup_{z\in\Omega} \|z\|_{M_{\mathrm{rel}}}.$$

   Here and below $\|A\|_2 := \sup_{v\neq 0} \|Av\|_2/\|v\|_2$ is the spectral norm (largest singular value); in particular, an unqualified $\|Q\|$ means $\|Q\|_2$. The Hessian norm alone is not a Lipschitz constant for the scalar readout H on an unbounded domain.

2. *Let $e(z) := \nabla H(z)$, $Q(z) := \nabla^2 H(z)$, and let $R(z) = R(z)^\top \succeq 0$. For $d(z) := e(z)^\top R(z) e(z)$,*

$$\nabla d(z) = 2Q(z)^\top R(z) e(z) + r_R(z), \qquad [r_R(z)]_i = e(z)^\top (\partial_i R(z)) e(z),$$

   and hence

$$L^x_{d,\mathrm{cert}} \geq \sup_{z\in\Omega} \|2Q(z)^\top R(z) e(z) + r_R(z)\|_{M_{\mathrm{rel}}^{-1}}.$$

   Thus a dissipation certificate generally depends on $\nabla^2 H$, R, and ∇R.

3. *For the port-power readout $P(z,u) := e(z)^\top B(z) u$, define $[b_B(z,u)]_i := e(z)^\top (\partial_i B(z)) u$. Then*

$$\nabla_z P(z,u) = Q(z)^\top B(z) u + b_B(z,u),$$

   and

$$L^x_{P,\mathrm{cert}} \geq \sup_{(z,u)\in\Omega\times U} \|Q(z)^\top B(z) u + b_B(z,u)\|_{M_{\mathrm{rel}}^{-1}}.$$

   State dependence of B and the declared input set U therefore cannot be omitted. For trapezoidal interval work at fixed input, the two endpoint constants are each $\Delta t L^x_{P,\mathrm{cert}}/2$ (Corollary F.4). For a linear electrical channel $o(z,u) = c^\top z + b(u)$, the global valid constant is $L^x_o = \|c\|_{M_{\mathrm{rel}}^{-1}}$.

For completeness, let a horizon contain $N_{\mathrm{pred}} \geq 1$ transitions, let $r_{1:N_{\mathrm{pred}}}$ denote its N_{pred} stage readouts, let $h_{0:N_{\mathrm{pred}}}$ denote its $N_{\mathrm{pred}} + 1$ endpoint readouts, and measure perturbations in ℓ_p, $1 \leq p \leq \infty$. Let p^* be the dual exponent, $1/p + 1/p^* = 1$ (with the usual endpoint conventions). The exact aggregator constants are as follows. For the ledger row, use the explicit signature

$$\mathcal{R}_{N_{\mathrm{pred}}}(h_0, h_{N_{\mathrm{pred}}}, d_{1:N_{\mathrm{pred}}}, w_{1:N_{\mathrm{pred}}}) := h_{N_{\mathrm{pred}}} - h_0 + \sum_{k=1}^{N_{\mathrm{pred}}} d_k - \sum_{k=1}^{N_{\mathrm{pred}}} w_k.$$

aggregator	*explicit signature*	$\kappa_{m,N_{\mathrm{pred}}}$ *in* ℓ_p
terminal	$h_{0:N_{\mathrm{pred}}} \mapsto h_{N_{\mathrm{pred}}}$	1
weighted linear	$r_{1:N_{\mathrm{pred}}} \mapsto \sum_{k=1}^{N_{\mathrm{pred}}} w_k r_k$	$\|w\|_{p^*}$
sum	$r_{1:N_{\mathrm{pred}}} \mapsto \sum_{k=1}^{N_{\mathrm{pred}}} r_k$	${N_{\mathrm{pred}}}^{1/p^*}$
rectangle integral	$r_{1:N_{\mathrm{pred}}} \mapsto \Delta t \sum_{k=1}^{N_{\mathrm{pred}}} r_k$	$\Delta t\, {N_{\mathrm{pred}}}^{1/p^*}$
mean	$r_{1:N_{\mathrm{pred}}} \mapsto {N_{\mathrm{pred}}}^{-1} \sum_{k=1}^{N_{\mathrm{pred}}} r_k$	${N_{\mathrm{pred}}}^{-1/p}$
maximum or minimum	$r_{1:N_{\mathrm{pred}}} \mapsto \max_k r_k$ *or* $\min_k r_k$	1
energy change	$h_{0:N_{\mathrm{pred}}} \mapsto h_{N_{\mathrm{pred}}} - h_0$	$2^{1/p^*}$
ledger residual	$\mathcal{R}_{N_{\mathrm{pred}}}$	$(2+2N_{\mathrm{pred}})^{1/p^*}$

If an equal-weight sum, rectangle integral, or mean consumes all $N_{\mathrm{pred}}+1$ endpoint values $r_{0:N_{\mathrm{pred}}}$ rather than N_{pred} stage values, every occurrence of N_{pred} in that row is replaced by $N_{\mathrm{pred}}+1$ (the maximum/minimum constant remains 1). The same $2^{1/p^*}$ bound applies after an absolute-value or positive-part map of the energy change. For trapezoidal quadrature on N_{pred} intervals,

$$\kappa_{m,N_{\mathrm{pred}}} = \Delta t\big(N_{\mathrm{pred}} - 1 + 2^{1-p^*}\big)^{1/p^*} \quad (p^* < \infty),$$

while for $p^* = \infty$ it is Δt when $N_{\mathrm{pred}} \geq 2$ and $\Delta t/2$ when $N_{\mathrm{pred}} = 1$. In particular, the sum has $\kappa = 1$ in ℓ_1, $\sqrt{N_{\mathrm{pred}}}$ in ℓ_2, and N_{pred} in ℓ_∞; if every stage error is bounded by the same scalar, its accumulated sum still grows linearly in N_{pred}. If the ledger accepts already-aggregated scalars $(h_0, h_{N_{\mathrm{pred}}}, D, W)$, its constant is instead $4^{1/p^*}$; a software reduction over additional keys or tensor entries must state that enlarged signature and coefficient vector.

All claims above concern differentiable readouts within one smooth mode. Contact, switching, and discrete support-count channels are excluded and require mode-conditional treatment. No bound on $\varepsilon^{\mathrm{map}}$ is asserted here.

Proof. For fixed u, convexity of Ω and the line-segment fundamental theorem of calculus give

$$o(z,u) - o(z',u) = \int_0^1 \nabla_z o(z' + t(z-z'),u)^\top (z-z')\, dt.$$

Cauchy–Schwarz in the dual pair $(\|\cdot\|_{M_{\mathrm{rel}}^{-1}}, \|\cdot\|_{M_{\mathrm{rel}}})$ proves (173). Direct differentiation gives the displayed energy, dissipation, and port-power gradients. The numerical bounds are outward interval enclosures of every operation in those symbolic gradients; therefore they bound the continuum boxes, not merely sampled points. Finally, Hölder's inequality gives $\|a\|_{p^*}$ for every linear aggregator with coefficient vector a, and the constants are sharp by duality. The maximum/minimum maps are 1-Lipschitz because $|\max r - \max s| \leq \|r-s\|_\infty \leq \|r-s\|_p$, with the same argument for the minimum. □

Corollary F.4 (Lipschitz full-objective margins)**.** *Let $\mathcal{J} : \mathcal{R} \subset \mathbb{R}^M \to \mathbb{R}$ be L_J-Lipschitz in ℓ_p on a set containing $\rho(a)$ and $\hat{\rho}(a)$ for every $a \in \mathcal{A}$, and define*

$$J(a) = \mathcal{J}(\rho(a)), \qquad \hat{J}(a) = \mathcal{J}(\hat{\rho}(a)), \qquad \Delta_J(a) := L_J \|\Delta_\rho(a)\|_p.$$

Under the channel envelope (168),

$$|\hat{J}(a) - J(a)| \leq \Delta_J(a). \tag{174}$$

Therefore the ranking and argmin conclusions in Theorem F.2(i)–(iii) hold verbatim for this full objective after using the displayed Δ_J.

In particular, the full objective in Eq. (20) is covered channel by channel as follows. For the terminal channel, set

$$\rho_G(a) := \Psi_G(x_{N_{\mathrm{pred}}}(a)), \qquad \hat{\rho}_G(a) := \Psi_G(\hat{x}_{N_{\mathrm{pred}}}(a)). \tag{175}$$

If the terminal readout has a uniform constant L_G^x satisfying Lemma F.3 on the declared operating region and a uniform readout-map error $\epsilon_G^{\mathrm{map}}$, then the terminal aggregator has $\kappa_{G,N_{\mathrm{pred}}} = 1$. Because this selector uses only the final coordinate, the direct (B1) bound gives the valid, sharper envelope

$$\Delta_G(a) := L_G^x \beta^a_{N_{\mathrm{pred}}} + \epsilon_G^{\mathrm{map}}. \tag{176}$$

The generic Proposition-F.1 ℓ_p envelope over a padded endpoint sequence is also valid but can be looser. For each normalized factor q, retain both endpoints of the interval readout in Eq. (19). Fix the candidate specification and step size, suppress these two arguments, and write

$$\begin{aligned} o_q(x^-, x^+, u) &:= \bar{\ell}_q\Big(\mathcal{O}^{\mathrm{phys}}_{\mathrm{target}}(x^-, x^+, u)\Big), \\ \hat{o}_q(z^-, z^+, u) &:= \bar{\ell}_q\big(\mathcal{O}_{\mathrm{target}}(z^-, z^+, u)\big). \end{aligned} \tag{177}$$

Here $\mathcal{O}_{\mathrm{target}}$ is the model readout defined in Methods; $\mathcal{O}^{\mathrm{phys}}_{\mathrm{target}}$ denotes its physical comparator. Let $z_\ell = \Pi_{\mathcal{S}_\kappa} x_\ell$ be the aligned true state, and let $\hat{z}_\ell$ denote the predicted state (written $\hat{x}_\ell$ in Methods). Assume that the predicted factor is smooth on a declared, mode-conditioned convex product region containing all aligned and predicted endpoint pairs. For each matched input, require uniformly on that region

$$|\hat{o}_q(z^-, z^+, u) - \hat{o}_q(\tilde{z}^-, \tilde{z}^+, u)| \le L_q^- \|z^- - \tilde{z}^-\|_M + L_q^+ \|z^+ - \tilde{z}^+\|_M, \tag{178}$$

$$|o_q(x^-, x^+, u) - \hat{o}_q(\Pi_{\mathcal{S}_\kappa} x^-, \Pi_{\mathcal{S}_\kappa} x^+, u)| \le \epsilon_q^{\mathrm{map}}. \tag{179}$$

The constants L_q^-, L_q^+ can be bounded by the corresponding partial gradient suprema, using Lemma F.3 in each endpoint variable and converting to the state-envelope metric M when needed. These are uniform assumptions, not consequences of the architecture. Define the rectangle-integrated factor channels by

$$\rho_q^{\mathrm{obj}}(a) := \Delta t \sum_{\ell=0}^{N_{\mathrm{pred}}-1} o_q(x_\ell(a), x_{\ell+1}(a), u_\ell), \tag{180}$$

$$\hat{\rho}_q^{\mathrm{obj}}(a) := \Delta t \sum_{\ell=0}^{N_{\mathrm{pred}}-1} \hat{o}_q(\hat{z}_\ell(a), \hat{z}_{\ell+1}(a), u_\ell). \tag{181}$$

For $1 \le p \le \infty$ and dual exponent p^*, the per-factor envelope is

$$\Delta_q(a) := \Delta t\, N_{\mathrm{pred}}^{1/p^*} \left\| \big(L_q^- \beta_\ell^a + L_q^+ \beta_{\ell+1}^a + \epsilon_q^{\mathrm{map}}\big)_{\ell=0}^{N_{\mathrm{pred}}-1} \right\|_p. \tag{182}$$

An endpoint-only factor has $L_q^- = 0$; interval work, losses, and accounting residuals generally require both terms, including β_0^a in the first interval. Thus, for the augmented consequence vector

$$\rho^{\mathrm{obj}}(a) := \big(\rho_G(a), (\rho_q^{\mathrm{obj}}(a))_q\big), \qquad \Delta_\rho^{\mathrm{obj}}(a) := \big(\Delta_G(a), (\Delta_q(a))_q\big),$$

the true objective and its predicted counterpart are exactly

$$J_w(a) = \rho_G(a) + \sum_q w_q \rho_q^{\mathrm{obj}}(a), \qquad \hat{J}_w(a) = \hat{\rho}_G(a) + \sum_q w_q \hat{\rho}_q^{\mathrm{obj}}(a). \tag{183}$$

Here $\hat{J}_w$ is the predicted score in Eq. (20), and J_w is its true-trajectory comparator. Thus Eq. (20) is covered explicitly, rather than being treated as a linear score in the original untransformed consequences. Its linear consumer has ℓ_p Lipschitz modulus $L_J = \|(1, w)\|_{p^*}$, so (174) applies with $\Delta_\rho^{\mathrm{obj}}$. The componentwise linear bound

$$|\hat{J}_w(a) - J_w(a)| \le \Delta_G(a) + \sum_q |w_q| \Delta_q(a) \tag{184}$$

is also available and can be sharper. Hard constraints are not absorbed into this scalar objective; they are handled by (B4) and Proposition F.5 below.

Proof. The Lipschitz hypothesis and (171) give

$$|\hat{J}(a) - J(a)| \le L_J \|\hat{\rho}(a) - \rho(a)\|_p \le L_J \|\Delta_\rho(a)\|_p.$$

Theorem F.2(i)–(iii) uses only such a pointwise objective-error envelope, so its arguments apply unchanged. The terminal channel follows from Proposition F.1 with selector constant $\kappa_{G,N_{\mathrm{pred}}} = 1$. For each interval factor, add and subtract $\hat{o}_q(z_\ell, z_{\ell+1}, u_\ell)$, where $z_k = \Pi_{\mathcal{S}_\kappa} x_k$. The map-error bound, the two endpoint Lipschitz bounds, and $\|z_k - \hat{z}_k\|_M \le \beta_k^a$ give

$$|o_q(x_\ell, x_{\ell+1}, u_\ell) - \hat{o}_q(\hat{z}_\ell, \hat{z}_{\ell+1}, u_\ell)| \le L_q^- \beta_\ell^a + L_q^+ \beta_{\ell+1}^a + \epsilon_q^{\mathrm{map}}.$$

Summing with rectangle weights and applying Hölder's inequality yields Eq. (182), since the dual norm of the weight vector is $\Delta t\, N_{\mathrm{pred}}^{1/p^*}$. Finally, the triangle inequality applied to Eq. (183) gives Eq. (184). □

Proposition F.5 (Robust feasibility and constrained regret)**.** *Let $\mathcal{A}$ be a nonempty finite candidate set. Assume the pointwise objective envelope*

$$|\hat{J}(a) - J(a)| \le \Delta_J(a) \tag{185}$$

and the constraint envelopes (172)*. The objective envelope may come from Theorem F.2 or from Corollary F.4. Define*

$$\mathcal{A}_{\mathrm{feas}} := \{a \in \mathcal{A} : g_j(a) \le 0 \text{ for every } j\}, \tag{186}$$

$$\widehat{\mathcal{A}}_{\mathrm{safe}} := \{a \in \mathcal{A} : \hat{g}_j(a) + \Delta_{g_j}(a) \le 0 \text{ for every } j\}. \tag{187}$$

Then every $a \in \widehat{\mathcal{A}}_{\mathrm{safe}}$ is truly feasible; equivalently,

$$\widehat{\mathcal{A}}_{\mathrm{safe}} \subseteq \mathcal{A}_{\mathrm{feas}}. \tag{188}$$

If $\widehat{\mathcal{A}}_{\mathrm{safe}} \neq \varnothing$ and

$$\hat{a}_{\mathrm{safe}} \in \arg\min_{a \in \widehat{\mathcal{A}}_{\mathrm{safe}}} \hat{J}(a), \qquad a^\star_{\mathrm{safe}} \in \arg\min_{a \in \widehat{\mathcal{A}}_{\mathrm{safe}}} J(a),$$

then the constrained regret relative to the best candidate in the same robust set satisfies

$$J(\hat{a}_{\mathrm{safe}}) - J(a^\star_{\mathrm{safe}}) \le 2 \max_{a \in \widehat{\mathcal{A}}_{\mathrm{safe}}} \Delta_J(a) \le 2 \max_{a \in \mathcal{A}} \Delta_J(a). \tag{189}$$

For comparison with the optimum over the complete true feasible set, assume there exists

$$a^\star_{\mathrm{feas}} \in \arg\min_{a \in \mathcal{A}_{\mathrm{feas}}} J(a) \tag{190}$$

satisfying the true optimum-retention margin

$$g_j(a^\star_{\mathrm{feas}}) \le -2\Delta_{g_j}(a^\star_{\mathrm{feas}}) \qquad \text{for every } j. \tag{191}$$

Then $a^\star_{\mathrm{feas}} \in \widehat{\mathcal{A}}_{\mathrm{safe}}$, and

$$J(\hat{a}_{\mathrm{safe}}) - J(a^\star_{\mathrm{feas}}) \le 2 \max_{a \in \widehat{\mathcal{A}}_{\mathrm{safe}}} \Delta_J(a). \tag{192}$$

Without membership of at least one true optimum in $\widehat{\mathcal{A}}_{\mathrm{safe}}$—for example, without (191)*—no $2\Delta_J$ regret bound relative to the optimum over all of $\mathcal{A}_{\mathrm{feas}}$ follows: safety tightening may exclude every true optimum. If $\widehat{\mathcal{A}}_{\mathrm{safe}}$ is empty, the proposition certifies no selectable candidate.*

Proof. If $a \in \widehat{\mathcal{A}}_{\mathrm{safe}}$, then for every j,

$$g_j(a) \le \hat{g}_j(a) + \Delta_{g_j}(a) \le 0,$$

which proves (188). Predicted optimality within the fixed set $\widehat{\mathcal{A}}_{\mathrm{safe}}$ and the two endpoint objective envelopes give

$$\begin{aligned} J(\hat{a}_{\mathrm{safe}}) - J(a^\star_{\mathrm{safe}}) &\le \hat{J}(\hat{a}_{\mathrm{safe}}) + \Delta_J(\hat{a}_{\mathrm{safe}}) - \hat{J}(a^\star_{\mathrm{safe}}) + \Delta_J(a^\star_{\mathrm{safe}}) \\ &\le \Delta_J(\hat{a}_{\mathrm{safe}}) + \Delta_J(a^\star_{\mathrm{safe}}), \end{aligned}$$

which proves (189).

Under (191), the symmetric constraint envelope yields

$$\hat{g}_j(a^\star_{\mathrm{feas}}) + \Delta_{g_j}(a^\star_{\mathrm{feas}}) \le g_j(a^\star_{\mathrm{feas}}) + 2\Delta_{g_j}(a^\star_{\mathrm{feas}}) \le 0.$$

Thus the true optimum belongs to the robust set. Together with $\widehat{\mathcal{A}}_{\mathrm{safe}} \subseteq \mathcal{A}_{\mathrm{feas}}$, this means the best true objective value on the two sets is the same, so (189) gives (192). □

Scope of the constants. The quantities L_{g_j} and L_J above are stated hypotheses (or algebraic consequences of a fixed linear consumer), and the Δ_m are conditional consequences of Proposition F.1 using the readout/aggregator hypotheses of Lemma F.3. The propositions do not themselves estimate these constants; Appendix F.3 instead constructs direct interval envelopes for an analytic electrical query. A normalized factor must have a denominator bounded away from zero and a uniform endpoint-gradient enclosures on its declared region before its L_q^- and L_q^+ can be used. Contact, switching, discrete support-count, and unmatched-mode channels remain mode-conditional as in Proposition F.1; this block does not turn sampled slopes or average errors into deterministic bounds.

F.1. From a query to a simultaneous decision certificate

The margin conditions become useful only when their error envelopes cover every candidate that can be selected. For the finite query set of Algorithm 1, the following corollary collects the initialization, rollout, and readout obligations in one statement. It uses the existing bounds to give one query-level conclusion.

Corollary F.6 (Simultaneous candidate certificate)**.** *Fix a nonempty finite candidate set $\mathcal{A}$ and horizon N_{pred}. For each $a \in \mathcal{A}$, assume the operating-region, alignment, and matched-mode hypotheses of Theorem E.2. Let $\epsilon_{\mathrm{lift}}^a$ bound the initial error* after $\mathrm{AssignState}_\kappa$*, including supplied new-component states and any reset or coordinate conversion. Suppose the candidate's one-step contraction factors and defects are bounded by $q_a \in (0,1]$ and $\bar{\delta}_a$, with $\bar{\delta}_a$ obtained as in Lemma E.1. Set*

$$\beta_k^a = q_a^k \epsilon_{\mathrm{lift}}^a + S_k(q_a)\bar{\delta}_a, \qquad 0 \le k \le N_{\mathrm{pred}}.$$

Under the channel and aggregation hypotheses of Proposition F.1, with the readout constants expressed in the state-envelope metric, define

$$\Delta_m(a) = \kappa_{m,N_{\mathrm{pred}}} \left\| (L_m^x \beta_k^a + \epsilon_m^{\mathrm{map}})_{k=0}^{N_{\mathrm{pred}}} \right\|_p .$$

When readout and state metrics differ, include the conversion factor of Remark E.7 in L_m^x. Use the actual endpoint or stage indices of each aggregator; unused coordinates have zero coefficients. For factors depending on both interval endpoints, use Eq. (182), which includes both β_ℓ^a and $\beta_{\ell+1}^a$. Let the objective and constraint envelopes $\Delta_J(a), \Delta_{g_j}(a)$ be those of Theorem F.2 or Corollary F.4. Define

$$\widehat{\mathcal{A}}_{\mathrm{safe}} = \{a \in \mathcal{A} : \hat{g}_j(a) + \Delta_{g_j}(a) \le 0 \text{ for all } j\}.$$

Then every retained candidate satisfies every stated constraint. If this set is nonempty, its predicted minimizer $\hat{a}$ satisfies

$$J(\hat{a}) - \min_{a \in \widehat{\mathcal{A}}_{\mathrm{safe}}} J(a) \le \Delta_J(\hat{a}) + \max_{a \in \widehat{\mathcal{A}}_{\mathrm{safe}}} \Delta_J(a). \tag{193}$$

Its ordering against a retained candidate b is certified whenever $\hat{J}(\hat{a}) + \Delta_J(\hat{a}) < \hat{J}(b) - \Delta_J(b)$. Comparison with the full truly feasible optimum additionally requires the optimum-retention condition of Proposition F.5.

These conclusions also hold on a common event of probability at least $1-\alpha$ if the required envelopes hold on that event. In particular, if candidate events E_a have failure probabilities at most α_a and $\sum_a \alpha_a \le \alpha$, all conclusions hold simultaneously with that probability. Independence between candidates is unnecessary. A shared initialization event that covers all assignments can instead be propagated directly, without allocating the same event separately to every candidate.

Proof. On the stated event, Theorem E.2 bounds each candidate trajectory at every step. Proposition F.1, with the two-endpoint extension in Corollary F.4 for interval factors, bounds each consequence; the objective and constraint maps give their stated envelopes. Thus $g_j(a) \le \hat{g}_j(a) + \Delta_{g_j}(a) \le 0$ for every retained candidate. If $a^\star$ minimizes the true score over the same retained set, predicted optimality gives

$$J(\hat{a}) - J(a^\star) \le \hat{J}(\hat{a}) - \hat{J}(a^\star) + \Delta_J(\hat{a}) + \Delta_J(a^\star) \le \Delta_J(\hat{a}) + \Delta_J(a^\star).$$

The interval-separation condition gives the ordering, and a union bound over the events E_a gives the simultaneous probabilistic claim. Because that event covers all candidates, selecting $\hat{a}$ using the same predictions does not require a new independence assumption. Any probabilistic bounds on model or readout errors must also be included in the common event; observer coverage alone does not certify those errors. □

An empty retained set gives no certified choice. Failure to certify a candidate does not establish that it violates a constraint. A certificate for sampled currents also does not exclude an intersample overload: that requires an additional continuous-time enclosure. For measurement-refreshed prediction, the simultaneous state envelope of Theorem E.4 can replace β_k^a; the subsequent readout and selection argument is unchanged.

F.2. Worst-component rather than average error

The RMS metric in Lemma E.1 controls average block error. To certify a local overload without a hidden block-count factor, one can propagate componentwise envelopes instead.

Proposition F.7 (Blockwise comparison envelope)**.** *For one fixed candidate, work in aligned coordinates and write the true and predicted steps as $x^+ = T_k(x)$ and $\hat{x}^+ = \widehat{T}_k(\hat{x})$. For each block i, use a fixed norm $\|\cdot\|_i$. Suppose throughout the common operating region that, for a nonnegative matrix P_k and vector d_k,*

$$\|T_{k,i}(x) - T_{k,i}(z)\|_i \le \sum_j (P_k)_{ij}\|x_j - z_j\|_j, \qquad \|T_{k,i}(z) - \widehat{T}_{k,i}(z)\|_i \le (d_k)_i.$$

If b_0 bounds the post-assignment block errors, the recursion

$$b_{k+1} = P_k b_k + d_k \tag{194}$$

bounds every block error simultaneously. If a local readout satisfies $|o_m(x) - \hat{o}_m(z)| \le \sum_i (\ell_m)_i\|x_i - z_i\|_i + \epsilon_m^{\mathrm{map}}$ for a nonnegative row ℓ_m, its error is at most $\ell_m b_k + \epsilon_m^{\mathrm{map}}$. If uniformly over a deployment family $\|P_k\|_\infty \le \gamma$, $\|d_k\|_\infty \le \bar{d}$, and $\|b_0\|_\infty \le \bar{b}_0$, then

$$\max_i \|x_{i,k} - \hat{x}_{i,k}\|_i \le \gamma^k \bar{b}_0 + \bar{d}\sum_{r=0}^{k-1}\gamma^r. \tag{195}$$

This bound is independent of the number of blocks at fixed horizon. It is uniform in horizon only if $\gamma < 1$ (apart from special zero-defect cases).

Proof. Insert $T_k(\hat{x}_k)$ between $T_k(x_k)$ and $\widehat{T}_k(\hat{x}_k)$ and apply the two block inequalities. Nonnegativity of P_k propagates the componentwise order by induction. Applying the readout inequality gives its envelope. Finally, $\|b_{k+1}\|_\infty \le \gamma\|b_k\|_\infty + \bar{d}$ and unrolling this scalar recursion proves (195). □

For a one-step map with at most $d_{\max}$ nonzero off-diagonal sensitivities per row, diagonal bound a_0, and off-diagonal bound a_1, one may take $\gamma = a_0 + d_{\max}a_1$. This condition concerns the *step map*, not merely the sparsity of the vector field: an exact flow or implicit solve may have dense sensitivities. Uniform row sums, local readout sensitivities, initialization bounds, and operating regions must be established rather than inferred from pH structure.

F.3. When the margins certify a decision

We test whether valid envelopes can resolve candidate decisions, rather than only contain observed errors. This is an analytic KNOWN-law check: the reference laws and an initial-state box are supplied. It does not calibrate a learned model or infer the box from partial observations.

Three DGUs form a chain, with unit capacitances, inductances, and resistances in their physical units. Initially, the bus voltages are $(22, 24, 26)$ V, the DGU currents are $(2, 2, 2)$ A, and the two line currents are zero; each node draws a known constant 2 A load. The nine candidates have source voltages

$$u_{c,s} = 26\mathbf{1} + c\mathbf{1} + s(-1, 0, 1)^\top \text{ V}, \qquad c \in \{-2, 0, 2\}, \quad s \in \{-1.5, 0, 1.5\}.$$

Each candidate keeps the graph fixed and rolls out 100 steps of 0.01 s. The maintained C-PHAST world-model interface calls its `CompositionalPHAST` core with four substeps. DGU resistance in the predictor is either the correct $1\,\Omega$ or a prescribed $1.1\,\Omega$; no parameters are fitted. The coordinatewise initial radius is $r \in \{0, 0.001, 0.01, 0.05, 0.2, 1\}$ V for voltage coordinates and A for current coordinates, shared by all candidates.

Table 18. Analytic certificate informativeness at the prescribed 1 A sampled line-current limit. Feasible, infeasible, and unresolved counts sum to nine; rankings count strictly separated score envelopes among 36 unordered pairs. Color distinguishes resolved feasible cases (green) from unresolved cases (orange), not a performance ranking.

Predictor R (Ω)	Radius r	Feasible	Infeasible	Unresolved	Rankings
1.0	0	6	3	0	33/36
1.0	0.001	6	3	0	33/36
1.0	0.01	6	3	0	33/36
1.0	0.05	6	0	3	29/36
1.0	0.2	0	0	9	0/36
1.0	1	0	0	9	0/36
1.1	0	6	3	0	32/36
1.1	0.001	6	3	0	32/36
1.1	0.01	6	3	0	30/36
1.1	0.05	6	0	3	24/36
1.1	0.2	0	0	9	0/36
1.1	1	0	0	9	0/36

The score is the existing voltage-regulation/current-sharing objective, averaged over samples $1{:}N_{\mathrm{pred}}$:

$$J(a) = \frac{1}{N_{\mathrm{pred}}} \sum_{k=1}^{N_{\mathrm{pred}}} \left[\left(\frac{\bar{V}_k - 24\,\mathrm{V}}{1\,\mathrm{V}} \right)^2 + \frac{1}{3} \sum_{i=1}^{3} \left(\frac{I_{i,k} - \bar{I}_k}{1\,\mathrm{A}} \right)^2 \right], \qquad \bar{V}_k = \frac{1}{3} \sum_i V_{i,k}, \quad \bar{I}_k = \frac{1}{3} \sum_i I_{i,k}.$$

The constraint limits the largest absolute sampled line current. We retain all four prescribed diagnostic limits, 0.25, 0.5, 1, and 2 A; these are not hardware ratings.

Enclosure. The independent reference dynamics are linear, $\dot{x} = Ax + Gu$, with $x = (V_1, I_1, V_2, I_2, V_3, I_3, I_{e,1}, I_{e,2})^\top$ and $u = (-I_{\mathrm{load},1}, V_{\mathrm{src},1}, \ldots, -I_{\mathrm{load},3}, V_{\mathrm{src},3})^\top$. Here each load is 2 A and the three source-voltage entries are $u_{c,s}$; the six-input forcing includes both, with units carried by G. We enclose the resulting 14×14 augmented exponential $\exp(\Delta t \left[\begin{smallmatrix} A & G \\ 0 & 0 \end{smallmatrix}\right])$ using 50-digit outward-rounded interval arithmetic, a 40-term Taylor polynomial, and the remainder $e^{\|Z\|_\infty} \|Z\|_\infty^{41}/41!$, where $Z = \Delta t \left[\begin{smallmatrix} A & G \\ 0 & 0 \end{smallmatrix}\right]$. Interval powers retain the signed initial-state sensitivity over the horizon. If $[E_k]$ encloses the state-transition block and $[c_k(a)]$ the reference trajectory from the box center, then every allowed initial state satisfies

$$x_{i,k}(a) \in [c_{i,k}(a)] + [-r, r] \sum_j |[E_{k,ij}]|.$$

Interval evaluation of the score and peak current gives uniform consequence enclosures. Their maximum deviations from the native predictor's scores give Δ_J and Δ_g, including the prescribed model mismatch and discretization error. This direct enclosure instantiates the decision-margin argument without requiring a contraction estimate for this electrical system.

With 10% resistance mismatch and $r = 0.05$, six candidates satisfy the 1 A limit for every state in the box, while three remain unresolved (Table 18). The selected retained candidate has $c = s = 0$; its peak-current envelope is $[0.8285, 0.9667]$ A (rounded outward). Its regret against the best candidate in the same retained set is at most 0.417 in the dimensionless score. The tighter limits, 0.25 and 0.5 A, certify all nine candidates infeasible for $r \leq 0.2$; the 2 A limit retains all nine for $r \leq 0.2$. At $r = 1$, every limit is unresolved. Thus validity alone does not imply an informative decision, and lack of certification does not imply infeasibility.

An independent adaptive ODE integration agrees with the nominal matrix flow within 1.6×10^{-12} in the reported physical coordinates. All 11,529 candidate–vertex diagnostic rollouts satisfy the asserted envelopes, orderings, feasibility implications, and regret bounds within 10^{-9} floating-point checking tolerance. The interval construction, not these finite tests, establishes the box-wide result. The certificate covers the sampled horizon only, not intersample overloads or repeated execution.

F.4. Comparing candidates under shared uncertainty

Separate score intervals discard the dependence induced by a common actual state. For example, costs $10 + 3\upsilon$ and $12 + 3\upsilon$, with $\upsilon \in [-1, 1]$, have overlapping intervals but differ by exactly -2. This section applies dependence-preserving range analysis (Stolfi & de Figueiredo, 2003) to the assignments and rewiring in a candidate query. Absolute feasibility remains governed by Corollary F.6.

Joint uncertainty after assignment. Fix the finite candidate family of Algorithm 1 and a joint physical uncertainty box, normalized as $\upsilon \in [-1,1]^m$. Each coordinate represents one uncertain physical quantity, reused across candidates only when it is the same quantity. Candidate-specific additions receive distinct coordinates. The box is a joint set, not an assumption of stochastic independence. For affine coordinate conversions and resets, gathering retained states and inserting supplied new states gives

$$x_0^a(\upsilon) = d_a + D_a\upsilon, \qquad d_a \in \mathbb{R}^{n_a}, \quad D_a \in \mathbb{R}^{n_a \times m}.$$

The center and radii are included in d_a, D_a; n_a may differ between candidates. Assume exact known affine sampled dynamics $x_{k+1}^a = F_{a,k}x_k^a + f_{a,k}$ over the fixed horizon, with prescribed inputs included in $f_{a,k}$ and all trajectories in the region of validity. Stacked affine readouts then have the form $r_a(\upsilon) = \bar{r}_a + T_a\upsilon \in \mathbb{R}^{p_a}$. For a fixed quadratic physical score, abbreviate

$$J_a(\upsilon) := J_w(a;\upsilon) = r_a^\top W_a r_a + v_a^\top r_a + c_{0,a}, \qquad W_a = W_a^\top \in \mathbb{R}^{p_a \times p_a}, \quad v_a \in \mathbb{R}^{p_a}. \tag{196}$$

All stage weights, normalizations, and terminal terms are fixed in this score. Peak-current constraints are screened separately, not treated as quadratic objectives.

Proposition F.8 (Assignment-aware candidate-difference enclosure)**.** *Under the preceding assumptions, each score is exactly $J_a = c_a + \ell_a^\top\upsilon + \upsilon^\top Q_a\upsilon$, where*

$$\begin{aligned} Q_a &= T_a^\top W_a T_a, \qquad \ell_a = T_a^\top(2W_a\bar{r}_a + v_a), \\ c_a &= \bar{r}_a^\top W_a\bar{r}_a + v_a^\top\bar{r}_a + c_{0,a}. \end{aligned} \tag{197}$$

For a quadratic $p(\upsilon) = c + \ell^\top\upsilon + \upsilon^\top Q\upsilon$ with $Q = Q^\top$, define the monomial enclosure

$$\begin{aligned} t(p) &= \sum_i |\ell_i| + 2\sum_{i<j}|Q_{ij}|, \\ \mathcal{E}(p) &= \left[c + \sum_i \min(0, Q_{ii}) - t(p),\ c + \sum_i \max(0, Q_{ii}) + t(p)\right]. \end{aligned} \tag{198}$$

Then the range of $J_a - J_b$ is contained in $\mathcal{E}(J_a - J_b)$, computed by subtracting the coefficients in Eq. (197). With interval subtraction on the right,

$$\mathcal{E}(J_a - J_b) \subseteq \mathcal{E}(J_a) - \mathcal{E}(J_b), \tag{199}$$

Proof. Affine propagation preserves the shared coordinates: if $x_k^a = t_k + T_k\upsilon$, then $t_{k+1} = F_{a,k}t_k + f_{a,k}$ and $T_{k+1} = F_{a,k}T_k$. Stacking affine readouts and expanding Eq. (196) gives Eq. (197). Each linear and off-diagonal monomial ranges in $[-1,1]$, whereas $\upsilon_i^2 \in [0,1]$; summing their signed ranges gives Eq. (198). For any common monomial range I and coefficients α, β, $(\alpha - \beta)I \subseteq \alpha I - \beta I$, since the right side permits different values of that monomial. Summing these inclusions over the same constant, linear, diagonal, and off-diagonal basis proves Eq. (199). □

A strictly negative upper endpoint certifies $J_a < J_b$ throughout the joint box; a strictly positive lower endpoint certifies the reverse. The inclusion compares the *same monomial representation*; a factored expression or tighter marginal optimizer can do better. For any valid individual score intervals $[L_a, U_a]$ and $[L_b, U_b]$, use

$$\mathcal{I}_{ab} = \mathcal{E}(J_a - J_b) \cap [L_a - U_b, U_a - L_b]. \tag{200}$$

Both intervals contain the true difference, so their intersection is valid and never wider than either. Containing zero leaves the ordering unresolved; an empty intersection indicates inconsistent enclosures. For any retained choice a, valid paired upper endpoints U_{ab}^{pair}, with $U_{aa}^{\mathrm{pair}} = 0$, also give

$$J_a(\upsilon) - \min_{b \in \widehat{\mathcal{A}}_{\mathrm{safe}}} J_b(\upsilon) \le \max_{b \in \widehat{\mathcal{A}}_{\mathrm{safe}}} U_{ab}^{\mathrm{pair}}, \tag{201}$$

because the left side equals $\max_b(J_a - J_b)$ over that same set.

Rewiring and residual errors. The quadratic term cancels only when $Q_a = Q_b$. Even with shared initial states and local laws, rewiring changes the response matrices T_a, T_b; the term $v^\top(Q_a - Q_b)v$ must therefore be retained. Shared fitted weights alone do not imply shared physical model error. For surrogate quadratic scores with certified residuals $|J_a - p_a| \le \rho_a$, a valid difference enclosure is $\mathcal{E}(p_a - p_b) + [-\rho_a - \rho_b, \rho_a + \rho_b]$. Nonlinear assignment, uncertain-parameter flow, and learned-law errors require such validated remainders; they are absent from the exact known-law case below. Numerical coefficients must also be outward enclosed through flow construction, coefficient assembly, and interval evaluation.

Known-law rewiring check. We use the three-DGU physical laws, initial state, nine source-voltage choices, quadratic regulation/current-sharing score, and 100-step horizon of Appendix F.3. All resistances are exactly $1\,\Omega$. Two prospective graphs use incidence matrices

$$B_{\mathrm{inc}}^{(1)} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}, \qquad B_{\mathrm{inc}}^{(2)} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The second graph moves one line's positive endpoint. Both graphs retain the same components and eight uncertain initial voltage/current coordinates, with identity assignment and reset. Each graph remains fixed during its counterfactual rollout; no switching transient is modeled. The 18 candidates yield 81 cross-graph comparisons before screening. We use 50-decimal outward-rounded interval arithmetic and the degree-40 exponential enclosure of Appendix F.3, preserving the signed initial-state dependence through the quadratic coefficient construction.

Table 19. **Shared uncertainty resolves additional rewiring comparisons.** Number of strictly certified cross-graph orderings after the same synthetic 2 A sampled-current screen. Radius r is in V for voltage coordinates and A for current coordinates. Separate raw intervals evaluate the reference score from state boxes; separate polynomial intervals subtract $\mathcal{E}(J_a), \mathcal{E}(J_b)$; paired bounds use Eq. (200). Bold green marks the largest count among nonempty comparisons, including ties. Zero eligible pairs means no admitted comparison, not successful certification.

Radius r	Retained	Eligible pairs	Separate raw	Separate polynomial	Paired
0	18	81	**81**	**81**	**81**
0.001	18	81	**81**	**81**	**81**
0.01	18	81	72	75	**79**
0.05	18	81	56	62	**74**
0.2	18	81	6	32	**60**
1	0	0	0	0	0

At $r = 0.05$, pairing resolves 12 more cross-graph comparisons than separate polynomial bounds, beyond the improvement from changing the marginal representation (Table 19). Direct and intersected counts coincide in this example.

Absolute feasibility is unchanged: the stricter 1 A screen retains only six chain candidates through $r = 0.05$, so admits no cross-graph comparison; at $r \ge 0.2$ it retains none. The 0.25 and 0.5 A screens retain none at any tested radius. At $r = 1$, none of the four screens admits a candidate.

The interval construction establishes the box-wide comparisons, conditional on the known laws and correct outward arithmetic. Twelve exact-rational regression tests and 756 finite reference paths check the implementation, but are not a substitute for those assumptions. This is an architecture-independent refinement of candidate comparison, not a trained-model superiority or learned-observer coverage result. Its baseline is the raw reference-score interval, not the predictor-centered envelope used in Appendix F.3.

G. Shared Observation and Evaluation Conventions

Section 4 evaluates different physical quantities under different observation budgets. This appendix collects their input contracts, metrics, and aggregation rules so that one-step prediction, open-loop takeover, and measurement-refreshed tracking can be interpreted separately. The study-specific protocols follow in the same order as the main results, beginning with electrical component learning in Appendix H.

G.1. Observation and Evaluation Protocols

Table 20 collects the observation, supplied-input, and target-output definitions for the main studies. For a trajectory of length T, let $\xi_t \in \mathbb{R}^{D_\xi}$ be the evaluated coordinate vector. One-step MSE is

$$\mathrm{MSE} = \frac{1}{(T-1)D_\xi} \sum_t \|\hat{\xi}_{t+1} - \xi_{t+1}\|^2.$$

Mechanical tables evaluate joint positions; electrical tables report voltage and current RMSE using the aggregation stated in their captions. For mechanical count transfer, takeover uses five observed context frames followed by N_{pred} open-loop predictions. Isaac's refreshed evaluation instead uses ten-frame context and re-anchors every five steps. The Isaac takeover exclusion rule is MSE $\geq 10^3$ per robot and seed; refreshed means retain all five seeds. Other benchmark failure counts and aggregation are specified with their tables. The coupling-power diagnostic is $\mathbb{E}[|(y^{\mathrm{port}})^\top \hat{u}|]$. Robot-arm, FHN, and canonical Isaac comparisons use 3, 3, and 5 seeds, respectively, unless a table states another count. The arm tables report population standard deviations across their three completed runs.

Table 20. **Benchmark observation and evaluation protocols.** The table separates what each model observes, what deployment information is supplied, what remains hidden, and what the reported evaluation establishes. The appendix gives simulator equations, physical parameters, full optimizer settings, and per-benchmark ablations.

Benchmark and split	Observed input	Supplied contract and hidden quantities	Query and evidence
Isaac transfer: 350 train + 50 validation ANYmal-D trajectories; 100 Go2 test trajectories	Relative joint-position history and normalized policy action; 5 context frames for takeover, 10 for refreshed evaluation	Fixed 4×3 leg layout and coarse target metadata supplied; momentum, floating-base state, contact response, applied torque, and the robot-specific command-to-torque map are not supplied to the canonical rollout model	N_{pred}=50 takeover and K=5 refreshed MSE; zero-shot fixed-layout embodiment transfer under a shared policy-command interface
Go2 push decision: 8 development scenarios; 20 held-out scenarios	Current root/joint telemetry, contact/slip geometry, declared friction, and nine supplied commands	Push label, push magnitude, and future simulator state hidden	Forecast 8 steps, choose once, execute 24 steps; survival, recovery, longest-survival ranking, and shortfall
Spot shadow mode: 897 train and 173 held-out windows from 6 episodes	2-s telemetry history plus numeric possible-next-behavior descriptor	No primitive label; replay-only future-derived descriptor fields are identified and must be supplied or predicted in deployment	Forecast 5-s typed consequences; Pearson/Spearman action-card agreement; no hardware command execution
Mechanical count transfer: arm trained at N=4 on 200 trajectories; 50 test trajectories generated per N, 20 evaluated for takeover per seed	L_h=5 position history and applied torque	Target part count, repeated block type, and chart supplied; momentum hidden in q-only rollout	N_{pred}=50 takeover MSE under zero-shot count change
FHN: 200 training trajectories at N=3; test $N \in \{2, 3, 4, 5, 6, 8\}$	Voltage/recovery histories and source current	Target cell count and chain graph supplied; local constitutive law fixed, partial, or learned according to regime	One-step and $N_{\mathrm{pred}} \in \{10, 50\}$ MSE; repeated excitable-cell recomposition
Nonlinear DGU: five independent fits, each with 32 train, 8 validation, and 16 test trajectories	Calibrated charge, flux, three node voltages, source current, and fixed applied inputs	Circuit structure supplied; numerical constitutive laws withheld; full-state calibration and evaluation	Five-second voltage/current RMSE; local learning beyond a fitted-linear control, not network transfer
Nonlinear electrical composition: five fits, 16 actual states and two candidate connections per fit	Full actual voltage/current state and supplied added-component state	Two four-to-five-node layouts, inputs, and line laws supplied; DGU laws learned. Component-only and source-network-calibrated fits reported separately	Two-second voltage/current RMSE, sampled peak-current error, and candidate ordering; fixed laws, no executed switch
Electrical partial sensing: five fits; source 3-node chain and 4-node star; four episodes per 5-node target layout	21 bus-voltage samples over 1 s, with 0 or 0.02 V measurement noise	Known source voltages, load currents, topology, and line coefficients; DGU and line currents, charge, and flux withheld at deployment; richer source calibration	0.5/2 s open-loop voltage/current RMSE on chain, star, and heterogeneous cycle; numerical and learned initialization under fixed dynamics

H. Electrical Component-Learning Protocols

Section 4.1 tests whether learned component laws improve prediction before they are composed into a network. Here we specify the data, fitting schedules, and failure accounting behind Table 3. The linear reconstruction below isolates the contribution of the fitted initialization; its separate network evaluation tests reuse without network retraining. Hidden-current observability and voltage-only network prediction are treated together in Appendix I.

Data and information. The simulated reference has capacitor charge $q_C/\mathrm{C} = 2.2[(v_C/\mathrm{V}) + (v_C/\mathrm{V})^3/3]$, inductance $1.8\,\mathrm{H}$, and resistance $1.2\,\Omega$. Initial capacitor voltage is uniform on $[-1.5, 1.5]\,\mathrm{V}$ and initial flux on $[-1, 1]\,\mathrm{Wb}$. The applied current into the capacitor node is $0.1\,\mathrm{A}$ and source voltage is $1\,\mathrm{V}$, both held constant. Each fit has 32 training trajectories and eight validation trajectories of 200 transitions, plus 16 test trajectories of 500 transitions, with $\Delta t = 0.01\,\mathrm{s}$. Calibration is

noiseless and fully observed: charge, flux, three node voltages, voltage-source current, and applied inputs are available. All learned models receive the same saved data, wiring, and component roles; the reference numerical laws are used only to generate and score trajectories. The linear control and C-PHAST warm start are fitted from those calibration data.

Training and evaluation. The five repetitions use fresh data, model, and minibatch random streams. Each trained arm receives 100,000 updates with batch size 128. Combined-loss C-PHAST uses its combined objective throughout; constraint-first C-PHAST and N-PHDAE use 25,000 measured-constraint updates followed by 75,000 combined-loss updates, each with its own losses. The C-PHAST arms start from identical installed weights and use the same minibatch sequence within each fit. C-PHAST uses Adam at 10^{-4}; N-PHDAE retains its Adam/cosine schedule starting at 10^{-4} and the pinned author networks and numerical implementation. The linear control receives no neural updates. We evaluate the fixed final checkpoints, without validation-based selection, on five-second open-loop test rollouts. Capacitor-voltage (e_3) and inductor-current RMSE are computed separately over all test paths and steps, excluding time zero. The measured source-current channel has the opposite sign to inductor current; this sign convention does not change its RMSE. If any path fails, the fit's all-path aggregate is unavailable rather than replaced by a successful-subset score.

Compositional pHNN component evaluation. We add the local model from the authors' compositional pHNN implementation (Neary & Topcu, 2023). Its Hamiltonian is a two-hidden-layer tanh network with 32 units per layer; dissipation is a learned constant positive-semidefinite matrix. The canonical skew matrix and input map are supplied. The model has 1,188 parameters, all optimized. Five fresh fits use the same source trajectories and 100,000 updates, with batch size 128 and Adam at 10^{-3}. The original loss predicts the next charge and flux; no voltage/current or constitutive loss is added. This differs from C-PHAST's data-fitted warm start and combined objectives.

The fixed final compositional pHNN checkpoints are also the component models used in Table 4A. For Table 3, each rollout starts from the measured test charge and flux, as do the C-PHAST rollouts. N-PHDAE retains those storage coordinates while solving for consistent algebraic values. Compositional pHNN uses its original RK4 transition at $\Delta t = 0.01$ s and the learned Hamiltonian gradient to read out capacitor voltage and inductor current. The physical-to-storage inversion used in the composition study is absent from this component test. The same 16 held-out trajectories, 500 future samples per trajectory, and all-path failure rule apply to every fit.

Combined-loss versus constraint-first training. Constraint-first training has no consistent effect on final accuracy. Relative to combined-loss C-PHAST, constraint-first C-PHAST's voltage RMSE changes by -0.77% to $+2.14\%$ (median -0.08%), and current RMSE by -17.26% to $+10.52\%$ (median $+0.82\%$). No equivalence margin was specified, so these results do not establish equivalence. Constraint-first training records 9.5–22.1% less fitting time across the five runs, but other workloads were present and the timings exclude validation/checkpoint diagnostics; they do not establish a controlled speedup.

These results compare complete learning procedures with different function families, initializations, losses, optimizers, and integrators. Equal calibration data and update counts do not match compute or isolate the cause of the cross-method gain. All final checkpoints replay, including the author initialization failure; separate physical-metric arithmetic and serial C-PHAST rollouts agree within the declared tolerances. The result establishes repeated predictive improvement beyond a linear approximation under this full-state, fixed-input protocol. It does not establish recovery of unique physical laws, generalization to new inputs, composed-network accuracy, partial-sensing superiority, or better reconfiguration decisions.

Linear-DGU initialization control. A reconstructed linear-DGU benchmark tests component fitting on noiseless, fully observed data with fixed source inputs and one reference parameter set. Each paired fit uses 30 training trajectories of 1,000 samples and ten evaluation trajectories of 10,000 samples at $\Delta t = 0.01$ s. Thus its approximately 100-second evaluation is separate from the five-second nonlinear study above. Each learner completes 100,000 updates in each of five independent paired fits using matched data. C-PHAST has lower voltage and current rollout root-mean-square error in all five fits than the authors' N-PHDAE implementation, both from measured starting states and after making the initial algebraic variables consistent with N-PHDAE's learned laws. Trainable parameter counts are 2,404 and 31,203, respectively. A data-fitted linear warm start already outperforms both N-PHDAE initialization modes; neural refinement further reduces its voltage and current errors by median paired values of 67% and 14%. This compares complete fitting procedures with different function families, initializations, objectives, integrators, and compute, rather than isolating the effect of architecture or parameter count. It is an isolated-component reconstruction, not a replay of published figures or evidence of composition, partial-sensing, or decision superiority.

Ten-DGU composition and numerical refinement. The separate composition experiment uses a complete graph of ten identical linear DGUs and 45 RL lines. Reference DGU parameters are $C = 2.2$ F, $L = 1.8$ H, and $R = 1.2\,\Omega$; these numerical laws are withheld from the learned component fits. The line resistances and inductances are independently sampled uniformly from $[0.1, 2]$ in ohms and henries, respectively, and supplied to both composed models. Source voltages are fixed at 1 V and load currents are sampled uniformly from $[0.1, 1]$ A and held constant. Each method's single local checkpoint is fitted on the same 30 full-state trajectories of 1,000 samples, with 100,000 updates and a 25,000/75,000 constraint-first schedule. This one-fit reconstruction is separate from the five-fit linear control above and the nonlinear experiment in Table 3.

At deployment, each method copies its own frozen checkpoint to all ten nodes and uses the supplied interconnection, without network fitting. Initial DGU charges, DGU fluxes, and line fluxes are sampled uniformly from $[-0.5, 0.5]$ in their respective units. N-PHDAE solves compatible algebraic initial values; PHAST uses the same initial storage values with its own constitutive readouts. The reference uses the authors' adaptive network integrator. Predictions are scored at 500 equally spaced times from zero to five seconds, excluding time zero; voltage RMSE pools the remaining 499 samples and all ten capacitor-voltage channels. Only one network realization, one initial condition, and one fitted checkpoint per method are evaluated.

The primary N-PHDAE prediction uses one RK4 step per output interval; PHAST uses its native network transition with two effective substeps. A fixed-checkpoint diagnostic doubles and quadruples those counts without training or reference-state feedback. N-PHDAE's voltage RMSE is 14.26693, 14.28139, and 14.28441 mV at the three resolutions; PHAST's is 0.475286, 0.236616, and 0.118077 mV. The corresponding known-law PHAST errors are 0.475253, 0.236579, and 0.118035 mV. The close agreement with the known-law control and decreasing PHAST error indicate a numerical limitation in this instance, rather than an additional nonlinear-learning gain. Different integrators and substep counts do not establish compute parity. This reconstructs the released-code composition of one reused checkpoint, not the paper's exact published trajectories or a network of ten separately trained component models.

H.1. Nonlinear composition and candidate comparison

Table 4 uses the candidate definition $a = (\kappa, u_{0:N_{\mathrm{pred}}-1})$ from Section 3.4. The actual four-DGU chain has edges $(0,1), (1,2), (2,3)$. Each candidate adds the same DGU and line, connecting the new unit to bus 0 or bus 2. Existing line resistances are $0.3, 0.4, 0.5\,\Omega$ and inductances are all 0.8 H; the added line has resistance $0.4\,\Omega$ and inductance 0.8 H. These line laws and both layouts are supplied. Numerical DGU laws remain withheld from the learned predictors.

For each of five fits, 16 simulated actual-system episodes supply a complete physical state after one second. Initial episode voltages are sampled from $[0.5, 1.3]$ V, DGU currents from $[-0.25, 0.35]$ A, and line currents from $[-0.1, 0.1]$ A. Loads in $[0.05, 0.2]$ A and source voltages in $[0.9, 1.2]$ V remain fixed within a candidate rollout. Each model maps the actual physical state to its own storage coordinates, retains those coordinates for existing components, and assigns the supplied state of the added DGU and line. No history estimator is used. Both candidate futures run for 200 steps at $\Delta t = 0.01$ s with fixed laws. The separately integrated physical reference is checked by comparing DOP853 and refined RK45 solutions; scores use the saved refined reference.

Panel A uses only the isolated-component fits. C-PHAST uses the constraint-first fits and explicit-midpoint coupling from Appendix A.4. Compositional pHNN retains the authors' local Hamiltonian model, network assembly, and RK4 transition. An external quadratic-energy factory supplies the known RL-line laws; the supplied incidence matrix defines their interconnection with the DGUs. Every DGU copy shares the same fitted Hamiltonian and dissipation parameters. For compositional pHNN, state entry solves the learned energy-gradient equations with a fixed zero start, an analytic Hessian, and the function-evaluation limit set to 100; the accepted physical residual is at most 10^{-7} in volts or amperes. Ten branches in fit 3 fail this declared procedure: eight involve the added DGU, and two an existing DGU. They remain failed outcomes; this does not show that an inverse is mathematically absent. No solver settings were changed after inspecting the outcomes. This panel reuses the roots previously examined in panel B and is a descriptive added-baseline comparison.

Panel B instead uses checkpoints with 10,000 additional calibration updates on three- and four-node source networks. C-PHAST uses the Euler-coupling transition in Appendix I.2 for both calibration and evaluation. Its two N-PHDAE variants share weights and retain the disclosed physical-state initialization and composed-training extensions. The original-integrator variant uses the authors' RK4 step. The corrected variant then solves the algebraic equations while preserving charge and flux, and carries that correction into the next step. The original-integrator trajectories are finite but all fail the 10^{-6} algebraic-residual check; their errors remain diagnostic. C-PHAST and corrected N-PHDAE pass the respective numerical

checks on all 160 branches. Residuals of different model equations are not accuracy ratios.

Trajectory RMSE excludes time zero and is first computed within each branch, then averaged over the 32 branches in a fit. Peak error is the absolute error in the maximum absolute line current across all lines and future samples. Candidate ordering compares these peaks; reference differences no larger than 2×10^{-6} A are unresolved. Panel B averages the five fit means; panel A retains all fit rows and labels the incomplete fit's conditional mean. The panel B primary criterion requires positive peak-error improvement in every fit, at least 0.01 mA mean improvement, and at least 10% relative improvement. It is not met: the corrected-baseline contrast improves in three of five fits, by 0.009993 mA (3.62%) on average. The secondary trajectory errors and ordering scores do not replace this primary result. Both panels evaluate full-state candidate prediction; neither executes a topology switch or certifies overload avoidance.

Effect of the coupling integrator. To separate numerical error from the learned laws, we compare explicit midpoint with Euler coupling on the same candidate bank, using the five constraint-first fits from panel A. Component parameters, initial storage states, inputs, local DGU updates, step durations, scoring times, and physical references are identical; only the coupling half-step changes. Both schemes qualify on all 160 trajectories. Midpoint reduces each of the four fit-level errors in all five fits. Median paired reductions are 11.0% for voltage RMSE, 7.1% for DGU-current RMSE, 32.5% for line-current RMSE, and 69.6% for sampled peak-current error. The corresponding numbers of improved individual trajectories are 139, 126, 120, and 149 out of 160: improvement of every fit mean does not imply improvement on every trajectory. Midpoint requires two coupling-field evaluations per half-step instead of one, with the same number of local DGU and implicit-solver calls. This comparison isolates the coupling discretization at fixed learned laws; it does not establish a convergence rate, a gain on fresh candidate networks, or improved decision performance.

H.2. Supplied-state comparison with original baselines

Table 5 reuses the fixed three-DGU network, calibration banks and 40 evaluation episodes (five fits $\times$ eight) of the CT-SUBNET comparison (Appendix I). Each calibration bank has 16 training and 8 validation trajectories with 40 permitted samples each, and provides physical and storage states and resistor drops; the fitting objectives use these labels differently. Wiring, ports and line laws are supplied, and the numerical DGU laws are withheld. Compositional pHNN and N-PHDAE are trained for 10,000 updates with their authors' objectives at pinned author commits; the C-PHAST fits are the 10,000-update fits of the CT-SUBNET comparison. All weights are frozen before evaluation.

Each episode starts at physical time 1 s and holds the recorded inputs for 200 steps of 0.01 s. Errors are scored at 40 times from 1.05 to 3.00 s, excluding the supplied time. The charge/flux condition supplies the eight true storage coordinates: three capacitor charges, three DGU fluxes and two line fluxes. The voltage/current condition supplies the corresponding true physical values, which each method converts through its learned laws. There is no observation history or measurement noise. Every planned forecast stays in its denominator, and any failure makes the five-fit mean unavailable.

Table 21. **N-PHDAE settings in the supplied-state comparison.** Completed / constraint-qualified forecasts out of 40 per supplied-state condition. A forecast is qualified when its maximum absolute algebraic residual is at most 10^{-6} over every returned state, including entry.

Arm	Setting	Charge/flux	Voltage/current
A	Saved setting: consistency solve at entry, then unprojected RK4	37 / 0	26 / 0
B	A with a bounded state-entry adapter	38 / 0	26 / 0
C	B with the authors' repeated-consistency option	33 / 33	26 / 26

N-PHDAE is evaluated in the three arms of Table 21. Arm A, the authors' saved setting, solves the algebraic states once at entry and then integrates with unprojected RK4. Arm B replaces the entry guess with a bounded scalar adapter whose search domains come from training data. Arm C additionally corrects the algebraic states after each step through the authors' repeated-consistency option; only this arm produces qualified forecasts, and the main text reports it. The residuals are in the model's mixed equation units, so qualification is a numerical check. This option differs from the study-specific per-step correction of Table 4B.

All seven arm-C failures with supplied charges and fluxes occur in fit 2. Four stop because the upstream solver reports insufficient progress although the terminal residual is near 10^{-16}, one stops with an unresolved residual of about 3.7×10^{-3}, and two fail during state entry. With supplied voltages and currents, all 21 compositional pHNN failures and all 14 N-PHDAE failures occur during state entry. Accepting the four near-zero-residual states and continuing their forecasts is untested.

With supplied voltages and currents, C-PHAST's five-fit mean current RMSE is 0.90 mA, the supplied-state diagnostic of Section 4.2.

I. Electrical partial-sensing protocol and observer ablations

Section 4.2 tests prediction when only bus voltages are measured at deployment. We first give the larger-network study's protocol, separating a structural ambiguity from estimation error, then specify the common data and network transition, then the numerical and learned initializers compared in Table 8. Keeping the dynamics fixed separates the effect of state inference from a change in the predictor; the final ablations compare observer objectives under the same evaluation protocol. Appendix I.5 then specifies the separate fixed-topology comparison with CT-SUBNET, which uses fresh models and a smaller calibration set.

Which hidden currents can voltage history reveal? A separate known-law qualification tests whether voltage history can determine hidden currents. It uses known linear DGU laws and observes all bus voltages on three-node networks; DGU and line currents are hidden. The finite-horizon state-to-voltage sensitivity has rank 8/8 on the tested tree, but rank 8/9 on a symmetric cycle with equal line decay rates. An explicit circulating-current perturbation leaves voltage trajectories equal to numerical precision: the maximum voltage difference is 4.44×10^{-16} V, despite initial branch-current differences of magnitude 0.577 A. This is a property of the declared symmetric fixture, not a claim that every cycle is unobservable. The learned-model evaluation in Section 4.2 keeps this information limitation separate from prediction error.

I.1. Measurements and evaluation

For a network with n DGUs and m lines, only the bus-voltage history is observed:

$$\tilde{v}_{i,k} = v_i(t_k) + \epsilon_{i,k}, \qquad \epsilon_{i,k} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_v^2). \tag{202}$$

Here $i = 1, \dots, n$, $k = 0, \dots, 20$, and $t_k = \tau_t - 1 + 0.05k$ seconds, where τ_t is the physical query time. We use $\sigma_v = 0$ or 0.02 V, the latter 2% of the 1 V reference. Noise affects the measurements only; the simulated dynamics and scoring targets are unchanged. DGU inductor currents and line currents are withheld, as are charge and flux. Source voltages, load currents, wiring, line resistance and inductance, and the assumed noise scale remain supplied. In particular, known load current is an external input, distinct from the unmeasured DGU and line currents.

Each estimate initializes one open-loop prediction with $\Delta t = 0.01$ s, scored over 50 and 200 steps. No future measurements enter the prediction; controls are constant within each episode. The same trajectory and noise draws are used for all initialization variants. Five independently fitted dynamics checkpoints, each calibrated on a three-node chain and four-node star, remain frozen at their final 10,000-update state (Appendix A.5.1). Each fit uses 16 training and eight validation trajectories per source graph, with 200 transitions per trajectory. New observers use the training trajectories only, with windows ending at steps 100, 125, 150, and 175: 64 windows per graph from 16 independent trajectories, not 64 independent trajectories. Source calibration includes phase, current, and voltage-rate labels. All observer fitting finishes before generating the fresh evaluation episodes. The target layouts are a five-node chain, star, and heterogeneous cycle, with four episodes per layout and fit. They are held out from observer training, not graph families unseen in all prior project experiments. Two fresh episodes per source layout and four per symmetric five-node cycle provide separate diagnostics.

Source and evaluation episodes sample initial bus voltages uniformly from $[0.5, 1.3]$ V, DGU currents from $[-0.25, 0.35]$ A, and line currents from $[-0.1, 0.1]$ A. Constant load currents lie in $[0.05, 0.2]$ A and source voltages in $[0.9, 1.2]$ V. Each line has inductance 0.8 H; resistances are equally spaced from 0.3 to $0.5\,\Omega$ in the declared edge order. The symmetric-cycle control instead uses $0.4\,\Omega$ on every line, so all line decay rates coincide. DGU reference laws are those of Appendix H. Evaluation simulates three seconds per episode: one second supplies the observation history and two seconds the forecast.

For each channel group, RMSE averages squared error over its coordinates and all future samples within an episode before taking the square root. We average these episode RMSEs within each fit, then report the mean and sample standard deviation across the five fit-level means. The primary average contains 12 target episodes per fit; neither source episodes nor the symmetric-cycle control is pooled into it. The two horizons share a rollout and are not independent trials.

I.2. Fixed dynamics and numerical initialization

Let $x = (q, \phi, \phi_e)$ collect DGU charge, DGU flux, and line flux. The DGU energy and readouts are those of Eq. (21); a line has $i_{e,k} = \phi_{e,k}/L_{e,k}$. This sensing study uses Euler coupling, held fixed across all observer comparisons. For incidence matrix A (negative at the sending bus, positive at the receiving bus), its coupling increment over duration τ is

$$\mathcal{E}_\tau(x) = (q + \tau A i_e,\ \phi,\ \phi_e - \tau A^\top v). \tag{203}$$

With $\delta = \Delta t/2$, let $\mathcal{L}_\delta$ apply one local input–damping–midpoint–damping–input pass from Algorithm 3 to each DGU. It also multiplies each line flux by $[(1 - \delta R_{e,k}/(4L_{e,k}))/(1 + \delta R_{e,k}/(4L_{e,k}))]^2$. Two repetitions of $\mathcal{E}_{\delta/2} \circ \mathcal{L}_\delta \circ \mathcal{E}_{\delta/2}$ define the network transition F used by every initializer. The coupling increment is explicit Euler; its mirrored placement does not make it exactly energy-preserving.

Numerical history estimation adjusts a state at the beginning of the observed interval. Let $h_{\mathrm{phys}}(x) = (v_1, i_1, \ldots, v_n, i_n, i_{e,1}, \ldots, i_{e,m})$ be the full physical readout, and write $y^{\mathrm{phys}}(\xi) = \mu + S_{\mathrm{prior}}\xi$. Here $\xi, \mu \in \mathbb{R}^{2n+m}$ and S_{prior} is the diagonal matrix of prior scales. The prior means are 1 V, 0.1 A, and 0 A for bus voltage, DGU current, and line current; their scales are 0.4 V, 0.3 A, and 0.2 A. The map $\mathrm{Storage}_\theta$ solves $\partial_{q_i} H_i = v_i$, $\partial_{\phi_i} H_i = i_i$ and sets $\phi_{e,k} = L_{e,k} i_{e,k}$. For history (50), use simulation indices $j = 0, \ldots, 100$ from physical time $\tau_t - 1$ to τ_t. Define

$$\begin{aligned} x_0(\xi) &= \mathrm{Storage}_\theta(y^{\mathrm{phys}}(\xi)), \\ x_{j+1}(\xi) &= F(x_j(\xi), u^{\mathrm{now}}_{t-100+j}), \quad j = 0, \ldots, 99, \\ \hat{\xi} &\approx \arg\min_\xi \left\{ \sum_{k=0}^{20} \frac{\|v_\theta(x_{5k}(\xi)) - \tilde{v}_k\|_2^2}{\max(\sigma_v, 0.002)^2} + \|\xi\|_2^2 \right\}, \\ \hat{x}_t &= \mathrm{Storage}_\theta\big(h_{\mathrm{phys}}(x_{100}(\hat{\xi}))\big). \end{aligned} \tag{204}$$

The noise floor is an inference weight, not extra noise added to clean measurements. The numerical fit returns $(\hat{\xi}, d_t) = \mathrm{HistoryFit}(h_t)$, where $d_t = (s_t, g_t, r_t)$ records optimizer success, reported optimality, and data-Jacobian rank. It uses `scipy.optimize.least_squares` with the trust-region reflective method, starts at $\xi = 0$, and allows 20 function evaluations. Its supplied central-difference Jacobian uses $\xi \pm 10^{-4} e_j$; function, step, and gradient tolerances are 10^{-8}. r_t counts singular values greater than 10^{-8} times the largest singular value of the noise-scaled data Jacobian, excluding prior rows. The state-only initializer uses this same fit's $\hat{\xi}$ in Eq. (204). Storage inversion starts at zero, caps each Newton update's infinity norm at 2, and allows 30 iterations with absolute residual tolerance 10^{-9}. A failed or singular conversion is recorded as a failure. All component parameters stay fixed. Finite forecasts retain d_t even when the optimizer has not converged. For a numerical-estimator decision query, the additional admission predicate is $s_t \wedge (g_t \le 10^{-4}) \wedge (r_t = 2n + m)$; the declared unobservable negative control is excluded. Failure causes abstention before candidate prediction, not removal of a finite forecast from this evaluation.

I.3. Learned electrical observer and losses

The alternative adapts PHAST's MAP-first temporal convolutional observer to voltage histories. For each bus independently, let $z_{0:20}$ be the smoothed voltage sequence:

$$\begin{aligned} z &= \arg\min_z \left\{ \sum_{k=0}^{20} \frac{(z_k - \tilde{v}_k)^2}{\bar{\sigma}_v^2} + \sum_{k=1}^{19} \frac{(z_{k+1} - 2z_k + z_{k-1})^2}{\Delta_o^4 \sigma_a^2} \right\}, \\ d_k &= (z_k - z_{k-1})/\Delta_o, \qquad k \ge 1, \\ \hat{v} &= z_{20}, \qquad \widehat{\dot{v}} = d_{20} + r_\eta(\tilde{v}_{0:20}, d_{0:20}, \sigma_v). \end{aligned} \tag{205}$$

Here $d_0 = 0$, $\Delta_o = 0.05\,\mathrm{s}$, $\bar{\sigma}_v = \max(\sigma_v, 10^{-6})\,\mathrm{V}$, and $\sigma_a = 1\,\mathrm{V\,s^{-2}}$. The residual r_η is a shared causal TCN with two 32-channel layers, kernel size 3, and dilations 1 and 2. The MAP-only ablation sets this residual to zero.

A shared linear estimator returns line current, $\hat{i}_{e,k} = w^\top(f_k/s_f)$, with elementwise feature normalization. The feature vector f_k contains the 21-sample histories of signed endpoint voltage difference, that difference times mean endpoint voltage, source-voltage difference, and external-current-injection difference. It also contains these histories multiplied by the line's $R_{e,k}/L_{e,k}$, and the signed endpoint degree difference. For any bus quantity a, the signed edge difference is $-A^\top a$.

The scale s_f is each feature's source-training root-mean-square value, floored at 10^{-6}. Initial weights minimize source line-current MSE plus $0.01\|w\|_2^2$.

Voltage and its rate are then converted to each DGU's charge–flux state through the frozen energy law:

$$\begin{aligned} \partial_{q_i} H_i(q_i, \phi_i) &= \hat{v}_i, \\ \nabla_{(q_i,\phi_i)} v_i^\top \begin{pmatrix} i_i + I_i^{\mathrm{ext}} + (A\hat{i}_e)_i \\ V_i^{\mathrm{src}} - v_i - D_i(q_i) i_i \end{pmatrix} &= \widehat{\dot{v}}_i, \qquad \hat{\phi}_{e,k} = L_{e,k}\hat{i}_{e,k}. \end{aligned} \tag{206}$$

Here $I_i^{\mathrm{ext}} = -I_i^{\mathrm{load}}$ and $i_i = \partial_{\phi_i} H_i$. The inputs in this lift are the actual-system u_t^{now}, not a candidate action; its sampled input histories are rows $5k$, $k = 0, \ldots, 20$, of U_t^{now} in Eq. (50). The local Newton solve uses at most 25 iterations, residual tolerance 10^{-8}, minimum Jacobian singular value 10^{-9}, and update cap 2. Initialization converts $(\hat{v}_i, 0.1\ \mathrm{A}, \hat{i}_e)$ through the storage map. Training differentiates through the converged local root by the implicit-function rule. No voltage-equals-charge or voltage-rate-equals-DGU-current approximation is used.

Rate-supervised training uses 600 Adam updates of $\mathcal{L}_{\mathrm{rate}} = \mathrm{mean}[(\widehat{\dot{v}} - \dot{v}^\star)^2]$, with batch size 64, learning rate 10^{-3}, and gradient clipping at 10. The star denotes source-training labels, not deployment measurements. For a channel group a, write $E_a = \mathrm{mean}[((\hat{a} - a^\star)/s_a)^2]$, where the mean is over samples and coordinates and s_a is its source endpoint population standard deviation, floored at 10^{-3} in the channel's units. Two matched continuations minimize

$$\begin{aligned} \mathcal{L}_{\mathrm{end}} &= \tfrac{1}{2}(E_\phi + E_{\phi_e}) + \tfrac{1}{3}(E_v + E_i + E_{i_e}), \\ \mathcal{L}_{\mathrm{pred}} &= \mathcal{L}_{\mathrm{end}} + \tfrac{1}{3} \sum_{a \in \{v,i,i_e\}} \mathrm{mean}_{b,\ell,c} \left(\frac{\hat{a}_{b,\ell,c} - a^\star_{b,\ell,c}}{s_a} \right)^2 . \end{aligned} \tag{207}$$

In the second term, b indexes batch samples, $\ell = 1, \ldots, 10$ future steps, and c the coordinates of that channel group; predictions use the native C-PHAST transition. The endpoint quantities in $\mathcal{L}_{\mathrm{end}}$ are evaluated at the query time. Both continuations use identical initial weights, batches, and noisy measurements for 300 updates. Learning rates are 2×10^{-4} for the TCN and 10^{-4} for line weights, with gradient clipping at 5. Each update uses four source windows at each noise level, alternating source graphs. The final checkpoint is used without target-based selection.

I.4. Results and limitations

Table 8 reports the main initialization comparison; Table 22 supplies the additional variants. Numerical history estimation has lower primary mean error than the learned predictive variant for all three channels, both horizons, and both noise levels. This is a comparison of the tested estimators, not a claim that learned observers cannot improve on numerical estimation. Matched endpoint-only and predictive training differ by at most 0.233% in the primary means. Adding the tested prediction loss does not resolve the voltage/current trade-off.

All 1,200 planned evaluations are finite. The numerical history solver reports success in 200/200 calls, although one exceeds the reported stationarity threshold 10^{-4}; its forecast is retained. Finite output and a successful termination flag therefore do not imply that every state estimate is stationary or accurate. A separate symmetric five-node cycle admits two different circulating-current states with identical voltage histories in the structural test, while their endpoint line currents differ by 0.0303 A. That control is excluded from the primary average and does not imply that every cycle has the same ambiguity.

The implementation checks exercise the actual `CompositionalPHAST` transition, local inversions, frozen-law gradients, relabeling, line orientation, and serialization. Independent recomputation checks all 1,200 saved forecast errors. One noisy 50-step query for each of the 15 saved observer checkpoints replays exactly. The study measures voltage-only prediction under known constant inputs and supplied topology; it does not test process noise, actuator noise, topology switching, or improved electrical decisions.

Table 22. **Additional electrical observer variants.** Same primary targets, noise, horizons, aggregation, and units as Table 8. Each row contains all 60 planned target cases. MAP only removes the temporal residual but retains the source-fitted line estimator. Endpoint training adds phase/port supervision without a prediction loss. No best-value highlighting is used.

σ_v (V)	Horizon (s)	Initialization	Voltage (mV)	DGU current (mA)	Line current (mA)
0	0.5	Rate-supervised	3.76 ± 0.431	123 ± 26.2	53.2 ± 10.1
		MAP only + line fit	0.353 ± 0.0521	81.8 ± 17.6	52.4 ± 9.89
		Endpoint-trained	9.90 ± 2.69	77.2 ± 5.88	57.8 ± 19.4
0	2	Rate-supervised	7.43 ± 1.09	84.0 ± 18.0	45.2 ± 8.98
		MAP only + line fit	1.66 ± 0.435	56.7 ± 12.2	38.6 ± 7.30
		Endpoint-trained	20.3 ± 5.30	50.9 ± 3.77	44.3 ± 11.6
0.02	0.5	Rate-supervised	15.8 ± 2.09	174 ± 25.5	59.6 ± 9.50
		MAP only + line fit	21.4 ± 2.01	221 ± 31.5	59.4 ± 9.72
		Endpoint-trained	17.8 ± 2.87	120 ± 10.3	57.1 ± 17.8
0.02	2	Rate-supervised	22.5 ± 3.58	117 ± 17.2	59.4 ± 9.30
		MAP only + line fit	36.7 ± 5.54	147 ± 20.9	64.8 ± 12.3
		Endpoint-trained	28.7 ± 5.04	78.6 ± 6.95	51.8 ± 8.22

I.5. Fixed-topology comparison with CT-SUBNET

Common data and deployment measurements. Table 7 uses five paired fits on the three-DGU chain with edges $(0, 1)$ and $(1, 2)$. Both methods receive the same 16 source training and eight validation trajectories per fit, sampled at 0.05 s from 0 through 1.95 s: 40 samples per trajectory. The 19 permitted labels comprise eight physical channels $(v_0, i_0, v_1, i_1, v_2, i_2, i_{e,0}, i_{e,1})$, eight measured charge/flux coordinates, and three signed resistor voltage drops. Source/load inputs are supplied and constant within an episode. Numerical DGU laws are withheld. The line inductances are 0.8 H and resistances are 0.3 and 0.5 Ω. C-PHAST uses these coefficients and the supplied incidence matrix explicitly; CT-SUBNET learns from fixed-topology input/output sequences without graph or coefficient conditioning.

These fits use no pretrained component checkpoints or four-node calibration data. The voltage-only deployment history follows Eq. (202), with 21 observations at 0, 0.05, . . ., 1 s, plus the six known source/load inputs. Neither currents nor storage measurements enter either history initializer. Predictions are scored at 1.05, 1.10, . . ., 3.00 s under the supplied future inputs. Training data are clean; noise standard deviations 0 and 0.02 V test robustness to measurement noise at deployment.

CT-SUBNET implementation. We use the author's `SUBNET_CT` in deepSI (commit `3851ebda`). The model has eight latent coordinates and two-hidden-layer, width-64 residual MLPs for its encoder, state derivative, and decoder. The encoder consumes 21 samples of three voltages and six inputs. The decoder predicts all 19 calibration channels, with no direct input feedthrough; only the eight physical channels are scored. Selecting the voltage history before normalization prevents the other labels from reaching the encoder during either training or deployment. This voltage-only encoder and auxiliary-output supervision adapt the author implementation to our information contract; they do not reproduce an original CT-SUBNET benchmark. The latent coordinates are not identified physical states.

We retain the author's state-derivative normalization and one RK4 step per 0.05 s sample, using float32 arithmetic. Training uses 10,000 Adam updates, learning rate 10^{-3}, batch size 16, and ten-sample prediction windows. The loss is normalized output MSE across all 19 channels; normalization is computed from training trajectories only. Source validation uses the 19 future samples available after the history window. We retain the final model, not the validation-best model. The encoder initializes the first predicted sample according to the author's past/future window convention; no extra integration step or future observation is added.

C-PHAST implementation. Each fresh fit uses the 2,407-parameter DGU family from Appendix H, shared over the three units. The linear initialization is fitted only from the permitted source measurements. Training uses 10,000 Adam updates, learning rate 10^{-4}, and batch size 16. The objective is next-physical-output MSE plus 0.01 times the current/voltage and resistor-drop consistency loss, with parameter regularization weighted by 10^{-8}. Each observed 0.05 s transition uses five 0.01 s network steps in float64 arithmetic. The final weights remain fixed during evaluation. Numerical history initialization uses Eq. (204), with the same priors, noise floor, and solver settings given above. A supplied-physical-state diagnostic bypasses history estimation. Thus the comparison matches available data and observations, while comparing different physical priors, objectives, integrators, and inference procedures; update counts do not imply equal computational cost or

convergence.

Evaluation and aggregation. After training, eight fresh episodes per fit use the initial-state and input distributions specified above. All 40 reference episodes pass timestep refinement with maximum physical-channel discrepancy below 10^{-6}. The same episode and voltage-noise draws are paired across methods. The primary outcome is two-second RMSE pooled over the three DGU currents and two line currents at 0.02 V noise. We first compute each episode's RMSE, then average eight episode errors within each fit, and report the mean and sample standard deviation of the five fit means. The SD includes differences in source data, fitted models, and test draws; it is not variation from initialization seeds alone. Clean histories, individual channel groups, and the 0.5 s horizon are secondary.

All ten training attempts complete their 10,000 updates. All 80 forecasts per method and all 80 supplied-state diagnostic forecasts are finite. One of the 80 numerical history estimates exceeds the reported stationarity threshold 10^{-4}; its forecast remains included. With supplied states, the two-second RMSEs are 1.19 ± 0.41 mV for voltage, 0.44 ± 0.06 mA for DGU current, 1.28 ± 0.48 mA for line current, and 0.90 ± 0.27 mA for pooled hidden current. These diagnostic values repeat across noise conditions because the initializer bypasses the noisy history.

The frozen protocol, sources, checkpoints, prediction arrays, and per-fit metrics are archived under the experiment directory:

`experiment_sandboxes/essential_baselines_20261010/sensing/matched_20261011/`, with results in `results/production_01/` and the final independent audit in `results/production_01_audit_v2/independent_audit.json`.

The saved-array audit verifies all 240 forecasts and recomputes the metrics without model execution. The checks also verify checkpoint reconstruction and invariance to changes in hidden history channels. The study establishes fixed-topology partial-sensing accuracy under richer calibration; it does not evaluate topology transfer, voltage-only identification, or candidate-selection safety.

J. Cross-Robot Evaluation Protocol and Rollout Diagnostics

Section 4.3 changes the robot body while keeping the four-leg block layout fixed. This appendix first specifies the robots, command semantics, and source-to-target split, then the artifacts and observation windows behind Table 9. The rollout diagnostics explain how open-loop and measurement-refreshed predictions differ when the source-trained model remains fixed.

J.1. Isaac Lab Benchmark Details

The source and target robots share q-only sensing and the same normalized policy-command shape, isolating *response mismatch* in the main text's deployment taxonomy. The downstream actuator semantics are not fixed, however, because the two robots use different action scales and PD gains.

Robots. We use ANYmal-D (ANYbotics) and Unitree Go2, which are commercially distinct quadrupeds with different mass distributions, link lengths, and actuator properties. Both are simulated in NVIDIA Isaac Lab (IsaacSim 5.1) at 50 Hz control frequency (decimation 4 from 200 Hz physics). Each robot has $N{=}12$ actuated joints organized as $s{=}4$ legs with $d{=}3$ joints per leg: hip abduction/adduction (HAA), hip flexion/extension (HFE), and knee flexion/extension (KFE).

Policy training. Locomotion policies are trained using RSL-RL on flat terrain with the default reward function (linear velocity tracking, angular velocity tracking, joint torque penalties). The canonical collection uses 1,500 PPO iterations and 4,096 parallel environments per robot (policy seed 42).

Data collection. We record relative joint angles q_t and normalized policy actions a_t from the trained policies deployed in Isaac Lab. The action is a desired joint-position offset, not measured applied torque. At the nominal controller level,

$$\tau_t^{\mathrm{nom}} = K_p\big(s_a \ \mathrm{clip}(a_t, -1, 1) - q_t\big) - K_d\dot{q}_t,$$

where s_a, K_p, and K_d differ between ANYmal-D and Go2. The canonical results use a_t as the model input and do not claim that $a_t^\top \dot{q}_t$ is physical power. We separately report action-semantic ablations using the scaled offset and nominal reconstructed torque; applied-torque evaluation requires recollection because the submitted trajectory files do not store

`robot.data.applied_torque`. We collect 500 trajectories per robot. The matched cross-robot split used in Table 9 trains on 350 ANYmal-D trajectories, validates on 50 ANYmal-D trajectories, and evaluates zero-shot on 100 Go2 trajectories, each with T=200 steps. Action noise is applied during data collection (Gaussian, σ=0.05 in normalized command coordinates) to diversify the training distribution. Joint velocity is retained for diagnostics, but the matched transfer models receive only the position history and action sequence; contact state is not supplied.

Cross-robot protocol. C-PHAST is trained on ANYmal-D data only and evaluated zero-shot on Go2 (no target-domain training data). Unlike the arm and FHN count-transfer studies in Appendix K, this experiment keeps the block layout fixed at 4 legs $\times$ 3 joints per leg while changing the body, actuator response, and contact regime. Because both robots expose the same 12-actuated-joint layout, fixed-dimension baselines are applicable here; unlike the variable-N benchmarks, this is not a shape-mismatch test. Rows in the matched transfer comparison are trained on ANYmal-D and tested on Go2 without target retraining.

J.2. Evaluation Artifacts and Rollout Diagnostics

Table 9 uses regenerated data and checkpoints under the collection protocol above; the original caches were not retained. Two frozen result artifacts record the matched training and refreshed evaluation, and the five-frame takeover reevaluation, respectively. An analysis script reconstructs the table from records keyed by model, robot, seed, and rollout mode. The refreshed artifact uses ten context frames and refreshes every five steps; its means and population standard deviations include all five seeds. Takeover summaries exclude per-robot seed errors at or above 10^3. Excluded counts on ANYmal-D/Go2 are $1/1$ for Shared-GRU, $0/1$ for DeepSets, and $1/3$ for Neural ODE, out of five seeds per robot; all other learned rows retain five. These are rollout-threshold exclusions, not training-failure counts. The two modes differ in context length and aggregation as well as measurement refresh, so their aggregate difference is not a controlled refresh-only effect.

The burn-in window is an information budget. The context supplies information for inferring velocity: with a single position frame, the observer initializes zero momentum and the rollout settles toward stance. Five-frame takeover gives Go2 MSE 0.073 against 0.152 for persistence; ten-frame takeover gives 0.078 against 0.136. In the ten-frame open-loop diagnostic, the predicted gait continues but its amplitude decays; lower coordinate MSE alone does not establish faithful gait continuation. With ten-frame context and measurements every five steps, the same trained model gives 0.015 against 0.025 for protocol-matched persistence; per-step refresh gives 0.001 against 0.003. Figure 9 traces the dependence on refresh frequency. These comparisons distinguish candidate prediction without future observations from tracking an actual trajectory with new measurements.

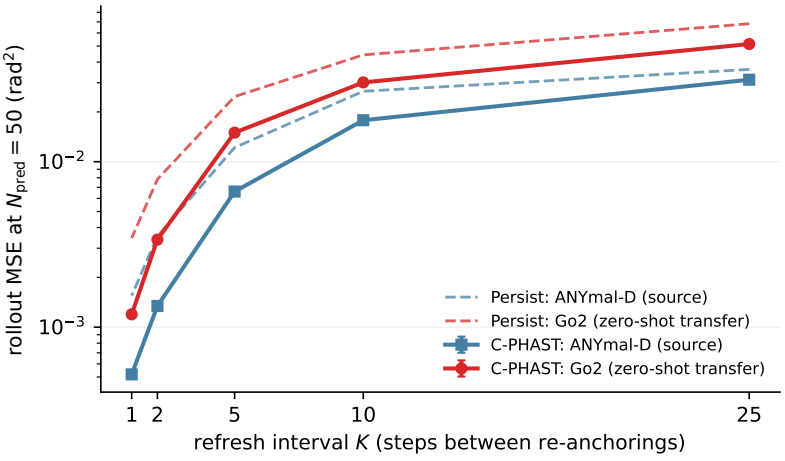


Figure 9. Rollout error at N_{pred}=50 as a function of the refresh interval K (canonical checkpoints, 3 seeds, mean $\pm$ 1σ). The same trained C-PHAST moves from bounded continuation under sparse re-anchoring to near interpolation under per-step refresh. Dashed curves apply the same refresh schedule to persistence; C-PHAST remains below them at every K.

Alternate training recipe. Training for 200 epochs with energy regularization and observer velocity supervision gives one-frame MSEs of 0.073 on ANYmal-D and 0.076 on Go2, but worsens Go2 prediction with five-frame takeover (0.149

versus 0.073) and refreshed observations (0.020 versus 0.015). Observer supervision also increases velocity-estimation error in this diagnostic. These results do not establish a general benefit of the auxiliary losses; the main comparison uses the matched 50-epoch position-loss recipe.

Rollout smoothing. An earlier diagnostic with 20-step context and an 80-step autoregressive horizon produced over-smoothed, near-stationary continuations for both C-PHAST and Shared-GRU, more strongly for C-PHAST. This is a limitation under that earlier protocol, not a measured jerk statistic for the current checkpoints; the current results retain protocol-matched persistence baselines.

K. Composition Across Subsystem Counts

Section 4.4 changes the number of repeated components, rather than only their physical response at a fixed layout. The planar arm isolates coupling and count transfer in a controlled mechanical system; the FitzHugh–Nagumo study tests coupled oscillators outside mechanics. Each protocol is kept with its supporting results and diagnostics.

K.1. Planar Arm Benchmark and Diagnostics

The planar-arm result in Table 10 asks whether a model trained on a 4-joint arm can be re-instantiated at new joint counts. The following protocol and coupling ablations establish what changes with the joint count and what is held fixed; the energy and mass-prior diagnostics then examine the assumptions behind that comparison.

K.1.1. Protocol and Schematic

This is the cleanest benchmark in the paper: it isolates repeated-joint count scaling without typed-layout, sparse-sensing, or embodiment complications. The robot arm is an N-joint planar manipulator with state $\boldsymbol{x} = (q, p) \in \mathbb{R}^{2N}$ where $q \in [-2.5, 2.5]^N$ are joint angles (with joint limits) and $p \in \mathbb{R}^N$ are momenta. Each joint has mass m_i=1, natural damping d_{nat}=0.1, and sinusoidal potential $V_i(q) = -\cos(q_i)$ (gravity). We simulate trajectories using semi-implicit Euler integration. Control inputs are per-joint sinusoids with random phases and frequencies, timestep $\Delta t = 0.02$s, trajectory length $T = 200$. Train: 200 trajectories, validate: 50, test: 50 per morphology $N \in \{2, 3, 4, 5, 6, 8\}$. The simulator injects inter-joint coupling through a fixed ground-truth skew matrix $\mathbf{C}_{\text{true}} \in \mathbb{R}^{N\times N}$: $u_{\text{coup}} = -\mathbf{C}_{\text{true}}\dot{q}$ with $\mathbf{C}_{\text{true}} = -\mathbf{C}_{\text{true}}^\top$. This term is power-preserving because $\dot{q}^\top u_{\text{coup}} = 0$. The learned model does not receive $\mathbf{C}_{\text{true}}$ in the Law-unknown regime; it learns a deployed skew coupling $\hat{\mathbf{C}}$ instead. Figure 10 summarizes this repeated-joint simulator and the power-preserving coupling used to generate the data.

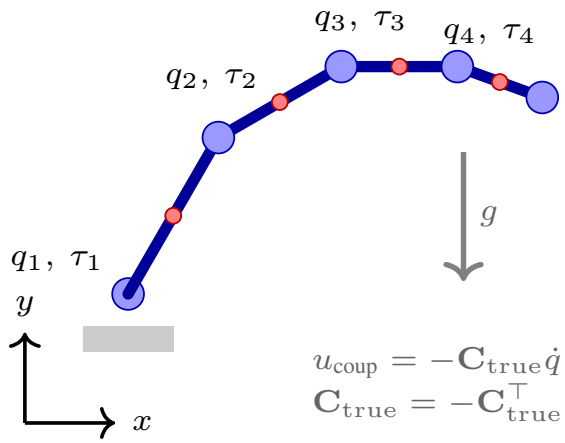


Figure 10. **Robot arm benchmark schematic.** An N-joint planar arm with joint-limited revolute joints $q \in [-2.5, 2.5]^N$, torques $u = \tau \in \mathbb{R}^N$, and q-only observations. Here $\mathbf{C}_{\text{true}}$ denotes the fixed simulator coupling used to generate data: $u_{\text{coup}} = -\mathbf{C}_{\text{true}}\dot{q}$ and $\mathbf{C}_{\text{true}} = -\mathbf{C}_{\text{true}}^\top$, so the coupling redistributes power across joints without injecting net energy.

K.1.2. One-Step Accuracy and Example Rollouts

The main text keeps the rollout-stability table because it carries the variable-N claim. Table 23 is included here as companion evidence: it confirms that the same compositional models can be instantiated at all tested N, but it is less diagnostic than the rollout comparison in Table 10.

Table 23. Compositional generalization on robot-arm dynamics (q-only). Compositional models can be instantiated at unseen N, whereas fixed-dimension baselines cannot. One-step MSE. Train on N=4, test on all N. Mean ± std over seeds. Best per column in **bold**.

Model	Params	N=2	N=3	N=4	N=5	N=6	N=8
							MSE ($\times 10^{-6}$), mean ± std
C-PHAST (known local physics)	13k	**3.17**±0.05	2.00±0.01	1.70±0.00	2.49±0.03	1.90±0.01	1.95±0.01
C-PHAST (partial local physics)	22k	3.19±0.07	**1.96**±0.02	**1.66**±0.04	**2.47**±0.02	**1.86**±0.02	**1.92**±0.02
C-PHAST (unknown local physics, one-step)	23k	8.14±0.11	6.27±0.05	5.63±0.04	6.95±0.07	5.93±0.05	6.18±0.05
C-PHAST (unknown local physics, rollout-aligned)	23k	14.4±1.4	10.2±1.4	9.30±1.4	11.4±1.4	9.50±1.4	10.3±1.3

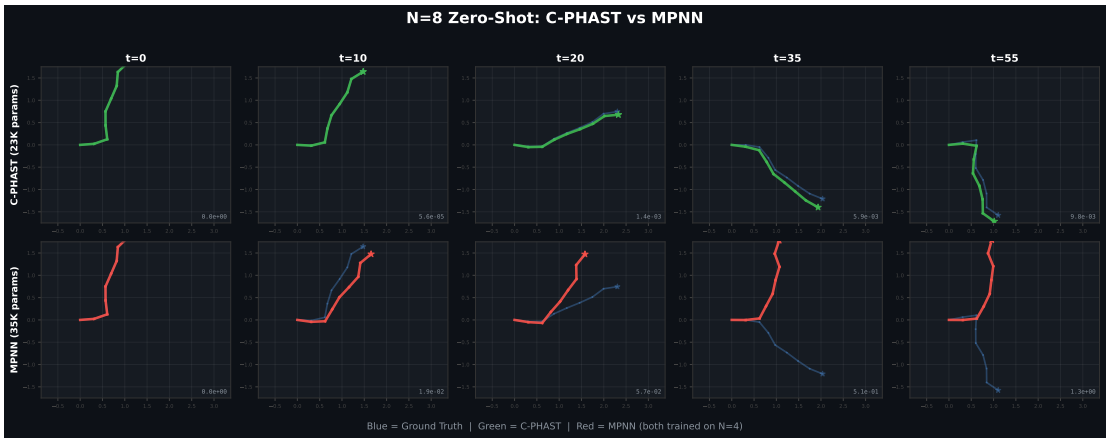


Figure 11. **Zero-shot transfer at N=8: C-PHAST vs MPNN** (both trained on N=4). This illustrative trajectory shows a hard extrapolation case: C-PHAST (green) stays close to ground truth with MSE 9.8×10^{-3} at t=55, while the locally shared MPNN baseline (red) departs from the reference with MSE 1.3 (130× gap). This single trajectory illustrates an error difference; it is not the seed-level divergence criterion used in the aggregate tables.

K.1.3. Coupling Parameterization

For Table 10, five observations of joint position initialize the latent state. The model then predicts 50 steps under the supplied torque inputs without further position measurements or observer calls. This infer-once evaluation, called takeover, uses 20 test trajectories per count and seed, with three seeds. The Known and Partial labels describe the local physics modules; they do not designate a supplied interconnection. The two local-law-unknown columns compare one-step and rollout-aligned training procedures. Besides rollout loss, these procedures differ in the observation history used during training and in their energy and passivity penalties. The dense coupling rule below explains what re-instantiating a learned model at another count does; the subsequent autoregressive ablation is a separate test with observer reuse.

Homogeneous count-transfer coupling parameterization The homogeneous count-transfer specialization indexes the deployed operator by the subsystem count s. In our implementation the coupling parameters θ_C are the entries of an unconstrained matrix $\mathbf{U}_{\theta_C} \in \mathbb{R}^{s_{\max}\times s_{\max}}$, from which we construct the homogeneous deployed coupling matrix

$$\hat{\mathbf{C}}^{(s)} \quad \text{for each deployed subsystem count } s.$$

Specifically,

$$\hat{\mathbf{C}}^{(s)} = \frac{1}{\sqrt{s}}\big((\mathbf{U}_{\theta_C})_{1:s,1:s} - (\mathbf{U}_{\theta_C})_{1:s,1:s}^{\top}\big), \tag{208}$$

which guarantees $\hat{\mathbf{C}}^{(s)} = -(\hat{\mathbf{C}}^{(s)})^\top$ by construction and allows evaluation at any $s \leq s_{\max}$ without changing parameters. Training at a source count s_0, however, supplies gradients only to the top-left $s_0 \times s_0$ block, so slicing to $s > s_0$ *evaluates* at the larger count but does not *learn* the new coupling entries; top-left slicing is moreover not permutation-equivariant across otherwise-equivalent repeated blocks. We therefore distinguish *local-block count transfer*—the shared per-block dynamics carrying over to more blocks, which the KNOWN/PARTIAL local-physics rows support strongly—from *learned-coupling count extrapolation*—predicting genuinely new inter-block couplings—for which a size-equivariant edge kernel $C_{ij} = f_\theta(\text{type}_i, \text{type}_j, \text{edge}_{ij})$ (with $C_{ji} = -C_{ij}$) would be the principled parameterization; the present dense-slice evidence for the latter is correspondingly weaker. We use the $1/\sqrt{s}$ normalization by default and ablate alternative normalizations in Appendix K.1.8.

For chain-structured systems, a nearest-neighbor template with $O(1)$ parameters is also available. Each subsystem exposes velocity port output $\hat{y}_i = \partial H_i / \partial p_i$; for multi-DOF subsystems we use $\hat{\mathbf{C}}^{(s)} \otimes I_d$ so the coupling remains defined at the subsystem level while the parameterization stays at $O(s^2)$ (dense) or $O(1)$ (chain).

Separate autoregressive coupling ablation. The coupling ablation asks which part of the compositional prior matters when the simulator contains the hidden skew inter-joint channel shown in Figure 10. All rows below are trained on the $N{=}4$ q-only arm and evaluated by instantiating the same learned template at each target joint count. The ablation compares three ways of handling the unknown inter-joint channel: no learned coupling, unconstrained neural coupling, and power-preserving skew coupling learned without access to $\mathbf{C}_{\text{true}}$. One-step error mainly measures local fit; a 50-step autoregressive rollout tests error accumulation after recomposition. Here each prediction uses the evolving prefix of predicted positions, including observer reuse. Unlike the infer-once takeover in Table 10, this protocol does not propagate a single initialized latent state throughout; neither protocol uses future measurements. The one-step ablation follows the expected local-fit pattern: Shared-GRU is best on this short-horizon metric, while the unknown-coupling C-PHAST variants are worse than the known/partial C-PHAST rows used for the main robot-arm result. We therefore tabulate the longer rollout stress below, where the limits of the unknown-coupling variants are most visible.

Table 24. **Robot-arm coupling ablation: 50-step autoregressive rollout MSE.** All C-PHAST variants use the LAW-UNKNOWN regime and differ in coupling parameterization; Shared-GRU and MPNN are learned baselines. Shared-GRU is strongest under this protocol. The physics-informed takeover results in Table 10 use a different rollout implementation and are not a matched comparison to these rows. Mean ± population standard deviation over three seeds.

Model	Params	N=2	N=3	N=4	N=5	N=6	N=8
Shared-GRU	59,602	**0.0083 ± 0.0009**	**0.0050 ± 0.0007**	**0.0056 ± 0.0009**	**0.0057 ± 0.0005**	**0.0115 ± 0.0051**	**0.0079 ± 0.0015**
MPNN	42,178	0.5069 ± 0.2053	60.3547 ± 65.9677	0.5207 ± 0.0500	15.6683 ± 19.5141	3.0587 ± 1.8844	1.1246 ± 0.3670
C-PHAST (no coupling)	23,243	0.0744 ± 0.0212	0.0526 ± 0.0109	0.0505 ± 0.0119	0.0477 ± 0.0139	0.0533 ± 0.0136	0.0464 ± 0.0101
C-PHAST (neural coupling)	23,391	0.0649 ± 0.0208	0.0528 ± 0.0190	0.0510 ± 0.0157	0.0458 ± 0.0186	0.0508 ± 0.0164	0.0476 ± 0.0153
C-PHAST (skew coupling)	23,387	0.0721 ± 0.0226	0.0516 ± 0.0114	0.0494 ± 0.0126	0.0464 ± 0.0144	0.0523 ± 0.0140	0.0458 ± 0.0104

K.1.4. ENERGY CONSISTENCY

Rollout MSE alone can hide physically implausible trajectories when errors remain small in position space. On the same unknown-coupling rollouts, the per-joint Hamiltonian diagnostic follows the same ordering as the rollout table: Shared-GRU has the smallest numerical error, MPNN is unstable, and the C-PHAST unknown-coupling variants remain bounded but do not match the known/partial-physics rows in Table 10. We use this as a boundary check rather than as a separate headline table.

K.1.5. Robot Arm Hyperparameters

Table 25. Hyperparameters for C-PHAST robot arm experiments.

Parameter	**Value**
Potential hidden dim	64
Number of PHAST substeps per step (L)	1
Coupling type	Skew-symmetric
Coupling structure	Dense (learn max-size skew matrix; slice to N)
Coupling normalization	$1/\sqrt{N}$
Observer (q-only)	Finite-diff + TCN residual
Observer clamp	$\|\hat{\dot{q}}\|_\infty \leq 6$
Canonicalizer	Identity
Batch size	64
Learning rate	10^{-3}
Weight decay	10^{-5}
Epochs	50
Timestep Δt	0.02s
Trajectory length	200
History length L_h	5
Rollout horizon N_{pred}	50

K.1.6. Fixed-Dimension Baselines

Table 26 contrasts fixed-dimension baselines, including non-compositional PHAST regimes, with C-PHAST at $N{=}4$. "—" indicates structural impossibility from shape mismatch, not poor performance.

Table 26. Fixed-dimension baselines vs C-PHAST on robot-arm dynamics (q-only). One-step MSE at $N{=}4$. Fixed-dimension methods cannot be evaluated at $N \neq 4$ due to hard-coded input projections. Mean $\pm$ std over seeds. Best per column in **bold**.

Model	**Params**	**MSE (N=4)**	**MSE (N=6)**	**MSE (N=8)**	**Generalizes?**
Fixed-dimension baselines (trained at N=4; not compositional)					
PHAST (known local physics)	4,039	$(1.660{\pm}0.007){\times}10^{-6}$	—	—	✗
PHAST (partial local physics)	14,547	$(1.670{\pm}0.002){\times}10^{-6}$	—	—	✗
PHAST (unknown local physics)	19,433	$(1.58{\pm}0.02){\times}10^{-6}$	—	—	✗
S5	17,732	0.0047 ± 0.0003	—	—	✗
LRU	34,372	0.0088 ± 0.0005	—	—	✗
LinOSS	17,732	0.0449 ± 0.0028	—	—	✗
GRU	39,428	0.0083 ± 0.0002	—	—	✗
Compositional architecture (per-joint weight sharing)					
C-PHAST (partial local physics)	21,977	$(1.66{\pm}0.04){\times}10^{-6}$	$(1.86{\pm}0.02){\times}10^{-6}$	$(1.92{\pm}0.02){\times}10^{-6}$	✓

K.1.7. Coupling Diagnostics

Beyond prediction error, we verify that the learned coupling behaves as a *power-preserving* interconnection. For skew-symmetric coupling, the coupling power residual $(y^{\text{port}})^\top \hat{u}$ should be numerically zero (Theorem 3.1).

Table 27. **Coupling diagnostics on robot-arm dynamics (q-only).** The coupling power residual $(y^{\text{port}})^\top \hat{u}$ should vanish for a skew interconnection (Theorem 3.1). The table reports mean and maximum absolute residuals over held-out trajectories and the relative skew-symmetry defect of $\hat{\mathbf{C}}$; values near machine precision verify that the learned interconnection preserves power rather than injecting energy.

N	$\mathbb{E}[\lvert(y^{\text{port}})^\top \hat{u}\rvert]$	$\max \lvert(y^{\text{port}})^\top \hat{u}\rvert$	$\lVert\hat{\mathbf{C}} + \hat{\mathbf{C}}^\top\rVert_F / \lVert\hat{\mathbf{C}}\rVert_F$
2	3.14×10^{-9}	5.96×10^{-8}	0
3	5.16×10^{-9}	1.24×10^{-7}	0
4	7.23×10^{-9}	1.19×10^{-7}	0
5	5.48×10^{-9}	9.00×10^{-8}	0
6	5.76×10^{-9}	1.02×10^{-7}	0
8	5.28×10^{-9}	9.00×10^{-8}	0

K.1.8. Coupling Normalization Ablation

In a separate periodic-joint arm study, we compare no normalization, $1/\sqrt{N}$, and $1/N$, with known per-joint dynamics and learned skew coupling. Training at $N = 4$ uses 500 trajectories of 100 steps, $\Delta t = 0.02$ s, width 32, two PHAST substeps, and a finite-difference plus TCN observer; 100 validation and 100 test trajectories are held out. Optimization uses 50 epochs, batch size 64, and learning rate 10^{-3}. At $N = 8$, all three one-step angular MSEs round to 0.0741 ± 0.0054. Their 50-step takeover angular MSEs are respectively 2.3644 ± 0.1012, 2.3637 ± 0.1009, and 2.3638 ± 0.1006 (means and population standard deviations over seeds 42–44). Takeover uses five observed frames and the subsequent torques on 32 test trajectories. Accuracy is thus insensitive to these scaling choices in this diagnostic; it does not establish size-uniform accuracy or uniquely favor the default $1/\sqrt{N}$ normalization.

K.1.9. Configuration-Dependent Coupling

Table 28 compares constant coupling $\hat{\mathbf{C}}$ with configuration-dependent coupling $\hat{\mathbf{C}}(q)$. The configuration-dependent variant achieves 1.3–3.3× lower takeover rollout error for $N \leq 4$ (interpolation), while constant coupling matches or outperforms for $N > 4$ (extrapolation).

Table 28. Configuration-dependent coupling $\hat{\mathbf{C}}(q)$ vs. constant $\hat{\mathbf{C}}$ on robot-arm dynamics (q-only, KNOWN regime). Takeover rollout MSE at N_{pred}=50. Train on N=4. Best per column in **bold**.

	Takeover Rollout MSE (N_{pred}=50, $\times10^{-3}$)					
N	2	3	4	5	6	8
$\hat{\mathbf{C}}$ (constant)	2.53±0.03	1.98±0.02	2.51±0.04	**1.81±0.02**	**1.74±0.02**	**1.74±0.03**
$\hat{\mathbf{C}}(q)$ (config-dep.)	**0.77±0.08**	**1.51±0.10**	**1.05±0.08**	2.83±0.21	1.78±0.08	2.14±0.10

K.1.10. Controlled Mass-Prior Corruption (Toy Stale-URDF Check)

Table 29 is intentionally a toy sanity check rather than the paper's main embodiment-mismatch claim. It perturbs only the planar-arm mass prior while keeping topology, sensing, and actuation fixed, so it should be read as a controlled misspecification probe for the KNOWN/PARTIAL regimes, not as a realistic fixed-layout sim-to-real workflow. Under this limited corruption, KNOWN/PARTIAL degrade gracefully (∼1.2× at ϵ=20%, 3.6–4.5× at ϵ=50%), and even at 50% mass error the physics-informed models still outperform LAW-UNKNOWN by ∼30×.

Table 29. Controlled mass-prior corruption on robot-arm dynamics (q-only). Takeover rollout MSE at N_{pred}=50 under mass perturbation ϵ. Best per row in **bold**.

	KNOWN		**PARTIAL**		**UNKNOWN**	
ϵ	N=4	N=8	N=4	N=8	N=4	N=8
0.00	$(2.46\pm0.04)\times10^{-3}$	$(1.65\pm0.03)\times10^{-3}$	$(2.48\pm0.06)\times10^{-3}$	$(1.66\pm0.05)\times10^{-3}$	$(2.87\pm0.04)\times10^{-1}$	$(2.61\pm0.03)\times10^{-1}$
0.05	$(2.26\pm0.02)\times10^{-3}$	$(1.47\pm0.01)\times10^{-3}$	$(2.26\pm0.02)\times10^{-3}$	$(1.47\pm0.02)\times10^{-3}$	$(2.87\pm0.04)\times10^{-1}$	$(2.61\pm0.03)\times10^{-1}$
0.10	$(2.24\pm0.03)\times10^{-3}$	$(1.45\pm0.01)\times10^{-3}$	$(2.26\pm0.03)\times10^{-3}$	$(1.46\pm0.02)\times10^{-3}$	$(2.87\pm0.04)\times10^{-1}$	$(2.61\pm0.03)\times10^{-1}$
0.20	$(2.92\pm0.06)\times10^{-3}$	$(2.08\pm0.04)\times10^{-3}$	$(2.94\pm0.06)\times10^{-3}$	$(2.09\pm0.05)\times10^{-3}$	$(2.87\pm0.04)\times10^{-1}$	$(2.61\pm0.03)\times10^{-1}$
0.50	$(8.77\pm0.20)\times10^{-3}$	$(7.45\pm0.18)\times10^{-3}$	$(8.78\pm0.19)\times10^{-3}$	$(7.45\pm0.16)\times10^{-3}$	$(2.87\pm0.04)\times10^{-1}$	$(2.61\pm0.03)\times10^{-1}$

K.2. Coupled FitzHugh–Nagumo Benchmark Details

An FHN unit is a compact two-state excitable circuit: the voltage-like state can spike, the recovery state pulls it back, and a diffusive link lets neighboring voltages influence one another. The benchmark uses that simple circuit family to test repeated-block recomposition outside robotics: copy the same excitable-cell block, then rebuild the chain graph for a new cell count. This appendix block gives the protocol behind Table 30; its scope is coupled excitable circuits rather than cardiac-field or bidomain readout modeling.

Single-cell dynamics. We use the standard FitzHugh–Nagumo dynamics

$$\dot{V} = V - \frac{V^3}{3} - W + I, \qquad \dot{W} = \epsilon(V + a - bW),$$

with the benchmark parameters used in the codebase: $\epsilon = 0.08$, $a = 0.7$, $b = 0.8$, and default stimulus $I = 1.0$ for the single-cell generator unless otherwise specified by the coupled-data routine. The state is (V, W), where V is the voltage-like coordinate and W is the recovery variable. C-PHAST represents each unit as one shared excitable-cell pH block; KNOWN fixes the analytical FHN terms, PARTIAL learns the nonlinear constitutive component within the pH scaffold, and LAW-UNKNOWN learns the block with a source-data initialization.

Port-Hamiltonian realization and energy ledger. Taking the (capacitive plus inductive) storage $H(V, W) = \frac{1}{2}V^2 + \frac{1}{2\epsilon}W^2$ (with C=1, L=$1/\epsilon$), the single-cell dynamics above yield the exact power balance

$$\dot{H} \;=\; \underbrace{\left(V^2 - \tfrac{1}{3}V^4\right)}_{\text{tunnel diode } R_1 \text{ (active)}} \;+\; \underbrace{V\,I}_{\text{stimulus port}} \;+\; \underbrace{a\,W}_{\text{source } E=-a} \;-\; \underbrace{b\,W^2}_{R_2 \text{ (passive, } \geq 0)}, \tag{209}$$

which we verified against the simulator to machine precision. The linear resistor $R_2 = b$ (matching the $-bW^2$ term above, with W the inductor current and $L = 1/\epsilon$) is the only passive term; the cubic tunnel diode R_1 contributes $V^2(1 - \frac{1}{3}V^2)$, which is *positive* (energy-injecting) for $|V| < \sqrt{3}$ and negative otherwise. Each cell is therefore an *active* element: it carries an active constitutive port rather than a positive-semidefinite dissipation matrix, and so does not lie in the passive $\dot{x} = (J - R)\nabla H$ class; the stimulus I and source $E = -a$ enter as external ports. Cells couple diffusively through the membrane voltage, whose network power is $\sum_i V_i \sum_j g_{ij}(V_j - V_i) = -\frac{1}{2}\sum_{ij} g_{ij}(V_i - V_j)^2 \leq 0$: this interconnect is the chain incidence wiring A_κ of Section 3.2 with a resistive link of conductance g as the interface, $-gA_\kappa A_\kappa^\top$, so it is passive and *dissipative* rather than power-preserving. The benchmark accordingly tests repeated-block *recomposition*, which is independent of whether the interconnect is skew or dissipative; (209) makes the block's active/passive accounting auditable rather than assumed.

Coupled-chain protocol. For the coupled benchmark, N identical units are connected in a chain with diffusive voltage coupling strength $g = 0.1$. The transfer axis is therefore concrete: the cell equations and coupling law stay fixed, while the number of excitable cells and the chain Laplacian change at deployment. Training uses N=3, 200 trajectories, 200 time steps, and $\Delta t = 0.1$. The coupling enters through explicit Euler half-steps around the local cell step; these runs predate the midpoint coupling update used in the electrical composition study (Section 4.1). The graph port applies the coupling to the membrane voltage V itself, so the dissipation statement above holds for the physical quadratic storage and is not implied for an arbitrary learned storage H_θ. At test time the same shared unit block is copied to $N \in \{2, 3, 4, 5, 6, 8\}$ and the chain Laplacian is rebuilt for the target count. The N-PHDAE baseline is trained as a single-cell model and composed over the same known chain graph. Reported values are means over seeds 42–44 using 100 training epochs.

The LAW-UNKNOWN initialization pools the three cells across the source training trajectories in coordinates $q = V$, $p = W/\epsilon$. Centered ridge regressions with penalty 10^{-4} fit finite-difference $\dot{q}$ from midpoint p, then $\dot{p}$ from midpoint displacement and estimated velocity. These fits initialize scalar mass, damping, and a quadratic potential with a learned residual; port and coupling contributions are not subtracted from the targets. A positive inverse-mass fit is accepted when $R^2 > 0.5$; otherwise, mass uses the ratio of momentum and finite-difference velocity standard deviations. Stiffness and damping are projected to nonnegative values, and scalar initializations are floored at 0.01. The supplied state scaling and chain coupling are unchanged. Target counts reuse the source-trained block, without retaining target-fitted block parameters.

Interpretation. The exact KNOWN C-PHAST rows are reference rows for integration and analytical-port correctness. The learned comparison is more conservative: N-PHDAE is the stronger learned raw forecaster at fixed N=3, while C-PHAST

PARTIAL and LAW-UNKNOWN demonstrate that the shared-block architecture recomposes without a size-dependent error increase. This is why the main text uses FHN as evidence for repeated-block transfer outside mechanics, not as a blanket superiority claim over N-PHDAE.

Table 30. **Coupled FHN: fixed-N hierarchy and repeated-block transfer.** *Top*: fixed $N{=}3$ accuracy across C-PHAST knowledge regimes, with N-PHDAE compose-and-predict as the learned electrical baseline. *Bottom*: C-PHAST transfer from $N{=}3$ to unseen neuron counts by copying the shared excitable-cell block and rebuilding the chain coupling. Mean over three seeds and 100 epochs. Dispersion is not uniformly negligible: at $N = 3$, PARTIAL has one-step MSE $(8.16 \pm 2.88) \times 10^{-6}$ and 50-step MSE $(1.287 \pm 0.361) \times 10^{-3}$ (population standard deviations). Bold entries mark the best value within each table block/metric.

Model ($N{=}3$)	**1-step MSE** ↓	$N_{\text{pred}}{=}10$ ↓	$N_{\text{pred}}{=}50$ ↓
C-PHAST KNOWN ($L{=}4$)	$\mathbf{5.5{\times}10^{-10}}$	$\mathbf{1.9{\times}10^{-8}}$	$\mathbf{3.1{\times}10^{-7}}$
C-PHAST KNOWN ($L{=}1$)	$4.0{\times}10^{-8}$	$9.5{\times}10^{-7}$	$5.8{\times}10^{-6}$
N-PHDAE (composed)	$8.0{\times}10^{-7}$	$1.4{\times}10^{-5}$	$5.8{\times}10^{-5}$
C-PHAST PARTIAL	$8.2{\times}10^{-6}$	$1.0{\times}10^{-3}$	$1.3{\times}10^{-3}$
C-PHAST LAW-UNKNOWN (data-initialized)	$2.0{\times}10^{-3}$	$7.9{\times}10^{-1}$	$1.2{\times}10^{1}$

	Compositional transfer: 1-step MSE ↓					
Model	$N{=}2$	$N{=}3$ (train)	$N{=}4$	$N{=}5$	$N{=}6$	$N{=}8$
C-PHAST PARTIAL	$\mathbf{6.6{\times}10^{-6}}$	$\mathbf{8.2{\times}10^{-6}}$	$\mathbf{7.5{\times}10^{-6}}$	$\mathbf{6.7{\times}10^{-6}}$	$\mathbf{6.7{\times}10^{-6}}$	$\mathbf{6.6{\times}10^{-6}}$
C-PHAST LAW-UNKNOWN (data-initialized)	$2.0{\times}10^{-3}$	$2.0{\times}10^{-3}$	$2.1{\times}10^{-3}$	$1.9{\times}10^{-3}$	$2.1{\times}10^{-3}$	$2.0{\times}10^{-3}$

L. Recovery Decision Protocols

Section 4.5 reports a directional warning sweep and a separate single-decision recovery test. The protocols below distinguish detecting an unsafe continuation from selecting and executing a recovery command. C-PHAST predicts consequences of supplied candidates; an external decision layer chooses what the frozen locomotion policy executes (Figure 2).

L.1. Directional Early Warning

The directional sweep uses the frozen Go2 RSL-RL policy and applies lateral pushes along $\pm x$ and $\pm y$ at five magnitudes per direction. Failure is the simulator-level predicate used for the push filmstrips: fall, support loss, or slip depending on the configured run. C-PHAST rolls the policy continuation forward and emits an unsafe-future signal when the predicted future leaves the calibrated physical envelope. The experiment therefore evaluates warning recall, false warnings, and lead time; it does not evaluate whether a different command recovers the robot.

Table 11 reports the directional results and the matched scalar-alarm comparison. The scalar and learned alarm baselines are calibrated leave-one-direction-out; C-PHAST uses its existing typed-veto signal. The IMU baseline thresholds measured body tilt, while the learned linear baseline uses measured state without candidate rollouts. These alarms are competitive at detecting failure, but they do not return the action-conditioned height, tilt, energy, support, contact, port, and residual forecasts exposed by C-PHAST. Offline diagnostic artifacts additionally expose modal surprise, joint residuals, block residuals, and interconnect power; these support the typed-residual interpretation but are kept out of the main table for readability. Figure 12 shows when the warning appears relative to simulator-level failure.

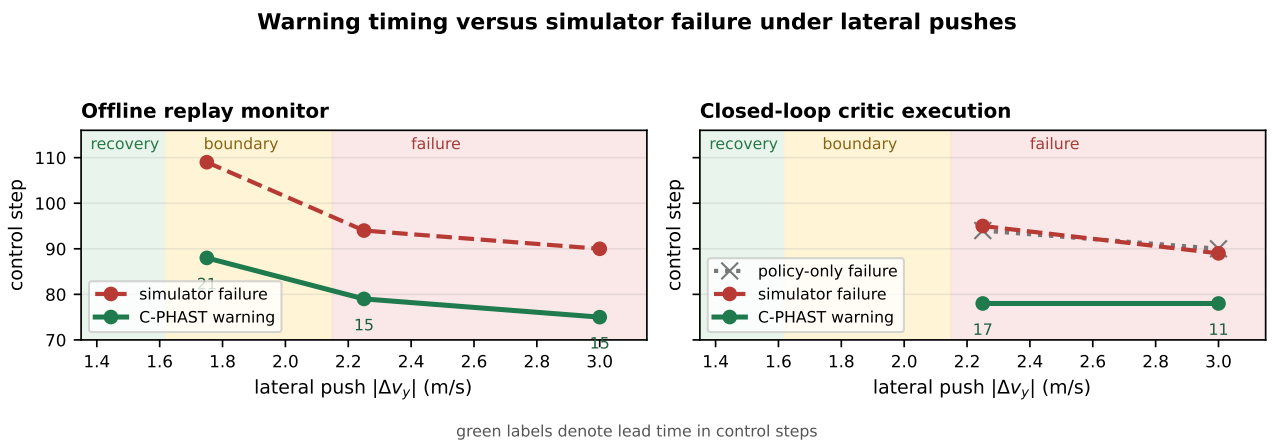


Figure 12. **Warning timing versus simulator-level failure under lateral pushes.** Left: offline replay asks whether C-PHAST consistency warnings appear before failure on logged policy rollouts. Right: online critic execution asks whether the model-predictive critic emits an unsafe-future signal before simulator-level failure. Green numbers denote lead time in control steps. The intended readout is temporal: warnings appear before failure; executed command selection is evaluated separately in Table 12.

L.2. Single-Decision Command Selection

The next experiment asks a different question: after the push has occurred, can C-PHAST's predicted consequences improve one recovery-command choice supplied to the frozen locomotion policy? The task is `Isaac-Velocity-Flat-Unitree-Go2-v0`. A body-velocity change of $\Delta v_y = \pm 2.75\,\mathrm{m/s}$ is applied at control step 75, and the command decision is made at step 76. Normal terrain uses static/dynamic friction $(1.0, 1.0)$; declared low-friction terrain uses $(0.25, 0.15)$.

At decision time, C-PHAST receives current root pose and velocity, joint position and velocity, contact and slip geometry, the declared terrain material, and the supplied command candidates. It does not receive the push label, push magnitude, or future simulator state. The candidate library has nine labels: nominal, hold, forward, backward, left, right, turn-left, turn-right, and default stance. Writing policy velocity commands as (v_x, v_y, ω_z), their values are nominal $(0.6, 0, 0)$, hold $(0, 0, 0)$, forward $(0.6, 0, 0)$, backward $(-0.6, 0, 0)$, left $(0, 0.6, 0)$, right $(0, -0.6, 0)$, turn-left $(0, 0, 0.75)$, and turn-right $(0, 0, -0.75)$. Default stance is a raw zero low-level action rather than a zero velocity command passed through the policy. Thus nominal and forward are distinct labels but share the same numerical velocity command in this experiment; hold and default stance are operationally different.

Single-decision predictor. The frozen runtime has one six-coordinate body block and four shared three-coordinate leg blocks. Its 36-dimensional state contains relative body translation, roll/pitch/yaw, joint positions, and their linear/angular or joint velocities; the velocity entries occupy latent momentum slots, without an independent physical-momentum calibration. Local potentials, mass and damping operators, and masked skew coupling are learned. A rank-six positive-semidefinite support correction conditions the response on 26 support/terrain features. It is a learned damping correction, not a contact solver.

The candidate velocity command is held while the frozen locomotion policy is reevaluated on predicted observations at each forecast step. The observation reconstruction retains the initial reconstruction residual and updates the previous-action channels. The resulting twelve policy actions enter the joint input slots, with six zero body inputs; default stance substitutes zero actions. Measured support and terrain context remain fixed. This command-conditioned input is not converted to measured actuator torque: its port-work readout is a model-coordinate feature, not calibrated work in joules.

The supplied checkpoint's training record contains 474 eligible transitions, split into 403 training and 71 validation transitions, and 80 training epochs. The documented source data comprise positive lateral pushes of 1.50, 2.25, and 3.00 m/s on two friction settings. The trainer fits standardized state prediction with optional multistep and support-response penalties; the retained checkpoint does not resolve their historical weights. Reproduction of the reported decision evaluation therefore uses the fixed released checkpoint, rather than assuming an exactly specified retraining recipe.

C-PHAST forecasts each candidate for eight model steps from the same measured state. The selected command is held

for 24 control steps at $0.02\,\mathrm{s}$ per step ($0.48\,\mathrm{s}$), after which the nominal command resumes. During execution, the policy responds to live observations with fixed parameters; default stance instead supplies raw zero actions during the 24-step interval. The command is selected once, without further replanning.

The survival score counts control steps completed without triggering the failure criterion, stopping at the first failure. A run with no failure receives the maximum score of 140. The 140-step evaluation includes an initial 20-step settling period during which failure checks are inactive. After settling, failure is checked after each control step: base height below $0.18\,\mathrm{m}$, absolute roll or pitch above $0.85\,\mathrm{rad}$, or simulator termination or truncation. Slip and support loss do not trigger failure in this experiment. Recovery means completing all 140 steps without triggering this failure criterion. Return to the nominal gait is not separately measured.

The development split contains seeds 58–59, both push signs, and both terrain settings: eight scenarios and $8 \times 9 = 72$ forced command branches. The held-out split contains seeds 60–64 under the same sign/terrain crossing: 20 scenarios and $20 \times 9 = 180$ forced branches. No held-out simulator outcome is used to fit the selector. Both selector variants use the same regularized pairwise head family, candidate identity, and current measured state. The reduced variant receives predicted motion/action summaries and maximum predicted energy growth; it is not strictly kinematics-only. The full variant adds 17 readouts: mean/maximum absolute energy-ledger residual; signed/absolute port-work and dissipation sums; total energy change; maximum support-response norm; and nine modal summaries (absolute modal-power sum/maximum, modal-floor sum, maximum damping coefficient and eigenvalue, minimum eigengap, inverse-gap sum, near-tie count, and feature count). These are model readouts, not extra future measurements. Both heads use pairwise logistic fitting with regularization strength 1.0. This external learned selector differs from the fixed weighted score in Eq. (20).

Table 12 reports two evaluations that should not be conflated. Panel A independently executes the command selected online by the frozen full head. Its mean survival score is 121.80 steps. Among fixed command choices for the 24-step interval, right has the highest mean score on the same 20 test scenarios, 116.55, giving a paired gain of 5.25 steps. To estimate uncertainty, we resample the five seeds with replacement, keeping each seed's four paired sign/terrain scenarios together. The trained selector and the right-command comparator remain unchanged across resamples. Enumerating all 5^5 resamples gives a 95% percentile bootstrap interval $[2.50, 7.05]$ steps. This interval does not include uncertainty from fitting the selector or choosing the comparator. Every online selected outcome matches the corresponding independently forced branch outcome ($20/20$), which verifies the execution ledger. Panel B ranks the exhaustive held-out branches: adding the pH-typed readouts increases longest-survival selection from $12/20$ to $14/20$ and reduces mean survival shortfall from 10.8 to 7.7 steps.

Candidate-library limitation. An earlier online library of policy continuation, held or scaled actions, and simple posture sequences did not improve recovery despite warning of impending failure; the supplied alternatives did not provide a recovery future.

M. Real-Robot Logs and Physical-Consequence Prediction

Section 4.6 tests the consequence interface on real Spot-with-arm telemetry. Unlike the simulator branches in Appendix L, each logged window contains only the future that was actually executed. We therefore specify the replay input contract before the prediction and objective-conditioned ranking results.

The Spot-with-arm evidence uses logged ROS bags collected in a lab environment under human teleoperation. C-PHAST is evaluated as a shadow model: it observes telemetry and possible-next-behavior descriptors, predicts typed physical consequences, and writes objective-conditioned action cards beside the teleoperated run. The data are converted into overlapping windows with a two-second history and a five-second future horizon. Each row supervises the future that was actually executed in the log. Thus Spot serves as the real-robot telemetry-to-action-card forecast: the log supplies physical supervision for the executed future, while branch execution and matched-state alternative rollouts are tested by the Isaac experiments. Table 31 gives the scale of this logged dataset and the history/future windows used by the reported result. Training episodes comprise static arm scanning (105 windows), fast traversal (130), slow traversal (258), and waypoint scanning (404). Testing uses a separate fast traversal (102) and trimmed yaw scan (71).

Table 31. **Spot-with-arm window dataset.** Four complete bags supply training windows and two separate bags supply test windows, after filtering unusable or calibration-test segments. No bag crosses the split.

Quantity	Value
usable teleoperated episodes	6
typed possible-next-behavior windows	1,070
training windows	897
held-out windows	173
training / held-out bags	4 / 2
history length	2 s
future-consequence horizon	5 s
window stride	1 s
reported query inputs	history + numeric behavior descriptors

Table 32 distinguishes this real-log study from the cross-robot and recovery studies in Appendices J and L. The Spot row is real-robot evidence for telemetry-to-action-card prediction under logged replay; the Isaac transfer and push rows carry the branchable PHASTCore rollout evidence.

Table 32. **Benchmark roles and hierarchy for the shared query interface.** Each artifact stresses a different part of the C-PHAST claim: simulated branch execution, typed physical-critic scoring, or real robot telemetry-to-action-card forecasting.

Experiment	Query mode	Hierarchy used	Real robot	Validated role
Isaac transfer	branchable q-only rollout	four 3-DOF leg PHASTCore rollout blocks (HAA/HFE/KFE); floating base excluded	no	cross-robot rollout accuracy and pH/compositional transfer
Push critic	full-state rollout plus critic	separately trained body-plus-legs C-PHAST runtime; root/joint telemetry, contact/slip geometry, and terrain context condition candidate forecasts	no	typed branch scoring, early warning, and single-decision command selection in simulation
Spot shadow mode	logged replay windows	interface blocks for base, leg support/contact, arm, camera/depth, and battery/power; target hierarchy for follow-up closed-loop hardware rollout	yes	real telemetry-to-action-card forecasting

Consequence predictors. The Spot hierarchy organizes telemetry and outputs, not compositional dynamics. Two direct predictors instantiate this interface. For objective-score evaluation, multioutput ridge regression maps 28 history summaries and ten numeric behavior descriptors to 38 consequence channels. Inputs and targets are standardized; regularized least squares uses strength 1.0 and an unpenalized intercept. Predicted channels are then aggregated into objective scores, rather than learning a different physical predictor for every objective. Panel B of Table 13 uses ridge regression throughout: its first four rows share the four-bag/two-bag split, including the comparator that predicts objective scores directly. The final row refits the consequence predictor on five bags and evaluates the sixth, recomputing standardization and factor scales within each training fold. Correlations are averaged equally over the six bags and five objectives.

The energetic-channel evaluation uses a separate neural ensemble. Two-layer history and descriptor encoders use SiLU activations and layer normalization; a joint multilayer perceptron receives both embeddings and their elementwise product. Sigmoid outputs enforce bounded channels, while softplus enforces nonnegative channels. Training minimizes weighted mean-squared error on scaled consequence targets, with seven channels weighted by 1.5; predictions are averaged over seeds 42–46. The archived launch and retained producer defaults specify width 32, AdamW with learning rate 0.002 and weight decay 0.001, batch size 128, and at most 700 epochs. Early stopping uses 20% of training-partition windows and patience 80; these overlapping validation windows are not independent episodes. Model checkpoints, resolved run settings, and stopping epochs were not retained with the exported ensemble predictions, so exact retraining is unverified. Neither predictor uses a PHASTCore transition or transferred dynamics weights.

The main subtlety is the Spot input contract. C-PHAST is not allowed to solve the task by memorizing primitive names: the reported model receives recent telemetry plus numeric behavior descriptors, not a one-hot primitive label. Table 33 separates four quantities that are easy to conflate: history observed before the query, behavior descriptors that a planner could provide, replay-only descriptor fields measured from the logged future, and consequence targets that must be predicted.

Table 33. **Spot input-contract audit.** The held-out-bag result tests the replay version of the planner-facing interface. A deployed confirm-to-execute planner would supply behavior descriptors from intended primitive parameters rather than measured future telemetry.

Quantity class	Examples in the Spot artifact	Use in this paper
Decision-time history	recent speed/contact, hand motion, gripper state, image/depth quality, TF/depth novelty summaries	observed before the query; valid model input
Command-like behavior descriptors	commanded base speed and turn rate; gripper opening; primitive prototype descriptor	intended planner inputs; logged values are used as replay proxies
Future-derived behavior descriptors	realized yaw change, realized hand speed, realized hand turn over the five-second future window	used only in logged replay; must be supplied by a primitive library or forecasted before live use
Predicted consequence targets	work, support/contact, battery draw, collision/contact proxy, sensing quality, depth novelty, residual-like terms	unavailable before execution; predicted by the typed consequence model
Visualization / metadata	behavior labels, episode type, trim markers	used for analysis and plots; not part of the main reported model input

This contract gives two related query types. The held-out objective-score task is logged-window consequence prediction: each dataset row uses the executed teleoperation future as supervision. The same-history action-card query is a model query: it fixes one observed history and swaps in four prototype behavior descriptors, namely fast base loop, slow base loop, move-and-scan, and static arm scan, estimated from the training episodes. That query demonstrates objective switching from one state-history context; Isaac experiments provide matched-state branch execution when actual alternatives must be run.

Consequences and objective scores. For the objective-score experiment, the regression predictor estimates consequence channels from history summaries and behavior descriptors. Table 34 gives their definitions and decision roles. The predicted channels are aggregated and scaled to form a factor vector, reused for five objectives: efficient motion, safe motion, mapping, balanced operation, and inspection. This is the separation illustrated in Figure 3, with a direct consequence predictor rather than a latent dynamics rollout.

Table 34. **Spot observable-to-factor examples.** These definitions specify telemetry-derived targets and operational proxies for the direct consequence predictor; the objective scores use its predictions. Objective scores scale each planner-facing factor by its training-split 10–90 percentile range before applying objective weights.

Factor family	Consequence definition	Decision role
Progress	future base path length in the horizontal plane	reward for traversal
Joint work	$\sum_t \sum_j \max(\tau_{j,t}\dot{q}_{j,t}, 0)\Delta t$; absolute work replaces each positive part by $\lvert\tau_{j,t}\dot{q}_{j,t}\rvert$; both in joules	physical cost
Battery draw	$\max(-\overline{I}_{\text{battery}}, 0)$, where the bar averages the summed battery current over the future horizon, in amperes	discharge-current cost
Support/contact	support-loss fraction, full-support fraction, contact-count deficit, foot/contact proxies	safety envelope and warning
Wrench/load	mean force and torque norms from available wrench or effort channels	contact/load cost
Collision/contact residual	hand or contact-envelope violation fraction; residual-like contact mismatch	veto or warning
Motion/aggression	speed, yaw motion, commanded speed and turn magnitude	blur risk, support risk, and progress estimate
Sensing quality	image frame/camera counts, sharpness, contrast, brightness, entropy, edge density	inspection quality
Depth/TF novelty	valid depth fraction, point count, observed/new voxel counts, novelty fraction, coverage rate	map gain

The reported Spot objectives apply soft weights to these factors (Table 35). A hard veto would require an explicit admission rule, rather than merely a larger penalty weight.

The five objectives used in the Spot result are fixed linear consumers of the planner-factor vector. Efficient motion rewards progress while penalizing work, battery draw, command aggression, support loss, and collision. Safe motion places stronger penalties on support loss, collision, force, and aggressive commands. Mapping rewards progress, yaw/hand motion, image quality, and depth novelty while penalizing speed, work, battery draw, support loss, and collision. Balanced operation trades progress, sensing motion, energy, support, force, and collision. Inspection emphasizes image and depth quality, stable viewpoint changes, hand motion, and hand turn, with penalties for speed, near-depth risk, support loss, collision, and work. Changing objectives therefore changes the weights applied to the same typed factor vector rather than retraining the dynamics or consequence model.

Table 35. **Objective weights for Spot action cards.** After quantile scaling, each objective is a transparent weighting of typed planner factors. Positive entries reward a predicted consequence, while negative entries penalize cost or risk.

Objective	Rewards	Penalties / veto pressure	Intended action card
Efficient	progress $+1.0$	support loss -1.0, collision -1.0, command aggression -0.8, work -0.5, energy/progress -0.4, battery draw -0.3	move if progress is worth the physical cost
Safe	progress $+0.2$	support loss -2.0, collision -1.5, force -0.6, command aggression -0.5	stay inside the support/contact envelope
Mapping	progress $+0.7$; yaw/hand motion $+0.6/+0.8$; image/depth novelty and quality $+0.3$–$+0.6$	support loss -1.0, collision -0.8, speed -0.5, work/battery -0.3	prefer informative viewpoints without unsafe motion
Balanced	progress $+0.7$; yaw/hand motion $+0.3/+0.4$	support loss -1.2, collision -0.8, force -0.6, work -0.4, energy/progress -0.25	trade progress, sensing, and physical risk
Inspection	image/depth quality $+0.4$–$+0.9$; hand motion $+1.0$; yaw motion $+0.8$; progress $+0.25$	support loss -1.2, collision -1.0, speed -0.8, near-depth risk -0.2, work -0.15	slow local scan with high visual/depth fidelity

Table 13 gives the compact real-robot evidence ledger in the main text. The appendix expands that ledger in five steps: Table 36 reports direct energetic-target agreement, Table 37 reports the held-out objective-score split, Table 38 tests leave-one-episode-out generalization, Table 39 checks which query inputs carry signal, and Table 40 compares typed readouts against direct scalar-objective training.

Table 36. **Direct energetic-target agreement on 173 held-out Spot windows.** Targets are computed from logged joint torque/velocity or battery telemetry and predicted by the five-seed typed query-model probe. They are telemetry-grounded physical quantities, not additional terms in the latent pH power identity.

Target	Pearson	R^2
Positive joint work	0.973	0.944
Absolute joint work	0.975	0.947
Locomotion positive work	0.971	0.939
Battery draw	0.943	0.878

Positive-work Pearson correlation is 0.975 on the 102 traversal windows and 0.069 on the 71 yaw-scan windows. Their RMSEs are similar (28.86 and 27.55 J), but the yaw-scan targets vary much less; pooled agreement therefore does not establish strong within-episode tracking in both behaviors.

Table 37. **Held-out objective-score agreement on Spot logs.** All rows use recent history plus numeric behavior descriptors without primitive identity labels. Brackets give 2.5th–97.5th percentiles from 5,000 bootstrap resamples of individual prediction–target pairs, with the model and normalization fixed. Because windows overlap, these intervals describe variation within the two test episodes, not 95% coverage for new-episode performance. Highest correlations are bold green; MAE scales differ across objectives.

Objective	Pearson	Spearman	MAE
Efficient	0.744 [0.682, 0.803]	0.753 [0.681, 0.811]	0.213
Safe	0.907 [0.873, 0.935]	0.875 [0.833, 0.903]	0.106
Mapping	0.919 [0.894, 0.939]	0.900 [0.861, 0.926]	0.370
Balanced	**0.948 [0.931, 0.962]**	**0.939 [0.901, 0.961]**	0.140
Inspection	0.884 [0.846, 0.914]	0.836 [0.767, 0.886]	0.540

Table 38. **Leave-one-episode-out Spot stress test.** Each of six folds trains on five episodes and holds out the sixth; metrics are then averaged equally across episodes. Unlike the pooled fixed-split result, every episode is evaluated out of sample. Both protocols hold out complete episodes, but differ in coverage and aggregation. Highest correlations are bold green; MAE scales differ across objectives.

Objective	Pearson	Spearman	MAE
Efficient	0.538	0.412	0.324
Safe	0.641	0.590	0.382
Mapping	**0.750**	**0.617**	0.318
Balanced	0.644	0.517	0.317
Inspection	0.686	0.590	0.473
Mean	0.652	0.545	0.363

Table 39. **Query-input ablation.** Both current history and behavior descriptors carry signal; primitive identity alone does not. Best per metric is shown in **bold green**.

Inputs	Mean Pearson	Mean Spearman	Mean MAE
history only	0.767	0.737	0.360
behavior descriptors only	0.780	0.778	0.334
history + behavior descriptors	**0.880**	**0.860**	**0.274**
primitive label only	-0.120	-0.054	0.740
history + descriptors + label	0.854	0.844	0.313

Table 40. **Typed readouts versus direct scalar objectives.** Typed consequences preserve scalar ranking quality while exposing the reason for the ranking. Best per metric is shown in **bold green**.

Training target	Mean Pearson	Mean Spearman	Mean MAE
typed consequences then score	**0.880**	0.860	**0.274**
direct scalar objective	0.871	**0.875**	0.330

For a fixed observed history, the ranked future changes when the objective changes. In the representative query in Figure 7, efficient and safe objectives choose a slow-loop candidate, while mapping, balanced, and inspection objectives choose a move-and-scan candidate. This is the operational reason to expose typed factors rather than a single learned score: a planner or operator can change the task objective while reusing the same consequence predictions.

N. From the Control Objective to the Physical Rollout

The main text uses two Hamiltonians for two different purposes. $\mathcal{H}_{\text{ctrl}}$ expresses the task objective together with the dynamics constraint. The physical H generates part of those dynamics. This appendix supplies the intermediate steps and then connects the learned vector field f_θ to the numerical transition $F_{\theta,\Delta t}$.

N.1. The objective and its costate

Fix one configuration κ and suppress its superscript in f. Here u denotes only adjustable inputs; prescribed loads are known forcing held fixed throughout the variation and suppressed from the arguments of f. Consider Equation (1) with a fixed horizon T, a fixed initial state, and a free terminal state. The differentiable functions ℓ_w and Ψ_G specify running and terminal costs. For this derivation there are no active state or endpoint constraints; inputs are initially taken in the interior of their admissible set. The normal minimization convention uses a positive unit multiplier on the objective.

Introduce the time-dependent costate $\lambda(t)$ and add the dynamics constraint to the objective. With the augmented action Lagrangian from Equation (2),

$$\begin{aligned}\mathcal{I}_{\text{ctrl}}[x,u,\lambda] &= \Psi_G(x(T)) + \int_0^T \mathcal{L}_{\text{aug}}(x,\dot{x},u,\lambda)\,dt \\ &= \Psi_G(x(T)) + \int_0^T \left[\ell_w(x,u) + \lambda^\top(f(x,u)-\dot{x})\right] dt \\ &= \Psi_G(x(T)) + \int_0^T \left[\mathcal{H}_{\text{ctrl}}(x,\lambda,u) - \lambda^\top\dot{x}\right] dt.\end{aligned} \tag{210}$$

On a feasible trajectory the added constraint term vanishes, leaving the original task cost. Varying λ therefore recovers $\dot{x} = f$. Varying all three functions gives

$$\begin{aligned}\delta\mathcal{I}_{\text{ctrl}} = {}& \nabla\Psi_G(x(T))^\top\delta x(T) \\ &+ \int_0^T \left[(\partial_x\mathcal{H}_{\text{ctrl}})^\top\delta x + (\partial_u\mathcal{H}_{\text{ctrl}})^\top\delta u + (f-\dot{x})^\top\delta\lambda - \lambda^\top\delta\dot{x}\right]dt.\end{aligned} \tag{211}$$

The last term contains the derivative of the state variation. Integration by parts transfers that derivative to the costate:

$$-\int_0^T \lambda^\top\delta\dot{x}\,dt = -\left[\lambda^\top\delta x\right]_0^T + \int_0^T \dot{\lambda}^\top\delta x\,dt.$$

The initial boundary term is zero because $\delta x(0) = 0$. Using $\partial_\lambda\mathcal{H}_{\text{ctrl}} = f$ now gives

$$\begin{aligned}\delta\mathcal{I}_{\text{ctrl}} = {}& \left[\nabla\Psi_G(x(T)) - \lambda(T)\right]^\top\delta x(T) \\ &+ \int_0^T \left[(\partial_x\mathcal{H}_{\text{ctrl}} + \dot{\lambda})^\top\delta x + (\partial_\lambda\mathcal{H}_{\text{ctrl}} - \dot{x})^\top\delta\lambda + (\partial_u\mathcal{H}_{\text{ctrl}})^\top\delta u\right]dt.\end{aligned} \tag{212}$$

Independent interior variations yield the state and costate equations in Equation (3) and $\partial_u\mathcal{H}_{\text{ctrl}} = 0$. The free terminal variation yields $\lambda(T) = \nabla\Psi_G(x(T))$: the terminal objective determines the costate's terminal value, rather than another physical boundary law. If inputs are restricted to $\mathcal{U}$, the stronger Pontryagin condition (Pontryagin et al., 1962; Liberzon, 2012) is

$$u^*(t) \in \arg\min_{v\in\mathcal{U}} \mathcal{H}_{\text{ctrl}}(x^*(t),\lambda^*(t),v) \quad \text{for almost every } t.$$

The interior first-variation calculation alone does not establish this global-in-input condition. Active state or terminal constraints require their corresponding multipliers and boundary conditions; the equations above are the smooth unconstrained-state case (Tedrake, 2024; Liberzon, 2012). The candidate feasibility tests in Appendix A.8 remain separate. No optimal-control solve is added to Algorithm 1.

N.2. The Legendre–Fenchel form of the control Hamiltonian

The control u need not equal the state rate $\dot{x}$. To take a transform in the state rate, first ask for the least running cost of producing a given rate v at state x:

$$L_{\text{ctrl}}(x,v) = \inf_{\substack{u\in\mathcal{U}\\ f(x,u)=v}} \ell_w(x,u). \tag{213}$$

An unattainable rate has cost $+\infty$. For fixed x, take $\mathcal{U}$ nonempty and $\ell_w(x,\cdot)$ finite and bounded below on $\mathcal{U}$. The Legendre–Fenchel conjugate in v then pairs the rate with a dual variable η (Liberzon, 2012):

$$\begin{aligned} L^*_{\mathrm{ctrl}}(x,\eta) &= \sup_{v}\big[\eta^\top v - L_{\mathrm{ctrl}}(x,v)\big] \\ &= \sup_{u\in\mathcal{U}}\big[\eta^\top f(x,u) - \ell_w(x,u)\big]. \end{aligned} \tag{214}$$

The second line follows by grouping controls according to the rate they produce: at each attainable v, subtracting the least cost is the same as maximizing $\eta^\top v - \ell_w$ over those controls.

This is the maximized control Hamiltonian. The main text instead keeps u as an argument and uses the minimum convention $\mathcal{H}_{\mathrm{ctrl}} = \ell_w + \lambda^\top f$. Their exact relation is

$$\boxed{\inf_{u\in\mathcal{U}} \mathcal{H}_{\mathrm{ctrl}}(x,\lambda,u) = -L^*_{\mathrm{ctrl}}(x,-\lambda).} \tag{215}$$

Thus $\eta = -\lambda$. Under this change of sign, the augmented action integrand becomes

$$\mathcal{L}_{\mathrm{aug}} = \eta^\top \dot{x} - \big[\eta^\top f(x,u) - \ell_w(x,u)\big].$$

The conjugacy is between the effective rate cost L_{ctrl} and the maximized Hamiltonian, not between the multiplier-augmented integrand and an unconstrained velocity transform. It requires neither an invertible map $u \mapsto f(x,u)$ nor an attained supremum; values can be extended-real. The variable η is dual to the full state rate, not mechanical momentum.

N.3. The physical Hamiltonian inside the dynamics

Now set aside the task cost and consider a conservative mechanical component. Let q be position, $v = \dot{q}$ velocity, and $K(q,v) = \frac{1}{2} v^\top M(q) v$ kinetic energy, with $M(q) \succ 0$. The mechanical Lagrangian is $L_{\mathrm{plant}}(q,v) = K(q,v) - V(q)$. Stationarity of the action $\mathcal{S}_{\mathrm{plant}}[q] = \int_0^T L_{\mathrm{plant}}(q,\dot{q})\,dt$ with fixed endpoint positions gives the Euler–Lagrange equation

$$\frac{d}{dt}\partial_v L_{\mathrm{plant}}(q,\dot{q}) = \partial_q L_{\mathrm{plant}}(q,\dot{q}).$$

Define the physical momentum $p = \partial_v L_{\mathrm{plant}} = M(q)v$. For fixed q, positive definiteness of M makes the Lagrangian strictly convex in v. Its Legendre–Fenchel supremum is attained uniquely at $v = M(q)^{-1}p$ (Liberzon, 2012):

$$\begin{aligned} H(q,p) &= \sup_{v}\big[p^\top v - L_{\mathrm{plant}}(q,v)\big] \\ &= \big[p^\top v - L_{\mathrm{plant}}(q,v)\big]_{v=M(q)^{-1}p} \\ &= \tfrac{1}{2} p^\top M(q)^{-1} p + V(q). \end{aligned} \tag{216}$$

Differentiating $p^\top v - L_{\mathrm{plant}}$ at the maximizing velocity cancels the velocity differentials because $\partial_v L_{\mathrm{plant}} = p$:

$$dH = v^\top dp - (\partial_q L_{\mathrm{plant}})^\top dq.$$

Thus $\partial_p H = v = \dot{q}$ and $\partial_q H = -\partial_q L_{\mathrm{plant}} = -\dot{p}$. In the state $x = (q,p)$,

$$\dot{x} = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \nabla H(x). \tag{217}$$

Here $H = K + V$ is stored energy, whereas the earlier ℓ_w is a chosen task cost. Mechanical stationary action is not minimization of that task cost. Adding dissipation $R \succeq 0$ and a collocated input port gives

$$f_\theta(x,u) = (J-R)\nabla H_\theta(x) + Bu, \qquad y^{\mathrm{port}} = B^\top \nabla H_\theta.$$

Substituting this model into the control Hamiltonian makes the distinction explicit:

$$\boxed{\mathcal{H}_{\mathrm{ctrl},\theta}(x,\lambda,u) = \ell_w(x,u) + \lambda^\top \underbrace{\big[(J-R)\nabla H_\theta(x) + Bu\big]}_{f_\theta(x,u)}.} \tag{218}$$

The physical momentum p is part of the mechanical state x; the costate λ has one entry for each state coordinate and belongs to the optimization problem. For a network, replace the component field above by the assembled field in Equation (11).

N.4. From state rates to predicted consequences

Fix a configuration and hold u constant over one step. The exact evolution of the learned continuous model satisfies

$$\dot{z}(\tau) = f_\theta^{(\mathcal{S}_\kappa)}(z(\tau), u), \quad z(0) = x, \quad z(\Delta t) = x + \int_0^{\Delta t} f_\theta^{(\mathcal{S}_\kappa)}(z(\tau), u)\, d\tau. \tag{219}$$

The numerical map $F_{\theta,\Delta t}^{(\mathcal{S}_\kappa)}(x, u)$ approximates $z(\Delta t)$. It is neither the instantaneous rate f_θ nor a second independently learned predictor. In $F = \mathrm{Compose}_\theta(\mathcal{S}_\kappa, \Delta t)$, `Compose` selects and connects the fixed component laws and constructs their declared numerical update. Calling F then advances a state. The mechanical and electrical schemes are specified in Appendices A.2 and A.4; the notation does not replace either with a new integrator.

Repeated steps give $\hat{x}_{0:N_{\mathrm{pred}}}(a)$. The readout converts successive states and inputs into $\rho_{1:N_{\mathrm{pred}}}(a)$, including instantaneous powers and interval work and loss. Equation (20) then evaluates the terminal goal and weighted consequence costs. For costs built from instantaneous state and input channels, the running sum is a quadrature approximation to the integral in Equation (1). Interval work or loss is already an integrated quantity; when used as a cost its normalization must account for the outer Δt in the score (for example, using interval loss divided by Δt as an average loss power). The declared consequence factors, rather than an assumed identification of energy with cost, specify the task.

Changing the weights w can rescore the same predicted futures. Changing the configuration κ assembles a different field and transition from the same component laws. Changing the numerical step size changes the approximation of those laws, not the laws themselves.